PART 6

OSCE: 58 practical stations

Each station gives the candidate instruction, the direction of thought, a full spoken model answer and the examiner's checklist.

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PART 6 — OSCE: 58 PRACTICAL & PROCEDURAL STATIONS

Format reminder: 10 stations, **2 minutes reading + 5 minutes inside**, 10 marks each, checklist-marked, equal weighting, examiner silent unless the task requires interaction.

How to work a 5-minute station. Say what you are doing as you do it. Work in a fixed order — *indication → preparation → procedure → completion → what I would record and warn about*. If you cannot complete the physical task, keep describing. If you finish early, stay and add the post-operative advice, the records, and the complications — those are almost always unclaimed checklist marks.

Read "Where the marks are" below before you start. The 58 stations that follow cover every learning outcome that Appendix A marks in the OS-P column, together with the practical skills listed in the FGDP training standards. They are deliberately weighted towards restorative and prosthodontic procedure, temporisation and communication, because that is where the blueprint places most of its procedural outcomes and where candidates most often under-prepare.

Each station below gives: the RCSEd-style station brief, the candidate instruction sheet, your **direction of thought** (what to do in the 2 minutes of reading), a **model answer**, a **10-point checklist** as an examiner would use it, and the **common pitfalls**.

Where the marks are in the OSCE

The OSCE is the component candidates most often misjudge. It is easy to assume that a *practical* examination in implant dentistry must be an examination in *surgery*. Read Appendix A carefully and it is clear that this is not so: the OS-P column is spread across assessment, radiographic interpretation, prosthodontic procedure, temporisation, communication with the dental team, post-operative care and the management of complications, and the FGDP training standards list the same range. Six points follow, and they should shape the way you prepare.

1. The circuit is weighted towards restorative and prosthodontic procedure, not surgery. Far more of the procedural learning outcomes concern impressions and scans, framework and crown fit, retention decisions, occlusal adjustment, temporisation and the handling of an existing restoration than concern flaps, osteotomies and sutures. Know the surgical stations — flap design, site preparation, suturing and minor augmentation all sit in the blueprint and can appear — but do not let them consume your revision. **If you can describe an osteotomy sequence fluently but cannot explain what you would do when a metal framework does not seat, you have prepared the wrong half of this component.**

2. Several stations are conversations, not silent tasks. The published sample station in Appendix D states that no interaction with the examiner is required, and that is true of some stations — but not of all. A station may ask you to **discuss** management, or to talk to a role-player representing a **dental technician**, a colleague or a patient. Prepare to talk *to a member of the team*, not only to perform in front of an observer. Rehearse saying what you want from the laboratory in the words you would actually use on a prescription.

2a. Expect to be handed the findings and questioned on them, rather than asked to obtain them. This is the single most common way a "clinical skill" station is actually run, and candidates prepare for the wrong version of it. You may be shown a **completed BPE chart, a set of probing depths, a radiographic report or a laboratory work ticket** and then asked, in quick succession: what does this tell you, what does each figure mean, what does it oblige you to do, what instrument produced it, describe that instrument, what force did you use, how do you know it was the right force. **Everything you would ordinarily do with your hands then has to be said accurately out loud** — the millimetre markings on a probe, the diameter of its ball tip, the force in grams or newtons, the calibration trick you use to judge it, the sequence of sites. That is a *harder* test than performing the task, not an easier one, because vague description is immediately visible where competent hands are not. For every instrument in your working week, be able to say what it is called, what is marked on it, how hard you press, and why.

3. Information may be released during a station, not all at the start. A station may hand you an additional piece of written information partway through — a radiographic finding, a result, a change in the history. This is deliberate, and it tests whether you can revise a differential when new data arrives. When it happens, **say out loud what it changes**: *"That changes my differential — with no bone loss, this is far more likely to be residual cement, a foreign body or a problem with the adjacent tooth than peri-implantitis."* Silence loses the mark; narration earns it.

4. Basic clinical skills are examined, and they are where marks are quietly lost. A station built on something you have done thousands of times — a BPE, a six-point chart, oral hygiene instruction, post-operative instructions — is not a gift. It is a station where **full marks are available and expected**, and where candidates lose them by working fast, silently and imprecisely. The examiner is marking a checklist that includes the instrument, the force, the sequence, the codes and what each finding obliges you to do next. Slow down and say the numbers.

5. Temporisation is a named topic and is chronically under-revised. The interim prosthesis over a graft, the provisional that shapes an emergence profile, and the thing the patient wears for the nine months in between all sit squarely in the blueprint. Be able to describe, compare and justify a **resin-bonded bridge, a spoon denture, an Essix retainer, an immediate partial denture and a screw-retained provisional** — indications, advantages, disadvantages, and the effect of each on a healing site.

6. The stations are drawn from ordinary practice, not from the literature. Everything in this component is something a general dentist with an interest in implants does, or should do, in a normal working week. That is the level. **If your answer needs a systematic review to justify it, you are pitching too high; if it does not include what you would record and what you would tell the patient, you are pitching too low.**

> KEY POINT
>
> **Revision priority for the OSCE, in order:** temporisation and provisionals → prosthetic retention decisions and communication with the technician → framework and restoration fit and how you verify it → post-operative care → periodontal and peri-implant assessment → radiographic interpretation and its effect on the replacement plan → complications of existing implant restorations → then the surgical stations.

How to work a station

Minute	What to do
Reading time	Read the instruction **twice**. Identify the verb — *carry out, describe, discuss, explain to the patient*. That verb tells you who you are talking to and what will be marked.
First 30 seconds	State your framework aloud before you start doing anything: *"I will check the medical history and consent, then work through X, Y and Z, and finish with what I would record and review."*
Body of the station	Narrate everything. Say the instrument, the force, the material, the reason. An examiner cannot award a mark for something you did silently.
If new information is given	Stop, read it, and say explicitly what it changes.
Last 30 seconds	Close with **records, patient instructions and review** — three marks that sit at the end of almost every checklist and are routinely left unclaimed when candidates run out of time.

How the five minutes is actually spent, and how to use the run sheets

The station model answers in this part are written to be **complete**, so that you can revise from them. They are not written to be **said** — several of them run to seven or eight minutes of unbroken speech, and you have five. That gap is where marks are lost, and closing it is a separate skill from knowing the material.

So **every station in this part carries its own run sheet**, printed immediately after the model answer and the checklist. Each one has the same four parts:

Running this station to time. How long the model answer actually takes at speaking pace, how much of it has to go, and — specifically for that station — how it goes.

The five-minute run sheet. A minute-by-minute table: what the examiner is doing in each block, and which checklist points must have been spoken by the time that block ends.

The compressed answer. What a strong candidate actually delivers inside the five minutes — the same content as the model answer, in about a third of the words, ordered for the clock. This is the artefact you rehearse.

Marking priority when the clock beats the candidate. The six points that carry the station, the four an examiner will accept late or on prompting, and the commonest way five minutes is wasted at that particular station.

KEY POINT

Use these as rehearsal scripts, not as reading. Set a timer for five minutes, work the compressed answer out loud against the clock, and stop dead when the timer goes. Then look at what you had not yet said. In almost every case it will be the last row of the table — **records, what you told the patient, and the review interval** — which is where three or four marks sit at every station in this examination.

OSCE 1 — Requesting a diagnostic wax-up

Blueprint: Diagnostic wax-ups (W + OS-P + SCE). This is RCSEd's own published sample station (Appendix D) — expect a version of it.

Station purpose: to test understanding of the practical steps needed for a diagnostic wax-up when planning an implant-retained prosthesis. **Equipment:** pre-operative photographs of the patient with and without the denture. **No interaction with the examiner.**

CANDIDATE INSTRUCTIONS

Venue: clinic. You are about to request a diagnostic wax-up to ascertain the most appropriate implant-retained tooth replacement option. The patient is a middle-aged edentulous patient. They are unhappy with the complete denture they have worn for 15 years and would like something fixed, but do not want to change or compromise the facial appearance, which they like. You have pre-operative photographs with and without the denture. Describe the practical steps you would take for requesting the diagnostic wax-up, including the request to the laboratory. You are not expected to go into details of the site of implant placement or the risks or benefits of implant treatment.

DIRECTION OF THOUGHT

The key clinical fact is *"does not want to change the facial appearance which they like."* The denture flange is providing lip support. A fixed prosthesis without a flange may not reproduce it. So the wax-up here is not decoration — it is the **diagnostic test of whether a fixed prosthesis is possible at all**. Structure: records → mounting → prescription → verification. End with what you would do with the result.

MODEL ANSWER

"I would start by explaining the purpose to the patient and taking consent, including consent for clinical photographs, which I already have. I would record the existing denture in detail because it is my reference for the appearance the patient likes.

Records: primary impressions in stock trays, or an intraoral scan, and cast a study model of both arches. I would then make special trays and take definitive impressions in an elastomeric material — addition silicone or polyether — and also **duplicate the existing denture** so the laboratory has the current tooth position and flange contour to work from.

I would take an interocclusal record at the patient's existing occlusal vertical dimension, having first checked the freeway space, and a **facebow record**, and mount the casts on a semi-adjustable articulator.

Clinically I would record: the midline, the incisal level at rest and on smiling, the smile line, the occlusal plane, the amount of lip support currently provided by the flange, the shade and the tooth mould.

My written laboratory prescription would identify the patient and me, give the date required, and ask specifically for a diagnostic wax-up of the planned teeth **set to reproduce the tooth positions and lip support of the existing denture**, at the recorded vertical dimension, and would ask the technician to indicate whether that tooth position can be achieved **without a labial flange**. I would ask for a duplicate in clear acrylic to serve as a radiographic guide, with radiopaque markers, and a vacuum-formed matrix or putty index for a chairside mock-up. I would state the materials and ask for the work to be returned disinfected, and I would keep the custom-made device statement in the notes.

When it returns, I would try the wax-up in the mouth as a trial denture or mock-up, check it with the patient in a mirror, assess lip support, speech and appearance, photograph it, and only then decide whether a fixed prosthesis is realistic. If the wax-up shows the appearance depends on the flange, I would discuss an implant-retained overdenture instead. I would record all of this in the notes."

EXAMINER CHECKLIST

1. Explains purpose/consent
2. Photographs and duplicate of existing denture
3. Primary then definitive impressions in an appropriate material (or scan)
4. Interocclusal record at recorded OVD
5. Facebow and articulator mounting
6. Records aesthetic parameters (midline, incisal level, smile line, lip support, shade)
7. **Written** laboratory prescription with specific instructions
8. Requests duplicate for radiographic guide ± matrix
9. Verification/try-in in the mouth with the patient
10. States the decision the wax-up will inform (flange vs no flange) and records it

Pitfalls. Reciting a generic impression sequence without engaging with the lip-support problem; forgetting the written prescription; forgetting to try the wax-up in the mouth.

Running this station to time

Time check. The model answer is 717 words, 5 minutes 7 seconds at 140 words per minute — and because this station carries **no interaction with the examiner**, there is no question to break the delivery and no prompt to rescue an omission: whatever is not said is simply not marked. About 250 words must go, by naming each record in a clause and spending the recovered time on the prescription and the flange decision.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Places the photographs with and without the denture in front of the candidate, indicates the instruction sheet, and then says nothing at all for the remaining five minutes.	Checklist 1 — purpose explained and consent taken, including for the clinical photographs. Before the sequence, the candidate must state the problem in one sentence: **the flange is providing the lip support the patient likes, so this wax-up is the test of whether a fixed prosthesis is possible at all**. That sentence frames everything that follows.
0:30–2:00	Silence.	Checklists 2, 3, 4 and 5, one clause each. **Duplicate of the existing denture** and the photographs as the reference for the appearance · primaries in stock trays or a scan, special trays, definitive impressions in **addition silicone or polyether** · interocclusal record at the **existing OVD** with the freeway space checked · **facebow** and mounting on a semi-adjustable articulator. Speed here is the mark; detail is not.
2:00–3:30	Silence.	Checklists 6 and 7. The aesthetic parameters recorded at the chair — midline, incisal level at rest and on smiling, smile line, occlusal plane, **lip support given by the flange**, shade, mould. Then the **written** prescription: identification, date required, wax-up set to reproduce the tooth position and lip support of the existing denture at the recorded OVD, and the decisive question — **can that position be achieved without a labial flange?**
3:30–4:30	Silence.	Checklists 8 and 9. **Duplicate in clear acrylic with radiopaque markers** as a radiographic guide, plus a vacuum-formed matrix or putty index for a chairside mock-up; materials stated, work returned disinfected, custom-made device statement kept. Then **try-in in the mouth** as a trial denture or mock-up — mirror, lip support, speech, appearance, photographed in the same views.

Time	The examiner does	What earns the mark
4:30–5:00	Silence; the bell.	Checklist 10 — the **decision the wax-up exists to inform**: flange or no flange, fixed prosthesis or implant-retained overdenture; then **records** (consent, prescription copy, try-in findings, photographs), **what the patient is told** about the flange, and the **review** at which the option is agreed.

The compressed answer — what a strong candidate says in five minutes

"This patient likes their face, and it is the **flange of the present denture that is producing that lip support** — so this wax-up is not decoration, it is the test of whether a fixed prosthesis is possible at all.

Consent and the reference. I explain the purpose and take consent, including for the clinical photographs I already have. The existing denture is my record of the appearance they like, so I describe it in the notes and **duplicate it** for the laboratory.

Records. Primary impressions in stock trays or an intraoral scan, study casts of both arches; special trays; definitive impressions in **addition silicone or polyether**. An **interocclusal record at the existing occlusal vertical dimension**, having first checked the freeway space, a **facebow record**, and both casts mounted on a semi-adjustable articulator.

Clinical measurements, recorded before the patient leaves: the midline, the incisal level at rest and on smiling, the smile line, the occlusal plane, the **lip support the flange is currently giving**, the shade and the mould.

The prescription is written, not spoken. It identifies the patient and me, gives the date required, and asks for a diagnostic wax-up **set to reproduce the tooth position and lip support of the existing denture** at the recorded vertical dimension, working from the duplicate. Then the instruction that matters: I ask the technician to state **whether that tooth position can be achieved without a labial flange**. I also ask for a duplicate of the wax-up in **clear acrylic with radiopaque markers** as a radiographic guide, and a vacuum-formed matrix or putty index for a chairside mock-up. I state the materials, ask for the work back disinfected, and keep the custom-made device statement in the notes.

When it returns, I try it in the mouth — as a trial denture or mock-up, not judged on the articulator. I look at it with the patient in a mirror, assess lip support, speech and appearance, and photograph it in the same views as the pre-operative photographs so the two compare directly.

Then the decision the whole exercise exists to make. If the tooth position and lip support can be reproduced **without a flange**, a fixed implant-retained prosthesis is realistic and I plan from that wax-up. If the appearance depends on the flange — and after fifteen years of resorption it often does — a fixed bridge will not give this patient the face they want, and I would discuss an **implant-retained overdenture**, which keeps the flange, before any implant is planned.

And I record all of it: the consent, the records taken, a copy of the written prescription, the try-in findings and the photographs, what I told the patient about the flange, and the review at which we agree which option to proceed with."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 7 written prescription with specific instructions · 10 the flange-versus-no-flange decision, stated and recorded · 9 try-in in the mouth with the patient · 2 photographs and duplicate of the existing denture · 6 aesthetic parameters, above all lip support · 4 interocclusal record at the recorded OVD.

Accept these late or on prompting: 1 purpose and consent · 3 primary then definitive impressions in an appropriate material · 5 facebow and articulator mounting · 8 duplicate for the radiographic guide, plus matrix.

The commonest way five minutes is wasted here: reciting a generic complete-denture impression sequence — stock trays, border moulding, wash impression, tray adhesive — for two minutes, as though the station were about impressions. The prescription then arrives at four minutes and the try-in, the flange decision and the records never arrive at all. The photographs were put on the table because the answer is about lip support, and with a silent examiner nobody will steer the candidate back.

OSCE 2 — Selecting a guide and transferring the plan to surgery

Blueprint: "Request the appropriate radiographic and surgical guides"; "Demonstrate an ability to select guides to optimize the planning and execute treatment"; "Explain transfer of planning information into the surgical procedure using appropriate guides" (W + OS-P + SCE).

EQUIPMENT

study casts with a wax-up of a missing UR2 and LR6, a clear acrylic radiographic guide, examples of non-limiting, partially limiting and fully limiting guides, a CBCT printout.

CANDIDATE INSTRUCTIONS

Two implant sites are planned: the UR2 and the LR6. Using the models and guides provided, describe the type of guide you would select for each and explain how the information from your planning is transferred into the surgical procedure.

DIRECTION OF THOUGHT

Two different sites, two different answers. Anterior = aesthetic, high precision of angulation and depth, tooth-supported partially limiting or fully guided. Posterior = anatomical risk (canal), fully guided or partially limiting plus direction indicator and radiograph. Always: guide derived **from the wax-up**, radiopaque markers, safety margin retained.

MODEL ANSWER

"The starting point for both is the diagnostic wax-up, because the guide is how I make the surgery restoratively driven.

I would duplicate the wax-up in clear acrylic and place radiopaque markers — gutta-percha or barium sulphate — in the planned tooth positions to make a **radiographic guide**. The patient wears this for the CBCT, which lets me relate the planned tooth position to the available bone.

For the **UR2** I would use a **tooth-supported guide**, which is the most accurate support. Because the aesthetic tolerance is small, I would use at least a partially limiting guide indicating the entry point and angle, or a fully limiting sleeve guide if I have merged the CBCT with an intraoral scan. My planning target is the shoulder about 3 mm apical to the planned mid-facial gingival margin and 1–2 mm palatal to the emergence of the adjacent teeth, angled through the cingulum so the screw channel emerges palatally.

For the **LR6** the critical constraint is the inferior alveolar canal. I would plan a length that leaves at least 2 mm of clearance, and use a tooth-supported guide with a drill stop, or a fully guided protocol. Even with a guide, I would keep the 2 mm safety margin because static guided surgery has a mean angular deviation of about 2 to 4 degrees and apical deviation of about 1.3 to 1.6 mm.

Transfer into surgery: the guide is seated and its fit verified on the teeth before the flap is raised; I check it is fully seated and stable, that the patient can open far enough for the drill and sleeve assembly, and that irrigation can reach the osteotomy. After the pilot drill I place a **direction indicator** and take a periapical radiograph at the LR6 to confirm angulation and depth before continuing. I record the guide type used and the final implant position.

Limitations I would state: a guide does not compensate for a poorly fitting guide or for lack of anatomical knowledge, sleeves restrict irrigation, and posterior access may be inadequate."

EXAMINER CHECKLIST

1. Guide derived from the wax-up
2. Radiographic guide with radiopaque markers
3. Tooth-supported preferred
4. Correct 3-D targets for the anterior site
5. 2 mm canal safety margin posteriorly
6. Names the levels of guidance (non-limiting/partially/fully limiting)
7. Verifies guide seating and access before drilling
8. Direction indicator + radiograph
9. States guided-surgery accuracy limits
10. Records what was used

Running this station to time

Time check. The model answer is 575 words, 4 minutes 6 seconds at 140 words per minute — but there are **two sites needing two different answers**, three guides to pick up and compare and a CBCT printout to refer to, which pushes honest delivery past six minutes. The fix is a **fixed 90 seconds per site** on one frame — support, level of guidance, the single controlling constraint — with the shared wax-up and radiographic-guide preamble compressed to two sentences.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Sets out the casts with the wax-up, the clear acrylic guide, the three examples of guidance and the CBCT printout: "Which guide for each site, and how does your planning get into the surgery?"	Checklist 1, and the organising sentence: **both guides derive from the diagnostic wax-up, because that is what makes the surgery restoratively driven rather than bone-driven**. Said in the first 15 seconds it earns the mark and buys permission to be brief about the hardware.
0:30–1:15	Silence; may hold up the clear acrylic guide.	Checklists 2 and 6. **Radiographic guide** — the wax-up duplicated in clear acrylic with **gutta-percha or barium sulphate markers** in the planned tooth positions, worn for the CBCT so the tooth position relates to the bone. Then the **three levels named in one clause each** while pointing at the examples: non-limiting, partially limiting, fully limiting.
1:15–2:30	Indicates the UR2.	Checklists 3 and 4. **Tooth-supported — the most accurate support**; small aesthetic tolerance, so partially limiting at minimum, fully limiting if the CBCT is merged with an intraoral scan. Then the targets, quickly: shoulder about **3 mm apical to the planned mid-facial gingival margin**, **1–2 mm palatal** to the emergence of the adjacent teeth, angled through the **cingulum** so the screw channel emerges palatally.
2:30–3:30	Indicates the LR6.	Checklists 5 and 9. The controlling constraint is the **inferior alveolar canal**: length planned to leave **at least 2 mm clearance**, tooth-supported guide with a drill stop or a fully guided protocol — and the margin is kept **even when guided**, because static guided surgery has a mean angular deviation of about **2 to 4 degrees** and apical deviation of about **1.3 to 1.6 mm**.
3:30–4:30	"So how does the plan actually get into the surgery?"	Checklists 7 and 8. Guide **seated and its fit verified on the teeth before the flap is raised** — fully seated, no rock, mouth opening sufficient for the drill-and-sleeve assembly, irrigation able to reach the osteotomy. **Direction indicator plus a periapical at the LR6** after the pilot drill. Limitations stated: a guide does not rescue a poor fit or absent anatomical knowledge, sleeves restrict irrigation, posterior access may be inadequate.
4:30–5:00	"Anything else?"	Checklist 10 — **records**: guide type and level of guidance used, the verification radiograph, the final implant position, system, diameter and length; **what the patient is told** (the guide is worn for the scan, and it assists but does not replace judgement); and **review**, with the guide retained until the case is restored.

The compressed answer — what a strong candidate says in five minutes

"**Both guides start from the diagnostic wax-up**, because the guide is how I make the surgery restoratively driven rather than driven by the shape of the bone.

The radiographic guide first. I duplicate the wax-up in clear acrylic and place **radiopaque markers — gutta-percha or barium sulphate — in the planned tooth positions**. The patient wears it for the CBCT, which relates the planned teeth to the available bone. Of the three on the table: a **non-limiting** guide shows the position only; a **partially limiting** guide fixes the entry point and angle for the pilot drill; a **fully limiting** guide carries sleeves that control the whole drill sequence in depth and angulation.

For the UR2, the constraint is aesthetics. I would use a **tooth-supported guide** — tooth support is the most accurate of the three supports — and because the tolerance here is a fraction of a millimetre, at least a **partially limiting** guide, or a **fully limiting** one if I have merged the CBCT with an intraoral scan. My planning targets are a shoulder about **3 mm apical to the planned mid-facial gingival margin**, **1 to 2 mm palatal** to the emergence of the adjacent teeth, and angulation through the **cingulum** so the screw channel emerges palatally and I am not forced into a cemented crown.

For the LR6, the constraint is the inferior alveolar canal. I plan a length that leaves **at least 2 mm of clearance**, and use a tooth-supported guide with a **drill stop**, or a fully guided protocol. I keep that 2 mm even when I am fully guided, because static guided surgery is not exact: mean angular deviation of about **2 to 4 degrees** and apical deviation of about **1.3 to 1.6 mm**. The safety margin is what absorbs that error.

Transferring the plan into the surgery. Before any flap, I seat the guide and **verify its fit on the teeth** — fully seated, no rocking, stable on finger pressure. I check the patient can open far enough for the drill and sleeve assembly, particularly at the LR6, and that irrigation can reach the osteotomy through the sleeve. I then work the sequence through the guide, and after the pilot drill at the LR6 I remove the guide, place a **direction indicator** and take a **periapical radiograph** to confirm angulation and depth before I go any deeper.

The limitations I would state plainly. A guide does not compensate for a guide that fits poorly, nor for not knowing the anatomy. Sleeves restrict irrigation and can raise the bone temperature. Posterior access is often the limiting factor, and I would rather convert to a partially limiting guide with a direction indicator and a radiograph than force a sleeve assembly I cannot fully seat.

Afterwards I record the guide type and level of guidance used at each site, the verification radiograph, and the final position, system, diameter and length of each implant. I tell the patient the guide is worn for the scan and again at surgery, that it makes the position more predictable but does not replace my own checks, and I keep it until the case is restored in case I need to re-verify."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 both guides derived from the wax-up · 6 the three levels of guidance named and matched to each site · 4 three-dimensional targets at the UR2 · 5 the 2 mm canal margin, kept even when guided · 7 guide seating, stability and access verified before drilling · 10 records the guide used and the final position.

Accept these late or on prompting: 2 radiographic guide with radiopaque markers · 3 tooth support preferred, with the reason · 8 direction indicator plus radiograph at the LR6 · 9 the guided-surgery accuracy figures.

The commonest way five minutes is wasted here: giving one generic answer for "an implant guide" instead of two site-specific ones. The station put a UR2 and an LR6 on the same cast because they are governed by different constraints — aesthetics and the canal — and a candidate who describes guided surgery in general terms, however fluently, collects neither the anterior targets nor the safety margin, and reaches the transfer-into-surgery half of the instruction with under a minute left.

OSCE 3 — Flap design and atraumatic elevation

Blueprint: "Evaluate the different flap designs and design the appropriate flap"; "importance of raising the flap atraumatically" (W + OS-P + SCE).

EQUIPMENT

a typodont/pig jaw model with an edentulous UL1 site showing a buccal concavity; scalpel handle, blades, periosteal elevators, retractors.

CANDIDATE INSTRUCTIONS

An implant is to be placed at the UL1 with simultaneous guided bone regeneration for a buccal dehiscence. Using the model and instruments provided, describe and, where possible, demonstrate your flap design and how you would raise the flap atraumatically.

DIRECTION OF THOUGHT

Aesthetic zone + GBR = need access, need to preserve papillae, need coronal advancement for tension-free closure, need incisions on sound bone away from the membrane.

MODEL ANSWER

"Design principles first: adequate access for the implant, the graft and the membrane; a base wider than the apex to preserve blood supply; incisions over sound bone and not over the graft or the membrane edge; preservation of the papillae because this is the aesthetic zone; and a design that allows tension-free primary closure.

I would use a **slightly palatally displaced crestal incision** across the edentulous UL1 site, extended intrasulcularly to the UL2 and UR1 with **papilla-sparing** incisions, and a **single vertical relieving incision** at the mesial line angle of the UL3 — placed at the line angle, not through a papilla, and away from where the membrane will sit. A second relieving incision distal to the UR1 only if I need more advancement.

I would incise firmly to bone in one stroke, then elevate a **full-thickness mucoperiosteal flap** using a sharp periosteal elevator, starting at the vertical incision and working against the bone rather than against the tissue, so the periosteum lifts cleanly. I would retract with a broad retractor resting on bone, never crushing the flap margin, handle the flap only at its base with tissue forceps, and keep it moist with saline-soaked gauze.

For tension-free closure over the membrane I would make a **periosteal releasing incision** at the base of the buccal flap, scoring the periosteum with the blade almost parallel to the flap and then blunt-dissecting, taking care not to perforate the flap. I would test with tissue forceps that the flap passively reaches 3–4 mm beyond the closure line without tension.

Closure: horizontal mattress sutures to evert and take the tension, with interrupted sutures between them for accurate margin adaptation, using 5/0 monofilament or ePTFE, with the suture line lying over sound bone rather than over the membrane."

EXAMINER CHECKLIST

1. States access/blood supply/sound-bone/tension-free principles
2. Palatally displaced crestal incision
3. Papilla preservation
4. Vertical relief at a line angle, away from the graft
5. Firm incision to bone
6. Sharp elevator working against bone
7. Broad retractor on bone; flap handled at the base; kept moist
8. Periosteal releasing incision described safely
9. Tests for passive closure
10. Appropriate suture material and technique

Running this station to time

Time check. The model answer is 484 words, 3 minutes 27 seconds at 140 words per minute — but the verb is *describe and, where possible, demonstrate*, so the words are spoken while the hands pick up a blade, elevate on a model and test a flap, which runs at perhaps two-thirds of reading speed. Realistically that is the whole five minutes with nothing spare, and the compression is to state each design principle in a **single clause before the blade is picked up**, then narrate continuously while working.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Indicates the model with the UL1 site and buccal concavity, the blades, elevators and retractors: "Describe and where possible demonstrate your flap design, and how you would raise it atraumatically."	Checklist 1, said **before touching an instrument**, one clause each: access for implant, graft and membrane · **base wider than the apex** for blood supply · incisions over **sound bone, not over the graft or membrane edge** · papillae preserved because this is the aesthetic zone · a design permitting **tension-free primary closure**. Five clauses, 30 seconds; they frame every subsequent choice.
0:45–2:00	Silent. Watches where the blade goes.	Checklists 2, 3 and 4, drawn out on the model with the blade held but not necessarily cutting. **Slightly palatally displaced crestal incision** across the UL1 site · extended **intrasulcularly with papilla-sparing incisions** to the UL2 and UR1 · a **single vertical relieving incision at the mesial line angle of the UL3** — at the line angle, not through a papilla, and away from where the membrane will sit; a second distal to the UR1 only if more advancement is needed.
2:00–3:15	Hands over the periosteal elevator and the retractor.	Checklists 5, 6 and 7, narrated as the hands work. **Firm incision to bone in one stroke · sharp periosteal elevator started at the vertical incision, worked against the bone** so the periosteum lifts cleanly · **broad retractor resting on bone**, never crushing the margin, flap held **only at its base** with tissue forceps, kept moist with saline-soaked gauze. Silent hands score nothing: the narration is the mark.
3:15–4:15	"You need to close this over a membrane. How?"	Checklists 8 and 9. **Periosteal releasing incision at the base of the buccal flap** — blade almost parallel to the flap, scoring periosteum only, then blunt dissection, without perforating the flap. Then the test: tissue forceps confirm the flap **reaches passively 3–4 mm beyond the closure line with no tension**.
4:15–5:00	"Close it, and tell me what happens afterwards."	Checklist 10 and the close — **5/0 monofilament or ePTFE**, **horizontal mattress to evert and take the tension with interrupted sutures between**, suture line lying **over sound bone, not over the membrane**; then **records** (flap design, releasing incision, suture material), **what the patient is told** (no denture pressure, no brushing over the site, report any exposure), and **review at one week**, sutures at 10–14 days over a graft.

The compressed answer — what a strong candidate says in five minutes

"**The principles first, then I will cut.** I need access for the implant, the graft and the membrane; a **base wider than the apex** so the blood supply is preserved; incisions over **sound bone and never over the graft or the membrane edge**; the **papillae preserved**, because this is the aesthetic zone and I cannot rebuild a lost papilla; and a design that will allow **tension-free primary closure** over a membrane.

The design. A **slightly palatally displaced crestal incision** across the edentulous UL1, which brings keratinised tissue buccally when I advance the flap. I extend it **intrasulcularly to the UL2 and the UR1 with papilla-sparing incisions**. I take a **single vertical relieving incision at the mesial line angle of the UL3** — at the line angle, not through the papilla, and deliberately **away from where the membrane will sit**. If I need more advancement I would add a second short relief distal to the UR1, and no more than that.

Raising it. I incise **firmly to bone in one stroke** — repeated shallow passes shred the margin. I start elevating at the vertical incision, where I have a free edge, with a **sharp periosteal elevator worked against**

the bone rather than against the tissue, so the periosteum lifts cleanly and completely. Full thickness, mucoperiosteal.

Handling it — this is the atraumatic part. I retract with a **broad retractor resting on bone**, never on the flap margin. I handle the flap **only at its base**, with tissue forceps, and never with pressure on the margin I am going to close. I keep it **covered with saline-soaked gauze** so it does not desiccate while I place the implant and the graft.

Getting tension-free closure. Because I am covering a membrane over a buccal dehiscence, the flap must advance. I make a **periosteal releasing incision at the base of the buccal flap**: blade almost parallel to the flap, scoring the periosteum only, then blunt dissection to develop the release without perforating the flap. Then I **test it** — I bring the flap across with tissue forceps and confirm it lies **3 to 4 mm past the closure line passively**, with no tension holding it there. If it will not, I release further; I do not close it under tension and hope.

Closure. 5/0 monofilament or ePTFE, not braided silk over a graft. **Horizontal mattress sutures first**, to evert the edges and carry the tension, with **simple interrupted sutures between them** for accurate margin adaptation, and the suture line lying **over sound bone rather than over the membrane**.

Afterwards. I record the flap design, the relieving incisions, the release performed and the suture material; I tell the patient not to wear anything that presses on the site, not to brush over the sutures, and to ring me if the wound opens or the membrane shows; and I **review at one week**, removing sutures at **10 to 14 days** because the site is grafted."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 the design principles stated before cutting · 2 palatally displaced crestal incision · 3 papillae preserved in the aesthetic zone · 7 atraumatic handling — retractor on bone, flap held at the base, kept moist · 8 periosteal releasing incision described safely · 9 tests for passive closure before suturing.

Accept these late or on prompting: 4 vertical relief at a line angle, away from the membrane · 5 firm incision to bone in one stroke · 6 sharp elevator worked against bone · 10 suture material and mattress-plus-interrupted technique.

The commonest way five minutes is wasted here: designing the flap in silence. A candidate who draws a beautiful incision line on the model with the blade tip, saying only "I'd come across here and up here", loses checklists 1, 4 and 7 outright, because the principle, the reason the relief sits at the line angle and the atraumatic handling all exist only in their head. The second waste is a lecture on flap classifications — envelope, triangular, trapezoidal — which is not what was asked and eats the minute the releasing incision needs.

OSCE 4 — Suturing (practical)

Blueprint: "Consider and justify the choice of suture material and technique used for wound closure" (W + OS-P + SCE).

EQUIPMENT

suturing pad or pig jaw with a raised flap over a simulated graft site; needle holders, tissue forceps, scissors; a selection of sutures (3/0 silk, 4/0 Vicryl, 5/0 ePTFE, 5/0 monofilament nylon).

CANDIDATE INSTRUCTIONS

A buccal flap has been raised and a bone graft placed with a collagen membrane. Select an appropriate suture and demonstrate closure. Explain your choice of material and technique as you work.

DIRECTION OF THOUGHT

Over a membrane → monofilament/ePTFE, not silk. Technique → horizontal or vertical mattress to take tension and evert, plus interrupted. Say the removal time.

MODEL ANSWER

"I would choose a **5/0 ePTFE or monofilament** suture on a reverse-cutting 3/8 circle needle. I would avoid braided silk here because it is multifilament and wicks bacteria into a grafted site, which risks membrane contamination.

I will place **horizontal mattress sutures** first to evert the wound edges and carry the tension, then **simple interrupted sutures** between them to adapt the margins precisely.

Technique: I pass from the **mobile buccal flap to the fixed palatal tissue**, entering perpendicular to the surface about 3 mm from the margin, taking equal bites of each side and equal depth, so the edges meet passively. I position the knot to one side of the incision line, and I tie so that the tissue is approximated but not blanched — tension causes ischaemia, necrosis and membrane exposure.

I would remove these sutures at **10 to 14 days** because the site is grafted, rather than the usual 7 to 10 days, and I would review the closure at one week for any sign of dehiscence."

EXAMINER CHECKLIST

1. Selects monofilament/ePTFE and justifies avoiding silk
2. Appropriate size (4/0–5/0)
3. Correct needle handling in the needle holder (2/3 from the tip)
4. Mobile to fixed
5. Perpendicular entry ~3 mm from the margin, equal bites
6. Mattress suture placed and named correctly
7. Interrupted sutures for adaptation
8. Knot placed off the incision line, secure throws
9. Approximation without blanching
10. States removal time and review

Running this station to time

Time check. The model answer is only 365 words, 2 minutes 36 seconds at 140 words per minute — which looks generous and is not. This is a **performance station**: the marks are in the hands, and threading a needle, placing a horizontal mattress and two interrupted sutures and tying them takes three to four minutes on a pad even when the fingers are practised, with speech running at perhaps half rate while the hands work. The rule here is not to cut words but to **front-load every justification into the first 45 seconds**, then let the hands run.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Indicates the pad or pig jaw with the raised flap over a simulated graft and membrane, and the four suture packets: "Select a suture and demonstrate closure. Explain your choice as you work."	Checklists 1 and 2, said **while opening the packet, not before**. **5/0 ePTFE or monofilament** on a reverse-cutting 3/8 circle needle, and the reason silk is rejected out loud: **multifilament, wicks bacteria into a grafted site and risks contaminating the membrane**. Naming the size 4/0–5/0 costs three words and is a whole mark.
0:45–1:30	Silent. Watches the hands.	Checklists 3 and 4. Needle **held in the needle holder about two-thirds from the tip**, on the tip of the beaks, not in the joint. First pass from the **mobile buccal flap to the fixed palatal tissue** — stated as a principle, because the mobile side is the one that must be moved.
1:30–3:00	Silent. May place a second packet within reach.	Checklists 5 and 6. **Perpendicular entry about 3 mm from the margin**, equal bites and equal depth on both sides so the edges meet passively. The **horizontal mattress placed and named as it is placed**, with its purpose said aloud — everts the edges and carries the tension off the margin, which is what protects a membrane.

Time	The examiner does	What earns the mark
3:00–4:00	"Carry on."	Checklists 7, 8 and 9. **Simple interrupted sutures between the mattresses** for accurate adaptation; **knot laid to one side of the incision line**, secure throws; and the tension test spoken as it is judged — **approximated but not blanched**, because tension causes ischaemia, necrosis and **membrane exposure**. If the hands are still working, keep talking: silent suturing scores nothing.
4:00–5:00	"When do these come out?"	Checklist 10 — **removal at 10 to 14 days because the site is grafted**, not the usual 7 to 10; then **records** (material, size, number and type of sutures), **what the patient is told** (do not pull at the sutures, no brushing over them, report any opening), and **review at one week for dehiscence**. Say all of this even mid-suture: it is the row candidates most often reach after the bell.

The compressed answer — what a strong candidate says in five minutes

"**Choosing first, while I open the pack.** I would use a **5/0 ePTFE or monofilament nylon** on a reverse-cutting three-eighths circle needle. I would **not use the braided silk** here: it is multifilament, it wicks bacteria along its length into a grafted site, and contamination of that membrane is what causes the graft to fail. Five-oh in the aesthetic zone gives me accurate adaptation with the least tissue trauma; four-oh would be acceptable posteriorly.

Loading the needle. I hold it in the needle holder about **two-thirds of the way back from the tip**, on the tips of the beaks rather than in the joint, so I do not bend or blunt it.

Direction of the pass. Always from the **mobile buccal flap to the fixed palatal tissue** — I move the side that moves, and the fixed tissue anchors it.

The bite. I enter **perpendicular to the surface, about 3 millimetres from the margin**, and take **equal bites and equal depth on each side**, so that the edges rise to meet each other passively rather than being dragged together.

The technique, and why. I place **horizontal mattress sutures first**. They evert the wound edges and, more importantly, they **carry the tension away from the margin itself** — over a graft and a membrane, the margin is the last place I want load. Then I place **simple interrupted sutures between them** for accurate adaptation of the edges, because the mattresses alone leave the margin slightly open.

Tying. Secure throws, and I lay the **knot to one side of the incision line**, not on it, so it does not sit in the wound and act as a plaque trap.

And the judgement that matters. I tie so the tissue is **approximated but not blanched**. If the edges blanch I have made them ischaemic; ischaemia becomes necrosis, and necrosis over a membrane becomes **membrane exposure and a failed graft**. If I cannot close it without blanching, the answer is not a tighter suture — it is to go back and release the flap further with a periosteal releasing incision.

Afterwards. I would remove these at **10 to 14 days rather than the usual 7 to 10, because the site is grafted** and I want the closure supported for longer. I record the material, the size, and the number and type of sutures placed. I tell the patient not to disturb them with their tongue, not to brush over them, to keep the rest of the mouth clean, and to ring me if anything opens or if they can see the membrane. And I **review at one week to check for dehiscence**, which is when it declares itself."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 monofilament or ePTFE selected and silk rejected with the reason · 6 mattress suture placed and correctly named · 9 approximated without blanching, with the consequence · 10 removal at 10–14 days over a graft, plus review · 4 mobile flap to fixed tissue · 7 interrupted sutures for margin adaptation.

Accept these late or on prompting: 2 size 4/0–5/0 stated · 3 needle held two-thirds from the tip in the needle holder · 5 perpendicular entry about 3 mm from the margin with equal bites · 8 knot laid off the incision line.

The commonest way five minutes is wasted here: suturing in silence. The hands absorb the candidate's whole attention, three neat sutures go in, and not one of checklists 1, 4, 9 or 10 has been said aloud — so a technically good closure scores four. The other waste is fighting the first knot: if the needle is badly loaded and the first pass tears, the candidate spends two minutes recovering and never reaches the removal time or the review. Take the material decision and its justification out of the way in the first 45 seconds, then talk continuously while the hands work.

OSCE 5 — Implant site preparation

Blueprint: "Demonstrate understanding of the principles of site preparation for the placement of dental implants" (W + OS-P + SCE).

EQUIPMENT

bone-analogue block or model, surgical motor (may be non-functional), a drill sequence, direction indicators, depth gauge, torque ratchet, irrigation.

CANDIDATE INSTRUCTIONS

Describe and demonstrate the preparation of an osteotomy for a 4.1 mm diameter, 10 mm long implant at the LR5 in D3 bone. Explain each step.

DIRECTION OF THOUGHT

Order: position from the guide → sequential drilling → temperature control → verification → under-preparation for D3 → insertion torque. Mention heat threshold and the 2 mm canal margin.

MODEL ANSWER

"I confirm the position from the surgical guide, not from the shape of the bone, and confirm the planned length allows at least 2 mm of clearance from the inferior alveolar canal.

I mark the entry point with a round bur or a precision drill through the guide. I then drill **sequentially**, from the pilot drill upwards in diameter, at the manufacturer's recommended speed — around 800 to 1500 rpm — with **copious sterile saline irrigation** and **intermittent, pumping pressure** so that irrigant reaches the cutting flutes and debris is cleared. I use sharp drills within their recommended number of uses. The reason is that bone necroses at **47 degrees for one minute**, and blunt drills, continuous pressure and inadequate irrigation are the main causes of overheating.

After the pilot drill I insert a **direction indicator**, check angulation mesiodistally and buccolingually and parallelism with the adjacent teeth, and take a periapical radiograph with the indicator in place because the canal is close.

I check the osteotomy with a depth gauge and probe the walls to confirm no perforation, particularly lingually where there may be a concavity.

Because this is **D3 bone**, I would **under-prepare** — stopping one drill short of the final diameter, or using an osteotome or densifying protocol — to improve primary stability.

I insert the implant at low speed, around 15 to 30 rpm, then finish by hand with a ratchet, monitoring insertion torque, aiming for 25 to 45 Ncm and avoiding torque above 50 Ncm which risks compression necrosis. I then record the system, diameter, length, site, lot number and insertion torque."

EXAMINER CHECKLIST

1. Position from the guide
2. Confirms safety margin from the canal
3. Sequential graded drilling
4. Correct speed
5. Copious sterile saline, intermittent pressure, sharp drills
6. States the 47 °C threshold

7. Direction indicator + radiograph
8. Checks walls/depth for perforation
9. Under-preparation for D3 justified
10. Insertion torque range and recording

Running this station to time

Time check. The model answer is 434 words, 3 minutes 6 seconds at 140 words per minute — but *describe and demonstrate* means the drill sequence is physically worked through on a block while the words are spoken, and hands changing drills, seating a direction indicator and fitting a ratchet run well below reading speed; the honest cost is over six minutes. The compression is a **fixed order narrated without pause** — position, drill, cool, verify, under-prepare, insert, record — with every number attached to the action rather than explained after it.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Indicates the bone block, the motor, the drill sequence, indicators, depth gauge, ratchet and irrigation: "Prepare an osteotomy for a 4.1 by 10 mm implant at the LR5 in D3 bone. Explain each step."	Checklists 1 and 2, before a drill is picked up. **Position confirmed from the surgical guide, not from the shape of the bone** — said explicitly, because the bone will invite the drill lingually. Then the planned 10 mm length checked to leave **at least 2 mm of clearance from the inferior alveolar canal**.
0:45–2:15	Silent. Watches the drilling hand and the irrigation.	Checklists 3, 4, 5 and 6, narrated while the hands work. Entry marked with a round bur or precision drill through the guide · **sequential graded drilling** from the pilot upwards · **800 to 1500 rpm** per the manufacturer · **copious sterile saline, intermittent pumping pressure, sharp drills within their use count** · and the reason in the same breath: **bone necroses at 47 °C for one minute**, with blunt drills, continuous pressure and poor irrigation as the three causes.
2:15–3:15	Silent, or "how do you know it is going where you want?"	Checklists 7 and 8. **Direction indicator after the pilot drill** — angulation checked mesiodistally and buccolingually and against the adjacent teeth, then a **periapical radiograph with the indicator in place because the canal is close**. Then the **depth gauge and probing of every wall for perforation, particularly lingually** where the concavity sits.
3:15–4:15	"This is D3 bone. Does that change anything?" — asked if not already volunteered.	Checklists 9 and 10, first half. **Under-preparation justified**: stop one drill short of the final diameter, or use an osteotome or densifying protocol, to gain primary stability in softer bone. Then insertion at **15 to 30 rpm**, finished by hand on the ratchet, **insertion torque 25 to 45 Ncm and not above 50 Ncm**, which risks compression necrosis.
4:15–5:00	"Finish off."	Checklist 10, second half — **records**: system, diameter, length, site, **lot number** and the insertion torque achieved, plus the verification radiograph; **what the patient is told** (that the implant is stable, or that it is not and healing will be longer); and **review**, with the healing period set by the stability actually obtained.

The compressed answer — what a strong candidate says in five minutes

"**I take the position from the surgical guide, not from the shape of the bone** — a soft D3 ridge will steer the drill lingually if I let it. And before I start I confirm my planned 10 millimetre length leaves **at least 2 millimetres of clearance from the inferior alveolar canal** on the CBCT.

Entry. I mark the entry point through the guide with a round bur or a precision drill, breaking the cortex where the plan says, not where the bone is easiest.

Drilling. Then **sequentially, from the pilot drill upwards in diameter**, at the manufacturer's speed — **around 800 to 1500 rpm** — with **copious sterile saline irrigation** and **intermittent, pumping pressure**, so that irrigant reaches the cutting flutes and the debris clears rather than packing them. I use **sharp drills within their recommended number of uses**. The reason for all of that is a single number: **bone is irreversibly damaged at 47 degrees for one minute**, and the three things that get me there are blunt drills, continuous pressure and inadequate irrigation.

Verification. After the pilot drill I place a **direction indicator** and check angulation mesiodistally and buccolingually, and parallelism with the adjacent teeth. Because the canal is close at the LR5 I take a **periapical radiograph with the indicator in place** before I go any deeper — that radiograph is far cheaper than a nerve injury.

Checking the osteotomy. I measure with the **depth gauge** and run a probe around every wall to confirm there is **no perforation**, paying particular attention **lingually**, where the submandibular concavity sits and where a perforation risks a floor-of-mouth bleed.

Adapting to the bone quality. This is **D3 bone** — thin cortex, low-density trabecular bone — so I would **under-prepare**: stop one drill short of the final diameter, or use an osteotome or a densifying protocol, to condense the walls and improve primary stability. In D1 I would do the reverse and prepare fully, with tapping, to avoid compression.

Insertion. I place the implant at **low speed, around 15 to 30 rpm**, and finish by hand with the ratchet, watching the **insertion torque**. I am aiming for **25 to 45 newton centimetres**; I would not exceed **50**, because excessive torque produces compression necrosis of the crest and crestal bone loss. If the torque comes out low I would not load it, and I would extend the healing period rather than accept a mobile implant.

Records. I write down the **system, diameter, length, site, lot number and the insertion torque achieved**, together with the verification radiograph and the drilling protocol used. I tell the patient what stability I obtained and what that means for the timing of the crown, and I set the review and healing interval by the stability I actually measured, not by the one I hoped for."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 position taken from the guide, not the bone · 2 the 2 mm canal safety margin confirmed · 5 copious saline, intermittent pressure, sharp drills · 6 the 47 °C for one minute threshold · 9 under-preparation for D3, justified · 10 insertion torque range and full recording including lot number.

Accept these late or on prompting: 3 sequential graded drilling · 4 drill speed in rpm · 7 direction indicator plus radiograph · 8 depth gauge and walls probed for perforation.

The commonest way five minutes is wasted here: treating it as a manual dexterity test and drilling in silence. Every drill change is dead air, and the marks in this station are the numbers — 47 degrees, 2 millimetres, 25 to 45 newton centimetres — none of which the hands can demonstrate. The second waste is a discourse on bone density classification: naming D1 to D4 and describing each is easy to talk about and worth nothing here, because the mark is for **what the candidate does differently in D3**, which is one sentence.

OSCE 6 — Bone defects: identification and management

Blueprint: "Describe the different types of bone defects and suggest appropriate means to manage these" (W + OS-P + SCE).

EQUIPMENT

four models showing (a) a buccal dehiscence around a placed implant, (b) a fenestration, (c) a knife-edge ridge 3 mm wide, (d) a vertical deficiency of 6 mm in the posterior mandible.

CANDIDATE INSTRUCTIONS

Identify the bone defect present on each model and state how you would manage it, including whether it is within the scope of a general dentist with a special interest in implant dentistry.

MODEL ANSWER

"**Model A — dehiscence.** Bone is missing from the crest apically along the implant face, communicating with the crest. If the implant is in the correct prosthetic position with good primary stability, I would treat this with **simultaneous guided bone regeneration**: decorticate the buccal plate to open marrow spaces, apply a composite graft of autogenous bone chips collected from the osteotomy with a slowly resorbing xenograft for volume stability, cover with a native collagen membrane extending 2–3 mm beyond the defect, and achieve **tension-free primary closure** with a periosteal releasing incision. I would use a **two-stage submerged** protocol and allow 4 to 6 months.

Model B — fenestration. A window of exposed implant with intact crestal bone. Same GBR approach; generally more predictable than a dehiscence because the crestal bone is intact.

Model C — knife-edge ridge, 3 mm. A horizontal deficiency. Options: a narrow-diameter implant if the prosthetic space allows; a **ridge split and expansion** with simultaneous placement and grafting, which is more predictable in the maxilla than the mandible; or **staged horizontal GBR** followed by placement at 6 to 9 months. I would judge which is within my competence — a straightforward simultaneous GBR is, an extensive staged augmentation may not be.

Model D — 6 mm vertical deficiency, posterior mandible. This is a **vertical** deficiency. Vertical augmentation is significantly less predictable than horizontal and has a higher complication rate. My GDP-level alternatives are a **short implant**, which the evidence shows performs comparably to a longer implant with augmentation over the medium term with fewer complications, or a non-implant option. If vertical augmentation is genuinely required, this is a **Complex** case on the SAC classification and I would refer."

EXAMINER CHECKLIST

1. 1–4 Names each defect correctly
2. GBR principles (decortication, graft, membrane, closure)
3. Composite graft rationale
4. Simultaneous vs staged decision criteria
5. Ridge split indications and maxilla/mandible difference
6. Short implants as an evidence-based alternative to vertical augmentation
7. Identifies the referral threshold explicitly

Running this station to time

Time check. The model answer is 425 words, 3 minutes 2 seconds at 140 words per minute — deceptively short, because there are **four models**, each needing a name, a management plan and a competence judgement, and the candidate must physically pick each one up and orientate it. That is a **fixed 60 seconds per model** with 60 seconds left for the referral decision; anything over a minute on the dehiscence is taken directly from the vertical defect, where the discriminating marks sit.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–1:00	Places the four models in a row and hands over model A: "Identify the defect on each and tell me how you would manage it, including whether it is within your scope."	Checklists 1, 5 and 6. **Dehiscence** — bone missing from the crest apically along the implant face, **communicating with the crest**. Then GBR in one sweep: **decorticate · autogenous chips from the osteotomy plus a slowly resorbing xenograft · native collagen membrane 2–3 mm beyond the defect · tension-free primary closure with a periosteal release**, submerged, 4 to 6 months. Said once here, referred back to thereafter.

Time	The examiner does	What earns the mark
1:00–1:45	Slides model B forward.	Checklists 2 and 7. **Fenestration** — a window of exposed implant with **intact crestal bone**, and therefore **more predictable than a dehiscence** for exactly that reason. Then the simultaneous-versus-staged criterion stated as a rule: **simultaneous if the implant is in the correct prosthetic position with good primary stability and the defect walls contain the graft; staged if either fails**.
1:45–2:45	Slides model C forward, may hand over a ruler.	Checklists 3 and 8. **Knife-edge ridge, 3 mm — a horizontal deficiency.** Three options in one clause each: **narrow-diameter implant** if the prosthetic space allows · **ridge split and expansion** with simultaneous placement and grafting, **more predictable in the maxilla than the mandible** because of the elastic cancellous core · **staged horizontal GBR** then placement at 6 to 9 months.
2:45–4:00	Slides model D forward: "And this one?"	Checklists 4 and 9. Named as a **vertical deficiency of 6 mm**, with the statement that **vertical augmentation is markedly less predictable than horizontal and carries a higher complication rate**. Then the GDP-level alternative: a **short implant**, which performs comparably to a longer implant with augmentation over the medium term with **fewer complications** — or a non-implant option.
4:00–5:00	"So which of these would you do yourself?"	Checklist 10 — the limit named explicitly: simultaneous GBR at A and B is within scope; C is a judgement, straightforward cases yes, extensive staged augmentation no; **D, if genuine vertical augmentation is required, is Complex on the SAC classification and I refer**. Then **records** (defect measured and photographed, decision and reasoning), **what the patient is told** (why this needs a colleague, and that a short implant may avoid it altogether), and **review**, with a plan if the referral is declined.

The compressed answer — what a strong candidate says in five minutes

Q: Model A? "A **buccal dehiscence** — bone missing from the crest apically along the face of the implant, **communicating with the crest**. If the implant is in the correct prosthetic position with good primary stability, I would treat it with **simultaneous guided bone regeneration**: **decorticate** the buccal plate to open the marrow spaces and bring in blood supply; a **composite graft** of autogenous chips collected from the osteotomy against the implant surface, layered with a **slowly resorbing xenograft** outside it for volume stability; a **native collagen membrane extending 2 to 3 millimetres beyond the defect margins**; and **tension-free primary closure** after a periosteal releasing incision. Two-stage, submerged, **4 to 6 months** before uncovering."

Q: Why the two materials rather than one? "The autogenous chips are the only **osteogenic** component — living cells — so they go against the implant surface where regeneration matters. The xenograft resorbs very slowly, so it is what still holds the buccal contour in two years' time. One provides the biology, the other the volume."

Q: Model B? "A **fenestration** — a window of exposed implant with the **crestal bone intact**. Same GBR approach, and **generally more predictable than a dehiscence** precisely because the crest is intact: the defect is contained and the closure is not sitting over the defect margin."

Q: When simultaneous and when staged? "**Simultaneous** when the implant can be placed in the correct prosthetic position with good primary stability and the defect walls will contain the graft. **Staged** when I would have to place the implant in the wrong position to get stability, when there is not enough bone

to stabilise it at all, or when the defect is non-contained. I do not compromise the position to achieve a simultaneous graft."

Q: Model C? "A **knife-edge ridge, about 3 millimetres wide — a horizontal deficiency**. Three options. A **narrow-diameter implant**, if the prosthetic space and the loading allow it. A **ridge split and expansion** with simultaneous placement and grafting, which is **more predictable in the maxilla than the mandible**, because the maxilla has an elastic cancellous core between the plates and the dense mandible tends to fracture the buccal plate. Or **staged horizontal GBR**, placing at 6 to 9 months."

Q: Model D? "A **6 millimetre vertical deficiency in the posterior mandible** — and vertical is the important word. **Vertical augmentation is significantly less predictable than horizontal and has a much higher complication rate**, with the inferior alveolar nerve immediately below. My general-practitioner-level alternative is a **short implant**: the evidence shows short implants perform comparably to longer implants with augmentation over the medium term, with **fewer complications** — or I consider a non-implant option altogether."

Q: Which of these would you do yourself? "A and B, simultaneous GBR — yes, that is within my competence. C is a judgement: a straightforward split or a contained horizontal graft, yes; an extensive staged augmentation, no. **D, if genuine vertical augmentation is required, is a Complex case on the SAC classification and I would refer it.** I would measure and photograph the defect and record the reasoning, explain to the patient why a colleague is doing that part and that a short implant may avoid it entirely, write the referral with the CBCT, and review them so they are not lost between us."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 10 the referral threshold named explicitly · 9 short implants as the evidence-based alternative to vertical augmentation · 4 the vertical deficiency named **as vertical**, with its predictability · 5 the GBR principles in sequence · 7 simultaneous versus staged criteria · 1 dehiscence named correctly.

Accept these late or on prompting: 2 fenestration named and contrasted with the dehiscence · 3 knife-edge ridge named as a horizontal deficiency · 6 composite graft rationale · 8 ridge split, and why the maxilla differs from the mandible.

The commonest way five minutes is wasted here: giving the full GBR protocol twice — once for the dehiscence and again, almost verbatim, for the fenestration. That duplication costs a minute and buys nothing, because the fenestration mark is for the **contrast** (intact crest, more predictable), not for a second recital. The other waste is drifting into the technique of vertical augmentation — tenting screws, titanium mesh, block grafts — when the mark on model D is for **not doing it**.

OSCE 7 — Guided bone regeneration: materials and barriers

Blueprint: "Differentiate between the different types of soft and hard tissue augmentation materials"; "role of barriers in bone augmentation and their application during surgery along with their limitations" (W + OS-P + SCE).

EQUIPMENT

samples/packaging of autogenous bone collector, DBBM xenograft, FDBA allograft, β-TCP alloplast, native collagen membrane, dPTFE membrane, titanium-reinforced membrane, tacks.

CANDIDATE INSTRUCTIONS

Identify the materials on the table. Explain how you would select between them for a buccal dehiscence around an implant at the UL1, and describe the role and limitations of the barrier membrane.

MODEL ANSWER

"These are, in turn, an autogenous bone collector, a deproteinised bovine bone mineral xenograft, a freeze-dried bone allograft, a beta-tricalcium phosphate alloplast, a native collagen membrane, a dense PTFE membrane, a titanium-reinforced membrane and fixation tacks.

The three mechanisms are **osteogenesis** — only autogenous bone, which contains living cells; **osteoinduction** — mediated by BMPs, principally autogenous and demineralised allograft; and **osteoconduction**, which all of them provide as a scaffold.

For a buccal dehiscence at the UL1 I would use a **composite graft**: autogenous chips collected from the osteotomy, placed against the implant surface for their osteogenic potential, layered with a **deproteinised bovine bone mineral** on the outside for its very slow resorption, which maintains buccal contour in the aesthetic zone. I would cover this with a **native collagen membrane** extending 2 to 3 mm beyond the defect margins, stabilised under the flap or with tacks.

The membrane's role is to **exclude fast-migrating epithelial and connective tissue cells** so that slower osteoprogenitor cells can populate the space. Its limitations are that a collagen membrane has **poor space maintenance and will collapse into a non-space-maintaining defect**, and that its resorption time is unpredictable — anywhere from about four weeks to six months. A dense PTFE membrane maintains space better and tolerates exposure, but must be removed. A titanium-reinforced membrane maintains space best, but that level of defect is a complex case which I would refer.

The commonest complication is **membrane exposure**, which follows tension at closure, a suture line over the membrane, denture pressure or smoking. If a collagen membrane exposes without infection I would manage it with chlorhexidine, no brushing and close review; an exposed PTFE membrane I would remove early.

Consent: I must tell the patient that the xenograft is **bovine in origin** and the collagen membrane porcine, and offer synthetic or autogenous alternatives if that is unacceptable to them."

EXAMINER CHECKLIST

1. Identifies all materials
2. Osteogenesis/induction/conduction defined
3. Composite graft rationale
4. DBBM chosen for volume stability
5. Membrane function stated correctly
6. Membrane extended beyond defect and stabilised
7. Collagen limitations (space maintenance, resorption)
8. Exposure management differs by membrane type
9. Recognises titanium-reinforced = refer
10. **Consent to material origin**

Running this station to time

Time check. The model answer is 464 words, 3 minutes 19 seconds at 140 words per minute — but there are **eight items to pick up and name** before a word of the actual question is answered, and the instruction then has three separate parts: identify, select, and the role and limitations of the barrier. Naming the materials must be done as a **single 40-second sweep along the table**, not one considered sentence per packet, or the membrane limitations and the consent mark are never reached.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Indicates the eight items laid out in order: "Identify these. Then tell me how you would select between them for a buccal dehiscence at the UL1, and the role and limitations of the barrier."	Checklist 1 — all eight named in one sweep, touching each: **autogenous bone collector · deproteinised bovine bone mineral xenograft · freeze-dried bone allograft · β-tricalcium phosphate alloplast · native collagen membrane · dense PTFE membrane · titanium-reinforced membrane · fixation tacks**. Forty seconds, no commentary.

Time	The examiner does	What earns the mark
0:45–1:30	Silent.	Checklist 2 — the three mechanisms, one clause each: **osteogenesis, living cells, autogenous bone only** · **osteoinduction, BMP-mediated, principally autogenous and demineralised allograft** · **osteoconduction, a scaffold, which all of them provide**. This is the classification that makes the selection argument work, so it must come before the selection, not after.
1:30–2:45	"So what would you use at the UL1?"	Checklists 3, 4 and 6. A **composite graft**: **autogenous chips from the osteotomy against the implant surface** for osteogenic potential, layered with **DBBM outside** for its very slow resorption, which is what **holds the buccal contour in the aesthetic zone** years later. Then the membrane **extended 2 to 3 mm beyond the defect margins and stabilised** under the flap or with the tacks on the table.
2:45–3:45	"And the membrane — what is it for, and what are its limits?"	Checklists 5, 7 and 9. Function stated precisely: **excludes fast-migrating epithelial and connective tissue cells so slower osteoprogenitor cells can populate the space**. Limits: collagen has **poor space maintenance and collapses into a non-space-maintaining defect**, and its **resorption is unpredictable, four weeks to six months**. dPTFE maintains space and tolerates exposure but must be removed; **titanium-reinforced implies a defect that is a complex case, which I would refer**.
3:45–4:30	"What goes wrong?"	Checklist 8. **Membrane exposure is the commonest complication**, following tension at closure, a suture line lying over the membrane, denture pressure or smoking — and the management **differs by membrane**: an exposed collagen membrane without infection is managed with chlorhexidine, no brushing and close review; an **exposed PTFE membrane is removed early**.
4:30–5:00	"Anything you must tell the patient?"	Checklist 10, the discriminating mark, plus the close — **consent to material origin**: the xenograft is **bovine**, the collagen membrane **porcine**, with **synthetic or autogenous alternatives offered** if that is unacceptable on religious, cultural or personal grounds. Then **records** (material, brand, lot number, quantity, and that origin was discussed), **what the patient is told** (no denture pressure, report any exposure), and **review at one week and 4 to 6 months**.

The compressed answer — what a strong candidate says in five minutes

Q: Identify these. "In order: an **autogenous bone collector**, a **deproteinised bovine bone mineral xenograft**, a **freeze-dried bone allograft**, a **beta-tricalcium phosphate alloplast**, a **native collagen membrane**, a **dense PTFE membrane**, a **titanium-reinforced membrane**, and **fixation tacks**."

Q: How do they differ in principle? "Three mechanisms. **Osteogenesis** — new bone from cells within the graft — is **autogenous bone alone**, because only it contains living cells. **Osteoinduction** — recruiting host cells through BMPs — is principally autogenous bone and demineralised allograft. **Osteoconduction** — acting as a scaffold for ingrowth — is provided by **all of them**, and it is all a xenograft or an alloplast provides."

Q: What would you use for a buccal dehiscence at the UL1? "A **composite graft**. **Autogenous chips collected from the osteotomy** with a filter, placed **directly against the exposed implant surface**, because that is where I want osteogenic and osteoinductive potential. Layered outside that, **deproteinised bovine bone mineral**, chosen specifically for its **very slow resorption** — in the aesthetic zone the graft has to still be holding the buccal contour in five years, and autogenous bone alone would resorb. Covered with a **native collagen membrane extended 2 to 3 millimetres beyond the defect margins in every direction** and stabilised, under the flap or with tacks, so it cannot move when I close."

Q: What is the membrane actually doing? "It is a **barrier that excludes the fast-migrating epithelial and connective tissue cells** from the defect space, so that the **slower osteoprogenitor cells** have time to populate it. Without it, soft tissue simply fills the space first."

Q: And its limitations? "A native collagen membrane has **poor space maintenance — it will collapse into a defect that is not itself space-maintaining**, and then it holds no space at all. Its **resorption time is unpredictable**, anywhere from about four weeks to six months, so I cannot rely on a barrier function for a fixed period. A **dense PTFE** membrane maintains space better and tolerates exposure without infection, but it is non-resorbable and must be removed. A **titanium-reinforced** membrane maintains space best of all — but a defect that needs one is a large, non-contained defect, and that is a **complex case I would refer** rather than treat."

Q: What is the commonest complication? "**Membrane exposure.** It follows tension at closure, a suture line lying over the membrane, denture pressure on the site, or smoking. Management depends on the material: a **collagen membrane exposed without infection** I would manage conservatively — chlorhexidine, no brushing over the site, no denture pressure, and close review, accepting some graft loss. An **exposed PTFE membrane I would remove early**, before the space beneath it becomes contaminated."

Q: Anything about consent? "Yes, and I would raise it before the day of surgery. The patient must be told that the **xenograft is bovine in origin** and the **collagen membrane porcine**, and I would **offer a synthetic alloplast or purely autogenous alternative** if that is unacceptable to them on religious, cultural or personal grounds. I record the material, brand, lot number and quantity used, that the origin was discussed and consented, what I have told the patient about avoiding pressure on the site and reporting any exposure, and I review at one week and again before uncovering at four to six months."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 10 consent to bovine and porcine origin, with alternatives offered · 5 the barrier function stated correctly · 7 collagen limitations — space maintenance and unpredictable resorption · 3 composite graft rationale · 9 titanium-reinforced membrane recognised as a referral · 8 exposure management differing by membrane type.

Accept these late or on prompting: 1 all eight materials identified · 2 osteogenesis, osteoinduction and osteoconduction defined · 4 DBBM chosen for volume stability in the aesthetic zone · 6 membrane extended beyond the defect and stabilised.

The commonest way five minutes is wasted here: treating it as a materials-science viva. A candidate who gives particle size, sintering temperature and resorption kinetics for each of the four graft materials in turn sounds impressive for two minutes, then has to compress the barrier — which is half the instruction — into thirty seconds, and never reaches consent to material origin, which is the mark the examiner most wants to hear. Name the eight, give the three mechanisms, then answer the clinical question.

OSCE 8 — Local anaesthesia and a drug interaction

Blueprint: "Apply an understanding of the range of local anaesthetics available, their side effects and the methods of using"; "Suggest appropriate management for drug interactions" — both marked W + OS-P + SCE.

EQUIPMENT

a selection of cartridges (lidocaine 2% with adrenaline 1:80,000; articaine 4% with adrenaline 1:100,000; prilocaine 3% with felypressin; mepivacaine 3% plain), a syringe, needles, and a medical history card.

CANDIDATE INSTRUCTIONS

You are to place a single implant at the LR5. The medical history card shows: 68-year-old male; propranolol 80 mg twice daily for essential tremor; warfarin (INR yesterday 2.4); simvastatin 40 mg. Select an appropriate local anaesthetic, justify your choice, and describe any modifications you would make to your management.

DIRECTION OF THOUGHT

Two traps: **adrenaline + non-selective beta-blocker**, and **warfarin**. Plus the LR5 site raises the mental foramen/anterior loop. Say maximum doses.

MODEL ANSWER

"Three issues arise.

First, **propranolol is a non-selective beta-blocker**. Adrenaline in the presence of unopposed alpha stimulation can produce a hypertensive response with reflex bradycardia. I would not regard adrenaline as absolutely contraindicated, but I would **limit the dose**, aspirate carefully to avoid intravascular injection, and inject slowly. A reasonable choice is **lidocaine 2% with adrenaline 1:80,000, limited to two cartridges**, or **prilocaine 3% with felypressin** if I want to avoid adrenaline entirely. Each 2.2 mL lidocaine cartridge contains 44 mg of lidocaine; the maximum is 4.4 mg per kilogram to 300 mg. Articaine 4% at 88 mg per cartridge has a maximum of 7 mg per kilogram.

Second, **warfarin with an INR of 2.4** measured within 24 hours. SDCEP guidance is to treat without interrupting the warfarin when the INR is below 4.0, using local haemostatic measures: an atraumatic technique, sutures, oxidised cellulose or collagen in the site, pressure with gauze, and tranexamic acid mouthwash if bleeding is troublesome. I would treat early in the day and early in the week, and give clear written post-operative bleeding instructions with an emergency number.

Third, **simvastatin**: if an antibiotic were indicated, I would avoid clarithromycin and erythromycin because of the risk of rhabdomyolysis, and avoid azole antifungals. If prophylaxis is indicated I would use a single 3 g dose of amoxicillin. I would also avoid NSAIDs for analgesia because of the warfarin, and use paracetamol.

Technique for the LR5: I would use **buccal and lingual infiltration** rather than an inferior alveolar nerve block. This limits the dose of adrenaline, avoids a block in an anticoagulated patient where a haematoma in the pterygomandibular space is undesirable, and leaves the patient able to report a sharp sensation if the drill approaches the mental or inferior alveolar nerve. I would also plan the implant at least 5 mm anterior to the radiographic mental foramen, or measure the anterior loop on CBCT.

I would record the agent, the number of cartridges, the batch number, the technique used and the patient's response."

EXAMINER CHECKLIST

1. Identifies the beta-blocker/adrenaline interaction
2. Practical management (limit dose, aspirate, slow injection, or alternative agent)
3. Quotes cartridge content and maximum dose
4. Applies SDCEP warfarin guidance (INR <4.0, treat)
5. Local haemostatic measures listed
6. Avoids NSAIDs
7. Statin–macrolide interaction identified
8. Chooses infiltration and justifies it
9. Mental foramen/anterior loop precaution
10. Records agent, dose, batch and response

Running this station to time

Time check. The model answer is 549 words, 3 minutes 55 seconds at 140 words per minute — which leaves barely a minute once the medical history card has been read aloud, a cartridge selected from four and the syringe assembled. There are **three drugs and a site-specific technique** to cover, so the shape is a **fixed 60 seconds per drug problem** with the last minute for the injection technique and the records; the numbers must be stated, not derived.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Hands over the medical history card and indicates the four cartridges: "Select a local anaesthetic for a single implant at the LR5, justify it, and tell me what you would modify."	Checklists 1 and 2. **Propranolol is a non-selective beta-blocker**; adrenaline with unopposed alpha stimulation can give a **hypertensive response with reflex bradycardia**. Not an absolute contraindication — **limit the dose, aspirate, inject slowly**. The candidate must physically **pick up a cartridge and name it**: lidocaine 2% with adrenaline 1:80,000, **two cartridges**, or **prilocaine 3% with felypressin** to avoid adrenaline entirely.
0:45–1:30	Silent; may turn the cartridge to show the label.	Checklist 3, and it must be quoted, not worked out. **2.2 mL lidocaine cartridge = 44 mg; maximum 4.4 mg/kg to 300 mg. Articaine 4% = 88 mg per cartridge; maximum 7 mg/kg.** Two sentences. A candidate doing arithmetic aloud here loses thirty seconds and usually still gets it wrong.
1:30–2:45	"And the warfarin?"	Checklists 4 and 5. **SDCEP: INR measured within 24 hours and below 4.0 — treat without interrupting the warfarin.** Then the haemostatic measures as a list: **atraumatic technique · sutures · oxidised cellulose or collagen in the socket · pressure with gauze · tranexamic acid mouthwash if troublesome**; treat **early in the day and early in the week**, with **written bleeding instructions and an emergency number**.
2:45–3:45	"Anything else on that card?"	Checklists 6 and 7. **Simvastatin** — if an antibiotic is indicated, **avoid clarithromycin and erythromycin because of rhabdomyolysis**, and avoid azole antifungals; **single 3 g amoxicillin** if prophylaxis is indicated. Then, back to the warfarin: **avoid NSAIDs for analgesia, use paracetamol** — a mark candidates lose by having moved on from the anticoagulant.
3:45–4:30	"How will you actually anaesthetise the LR5?"	Checklists 8 and 9. **Buccal and lingual infiltration, not an inferior alveolar nerve block**, with all three reasons: it **limits the adrenaline dose**, it **avoids a pterygomandibular haematoma in an anticoagulated patient**, and it **leaves the patient able to report a sharp sensation** if the drill approaches the nerve. Then the site precaution: implant planned **at least 5 mm anterior to the radiographic mental foramen**, or the **anterior loop measured on CBCT**.
4:30–5:00	"Anything to record?"	Checklist 10 — **records**: agent, number of cartridges, **batch number**, technique used and the patient's response, plus the INR and its date; **what the patient is told** (bleeding instructions in writing, what to take for pain and what not to, when to ring); and **review**, with a same-day telephone call and the sutures checked at one week.

The compressed answer — what a strong candidate says in five minutes

Q: What do you notice on this card? "Three things: **propranolol**, a non-selective beta-blocker; **warfarin with an INR of 2.4 taken yesterday**; and **simvastatin**. And the site itself — the LR5 sits near the mental foramen."

Q: Deal with the propranolol. "Adrenaline in a patient on a **non-selective beta-blocker** acts on unopposed alpha receptors, which can produce a **hypertensive response with reflex bradycardia**. I would not call adrenaline absolutely contraindicated, but I would **limit the dose, aspirate carefully to avoid an**

intravascular injection, and inject slowly. My choice would be **lidocaine 2% with adrenaline 1:80,000, limited to two cartridges** — or **prilocaine 3% with felypressin** if I wanted to avoid adrenaline altogether."

Q: What is in a cartridge? "A 2.2 millilitre cartridge of **lidocaine 2% contains 44 milligrams**; the maximum is **4.4 milligrams per kilogram, to 300 milligrams**. **Articaine 4% is 88 milligrams a cartridge**, maximum **7 milligrams per kilogram**. Two cartridges of lidocaine is well inside the dose limit for this man; the limit I am imposing is for the adrenaline, not the lidocaine."

Q: And the warfarin? "The **INR is 2.4, measured within 24 hours**. **SDCEP guidance is to treat without interrupting the warfarin where the INR is below 4.0**, using local haemostatic measures: an **atraumatic technique**, **sutures**, **oxidised cellulose or collagen in the site**, **pressure with gauze**, and **tranexamic acid mouthwash** if bleeding is troublesome. I would treat **early in the day and early in the week**, so there is time to manage a bleed in working hours, and give **written post-operative bleeding instructions with an emergency number**. I would not stop the warfarin and I would not ask the physician to."

Q: Does the simvastatin matter? "It matters for what I prescribe. If an antibiotic were indicated I would **avoid clarithromycin and erythromycin**, because inhibition of statin metabolism risks **rhabdomyolysis**, and I would avoid azole antifungals for the same reason. If prophylaxis is indicated, a **single 3 gram dose of amoxicillin**. And separately — because of the warfarin — I would **avoid NSAIDs for analgesia and use paracetamol**."

Q: What technique at the LR5? "**Buccal and lingual infiltration, not an inferior alveolar nerve block.** Three reasons. It **limits the total adrenaline dose**, which is the point of the beta-blocker precaution. It **avoids a block in an anticoagulated patient**, where a haematoma in the pterygomandibular space is genuinely undesirable. And it **leaves the patient able to report a sharp sensation** if the drill approaches the mental or inferior alveolar nerve, which is a safety feature I do not want to give away."

Q: And the site itself? "I would plan the implant **at least 5 millimetres anterior to the radiographic mental foramen**, or **measure the anterior loop on the CBCT** and work from that, because the loop can extend several millimetres anterior to the foramen and is not visible on a panoramic."

Q: Records? "I record the **agent, the number of cartridges, the batch number, the technique used and the patient's response**, together with the **INR and the date it was taken**. I give the patient the written bleeding instructions and the emergency number, tell them to take **paracetamol and not ibuprofen**, and explain that they should ring if bleeding does not settle with twenty minutes of firm pressure. I **telephone the same evening** and review the closure at one week."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 the beta-blocker–adrenaline interaction identified · 4 SDCEP applied — INR below 4.0, treat without interrupting warfarin · 2 practical management: limit the dose, aspirate, inject slowly, or an alternative agent · 8 infiltration rather than a block, justified · 6 NSAIDs avoided, paracetamol used · 10 agent, dose, batch and response recorded.

Accept these late or on prompting: 3 cartridge content and maximum doses quoted · 5 the local haemostatic measures listed · 7 the statin–macrolide interaction · 9 mental foramen and anterior loop precaution.

The commonest way five minutes is wasted here: a lecture on the pharmacology of adrenaline and beta-adrenergic receptors. The mark is for the **interaction and what is done about it**, not for the mechanism explained at length — and the candidate who explains it beautifully arrives at the warfarin at three minutes, gives SDCEP in a rush and never reaches the infiltration decision, the anterior loop or the records. The other failure is stopping after the two obvious traps: **simvastatin is on that card deliberately**, and saying nothing about it reads as not having seen it.

OSCE 9 — Infection control and aseptic technique

Blueprint: "Show an awareness of the importance of infection control and aseptic technique"; "Discuss maintenance of infection control and the aseptic technique" (W + OS-P + SCE).

EQUIPMENT

a surgical trolley with a mix of appropriate and inappropriate items (sterile drapes, gown, gloves, sterile saline, an air-turbine handpiece, a non-sterile bur block, an implant in a damaged sterile pouch with a past expiry date, chlorhexidine mouthwash, steel curettes).

CANDIDATE INSTRUCTIONS

Prepare the trolley for implant surgery. Identify any items that are inappropriate and explain why. Describe your aseptic technique.

MODEL ANSWER

"Working through the trolley: I would **remove the air-turbine handpiece** — an air-driven handpiece risks driving air into the tissues and causing surgical emphysema and contamination; implant surgery requires an **electric surgical motor with sterile tubing**. I would remove the **non-sterile bur block**. I would **discard the implant** — the pouch is damaged and the expiry date has passed, so sterility cannot be assumed, and I would record that. I would remove the **steel curettes** from the implant tray, because steel scratches the titanium oxide layer; titanium, PEEK or carbon-fibre instruments are used on implant surfaces.

Environment: a designated clean surgery, surfaces cleaned and disinfected, doors closed, minimal traffic, only the equipment I need on the field.

Instruments: decontaminated and sterilised in line with HTM 01-05 — cleaning in a washer-disinfector or ultrasonic bath, inspection, packaging, a **vacuum type B autoclave** for hollow and lumened instruments, correct storage, and retained cycle records.

Team: surgical scrub, sterile gown, sterile gloves, mask, cap and eye protection for the operator and assistant.

Patient: **chlorhexidine 0.2% mouthrinse for one minute**, perioral skin disinfection and sterile drapes.

Field: sterile saline irrigation, single-use or use-limited drills, the implant handled only with its sterile carrier and never with gloved fingers or non-titanium instruments.

Finally: sharps handled with a re-sheathing device, an instrument and swab count, correct waste segregation, and recording of implant and graft batch numbers for traceability."

EXAMINER CHECKLIST

1. Rejects air-turbine and states why
2. Rejects non-sterile items
3. Rejects the damaged/expired implant and records it
4. Rejects steel instruments for the implant surface
5. Environment control
6. HTM 01-05 / vacuum autoclave and cycle records
7. Scrub, gown, gloves, mask, eye protection
8. Patient chlorhexidine rinse and draping
9. Sterile saline and implant handling
10. Sharps, waste, counts and batch traceability

Running this station to time

Time check. The model answer is 398 words — 2 minutes 51 seconds at 140 words a minute — which looks generous until you remember that your hands are on the trolley: lifting four wrong items off it and giving a reason for each takes far longer than reading the same sentence aloud. Budget **90 seconds for the four rejections, one clause of reason each**, and keep the remaining three minutes for the aseptic chain, from the room to the batch numbers.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Indicates the loaded trolley: "Prepare this trolley for implant surgery, identify anything inappropriate and explain why, then describe your aseptic technique."	Nothing scores yet, but the candidate must **put a hand on the trolley immediately** and state the order in one sentence — "I will take off what does not belong, then run the technique from the room through to the lot numbers." A definition of asepsis delivered before anything is touched is 30 seconds of nothing.

Time	The examiner does	What earns the mark
0:30–2:00	Watches in silence while items come off.	Checklists 1, 2, 3 and 4, **one clause of reason each**. Air-turbine off — **surgical emphysema and contaminated exhaust into the wound**; replaced by an **electric surgical motor with sterile irrigation tubing**. Non-sterile bur block off. The implant in the **damaged, out-of-date pouch discarded, and the discard recorded and reported**. **Steel curettes off the implant tray** — steel scratches the titanium oxide layer; titanium, PEEK or carbon fibre only.
2:00–3:15	"Describe your aseptic technique."	Checklists 5, 6 and 7. **Environment** — designated clean surgery, surfaces disinfected, doors closed, minimal traffic, only what is needed on the field. **Instruments** — washer-disinfector or ultrasonic, inspection, packaging, a **vacuum type B autoclave for hollow and lumened items under HTM 01-05, with cycle records retained**. **Team** — surgical scrub, sterile gown and gloves, mask, cap and eye protection, for the assistant as well as the operator.
3:15–4:15	"And the patient, and the field itself?"	Checklists 8 and 9. **Chlorhexidine 0.2% rinse for one minute**, perioral skin disinfection, sterile drapes. Then **sterile saline irrigation**, single-use or use-limited drills with a cycle count, and the implant **handled only by its sterile carrier** — never with a gloved finger, never with a non-titanium instrument.
4:15–5:00	"Anything to finish with?"	Checklist 10 — **sharps** into a re-sheathing device at the point of use, an **instrument and swab count before closure**, **waste segregated**, and **implant, graft and membrane lot numbers recorded for traceability**; then **what the patient is told** (a fresh implant was opened because the first pouch was compromised) and the **follow-up loop** — device incident reported, cycle record filed against the case.

The compressed answer — what a strong candidate says in five minutes

"**Four things come off this trolley before anything else.**

The air-turbine handpiece. An air-driven handpiece exhausts into the surgical field and can drive air into the tissues — surgical emphysema — so it is never used for a bone cut. I would use an **electric surgical motor with a reduction handpiece and sterile irrigation tubing**, torque and speed set at the console.

The bur block. Not sterile, so it does not belong on a sterile field.

This implant. The pouch is damaged and the expiry date has passed, so sterility cannot be assumed. It is discarded, a replacement opened, and I **record the batch number and the reason for the discard** and report it to the manufacturer as a device incident. I would not use it and I would not re-sterilise it.

The steel curettes. Steel scratches the titanium oxide layer and leaves metal debris at the interface. Anything that touches an implant surface is **titanium, PEEK or carbon fibre**.

What stays — the sterile drapes, gown, gloves, saline and the chlorhexidine — I will use.

The technique starts with the room. A designated clean surgery, surfaces cleaned and disinfected before the list, doors closed, traffic limited to operator, assistant and patient, and only the equipment I actually need laid out.

Instruments are decontaminated and sterilised to **HTM 01-05** — washer-disinfector or ultrasonic bath, inspection under magnification, packaging, then a **vacuum type B autoclave**, which is what hollow and lumened items such as the surgical handpiece and the irrigation line require. Sterile storage, pouch integrity and expiry checked at the point of use, and **cycle records retained**.

The team scrub — sterile gown and gloves, mask, cap and eye protection. The assistant scrubs too; an unscrubbed assistant handling the field is the commonest breach in practice.

The patient has **chlorhexidine 0.2% mouthrinse for one minute**, perioral skin disinfection, and sterile drapes so only the operative field is exposed.

On the field, irrigation is **sterile saline** — never tap water — copious, internal and external, to keep the osteotomy cool. Drills are single-use or use-limited and I keep count of their cycles. The implant comes out of its pouch **in its own sterile carrier and goes straight into the site**; I do not touch it with a gloved finger or lay it on the drape.

At the end, sharps go into a re-sheathing device and are disposed of at the point of use; I take an **instrument and swab count before closing**; waste is segregated; and the **implant, graft and membrane lot numbers go into the notes and onto the patient's implant card**, so the device is traceable. I would tell the patient that the first implant was discarded because its packaging was compromised, and record that I had told them."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 air-turbine rejected, with emphysema and the electric motor named · 3 damaged and expired implant discarded **and recorded** · 4 steel rejected for the titanium surface · 6 HTM 01-05, vacuum type B and cycle records · 9 sterile saline and the implant handled only by its carrier · 10 sharps, counts, waste and lot-number traceability.

Accept these late or on prompting: 2 the non-sterile bur block · 5 environment and traffic control · 7 scrub, gown, mask and eye protection for both operator and assistant · 8 chlorhexidine rinse and draping.

The commonest way five minutes is wasted here: opening with a textbook account of the chain of infection while the trolley sits untouched — the station was laid out with four deliberate errors on it, and a candidate who reaches them at three minutes forfeits marks that were within arm's length. The second failure is removing the expired implant without saying it is recorded and reported: spotting it is only half of checklist 3.

OSCE 10 — Post-operative care and advice

Blueprint: "Suggest appropriate and adequate postoperative care and advice including the use of post-operative painkillers and antibiotics" (OS-P + SCE).

EQUIPMENT

a simulated patient or the examiner acting as a patient; a blank post-operative instruction sheet.

CANDIDATE INSTRUCTIONS

You have just placed an implant at the UL1 with a bone graft and a collagen membrane. Give the patient their post-operative instructions.

DIRECTION OF THOUGHT

This is a **spoken** station: plain language, no jargon, structured, ending with warning signs and a contact number. Cover: bleeding, swelling, pain, medication, hygiene, diet, denture, activity, what is normal, what is not, review.

MODEL ANSWER

As spoken to the patient

"Everything went as planned. Here is what to expect and what I need you to do — I'll also give you this in writing.

Bleeding: some oozing today is normal. If it bleeds, bite firmly on the gauze I've given you for 20 minutes without checking it. Don't rinse today.

Swelling and bruising: these are normal and are usually worst around two to three days, then settle. Use a cold pack on the outside of the lip, 20 minutes on and 20 minutes off, for the first day, and sleep with an extra pillow.

Pain: take ibuprofen 400 mg with food, up to four times a day, and paracetamol 1 gram up to four times a day — you can take them together. Take the first dose now, before the numbness wears off. Avoid aspirin.

Antibiotics/mouthwash: [if prescribed] finish the whole course. Start the chlorhexidine mouthwash tomorrow, twice a day for two weeks — hold it in your mouth, don't swill vigorously.

Cleaning: brush all your other teeth normally, but **do not brush over the wound** until I tell you.

Diet: soft food, nothing hot today, and chew on the other side. Keep well hydrated.

Denture: leave your denture out for the next 48 hours; then bring it in and I'll adjust and reline it so it does not press on the graft.

Activity: no smoking at all — smoking is the biggest single thing that will make this graft fail. No alcohol and no strenuous exercise for two to three days.

What's not normal — contact me if: the bleeding doesn't stop with pressure; the pain gets worse after day three rather than better; you get a temperature or the swelling spreads; you feel something rough or a piece of material coming through the gum; or you have any numbness that hasn't gone by tomorrow. If you have **difficulty swallowing or breathing, that's an emergency — call 999**.

Here is my number and the out-of-hours number. I'll see you in a week to take the stitches out."

EXAMINER CHECKLIST

1. Bleeding advice
2. Swelling/cold pack/head elevation
3. Specific analgesia with doses and pre-emptive timing
4. Antibiotic/chlorhexidine instructions
5. Hygiene — no brushing over the wound
6. Diet
7. Denture management for a grafted site
8. Smoking, alcohol, exercise
9. Warning signs including the airway emergency
10. **Written** instructions, contact number and review appointment

Running this station to time

Time check. The model answer is 509 words — 3 minutes 38 seconds of pure speech at 140 words a minute — but this is spoken to a patient who interrupts, and a candidate who greets, hands over the sheet, checks understanding and answers one question will pass five minutes comfortably. Nothing can be dropped, because every heading is an instruction the patient needs; the compression is **one sentence per heading and no physiology at all**.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Sits as the patient, an implant with a graft and membrane just placed at the UL1: "So how did it go? What happens now?"	Checklists 1 and 2. One sentence that it went to plan and that everything will also be in writing. **Bleeding** — bite firmly on the gauze for **20 minutes without checking**, no rinsing today. **Swelling and bruising** — worst at **two to three days**, cold pack on the outside of the lip **20 minutes on and 20 off**, extra pillow tonight.
0:45–1:45	"What can I take for the pain?"	Checklist 3, the highest-value minute here. **Ibuprofen 400 mg with food up to four times a day and paracetamol 1 g up to four times a day, taken together**, with the **first dose now, before the numbness wears off**; avoid aspirin. Then checklist 4 — finish the whole antibiotic course, chlorhexidine from **tomorrow, twice daily for two weeks**, held not swilled.

Time	The examiner does	What earns the mark
1:45–3:00	"Can I clean it? Can I eat? What about my denture?"	Checklists 5, 6 and 7. **Brush everything else normally, nothing over the wound** until told. Soft food, nothing hot today, chew on the other side, drink plenty, no straws. Then the graft-specific mark: the **denture stays out completely for 48 hours, then comes in to be eased and relined, and is not worn until that is done** — pressure over a graft is a leading cause of failure.
3:00–4:15	"When can I have a cigarette?"	Checklists 8 and 9. **No smoking at all**, named as the single biggest thing that will make this graft fail; no alcohol; no strenuous exercise for two to three days. Then the warning signs, in the patient's words: bleeding that will not stop, pain **worse** after day three, temperature or spreading swelling, **something rough or gritty coming through the gum** — the graft or membrane exposing — any numbness still present tomorrow, and **difficulty swallowing or breathing means 999**.
4:15–5:00	"Is that everything?"	Checklist 10 — the **written sheet handed over**, the practice number and the **out-of-hours number**, **review in a week for the sutures** and a call this evening, then a check of understanding: "Tell me what you will do tonight if it bleeds." Records: that instructions were given, given in writing, and understood.

The compressed answer — what a strong candidate says in five minutes

"That all went to plan — the implant is in, with a graft and a small membrane over it. Let me tell you what to expect and what I need you to do, and you will have all of it in writing as well.

Bleeding. Some oozing today is normal, and it looks worse than it is because it mixes with saliva. If it bleeds, bite firmly on the gauze for **twenty minutes without checking it**. Do not rinse at all today.

Swelling and bruising. Both are normal, usually worst at **two to three days**, then they settle. Hold a cold pack against the outside of your lip, **twenty minutes on and twenty minutes off**, for the first day, and sleep with an extra pillow tonight.

Pain. Take **ibuprofen four hundred milligrams with food, up to four times a day**, and **paracetamol one gram, up to four times a day**. You can take them together, and together they work better than either alone. **Take your first dose now, before the numbness wears off** — do not wait for it to start hurting. Please avoid aspirin.

Your medicines. Finish the whole course of antibiotics. Start the mouthwash **tomorrow, twice a day for two weeks** — hold it in your mouth gently, do not swill it hard.

Cleaning. Brush all your other teeth normally, but **do not brush over the wound** until I tell you it is safe. Keeping the rest of your mouth clean matters more than usual just now.

Eating. Soft food, nothing hot today, and chew on the other side. Drink plenty, but **no straws**.

Your denture. Leave it out completely for the **next forty-eight hours**. Then bring it to me and I will ease it and put a soft lining in it. **Please do not wear it until I have done that** — pressure on a graft is one of the main reasons it fails. Leave it out at night from now on.

Smoking, alcohol and exercise. No smoking at all — smoking is the single biggest thing that will make this graft fail. No alcohol, and no hard exercise or heavy lifting for two to three days, because that can restart the bleeding.

Ring me if the bleeding does not stop after twenty minutes of firm pressure; if the pain gets **worse** after day three instead of better; if you get a temperature or the swelling starts spreading; if you feel something rough or gritty coming through the gum, because that can be the graft or the membrane showing and I need to see it; or if any numbness in your lip is still there tomorrow morning.

And this one is an emergency — call 999 if you get difficulty swallowing or any difficulty breathing.

Here is the written sheet with all of that on it, my number and the out-of-hours number. I will see you **in a week to take the stitches out**, and I will ring you this evening. Before you go — just tell me what you would do tonight if it started bleeding?"

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 3 analgesia by drug, dose, frequency and pre-emptive timing · 9 warning signs including exposure of the graft and the airway 999 emergency · 10 written sheet, contact number and review · 7 denture out 48 hours, then eased and relined before wear · 8 no smoking, named as the biggest cause of graft failure · 1 bleeding advice with a stated duration of pressure.

Accept these late or on prompting: 2 swelling, cold pack and head elevation · 4 antibiotic and chlorhexidine instructions · 5 no brushing over the wound · 6 diet.

The commonest way five minutes is wasted here: explaining rather than instructing — osseointegration, why a membrane was used, how bone grafts remodel. The patient did not ask and it is not on the checklist, and the minute it costs is the minute in which the airway warning, the written sheet and the review appointment were supposed to be said. The other loss is generic "take painkillers as needed": checklist 3 wants two drugs, two doses, two frequencies and the first dose taken now.

OSCE 11 — The surgical record

Blueprint: "Demonstrate understanding of the role of clear, accurate, contemporaneous record keeping" (OS-P + SCE).

EQUIPMENT

a blank clinical record proforma, an implant packaging label with lot number, a graft box, a membrane packet.

CANDIDATE INSTRUCTIONS

Complete the clinical record for the implant placement you have just performed. Say aloud what you are writing and why.

MODEL ANSWER

"I would record, contemporaneously: date and time; the operator and assistant; that the medical history was checked and unchanged; **that consent was confirmed and the risks re-stated on the day**; the local anaesthetic — agent, concentration, vasoconstrictor, number of cartridges, technique, batch number; that aseptic technique was used; the flap design; the site; the drilling sequence; the implant **system, diameter, length and lot number, with the label stuck into the notes**; the **insertion torque** and, if measured, the ISQ; whether a cover screw or healing abutment was placed; any augmentation — **graft material, brand and lot number, membrane type and lot number**; suture material and technique and the number of sutures; any complication and how it was managed; the post-operative instructions given, **and that they were given in writing**; medication prescribed with doses; the review arrangements; and my signature.

I would also give the patient an **implant record card** with the system, diameter, length and lot number so that any future clinician can identify and service the implant. That is both good practice and a medical device traceability requirement.

Records must be contemporaneous — if I have to add anything later I would date and sign the addition rather than alter the original entry."

EXAMINER CHECKLIST

1. Date/operator/assistant
2. MH checked, consent confirmed on the day
3. LA details including batch
4. Flap and site
5. Implant system, diameter, length, **lot number**
6. Insertion torque ± ISQ
7. Graft and membrane with lot numbers
8. Sutures and closure

9. Post-operative instructions given in writing, medication, review
10. Patient implant card and the principle of contemporaneous, unaltered records

Running this station to time

Time check. The model answer is 335 words — 2 minutes 24 seconds at 140 words a minute, the shortest in this part, and the only station where the clock is not the enemy. The enemy is silence: the instruction is *say aloud what you are writing*, and an entry made quietly scores nothing. Spend the whole five minutes at **dictation pace, reading the lot numbers off the label, the box and the packet in front of you** rather than from memory.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Puts the blank proforma, the implant label, the graft box and the membrane packet in front of the candidate: "Complete the record for the placement you have just done. Say aloud what you are writing and why."	Checklists 1 and 2. The candidate **picks up the pen and starts writing while speaking** — one sentence that the entry is contemporaneous, made now and not at the end of the list — then dictates **date and time, operator and assistant by name**, that the **medical history was checked and unchanged**, and that **consent was confirmed and the risks re-stated on the day**, not merely signed a fortnight ago.
0:45–2:00	Silent. Lets the candidate run the sequence.	Checklists 3 and 4, spoken as figures. **Local anaesthetic** — agent, concentration, vasoconstrictor ratio, **number of cartridges, technique and batch number**. Then that aseptic technique was used, the **flap design**, the **site in FDI notation**, and the drilling sequence with the final drill diameter.
2:00–3:15	May pick up the implant label and slide it across.	Checklists 5 and 6, the core of the station. The candidate **reads aloud from the label** — system, platform, **diameter and length in millimetres, and the lot number** — and **sticks the peel-off label into the notes** rather than transcribing it. Then the **insertion torque in newton centimetres**, the **ISQ if measured**, and whether a cover screw or a healing abutment was placed, with its height.
3:15–4:15	Slides the graft box and the membrane packet across.	Checklists 7 and 8. **Graft material, brand and lot number read off the box**; **membrane type and lot number read off the packet**; both with the site and volume placed. Then **suture material, technique and number of sutures**, the closure achieved, and any complication with how it was managed — recorded even when nothing went wrong, stated explicitly.
4:15–5:00	"Anything else?"	Checklists 9 and 10 — the marks candidates lose. **Post-operative instructions given, and given in writing**, medication prescribed with doses, and the **review arrangements**. Then the **patient implant card** handed over with system, diameter, length and lot number, and the principle stated: records are **contemporaneous and never altered** — a later addition is dated, timed and signed as an addition. Signature at the end.

The compressed answer — what a strong candidate says in five minutes

"I am writing this now, at the chairside, because the record has to be contemporaneous — and I will read the numbers off the packaging rather than from memory.

Date and time, today's date, fourteen twenty. **Operator**, myself; **assistant**, the nurse by name. **Medical history checked and unchanged. Consent confirmed on the day and the risks re-stated** — I write that explicitly, because a consent form signed at the planning visit does not record what was said this morning.

Anaesthetic. Articaine four per cent with adrenaline one in one hundred thousand, two point two millilitres, two cartridges, buccal infiltration with palatal infiltration; **batch number read off the cartridge box** and written in.

Technique. Aseptic technique used throughout. A crestal incision at the **UL1** with a single relieving incision distal to the UL2, full-thickness mucoperiosteal flap raised. Sequential osteotomy under copious sterile saline irrigation, final drill three point five millimetres.

The implant — and I take this straight from the label. The system, platform and connection; **four point one millimetres by ten millimetres**; **lot number, which I read out rather than recall — and I stick the peel-off label into the notes here**, because a transcribed number can be transcribed wrongly. **Insertion torque, thirty-five newton centimetres.** ISQ measured at seventy-two. **Cover screw placed**, submerged healing.

Augmentation. Deproteinised bovine bone mineral, small particle, half a gram, **brand and lot number from the box**, placed buccally to correct a two-millimetre dehiscence. **Resorbable collagen membrane**, type and **lot number from the packet**, trimmed and placed over the graft, tucked under the flap margins.

Closure. Periosteal releasing incision to achieve tension-free primary closure. Four interrupted five-oh polypropylene sutures. **No complication** — and I write that, rather than leave it blank, because a blank space records nothing.

Then the part that is most often missing. Post-operative instructions given verbally and in writing, sheet number recorded, and the patient repeated them back. Analgesia advice given; amoxicillin three grams pre-operatively; chlorhexidine 0.2 per cent, twice daily for two weeks. **Review in seven days for suture removal**, and a telephone call this evening.

The patient's implant card, completed with the system, diameter, length and lot number, and handed to her — so any clinician in ten years can identify and service this implant. That is good practice and a **medical device traceability requirement**.

Then I sign and print my name. If I have to add anything later, I add it as a **dated and signed addition** — I do not alter, overwrite or squeeze anything into the original entry, because an altered record is worth less than no record at all."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 5 implant system, diameter, length and **lot number, with the label in the notes** · 7 graft and membrane brands with lot numbers · 2 medical history checked and consent confirmed **on the day** · 6 insertion torque, with ISQ if measured · 10 patient implant card and contemporaneous, unaltered records · 9 post-operative instructions given **in writing**, medication and review.

Accept these late or on prompting: 1 date, time, operator and assistant · 3 local anaesthetic detail including batch · 4 flap design and site · 8 suture material, technique and number.

The commonest way five minutes is wasted here: writing in silence. The candidate produces a beautiful entry, hands it over at 4:30, and has scored almost nothing, because the examiner marks what was said. The second loss is reciting the headings — "I would record the implant details" — instead of speaking the actual figures off the label and the box that were deliberately placed on the table for that purpose.

OSCE 12 — Abutment selection

Blueprint: "Select the appropriate abutment to restore the implant" (W + OS-P + SCE).

> **EQUIPMENT**
>
> models of (a) an ideally placed UR1 implant with 3 mm of soft tissue, (b) a buccally inclined UL2 implant, (c) an LR6 implant with 2 mm of soft tissue; a tray of abutments — stock straight titanium, angled titanium, Ti-base, custom CAD-CAM, zirconia, multi-unit; and an angulated screw channel abutment.

CANDIDATE INSTRUCTIONS

For each of the three implants, select an appropriate abutment and justify your choice.

MODEL ANSWER

"I decide in a fixed order: retention type, level, angulation, height and margin position, material, and stock versus custom.

(a) UR1, ideally placed, 3 mm of soft tissue. I would restore this **screw-retained** at implant level, using a **custom CAD-CAM abutment or a Ti-base with a milled ceramic superstructure**. My reasons: the emergence profile through 3 mm of tissue needs to be customised to support the papillae and the mid-facial margin; screw retention removes any cement risk at that depth; and a Ti-base gives a titanium-to-titanium connection at the implant with an aesthetic ceramic above. If the tissue were thin I would avoid a plain titanium abutment because of grey show-through.

(b) UL2, buccally inclined. The screw channel would emerge through the labial surface. My options are an **angulated screw channel abutment**, which lets me keep screw retention with the access hole moved palatally — my preferred solution — or a **cement-retained restoration on an angled abutment**, in which case I would keep the margin at or no more than 1 mm below the mucosal margin, use a minimal volume of radiopaque cement, and take a radiograph to check for excess.

(c) LR6, 2 mm of soft tissue, posterior. A **stock or custom straight titanium abutment, screw-retained**, is appropriate. Aesthetics are not critical; titanium gives maximum strength for posterior loading; screw retention gives retrievability, which matters most in the site with the highest mechanical complication rate.

In all three I would confirm that the abutment gives adequate restorative space — about 5 to 6 mm for a screw-retained crown and 7 to 8 mm for cement retention — and I would torque to the manufacturer's value and re-torque after ten minutes."

EXAMINER CHECKLIST

1. States a systematic decision order
2. Screw retention preferred where angulation allows
3. Custom abutment for emergence profile in the aesthetic zone
4. Ti-base rationale/avoids zirconia-on-titanium interface
5. Grey show-through with thin tissue considered
6. ASC abutment identified for the inclined implant
7. If cementing — margin position, radiopaque cement, minimal volume, radiograph
8. Titanium chosen for posterior strength
9. Restorative space requirements quoted
10. Torque and re-torque at 10 minutes

Running this station to time

Time check. The model answer is 449 words — 3 minutes 12 seconds at 140 words a minute — but there are three models to pick up, an abutment tray to choose from three times, and a "why?" after each, so the honest delivery is nearer seven minutes. The fix is a **fixed 60 seconds per model** — abutment named, one sentence of justification, one caveat — after 30 seconds spent stating the decision order, leaving the last minute for restorative space and torque.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Sets out the three models and the abutment tray: "For each of these implants, select an appropriate abutment and justify your choice."	Checklist 1. The candidate states the **decision order in one sentence before touching a model** — retention type, implant or abutment level, angulation, height and margin position, material, stock or custom — and adds the governing preference, **screw retention wherever angulation allows** (which pre-earns checklist 2 and buys permission to be brief three times over).

Time	The examiner does	What earns the mark
0:30–1:45	Pushes model (a) forward — the ideally placed UR1 with 3 mm of soft tissue.	Checklists 3, 4 and 5. **Custom CAD-CAM abutment or a Ti-base with a milled ceramic superstructure, screw-retained**, because the emergence through 3 mm of tissue must be customised to support the papillae and the mid-facial margin. **Ti-base rationale** — a titanium-to-titanium connection at the implant, avoiding zirconia bearing directly on the titanium platform. And **grey show-through** named as the reason plain titanium is avoided under thin tissue.
1:45–2:45	Pushes model (b) forward — the buccally inclined UL2 — and picks up the ASC abutment: "What is this for?"	Checklists 6 and 7. The screw channel would emerge **through the labial surface**, so an **angulated screw channel abutment** keeps screw retention with the access moved palatally — stated as the preferred solution. The cement alternative is given with its four conditions in one clause: **margin at or no more than 1 mm below the mucosa, minimal volume, radiopaque cement, radiograph to check for excess**.
2:45–3:45	Pushes model (c) forward — the LR6 with 2 mm of soft tissue.	Checklist 8, with checklist 2 restated. **Stock or custom straight titanium, screw-retained.** Aesthetics are not critical, titanium gives maximum strength under posterior load, and retrievability matters most in the site with the highest rate of screw loosening and ceramic fracture. Thirty seconds is enough here; the mark is the choice and the reason, not a survey of alternatives.
3:45–5:00	"How much space do you need, and how would you fit it?"	Checklists 9 and 10. **5 to 6 mm for a screw-retained crown, 7 to 8 mm for cement retention**, measured from platform to opposing occlusal plane. Then **torque to the manufacturer's value with a calibrated wrench and re-torque after ten minutes** for embedment relaxation; then **records** (abutment type, material, batch and torque value), **what the patient is told** (which components are in, and the implant card), and the **review** with a baseline radiograph.

The compressed answer — what a strong candidate says in five minutes

Q: How do you approach a choice like this? "In a fixed order, every time: **retention type, level, angulation, height and margin position, material, then stock against custom**. And a governing preference — **screw retention wherever the angulation allows it**, and always where the margin would sit more than about a millimetre below the mucosa."

Q: Take the first model, the UR1. "Ideally placed, 3 mm of tissue, aesthetic zone. I would restore it **screw-retained at implant level on a custom CAD-CAM abutment, or a Ti-base with a milled ceramic superstructure**. The emergence profile through 3 mm of tissue has to be customised to support the papillae and the mid-facial margin — a stock abutment cannot do that. And screw retention removes the cement risk entirely at that depth, where excess would be undetectable."

Q: Why a Ti-base rather than full zirconia onto the implant? "Because it gives a **titanium-to-titanium connection at the implant interface**. Zirconia bearing directly on a titanium platform is a harder material against a softer one, with wear of the platform and microgap concerns; the Ti-base keeps the metal-to-metal joint and puts the ceramic above it."

Q: Anything about the tissue itself? "Yes — if the tissue were **thin, I would avoid a plain titanium abutment because of grey show-through** at the margin. That is an argument for a ceramic superstructure on the Ti-base, or for a zirconia abutment above a titanium base."

Q: The UL2 is buccally inclined. "Then the screw channel emerges **through the labial surface**, which is unrestorable as a screw-retained crown. My preferred answer is this — the **angulated screw channel**

abutment — which redirects the access up to about 25 degrees and lets me keep screw retention with the hole moved palatally. My alternative is a **cement-retained crown on an angled abutment**, and if I do that I keep the **margin at or no more than 1 mm below the mucosal margin**, use a **minimal volume of radiopaque cement**, seat it on a replica extra-orally to express excess, and **take a radiograph to check for residual cement**."

Q: And the LR6? "Two millimetres of tissue, posterior, aesthetics not critical. A **stock or custom straight titanium abutment, screw-retained**. Titanium gives maximum strength for posterior loading, and screw retention gives retrievability — which matters most at the molar, the site with the highest mechanical complication rate."

Q: How much space do these need? "Measured from the implant platform to the opposing occlusal plane, about **5 to 6 mm for a screw-retained crown and 7 to 8 mm for cement retention** — cement needs more because of the abutment height required for retention. I confirm that on the articulated casts before I prescribe, not at the fit."

Q: And fitting it? "Clean the internal connection, seat fully, **confirm seating radiographically before torquing**, then **torque to the manufacturer's value with a calibrated wrench and re-torque after ten minutes** to compensate for embedment relaxation. I record the abutment type, material, batch number and torque value, give the patient the implant card and site-specific hygiene instruction, take a **baseline periapical**, and review."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 a stated decision order · 3 custom abutment for emergence in the aesthetic zone · 6 ASC abutment identified for the inclined implant · 7 the four cement conditions if cementing · 9 restorative space quoted, 5–6 mm against 7–8 mm · 10 torque and re-torque at ten minutes.

Accept these late or on prompting: 2 screw retention preferred where angulation allows · 4 Ti-base rationale and the zirconia-on-titanium interface · 5 grey show-through with thin tissue · 8 titanium chosen for posterior strength.

The commonest way five minutes is wasted here: touring the abutment tray. A candidate who names and characterises all seven components before addressing model (a) spends two minutes on material science and reaches the LR6 at 4:40, losing the space figures and the torque protocol. The other failure is answering (b) with "cement-retained" alone: the ASC abutment was put on the tray to be picked up, and not picking it up reads as not recognising it.

OSCE 13 — Screw- versus cement-retained: decision and steps

Blueprint: "Show an understanding of the different steps involved in the provision of a screw retained prosthesis"; "difference between screw and cement retained restoration and their application" (W + OS-P + SCE).

CANDIDATE INSTRUCTIONS

Explain the differences between screw- and cement-retained implant restorations and describe the clinical and laboratory steps for providing a screw-retained crown on an LR5 implant.

MODEL ANSWER

"The essential difference is **retrievability**. A screw-retained crown can be removed and replaced without damage, which matters because implant restorations require maintenance — screws loosen, ceramic chips, and biological problems need access. A cement-retained crown is difficult or destructive to remove.

The second key difference is **cement**. Residual submucosal cement is a well-documented cause of peri-implant disease, and screw retention eliminates that risk entirely.

Screw retention also needs **less restorative space** — about 5 to 6 mm compared with 7 to 8 mm — and demands a **passive framework**, because there is no cement layer to absorb misfit.

Cement retention has two advantages: it can compensate for unfavourable implant angulation where the screw channel would emerge through an aesthetic or occlusal surface, and it gives an unin-

terrupted occlusal surface. Angulated screw channel abutments have reduced the first advantage considerably.

My preference is screw retention wherever angulation allows, and always where the margin would sit more than about 1 mm below the mucosa.

Steps for the screw-retained crown at LR5: 1. Confirm osseointegration clinically and radiographically, and check the peri-implant tissues. 2. Remove the healing abutment; assess the emergence profile and soft tissue. 3. Take an **implant-level impression** — a closed-tray coping is acceptable for a single unit, an open tray if access allows — in polyether or addition silicone in a custom tray; verify the coping is fully seated, radiographically if it is deep. 4. Record the opposing arch, the intercuspal relationship and the shade; take photographs. 5. Written laboratory prescription: implant system, platform, connection, **screw-retained crown**, material, shade, contact points, emergence and occlusal scheme. 6. At try-in: verify complete seating clinically and radiographically, check proximal contacts with floss, check emergence and cleansability, check the occlusion. 7. Fit: clean the internal connection, seat the crown, **hand-tighten, radiograph to confirm seating, then torque to the manufacturer's value**, check the occlusion is light in intercuspation and free in excursions, **re-torque after 10 minutes**. 8. Seal the screw access with PTFE tape over the screw head and composite over that, so the screw can be retrieved. 9. Give hygiene instruction specific to the site, take a **baseline periapical radiograph**, record everything, and arrange review and maintenance."

EXAMINER CHECKLIST

1. Retrievability
2. Cement risk
3. Space requirement difference
4. Passive fit requirement
5. Angulation/aesthetics advantage of cement, and ASC as the alternative
6. States a preference with justification
7. Correct impression sequence and verification
8. Laboratory prescription content
9. Torque, radiographic verification, re-torque at 10 min, light occlusion
10. PTFE + composite seal, baseline radiograph, hygiene and review

Running this station to time

Time check. The model answer is 495 words — 3 minutes 32 seconds at 140 words a minute — but it carries a five-part comparison *and* a nine-step clinical sequence, and every one of those steps invites a "why?", so the honest delivery is past six and a half minutes. The compression is a **90-second cap on the comparison — four differences, one clause each, then the stated preference** — with the steps run at roughly ten seconds apiece, grouped into impression, prescription, try-in and fit.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:20	"Explain the difference between screw- and cement-retained restorations, then describe the steps for a screw-retained crown on an LR5."	Checklist 1, immediately. **Retrievability** named as *the* essential difference in the first sentence, with the reason — implant restorations need maintenance: screws loosen, ceramic chips, biological problems need access. Do not begin with a list of advantages and disadvantages.
0:20–1:30	Silent. Lets the comparison run.	Checklists 2, 3, 4 and 5, **one clause each**. **Residual submucosal cement** as a documented cause of peri-implant disease, eliminated entirely by screw retention. **Space** — 5 to 6 mm against 7 to 8 mm. **Passive fit** demanded by screw retention because there is no cement layer to absorb misfit. Cement's two advantages — **unfavourable angulation and an uninterrupted occlusal surface** — with **ASC abutments having largely removed the first**.

Time	The examiner does	What earns the mark
1:30–2:00	"So which would you use?"	Checklist 6. A **stated preference with justification**, not a survey: screw retention wherever the angulation allows, and **always where the margin would sit more than about 1 mm below the mucosa**, because excess cement at that depth cannot reliably be detected or removed.
2:00–3:15	"Now the steps at the LR5."	Checklists 7 and 8, run fast. Integration confirmed **clinically and radiographically**, tissues checked, healing abutment out, emergence assessed. **Implant-level impression** — closed tray acceptable for a single unit, open tray if access allows — in **polyether or addition silicone in a custom tray**, with the coping **verified fully seated, radiographically if deep**. Opposing arch, interocclusal record, shade and photographs. Then the **written prescription** — system, platform, connection, **screw-retained**, material, shade, contacts, emergence and occlusal scheme.
3:15–4:30	"And the fit appointment?"	Checklist 9. Try-in seating verified clinically and radiographically, contacts by floss, emergence and cleansability, occlusion. Then at fit: connection cleaned, crown seated, **hand-tightened, radiographed to confirm seating, then torqued to the manufacturer's value**; occlusion **light in intercuspation and free in excursions**; **re-torqued after ten minutes**.
4:30–5:00	"Finish off."	Checklist 10 — **PTFE tape over the screw head then composite** so the screw stays retrievable, with the occlusion re-checked afterwards; a **baseline periapical radiograph**; **records** (abutment, torque value, batch numbers, occlusal scheme); **what the patient is told** — site-specific hygiene, the implant card, that the crown can be unscrewed if needed; and the **maintenance recall interval**.

The compressed answer — what a strong candidate says in five minutes

Q: What is the essential difference? "**Retrievability.** A screw-retained crown comes off and goes back on without damage. That matters because implant restorations require maintenance over a lifetime — screws loosen, ceramic chips, and biological problems need access to the platform. A cement-retained crown is difficult or destructive to remove, so a small problem becomes a remake."

Q: And the second difference? "**Cement.** Residual submucosal cement is a well-documented cause of peri-implant mucositis and peri-implantitis, and it is hard to detect and harder to remove once the margin sits deep. **Screw retention eliminates that risk entirely.**"

Q: Any technical differences? "Two. Screw retention needs **less restorative space — about 5 to 6 mm against 7 to 8 mm** — because it does not need abutment height for retention. And it demands a **passive framework**, because there is no cement layer to absorb a misfit; the stress goes straight into the components and the bone."

Q: What does cement retention offer? "It compensates for **unfavourable angulation** where the screw channel would emerge through an aesthetic or occlusal surface, and it gives an **uninterrupted occlusal surface**. But **angulated screw channel abutments have largely removed the first advantage**, which is why the balance has shifted."

Q: Your preference? "**Screw retention wherever the angulation allows**, and always where the margin would sit more than about a millimetre below the mucosa. I would accept cement only with a supragingival or just-subgingival margin, radiopaque cement, minimal volume, and a radiograph."

Q: Take me through the LR5. "First I confirm **osseointegration clinically and radiographically** and that the peri-implant tissues are healthy. I remove the healing abutment and assess the emergence profile and the soft tissue.

Then the **implant-level impression**. For a single unit a **closed-tray coping is acceptable**; I would use an **open tray if access allows**, in a **custom tray**, with **polyether or addition silicone** — polyether for its rigidity and dimensional stability. I **verify the coping is fully seated**, and if it sits deep I confirm that **radiographically** rather than by feel. I record the opposing arch, the interocclusal relationship and the shade, and I take photographs."

Q: What goes to the laboratory? "A **written prescription**: implant system, platform and connection, that it is a **screw-retained crown**, the material, the shade, the contact points, the emergence profile and the occlusal scheme. Not 'please make a crown'."

Q: Try-in and fit? "At try-in I verify **complete seating clinically and radiographically**, check the proximal contacts with floss, check the emergence is cleansable with an interdental brush, and check the occlusion.

At fit I clean the internal connection, seat the crown, **hand-tighten and take a periapical to confirm seating before I torque anything** — a wedge-shaped gap means it is not seated and I would never drive it home with the wrench. Then **torque to the manufacturer's value with a calibrated wrench**, check the occlusion is **light in intercuspation and free in excursions**, and **re-torque after ten minutes** for embedment relaxation."

Q: And to finish? "**PTFE tape over the screw head, composite over that**, so it remains retrievable, then re-check the occlusion because the composite adds height. A **baseline periapical**, photographs, and records of the abutment, torque value, batch numbers and occlusal scheme. The patient gets their implant card, site-specific hygiene instruction — interdental brush, single-tufted brush — and a **maintenance recall**."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 retrievability named as the essential difference · 2 residual cement as a cause of peri-implant disease · 6 a stated preference with justification · 9 hand-tighten, radiograph, torque, light occlusion, re-torque at ten minutes · 7 implant-level impression with verified seating · 10 PTFE and composite seal, baseline radiograph, hygiene and recall.

Accept these late or on prompting: 3 the space difference, 5–6 mm against 7–8 mm · 4 the passive fit requirement · 5 cement's angulation advantage and the ASC alternative · 8 the content of the laboratory prescription.

The commonest way five minutes is wasted here: treating the comparison as the whole station. The verb is *explain* **and** *describe the steps*, and the steps carry four of the ten marks; a candidate who spends three minutes weighing screw against cement arrives at the impression with 90 seconds left and never reaches the torque protocol or the access seal. Ninety seconds on the comparison, then move to the LR5.

OSCE 14 — Implant impression (practical)

Blueprint: "Apply knowledge of the different impression techniques used when restoring implants"; "Justify the choice of impression material and technique" (W + OS-P + SCE).

EQUIPMENT

a model with three implant analogues in the lower left quadrant (one divergent); open- and closed-tray copings, guide screws, a custom tray with windows, stock trays, impression materials, pattern resin, dental floss.

CANDIDATE INSTRUCTIONS

Take, or describe taking, a definitive impression of these three implants. Justify your choice of technique and material.

MODEL ANSWER

"With three implants, one of which is divergent, I would use an **open-tray (pick-up) technique with splinted copings**, because the closed-tray technique relies on re-seating the copings into the impression, and with divergent implants that re-seating is unreliable.

Steps: remove the healing abutments and clean the connections. Seat the **open-tray impression copings**, confirm each is fully seated by feel and, because these are deep, with a **periapical radiograph**. Splint the copings with **pattern resin** — I would section the resin once set and re-join it, to compensate for polymerisation shrinkage — or use floss and light-cured composite.

Check the **custom tray** for fit and that the guide screws project freely through the windows without contacting the tray. Apply the correct tray adhesive and allow it to dry.

Material: **polyether**, because it is rigid, hydrophilic and dimensionally stable, which suits splinted copings. Addition silicone of heavy and light body is an acceptable alternative — but I would not use latex gloves with addition silicone, because sulphur inhibits the set.

Load the tray, syringe material around each coping, seat, and hold still until fully set. Then **unscrew the guide screws completely** before removing the impression, remove the impression, attach the analogues, and check that each coping is fully embedded and undistorted.

Also record: the opposing arch, an interocclusal record on rigid bases, the shade with photographs, and a facebow if the occlusal scheme is being altered. Disinfect the impression and send it with a written prescription."

EXAMINER CHECKLIST

1. Chooses open tray and justifies it
2. Splinting with sectioned/re-joined resin
3. Verifies coping seating radiographically
4. Custom tray with windows, checks screw clearance
5. Adhesive applied and dried
6. Selects polyether/PVS with justification
7. Latex/PVS incompatibility noted
8. Unscrews guide screws before removal
9. Attaches analogues and inspects
10. Opposing arch, records, disinfection, prescription

Running this station to time

Time check. The model answer is 392 words — 2 minutes 48 seconds at 140 words a minute — but that figure is misleading, because this is a **practical** station and narrating while the hands work is far slower than reading aloud: seating three copings, checking each is home, splinting with pattern resin, sectioning and re-joining it, and painting and drying tray adhesive cannot be done at speaking speed. Budget six minutes of real work into five by **justifying once at the start, then giving each step a single clause while the hands keep moving**.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Indicates the model with three analogues, one divergent, and the two sets of copings: "Take, or describe taking, a definitive impression of these three implants. Justify your technique and material."	Checklist 1, before anything is picked up. **Open-tray pick-up technique with splinted copings**, with the reason stated in the same breath — the **closed-tray technique relies on re-seating the copings back into the set impression, and with a divergent implant that re-seating is unreliable**. The candidate should then reach for the **open-tray copings**, not the closed-tray ones.
0:30–1:45	Watches the hands. May say: "How do you know they are seated?"	Checklists 3 and 4. Healing abutments off, connections irrigated and inspected, **open-tray copings seated and each confirmed home by feel and by a periapical radiograph**, because these are deep and a coping that is not fully seated reproduces the wrong position. Then the **custom tray tried in** and the **guide screws confirmed projecting freely through the windows without touching the tray**.

Time	The examiner does	What earns the mark
1:45–3:00	Silent. Hands over pattern resin and floss.	Checklists 2 and 5. **Splinting** — floss between the copings, pattern resin applied in small increments, then **sectioned once set and re-joined**, with the reason given in one clause: **polymerisation shrinkage**. Then **tray adhesive applied to the tray and allowed to dry fully** — said out loud, because it is invisible and unmarked otherwise.
3:00–4:30	"Which material, and why?" Then silence while the impression is taken.	Checklists 6, 7, 8 and 9. **Polyether** — rigid, hydrophilic, dimensionally stable, which is what splinted copings need; **addition silicone heavy and light body** an acceptable alternative, **but not with latex gloves, because sulphur inhibits the set**. Load, syringe around each coping, seat, hold still to full set. Then **unscrew all three guide screws completely before removal**, remove, **attach the analogues** and inspect every coping for full embedment and undistorted resin.
4:30–5:00	"Anything else before it goes?"	Checklist 10 — **opposing arch impression, an interocclusal record on rigid bases, shade with photographs, and a facebow if the occlusal scheme is being altered**; then **disinfection** to the manufacturer's protocol, the **written laboratory prescription**, **what the patient is told** (a healing abutment replaced, the crown at the next visit, ring if anything feels sharp), and the **review** — the try-in appointment booked.

The compressed answer — what a strong candidate says in five minutes

"**Before I touch anything, the choice and the reason.** Three implants, one of them divergent, so I would take an **open-tray pick-up impression with the copings splinted**. A closed-tray impression relies on unscrewing the copings, removing the impression, and **re-seating each coping back into its own space** — and with a divergent implant that re-seating is unreliable, so the cast reproduces the wrong angulation. In an open-tray technique the copings come out **in** the impression and never move.

So — healing abutments out, connections irrigated and inspected for tissue or debris.

Copings on. These are the **open-tray copings** with the long guide screws. I seat each one, tighten by hand, and confirm it is fully home. These sit deep, so I do not accept feel alone — I take a **periapical radiograph** of each. A coping that is a fraction unseated makes an impression that is accurate to nothing.

Tray next. The **custom tray with windows**: I try it in and confirm that all three guide screws **project freely through the windows without contacting the tray**, because a screw touching the tray lifts the impression and distorts it. I adjust the windows if needed.

Splinting. I connect the copings with **floss and then pattern resin**, applied in small increments to limit the shrinkage. Once set I **section the resin between each pair and re-join the sections with a small amount of fresh resin**, which compensates for polymerisation shrinkage. Floss and light-cured composite does the same job.

Adhesive. Correct adhesive for the material, painted onto the tray, and **allowed to dry fully** before I load it.

Material — polyether. It is **rigid**, so it holds the splinted copings without flexing on removal; it is **hydrophilic**, so it copes with a moist sulcus; and it is **dimensionally stable**. Addition silicone in heavy and light body is an acceptable alternative — but **I would not handle it in latex gloves, because sulphur inhibits the set** and you get an unset layer exactly where the detail matters.

Taking it. Syringe material around each coping and into the sulcus, load and seat the tray, and hold it absolutely still until it is fully set — not just until it feels firm.

Then, and this is the step that is forgotten: I unscrew all three guide screws completely before I remove the impression. If I pull on a tray with a screw still engaged I fracture the impression or damage the implant.

Out of the mouth, I inspect it: each coping fully embedded, none rotated, the resin intact and undistorted, no voids at the margins. Then I **attach the analogues**, hand-tight, holding the coping so it does not spin.

Finally, the opposing arch, an **interocclusal record on rigid bases**, the shade with photographs, and a facebow if I am altering the occlusal scheme. I replace the healing abutments and tell the patient what to expect. The impression is **disinfected** to the manufacturer's protocol and sent with a **written prescription**, and the try-in is booked."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 open tray chosen **and justified by the divergence** · 2 splinting with resin sectioned and re-joined for shrinkage · 3 seating verified radiographically · 8 guide screws unscrewed completely before removal · 6 polyether selected with its properties · 10 opposing arch, records, disinfection and written prescription.

Accept these late or on prompting: 4 custom tray with windows and screw clearance · 5 adhesive applied and dried · 7 the latex and addition silicone incompatibility · 9 analogues attached and the impression inspected.

The commonest way five minutes is wasted here: stopping to explain. A candidate who lifts each coping, sets it down, and delivers a paragraph on impression material chemistry before seating the next one is at the tray try-in when the bell goes. Justify the technique once at the start, then keep the hands moving and give every subsequent step a single clause. The other loss is silent working — three minutes of neat, wordless splinting scores nothing.

OSCE 15 — Assessing restorative space and freeway space

Blueprint: "Demonstrate an understanding of the importance of interocclusal and intra-occlusal space in relation to the planning of tooth replacement" (SCE, with practical assessment).

EQUIPMENT

a patient/manikin, Willis gauge, dividers, ruler, indelible pencil, articulated study casts of an edentulous patient.

CANDIDATE INSTRUCTIONS

Demonstrate how you would measure the freeway space and determine whether there is adequate restorative space for a mandibular implant overdenture with stud attachments.

MODEL ANSWER

"**Freeway space is the rest vertical dimension minus the occlusal vertical dimension**, and is normally 2 to 4 mm.

To measure it: I seat the patient upright with the head unsupported and the lips relaxed. I mark two points, one on the tip of the nose and one on the chin, with an indelible pencil.

I record the **occlusal vertical dimension** with the patient in maximum intercuspation, or with the dentures in occlusion, measuring between the marks with a Willis gauge or dividers.

I then establish the **rest position**: I ask the patient to relax, moisten the lips, swallow and let the jaw drop naturally; or I use phonetics — asking them to repeat 'Emma' or count from fifty to sixty and stop, then measure. I repeat the measurement three times and take the mean, because the rest position is variable.

The difference between the two is the freeway space. If it is less than about 2 mm, there is little room; if it is greater than 4 mm, the vertical dimension may have collapsed and there may be more space available than the current denture suggests.

For a **stud-retained mandibular overdenture** I need approximately **8 to 10 mm of restorative space** measured from the implant platform to the opposing occlusal plane — that accommodates the attachment housing, the acrylic thickness around it and the denture tooth. A bar would need 12 to 15 mm.

I would confirm this on **articulated study casts**, measuring from the planned platform level to the opposing occlusal plane, and I would check whether alveoloplasty at the time of surgery is needed to create space, or whether the vertical dimension can be increased within the freeway space. I would record the measurements and the method used."

EXAMINER CHECKLIST

1. Defines freeway space correctly
2. Correct patient positioning
3. Two marks and a measuring device
4. Measures OVD
5. Establishes rest position by relaxation/swallowing and/or phonetics
6. Repeats and averages
7. States normal range 2–4 mm
8. Quotes stud (8–10 mm) and bar (12–15 mm) space requirements
9. Confirms on articulated casts
10. States remedies for inadequate space (alveoloplasty, altering OVD within the freeway space) and records findings

Running this station to time

Time check. The model answer is 440 words — 3 minutes 9 seconds at 140 words a minute — but this station is measured, not merely described: the patient has to be seated and relaxed, the rest position established three separate times and averaged, and the casts measured afterwards, which is well over six minutes at a comfortable pace. The compression is to **say every figure out loud the moment the gauge reads it**, and to describe each manoeuvre in one clause, because the marks here are in the spoken numbers rather than in the account of the technique.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Hands over the Willis gauge and indicates the patient or manikin: "Show me how you would measure the freeway space and decide whether there is room for a mandibular implant overdenture with stud attachments."	Checklists 1 and 2. The definition in one sentence — **freeway space is the rest vertical dimension minus the occlusal vertical dimension, normally 2 to 4 mm** — then the patient **seated upright, head unsupported, Frankfort plane horizontal, lips relaxed**, which is stated as it is done, not afterwards.
0:40–1:40	Watches the hands. Says nothing.	Checklists 3 and 4. **Two indelible marks, one on the tip of the nose and one on the chin**, and a **Willis gauge or dividers** identified by name. Then the **OVD recorded in maximum intercuspation or with the dentures in occlusion, and the figure spoken aloud** — "sixty-two millimetres" — not merely obtained.
1:40–3:00	May say: "How do you know they are at rest?"	Checklists 5 and 6, the technical core. Rest position established **two ways** — relax, moisten the lips, swallow and let the jaw drop; and **phonetically**, repeating "Emma" or counting fifty to sixty and stopping. Then the discriminating mark: **three measurements taken and the mean stated out loud** — "sixty-five, sixty-five, sixty-six; mean sixty-five point three" — because the rest position is variable.

Time	The examiner does	What earns the mark
3:00–4:00	"So what is it, and is there room?"	Checklists 7 and 8. The **subtraction spoken**: "sixty-five point three minus sixty-two, a freeway space of three point three millimetres — within the normal 2 to 4 mm." Then the requirement quoted from the platform to the opposing occlusal plane: **8 to 10 mm for a stud-retained overdenture, 12 to 15 mm for a bar**, with what that space accommodates — housing, acrylic thickness around it, denture tooth.
4:00–5:00	"Show me on the casts." Then: "And if there isn't enough?"	Checklists 9 and 10. **Measured on the articulated study casts with dividers, from the planned platform level to the opposing occlusal plane, and the figure spoken.** Then the remedies — **alveoloplasty at placement, a lower-profile attachment, or increasing the OVD within the available freeway space** — with the limit named. Finally **records** (every figure, the method and the device used), **what the patient is told** (why the denture may need to be remade rather than adapted), and the **review** before surgery.

The compressed answer — what a strong candidate says in five minutes

"**Freeway space is the difference between the rest vertical dimension and the occlusal vertical dimension**, and it is normally **2 to 4 millimetres**. I will measure both and say the numbers as I get them.

Position first. I sit the patient upright, head unsupported against nothing, Frankfort plane horizontal, lips together and relaxed. If the head is supported the mandible postures differently and every figure after this is wrong.

Two marks. One on the tip of the nose, one on the point of the chin, in indelible pencil — and the same two marks for every measurement, because moving them by a millimetre changes the answer by a millimetre.

The occlusal vertical dimension. I ask the patient to close fully into maximum intercuspation — or, with dentures, into occlusion — and measure between the marks with the **Willis gauge**. **That reads sixty-two millimetres. OVD, sixty-two.**

Now the rest position, which is the harder half. I ask the patient to relax, moisten the lips, swallow, and let the jaw drop naturally, and I measure without them knowing exactly when. I also check it **phonetically** — asking them to repeat 'Emma', or to count from fifty to sixty and stop — and measure at the point they stop, because the mandible returns to rest.

And I do it three times, because the rest position is variable. Sixty-five. Sixty-five. Sixty-six. Mean, sixty-five point three.

So the freeway space is sixty-five point three minus sixty-two — three point three millimetres, which sits within the normal range. If it were **under 2 millimetres** I would have almost no room to work in; if it were **over 4** I would suspect the vertical dimension had collapsed, and there is then more space available than the present denture suggests.

Now the restorative space, which is a different measurement. For a **stud-retained mandibular overdenture** I need approximately **8 to 10 millimetres, measured from the implant platform to the opposing occlusal plane** — that has to accommodate the attachment housing, an adequate thickness of acrylic all around it, and the denture tooth. Anything less and the acrylic fractures over the housing. A **bar-retained overdenture needs 12 to 15 millimetres**, because the bar, the clip and its housing sit above the platform.

I confirm that on the articulated casts, with dividers, from the planned platform level — which is the crest less the intended sink depth — to the opposing occlusal plane. **On these casts that measures nine millimetres, so a stud attachment is feasible and a bar is not.**

If there were not enough space, I have three options. **Alveoloplasty at the time of placement**, to lower the platform and gain the millimetres surgically. A **lower-profile attachment system**. Or **increasing the occlusal vertical dimension**, but only **within the freeway space** and never reducing it below about 2 millimetres, because a patient closing onto no freeway space gets soreness, muscle pain and a denture they cannot tolerate.

I record all of it — the OVD, the three rest readings and the mean, the calculated freeway space, the cast measurement, the device used and the method — because the next clinician needs to know how the figure was obtained. I explain to the patient why the existing denture may need to be remade rather than adapted, and I review these measurements against the surgical plan before we proceed."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 freeway space defined as RVD minus OVD · 5 rest position established by relaxation or swallowing **and** phonetics · 6 three measurements taken and averaged · 8 8–10 mm for studs and 12–15 mm for a bar · 9 confirmed on articulated casts, platform to opposing occlusal plane · 10 remedies named and the figures recorded.

Accept these late or on prompting: 2 upright, unsupported, lips relaxed · 3 two indelible marks and a named measuring device · 4 the OVD recorded in intercuspation · 7 the normal range of 2 to 4 mm.

The commonest way five minutes is wasted here: measuring in silence and then reporting a conclusion — "there is adequate space." The examiner cannot mark a figure that was never said, and checklists 4, 6 and 9 are all figures. The second failure is conflating the two measurements: freeway space and restorative space are different quantities measured between different landmarks, and a candidate who offers the 8 to 10 mm requirement as though it were the freeway space has lost both marks at once.

OSCE 16 — Fitting the prosthesis

Blueprint: "Demonstrate the fitting of a prosthesis" (OS-P); "Deliver a satisfactory restoration to the patient" (OS-P + SCE).

EQUIPMENT

a model with an implant analogue, a completed screw-retained crown, torque wrench, articulating paper of two thicknesses, shimstock, floss, PTFE tape, composite.

CANDIDATE INSTRUCTIONS

Fit this screw-retained crown. Describe and demonstrate each check you make.

MODEL ANSWER

"Before I start: confirm the patient's identity and the work returned matches the prescription, and check the peri-implant tissues are healthy.

1. Remove the healing abutment, irrigate and inspect the internal connection.
2. Seat the crown by hand and confirm it is fully seated — visually, by probing the margin, and **radiographically** with a paralleling periapical, because a wedge-shaped gap on the radiograph is the sign of a misfit. **I would never force a non-seating restoration with the torque wrench** — that causes screw fracture, bone loss and pain.
3. Check **proximal contacts** with floss — they should offer resistance but allow the floss to pass, and I check the contact is cleansable.
4. Check **emergence and contour** — the restoration must be cleansable with an interdental brush; an over-contoured or ridge-lap emergence is a future peri-implantitis risk.
5. Check the **occlusion**: in intercuspation the implant crown should hold shimstock only under heavy clenching and not under light tapping — that is, about 25 to 30 micrometres lighter than the adjacent teeth. In excursions there should be **no contact on the implant** where the natural canine can guide. I check working and non-working sides.
6. Check aesthetics with the patient using a mirror, in daylight if possible, and confirm they are happy **before** final torquing.
7. **Torque** to the manufacturer's value with a calibrated torque wrench, and **re-torque after ten minutes** to compensate for embedment relaxation.
8. Seal the access: **PTFE tape over the screw head** so it can be retrieved, then composite, then re-check the occlusion because the composite adds height.
9. Give **site-specific hygiene instruction** — interdental brush of the correct size, single-tufted brush, superfloss.

10. Take a **baseline periapical radiograph**, photograph, record the abutment, torque value, batch numbers and occlusal scheme, give the patient their implant card, and arrange the maintenance recall."

EXAMINER CHECKLIST

1. Checks work against prescription; assesses tissues
2. Verifies complete seating clinically **and radiographically**
3. States never to force with torque
4. Proximal contacts
5. Contour/cleansability
6. Light occlusion in ICP, none in excursion
7. Patient approval of aesthetics before torquing
8. Torque + re-torque at 10 min
9. PTFE + composite, re-check occlusion
10. Baseline radiograph, hygiene, records, recall

Running this station to time

Time check. The model answer is 433 words — 3 minutes 5 seconds at 140 words a minute — but the instruction is *describe and demonstrate*, so the hands are working: a radiograph taken before torquing, floss through two contacts, two thicknesses of articulating paper, shimstock in three positions, a torque wrench, then PTFE and composite. That is well over six minutes of real chairside time, and it fits into five only by giving **each of the ten checks a single narrated clause**, with the justification kept for three places — the refusal to force, the shimstock figure, and the PTFE.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Indicates the model, the crown and the tray: "Fit this screw-retained crown. Describe and demonstrate each check you make."	Checklist 1, before the healing abutment comes off. **Patient identity confirmed, the returned work checked against the written prescription** — right system, right platform, right shade, screw-retained as prescribed — and the **peri-implant tissues assessed** for inflammation, bleeding and probing depth. A candidate who starts by unscrewing has already lost a mark.
0:30–1:30	Hands over the periapical, or asks: "How do you know it is seated?"	Checklists 2 and 3, the two that carry the station. Healing abutment removed, connection **irrigated and inspected**; crown seated **by hand**; seating confirmed **visually, by probing the margin circumferentially, and radiographically with a paralleling periapical**, because a **wedge-shaped radiolucent gap** is the sign of misfit. Then said explicitly: **"I would never force a non-seating restoration with the torque wrench"** — screw fracture, marginal bone loss, pain.
1:30–2:30	Slides the floss across. Silent.	Checklists 4 and 5. **Floss through both proximal contacts** — resistance, but it passes and does not shred — and the contact confirmed **cleansable**. Then **emergence and contour** checked against an interdental brush of the correct size, with the reason in one clause: an **over-contoured or ridge-lap emergence is a future peri-implantitis risk** because it cannot be cleaned.

Time	The examiner does	What earns the mark
2:30–3:30	Hands over both thicknesses of paper and the shimstock. May ask: "How light is light?"	Checklists 6 and 7. **Shimstock held only under heavy clenching and released under light tapping — about 25 to 30 micrometres lighter than the adjacent teeth** — because there is no periodontal ligament to allow intrusion. **No implant contact in excursions** where the natural canine can guide, checked on **working and non-working sides and in protrusion**. Then **aesthetics shown to the patient in a mirror, in daylight, and their approval obtained before final torquing**.
3:30–4:30	"Go ahead and fit it."	Checklists 8 and 9. **Torque to the manufacturer's value with a calibrated wrench**, then **re-torque after ten minutes** for embedment relaxation, said with the reason. **PTFE tape over the screw head** so the screw can be retrieved, composite over that — and the **occlusion re-checked afterwards, because the composite adds height**.
4:30–5:00	"Anything to finish?"	Checklist 10 — **baseline periapical radiograph** and photographs; **records** (abutment, torque value, batch numbers, occlusal scheme); **what the patient is told** — site-specific hygiene with an interdental brush of a named size, single-tufted brush and superfloss, the **implant card** handed over, and that the crown can be unscrewed if it ever needs attention; then the **maintenance recall interval**.

The compressed answer — what a strong candidate says in five minutes

"**Before the healing abutment comes off**, I confirm this is the right patient, check the work against my **written prescription** — correct system and platform, correct shade, screw-retained as I asked — and assess the peri-implant tissues: no bleeding on probing, no suppuration, healthy keratinised cuff.

Healing abutment out, connection irrigated and inspected for tissue tags or debris.

I seat the crown by hand only. I check it visually, run a fine probe right round the margin, and then take a **paralleling periapical**, because at this depth the eye and the probe are not enough — the sign of a misfit is a **wedge-shaped radiolucent gap** between crown and platform. **If it is not fully seated, I do not force it home with the torque wrench.** Torquing a non-seated restoration fractures screws, causes marginal bone loss and hurts; I would find the cause instead — soft tissue over the platform, a proximal contact that is too tight, or a laboratory error.

Contacts. Floss through each contact: it should meet resistance but pass and not shred. If it shreds, the contact or the finish is wrong. And I check the contact is **cleansable**, not just present.

Contour and emergence. I check an **interdental brush of the correct size passes** on both sides. An over-contoured or ridge-lap emergence looks fine on the day and is a peri-implantitis risk for the next twenty years.

Occlusion, and it is deliberately light. In intercuspation the implant crown should hold **shimstock only under heavy clenching, and let it pull through under light tapping** — about **25 to 30 micrometres lighter than the adjacent teeth**. That is because there is no periodontal ligament: the natural teeth intrude under load and the implant does not, so an implant crown built to the same height takes a rising share of the force. **In excursions there should be no contact on the implant at all** where the natural canine can guide — I check **working side, non-working side and protrusion**, not just tapping.

Then the patient. I give them a mirror, in daylight if I can, and I get their **approval of the appearance before I torque it**, because after torquing and sealing, a change is a remake.

Torque. To the **manufacturer's value with a calibrated wrench** — I would not guess it by hand — and I **re-torque after ten minutes**, which compensates for embedment relaxation as the microscopic surface irregularities flatten and the preload drops.

Sealing the access. PTFE tape over the screw head, so the screw can be found and retrieved cleanly, then composite over that. Then I **re-check the occlusion**, because the composite has added height and a crown that was 30 micrometres light can now be the first contact in the arch.

Finally. A **baseline periapical** against which every future radiograph is compared, and photographs. I record the abutment, the **torque value**, the batch numbers and the occlusal scheme. I give **site-specific hygiene instruction** — an interdental brush of the size I have just checked, a single-tufted brush and superfloss — hand over the **implant card**, tell the patient the crown is retrievable if it ever needs attention, and arrange the **maintenance recall**."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 2 seating verified clinically **and radiographically** · 3 states explicitly that it must never be forced with the torque wrench · 6 light in ICP with the shimstock figure, and free in excursions · 8 torque to the manufacturer's value and re-torque at ten minutes · 9 PTFE and composite with the occlusion re-checked afterwards · 10 baseline radiograph, hygiene, records and recall.

Accept these late or on prompting: 1 work checked against the prescription and tissues assessed · 4 proximal contacts by floss · 5 contour and cleansability · 7 patient approval of aesthetics before torquing.

The commonest way five minutes is wasted here: picking up the torque wrench early. A candidate who seats, torques and then starts checking has forfeited checklists 2, 3 and 7 in one movement — the radiograph, the refusal to force, and the patient's approval all happen *before* the wrench. The other loss is checking the occlusion once: the composite over the access hole adds height, and a candidate who does not go back to the articulating paper after sealing loses the second half of checklist 9.

OSCE 17 — The mis-fitting framework

Blueprint: "Describe how to manage a mis-fitting prosthesis or framework" (OS-P).

EQUIPMENT

a model with four implant analogues and a cast framework that does not seat passively; a periapical radiograph showing a wedge-shaped gap; probes, screws.

CANDIDATE INSTRUCTIONS

This framework does not seat. Demonstrate how you would confirm and manage the misfit.

MODEL ANSWER

"First I confirm the misfit objectively rather than assuming it.

- **Visual and tactile:** inspect each interface and run a fine probe around the margin.
- **The one-screw or Sheffield test:** tighten only the most distal screw and examine the fit at the opposite end; a visible gap or rocking confirms the framework is not passive. I repeat this from the other end.
- **Radiographic:** a paralleling periapical at the affected abutment — a wedge-shaped radiolucent gap between framework and implant confirms it.
- **Patient symptoms:** pressure or pain as the screws are tightened is a warning sign.

Then I identify the cause: soft tissue or bone interference over the platform; a component not fully seated; an incorrectly oriented analogue; impression distortion; an inaccurate cast; laboratory distortion during firing; or simply the wrong component.

I do not force it. Torquing a non-seating framework causes screw fracture, marginal bone loss and pain.

Management according to cause: remove any tissue or bone interference and re-try; if the framework itself is distorted, **section it at the point of misfit and re-join it** by soldering or laser welding, then

re-verify; if the error is in the impression or the cast, take a **new impression with splinted open-tray copings, or make a verification jig**, and have the framework remade.

I would explain the position to the patient honestly, apologise for the delay, provide or adjust a provisional in the meantime, and record everything — and I would discuss the case with the technician to find out where the error arose so it doesn't recur. I would not criticise the laboratory in front of the patient."

EXAMINER CHECKLIST

1. Visual and tactile check
2. **One-screw/Sheffield test** named and performed
3. Radiographic confirmation and the wedge-shaped gap
4. Patient symptoms considered
5. Lists possible causes
6. States explicitly that the framework must not be forced, with the consequences
7. Removes interference and re-tries
8. Section and re-join as a remedy
9. New splinted impression/verification jig and remake
10. Communication with patient and technician, provisional, records

Running this station to time

Time check. The model answer is 402 words, 2 minutes 52 seconds at 140 words per minute — comfortable on paper, but the verb is *demonstrate*: screws have to be turned, a fine probe run round four interfaces, the test repeated from the other end, and the periapical read. Handling doubles the honest running time, so the causes and the remedies must each be delivered in a single clause, not a paragraph.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Sets out the model with four analogues, the framework, screws, probes and the periapical: "This framework does not seat. Demonstrate how you would confirm and manage the misfit."	Nothing is marked yet, but the candidate must **state the organising principle in one sentence** — "I confirm the misfit objectively before I assume it, and whatever I find I do not force it" — then seat the framework by hand and **hand-tighten one screw only**. Saying "I do not force it" this early protects checklist 6 if the clock beats them later.
0:30–2:00	Silence. Slides the periapical across halfway through.	Checklists 1, 2, 3 and 4, **performed and narrated together**. Fine probe run round *every* interface with the eye at the level of the joint; the **one-screw or Sheffield test named aloud**, the most distal screw tightened, the far end examined, then **repeated from the other end**, because a framework can fit from one direction and not the other; the periapical described as a **wedge-shaped radiolucent gap**; pressure or pain on tightening named as a warning that stops the candidate.
2:00–3:00	"So it does not fit. What now?"	Checklists 6 then 5, in that order. **"I do not force it"** with the consequences — screw loosening and fracture, marginal bone loss, pain — then the causes in one sweep: soft tissue or bone over a platform, a component not fully seated, a rotated or wrong analogue, impression distortion, an inaccurate cast, laboratory distortion during casting or firing.

Time	The examiner does	What earns the mark
3:00–4:15	"And what do you actually do about each of those?"	Checklists 7, 8 and 9, escalating. Remove the interference and re-try; **section at the point of misfit, lute the sections intra-orally, return for soldering or laser welding, and re-verify with the same tests**; and if the error lies in the impression or the cast, a **verification jig, a new splinted open-tray impression and a remade framework**. Explicitly refuses to grind the intaglio.
4:15–5:00	"Anything else?"	Checklist 10 — **what the patient is told** (an honest explanation of the delay, an apology, the provisional adjusted or replaced so they are not inconvenienced), the **discussion with the technician** to find where the error arose, without criticising the laboratory in front of the patient, and the **record**: which tests, from which end, the radiographic finding, the cause identified, the decision taken and the re-try appointment.

The compressed answer — what a strong candidate says in five minutes

"I will confirm this objectively before I assume anything — and whatever I find, **I will not force it**.

Seating it. I hand-tighten one screw only, and I look first. **Visually and tactilely**, with my eye down at the level of the joint and a fine probe run right round each of the four interfaces — a probe that catches at one margin and not the others tells me where the problem is.

The one-screw test. I tighten only the **most distal screw** and examine the fit at the opposite end. A visible gap, or rocking under finger pressure, means the framework is not passive. Then I **repeat the test from the other end**, because a distorted framework can appear to seat from one direction and not from the other.

Radiographically. A paralleling periapical at each abutment. On this film there is the classic sign — a **wedge-shaped radiolucent gap** between the framework and the implant platform, widening away from the tightened screw.

And by the patient. Pressure, tightness or pain as I tighten is a warning sign, not something to work through. I stop.

So it is confirmed, and I do not force it. Torquing a non-passive framework transmits static stress into the implants and components: screw loosening, screw or abutment fracture, marginal bone loss, and pain.

The causes, quickly. Soft tissue or bone lying over a platform; a component not fully seated; the wrong analogue, or one rotated in the cast; impression distortion from unsplinted copings, a coping that moved, or a flexible tray; an inaccurate or damaged cast; or laboratory distortion during casting, soldering or ceramic firing — a framework can be passive before firing and not afterwards.

The remedies, in escalating order. First, remove any tissue or bone interference and re-try; that costs nothing and solves a proportion of them. Second, if the framework is accurate in its parts but distorted overall, **section it at the point of misfit, lute the sections together intra-orally with pattern resin, and return it for soldering or laser welding** — then re-verify with exactly the same tests, from both ends. Third, if the error is in the impression or the cast, I make a **verification jig, take a new splinted open-tray impression, and have the framework remade**. What I do not do is grind the intaglio until it drops in; that produces a framework that looks seated and is not.

Then the professional part. I explain the position to the patient honestly and apologise for the delay, and I adjust or replace the provisional so they leave comfortable and are not inconvenienced. I **discuss the case with the technician** to establish where the error arose so it does not recur — but I would not criticise the laboratory in front of the patient.

And I record it: which tests I performed and from which end, the radiographic finding, the cause I identified, the decision taken, what I told the patient, and the appointment for the re-try and re-verification."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 2 one-screw/Sheffield test performed **and repeated from both ends** · 6 states explicitly that it must not be forced, with the consequences · 3 radiographic

confirmation and the wedge-shaped gap · 8 section and re-join as the standard remedy · 9 verification jig, new splinted impression and remake · 10 patient communication, provisional, technician and records.

Accept these late or on prompting: 1 visual and tactile check at each interface · 4 patient symptoms as a warning sign · 5 the systematic list of causes · 7 removing interference and re-trying.

The commonest way five minutes is wasted here: reciting the causes list as a lecture before the misfit has actually been demonstrated. The framework, the screws and the probe were put on the table to be handled, and a candidate who talks for two minutes without tightening a single screw forfeits the Sheffield test — the one mark the whole station is built around. Perform first, list second, and keep each cause to four words.

OSCE 18 — Writing the laboratory prescription

Blueprint: "Prescribe an appropriate type of restoration from the laboratory"; "Evaluate the laboratory techniques involved in implant reconstruction" (W + OS-P + SCE).

EQUIPMENT

a blank laboratory prescription form, a working cast with an implant analogue at the UR1, photographs, a shade tab.

CANDIDATE INSTRUCTIONS

Complete the laboratory prescription for a definitive restoration of the UR1 implant. Explain your entries.

MODEL ANSWER

"Header: **patient identifier**, my name, practice address and GDC number, the date sent and the **date required**.

Work required: '**UR1 — definitive implant crown, screw-retained**' — then the implant details: system, platform diameter, connection type, and the analogue supplied.

Abutment: 'custom CAD-CAM titanium base with a milled lithium disilicate superstructure' — or 'custom titanium abutment', with the **margin to follow the gingival scallop and lie no more than 1 mm subgingival**, and the emergence profile to be copied from the provisional restoration/customised impression coping supplied.

Materials: framework and veneering material specified; for a bruxist I would specify monolithic zirconia.

Aesthetics: **shade** — with the shade tab photographed in the mouth alongside the adjacent teeth — plus stump shade, characterisation, translucency, surface texture and any notes on the incisal edge and mamelons; photographs attached.

Form and function: contact points to be light and cleansable; contour to permit an interdental brush; **occlusion — light contact in intercuspation and no contact in excursions**; and whether a bisque try-in is required.

Administration: state that the impression and cast are **disinfected**; request that the work be returned disinfected; request the **custom-made device statement** for my records; give my contact number in case the technician has questions; and record the laboratory's details and, on return, the batch numbers.

I would keep a copy of the prescription in the notes."

EXAMINER CHECKLIST

1. Patient and clinician identifiers, dates
2. Precise description of the work and tooth notation
3. Implant system, platform, connection
4. Abutment type, material and **margin position**
5. Emergence profile instruction

6. Materials specified
7. Shade and aesthetic detail with photographs
8. Occlusal instruction
9. Try-in requirement
10. Disinfection statement, custom-made device statement, copy retained

Running this station to time

Time check. The model answer is 352 words, 2 minutes 31 seconds at 140 words per minute — the shortest in this part, and the trap is that the candidate has to **write** as well as speak, and writing is far slower than talking. Ten headed entries filled in legibly will not fit in five minutes at conversational pace; the fix is **note form in the box, one clause of justification aloud**, with full explanation reserved for three entries only.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Puts the blank prescription form, the working cast with the UR1 analogue, the photographs and the shade tab in front of the candidate: "Complete the laboratory prescription for a definitive restoration of the UR1 implant. Explain your entries."	Checklist 1, written while speaking. **Patient identifier, clinician name, practice address, GDC number, date sent and date required** — filled in, not discussed. The candidate should state the organising principle once: *the prescription is a written instruction the technician must be able to work from without telephoning me*, which frames everything that follows.
0:45–1:45	Silence. Lets the candidate work down the form.	Checklists 2 and 3. The work described precisely and in **FDI notation** — "**UR1, definitive implant crown, screw-retained**" — not "crown please"; then the **implant system, platform diameter, connection type** and confirmation that the correct **analogue** is supplied with the cast.
1:45–3:00	"Tell me about the abutment entry."	Checklists 4, 5 and 6, and the first of the three entries worth explaining. **Custom CAD-CAM titanium base with a milled lithium disilicate superstructure**, or a custom titanium abutment; the **margin to follow the gingival scallop and lie no more than 1 mm subgingival**, with the reason — cement retrievability and cleansability; the **emergence profile to be copied from the customised impression coping or the provisional supplied**; and framework and veneering materials named, **monolithic zirconia for a bruxist**.
3:00–4:00	"And the aesthetic and functional instructions?"	Checklists 7, 8 and 9. **Shade recorded with the tab photographed in the mouth beside the adjacent teeth**, plus stump shade, characterisation, translucency, surface texture and incisal notes, photographs attached; contacts light and cleansable, contour to admit an interdental brush; **occlusion — light contact in intercuspation, no contact in excursions**; and whether a **bisque try-in** is required stated explicitly, one way or the other.

Time	The examiner does	What earns the mark
4:00–5:00	"Anything left on the form?"	Checklist 10 — the administrative block candidates skip: **the impression and cast are disinfected and the work is to be returned disinfected**, the **custom-made device statement requested for the records**, a contact number for queries, the laboratory's details, and the **batch and lot numbers recorded on return**. Then **a copy of the prescription kept in the notes**, the patient told the return date and what the try-in appointment involves, and the work **checked against the prescription on receipt** before the patient is seated.

The compressed answer — what a strong candidate says in five minutes

"I will fill this in as I talk. The test of a prescription is whether the technician can work from it **without ringing me**.

Header. Patient's name and identifier, my name, the practice address, my **GDC number**, the date sent and the **date required** — the date required matters because a bisque try-in has to be built into the timetable.

Work required. '**UR1 — definitive implant crown, screw-retained**', in FDI notation. Then the implant details: the **system, the platform diameter and the connection type**, and I confirm the correct **analogue** is in the cast I am sending.

Abutment. '**Custom CAD-CAM titanium base with a milled lithium disilicate superstructure**' — or a custom titanium abutment if the tissue is thin. The instruction that matters here is the **margin: to follow the gingival scallop and to lie no more than 1 millimetre subgingival**. That is not aesthetics, it is so that any cement can be reached and removed, and so the margin can be cleaned.

Emergence profile. '**Copy the emergence profile from the customised impression coping supplied**' — the provisional has been shaping that tissue for weeks, and the definitive crown must reproduce it, not invent a new one.

Materials. Framework and veneering material specified. If this patient were a bruxist I would prescribe **monolithic zirconia** and say so on the form, with the reason.

Aesthetics. The **shade**, with the tab photographed in the mouth alongside the adjacent teeth so the technician sees my lighting, not just my letter. Then stump shade, characterisation, translucency, surface texture, and a note on the incisal edge and mamelons. Photographs attached and referenced by number.

Form and function. Contact points light and cleansable; contour to allow an **interdental brush** to pass; and the occlusion written out — **light contact in intercuspation and no contact in any excursion**, so that the implant is not the guiding unit. And I state whether I want a **bisque try-in**: on a single upper central with a demanding shade, yes.

Administration. I state that the **impression and cast have been disinfected**, and ask for the work to be **returned disinfected**. I request the **statement of manufacture for the custom-made device** for my records. I give my direct number for queries, record the laboratory's name and address, and on return I record the **batch and lot numbers** of the materials used.

And finally, records. A **copy of the prescription goes in the notes** — it is a legal instruction and my evidence of what I asked for. I tell the patient the date the work is due, that there may be a try-in before the crown is finished, and what that appointment involves. When the work comes back I **check it against my own prescription on the cast before the patient is in the chair**, because that is the point at which an error is still cheap to correct."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 2 precise work description in FDI notation · 3 implant system, platform and connection · 4 abutment type, material and **margin position with the reason** · 5 emergence profile copied from the coping or provisional · 8 occlusal instruction written out · 10 disinfection both ways, custom-made device statement and copy retained.

Accept these late or on prompting: 1 identifiers and dates · 6 materials specified, monolithic zirconia for a bruxist · 7 shade and aesthetic detail with photographs · 9 bisque try-in stated one way or the other.

The commonest way five minutes is wasted here: treating it as a viva about crown materials. Two minutes on lithium disilicate versus zirconia leaves the form half empty, and the administrative block — disinfection, the custom-made device statement, the retained copy — is never written, which is three marks and the only part of the prescription with a legal weight. Write in note form, justify only the margin, the emergence profile and the occlusion.

OSCE 19 — Digital workflow and guided surgery

Blueprint: "Demonstrate understanding of the role of digital techniques in restoring implants" (W + OS-P + SCE).

EQUIPMENT

an intraoral scanner (or images of scans), scan bodies, a printed surgical guide, a CBCT/STL merge on screen.

CANDIDATE INSTRUCTIONS

Explain the digital workflow from planning through to the definitive restoration for a single implant, and state the limitations of each stage.

MODEL ANSWER

"The chain is: **intraoral scan → digital wax-up → CBCT → merge of the DICOM and STL data → virtual implant planning → guide design and manufacture → guided placement → scan body impression → CAD design → milled or printed restoration.**

Stage by stage:

Scan and digital wax-up. I scan both arches and the bite, and design the planned tooth position digitally — the digital equivalent of the diagnostic wax-up. This keeps the plan restoratively driven. Limitation: **accuracy falls across long edentulous spans**, and deep subgingival margins scan poorly.

CBCT and merge. The CBCT is justified individually and I report the whole volume. The merge relies on matching common reference points — usually teeth. Limitation: **metal artefact** degrades the merge, and in a fully edentulous patient there are no teeth to match, so a radiographic guide with markers is needed.

Virtual planning. I place the implant virtually with correct 3-D position, respecting 1.5 mm to teeth, 3 mm to implants, 2 mm of buccal bone and 2 mm from the canal.

Guide manufacture and surgery. Tooth-supported guides are most accurate. Limitations: mean angular deviation of about 2 to 4 degrees, so the **2 mm safety margin is retained**; sleeves restrict irrigation, so heat control needs attention; and posterior access may be inadequate for the drill-and-sleeve assembly.

Restorative digital stage. A **scan body** is screwed to the implant and scanned, transferring the implant position digitally, and the abutment and crown are designed and milled. Limitation: **scan body wear or incomplete seating introduces error**, and for multiple implants across a long span a conventional splinted impression may still be more accurate.

Overall advantages: patient comfort, immediate re-scanning, archiving, communication, restoratively driven planning. Overall limitations: cost, learning curve, and the fact that **a guide replaces neither anatomical knowledge nor surgical judgement**."

EXAMINER CHECKLIST

1. States the full workflow in order
2. Digital wax-up as the restorative driver
3. CBCT justification and full-volume reporting
4. DICOM/STL merge and its reference points
5. Metal artefact and edentulous merging limitation
6. Correct 3-D planning parameters
7. Guide support types and accuracy figures

8. Retains the 2 mm safety margin despite guidance
9. Scan body use and its error sources
10. Balanced statement of advantages and limitations

Running this station to time

Time check. The model answer is 443 words, 3 minutes 10 seconds at 140 words per minute — but the instruction asks for a workflow **and** a limitation at every stage, which is effectively eighteen items, and a candidate who separates them into two passes will deliver the limitations after the bell. The fix is a **fixed 45 seconds per stage, with the limitation named in the same breath as the stage**, and the whole chain stated once at the start as a spine.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Indicates the scanner, the scan bodies, the printed guide and the merged CBCT/STL on screen: "Explain the digital workflow from planning through to the definitive restoration for a single implant, and state the limitations of each stage."	Checklist 1 — the **whole chain named in order, in one breath**, before any stage is expanded: intraoral scan, digital wax-up, CBCT, DICOM/STL merge, virtual planning, guide design and manufacture, guided placement, scan body impression, CAD design, milled or printed restoration. Thirty seconds spent here buys permission to be brief afterwards.
0:40–1:50	Silence, then: "And the limitations of those first steps?"	Checklists 2, 3, 4 and 5. **Digital wax-up as the restorative driver** — the plan is prosthetically led, not bone-led; **CBCT justified individually and the whole volume reported**; the **merge matched on common reference points, usually teeth**; and the two limitations that matter — **accuracy falls across long edentulous spans and deep subgingival margins**, and **metal artefact degrades the merge**, with a radiographic guide carrying markers needed in the fully edentulous patient.
1:50–3:10	"How do you place it virtually, and how accurate is the guide?"	Checklists 6, 7 and 8. The 3-D parameters as numbers — **1.5 mm to a tooth, 3 mm between implants, 2 mm of buccal bone, 2 mm from the canal**; **tooth-supported guides most accurate**, then mucosa-supported, then bone-supported; **mean angular deviation of about 2 to 4 degrees**; and the decisive statement that the **2 mm safety margin is retained despite the guide**, not consumed by it. Sleeves restrict irrigation, and posterior access may not admit the drill-and-sleeve assembly.
3:10–4:10	"And the restorative stage?"	Checklist 9. The **scan body** screwed to the implant and scanned to transfer the position digitally, then abutment and crown designed and milled or printed. Error sources named specifically: **wear or damage to the scan body, and incomplete seating**; and the honest concession that **for multiple implants across a long span a conventional splinted open-tray impression may still be more accurate**.

Time	The examiner does	What earns the mark
4:10–5:00	"So overall — is it better?"	Checklist 10 and the close. A **balanced** answer: comfort, immediate re-scanning, archiving, communication and restoratively driven planning against cost, learning curve and the fact that **a guide replaces neither anatomical knowledge nor surgical judgement**. Then **records** (the CBCT report written in the notes with the justification, the plan and the STL files archived, the guide's fit and sterilisation checked and recorded), **what the patient is told** (the dose, and that the plan may be converted to a flap intra-operatively), and **review** at the planned surgical and restorative stages.

The compressed answer — what a strong candidate says in five minutes

Q: Take me through the workflow. "**Intraoral scan, digital wax-up, CBCT, merge of the DICOM and STL data, virtual implant planning, guide design and manufacture, guided placement, scan body impression, CAD design, milled or printed restoration.** I will take each stage with its limitation attached."

Q: The scan and the wax-up. "I scan both arches and the bite, and design the planned tooth position digitally — the digital equivalent of the diagnostic wax-up, and the reason the whole plan stays **restoratively driven** rather than driven by where the bone happens to be. Its limitation is accuracy: scanning **drifts across long edentulous spans**, because there are no landmarks for the software to stitch to, and **deep subgingival margins scan poorly** because the scanner only records what it can see."

Q: The CBCT and the merge. "The CBCT is **justified individually**, and I report the **whole volume**, not only the site — that report goes in the notes. The merge works by matching common reference points, in a dentate patient the teeth. Its limitation is **metal artefact**: a heavily restored arch degrades the surfaces the software matches on. And in a **fully edentulous** patient there are no teeth to match at all, so I need a radiographic guide with fiducial markers, or a dual-scan protocol."

Q: The virtual plan. "I position the implant in three dimensions to the restorative plan: **at least 1.5 mm from an adjacent tooth, 3 mm from an adjacent implant, 2 mm of buccal bone, and 2 mm clear of the inferior alveolar canal**, with the emergence under the planned tooth."

Q: The guide. "**Tooth-supported guides are the most accurate**; mucosa-supported guides depend on the tissue and need fixation pins; bone-supported guides need a flap. The published **mean angular deviation is about 2 to 4 degrees**, with linear deviation of a millimetre or more at the apex — which is why the **2 mm safety margin is retained and not spent**. A guide narrows the distribution; it does not abolish it. Practically: the **sleeve restricts irrigation**, so I use copious external irrigation and intermittent drilling for heat control, and I check **mouth opening and posterior access** for the drill-and-sleeve assembly before I order it."

Q: The restorative stage. "A **scan body** is screwed to the implant and scanned; that transfers the implant's three-dimensional position and rotation into the design software, and the abutment and crown are designed and milled or printed. Its errors are mechanical: a **worn, damaged or autoclaved-out-of-shape scan body**, or one **not fully seated** — which I confirm radiographically if I am in any doubt. And I would say honestly that for **multiple implants across a long span** a conventional **splinted open-tray impression** is still at least as accurate."

Q: Overall? "Advantages: patient comfort, immediate re-scanning if something fails, archiving and communication, and a genuinely restoratively driven plan. Limitations: cost, a real learning curve, and the important one — **a guide replaces neither anatomical knowledge nor surgical judgement**. I plan every case so that I could complete it freehand if the guide did not seat.

Records and follow-up: the CBCT report and its justification written in the notes, the plan and the STL and DICOM files archived, the guide checked for fit on the cast and sterilised before use, and that recorded. I tell the patient what the scan dose is and why it is justified, and that if the guide does not seat accurately on the day I will raise a flap and place freehand rather than proceed blind."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 the full workflow stated in order · 2 digital wax-up as the restorative driver · 6 the 3-D planning parameters as numbers · 8 the 2 mm safety margin retained

despite guidance · 9 scan body use and its error sources · 10 balanced advantages and limitations, with records.

Accept these late or on prompting: 3 CBCT justification and whole-volume reporting · 4 the merge and its reference points · 5 metal artefact and the edentulous merge · 7 guide support types and the deviation figures.

The commonest way five minutes is wasted here: describing the software rather than the clinical chain — naming planning packages, scanner brands and printer types. The examiner is testing whether the candidate knows where each stage can fail, not what it is called. The second failure is running the whole workflow first and promising the limitations afterwards; the bell arrives at the scan body, and the four limitation marks are never scored.

OSCE 20 — Peri-implant assessment and maintenance

Blueprint: "Identify appropriate maintenance care regimes"; "Assess treatment outcomes and demonstrate understanding of the parameters used to monitor outcomes" (W + SCE, assessed practically).

EQUIPMENT

a manikin with an implant-supported crown at the LR6, a plastic/titanium probe, a steel probe (distractor), interdental brushes of various sizes, single-tufted brush, superfloss, a periapical radiograph.

CANDIDATE INSTRUCTIONS

Carry out a peri-implant assessment on this implant and give the patient appropriate maintenance advice.

MODEL ANSWER

"I would first update the medical history, ask about smoking and about any symptoms — soreness, bleeding, bad taste, discharge, or a feeling of movement.

Assessment: - **Plaque and bleeding scores** for the whole mouth, because peri-implant health depends on overall plaque control. - **Peri-implant probing at six sites** with a light force of about 0.2 to 0.25 newtons — the force at which the probe tip just blanches my fingernail. I would use a plastic or titanium probe. (*Selecting the plastic probe over the steel one is a checklist item.*) Probing implants is safe and essential; the old advice not to probe is obsolete. - Record **bleeding on probing** and **suppuration**, probing depths, **mucosal recession** and the **width of keratinised mucosa**. - Assess **mobility** — and if there is any doubt, I would remove the restoration to distinguish a loose prosthesis from a mobile implant, because a mobile implant means loss of osseointegration. - Assess the **prosthesis**: fit, contour, cleansability, screw integrity, wear, chipping, occlusion, and check for residual cement. - Compare a standardised **periapical radiograph** with the baseline taken at fitting. I would only take a new radiograph if there were a clinical indication — increased probing depth, bleeding, suppuration or symptoms — because the exposure must be justified.

Diagnosis using the 2017 criteria: health, mucositis if there is bleeding on gentle probing without bone loss, and peri-implantitis if there is bleeding or suppuration with increased probing depth and bone loss beyond initial remodelling.

Maintenance advice: professional biofilm removal with titanium or plastic instruments and glycine air-polishing, then **site-specific hygiene instruction** — I would size an interdental brush to fit the embrasure and demonstrate it in the mirror, add a single-tufted brush around the emergence and superfloss under the pontic area if there is one, and recommend a powered brush.

Recall interval: risk-based — three to four months for this patient if there is a history of periodontitis or smoking, otherwise three to six months. I would record the findings, the advice given, and the next review."

EXAMINER CHECKLIST

1. Medical history and symptoms updated

2. Whole-mouth plaque and bleeding scores
3. **Selects the plastic/titanium probe**
4. Probes 6 sites with gentle force, states the force
5. Records BOP, suppuration, recession, keratinised mucosa
6. Assesses implant vs prosthesis mobility
7. Assesses prosthesis fit, contour, cleansability, cement
8. Compares with baseline radiograph; justifies any new exposure
9. Applies the 2017 diagnostic criteria
10. Individualised hygiene demonstration and risk-based recall, recorded

Running this station to time

Time check. The model answer is 492 words, 3 minutes 31 seconds at 140 words per minute — but the verb is *carry out*: six sites have to be probed, the prosthesis examined, the radiograph compared and an interdental brush sized and demonstrated, and hands are always slower than speech. Roughly 150 words must go, and they go by **naming each finding as it is recorded rather than explaining why it is recorded**, keeping justification for the probe, the force and the radiograph.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Presents the manikin with the LR6 implant crown and a tray carrying **both** probes, brushes, superfloss and a periapical: "Carry out a peri-implant assessment on this implant and give the patient appropriate maintenance advice."	Checklists 1 and 3, and 3 must happen **now**. History updated in one sentence — medical changes, **smoking**, and symptoms: soreness, bleeding, bad taste, discharge, a feeling of movement. Then the hand goes to the **plastic or titanium probe**, the steel probe is pushed aside, and **the reason is said out loud**. A candidate who picks up the right probe silently at 2 minutes may not be credited for it.
0:45–2:00	Silence. Watches the hands.	Checklists 2, 4 and 5. **Whole-mouth plaque and bleeding scores first**, because peri-implant health follows overall plaque control. Then **six sites round the implant** — mesiobuccal, buccal, distobuccal and the three palatal or lingual equivalents — at **0.2 to 0.25 newtons, calibrated until the tip just blanches the thumbnail**, stated as a number. Findings called out as they are taken: **probing depths, bleeding on probing, suppuration, mucosal recession, width of keratinised mucosa**.
2:00–3:00	"What else would you examine?"	Checklists 6 and 7. **Mobility**, with the discriminating clause — if there is any doubt, **remove the restoration to distinguish a loose prosthesis from a mobile implant**, because a mobile implant means lost osseointegration and cannot be treated. Then the **prosthesis**: fit, contour, cleansability, screw integrity, wear, chipping, occlusion, and an explicit look for **residual cement**.
3:00–3:50	Hands over the periapical, then: "What is your diagnosis?"	Checklists 8 and 9. The film **compared with the baseline taken at fitting**, not read in isolation, with a new exposure taken **only on a clinical indication** — increased depth, bleeding, suppuration or symptoms — because it must be justified. Then the **2017 criteria**: health; **mucositis** — bleeding on gentle probing without bone loss; **peri-implantitis** — bleeding or suppuration with increased probing depth and bone loss beyond initial remodelling.

Time	The examiner does	What earns the mark
3:50–5:00	"What advice would you give, and when will you see her?"	Checklist 10. **Professional biofilm removal** with titanium or plastic instruments and glycine air-polishing; then **site-specific instruction** — an interdental brush **sized to the embrasure and demonstrated in the mirror in the patient's own hand**, a single-tufted brush at the emergence, superfloss under a pontic, a powered brush. Then **records** (the six depths, bleeding, suppuration, recession, keratinised width, the advice given), **what the patient is told**, and a **risk-based recall — three to four months with a history of periodontitis or smoking, otherwise three to six**.

The compressed answer — what a strong candidate says in five minutes

"**Before I touch her I update the history** — any change in health or medication, whether she still smokes, and whether she has had soreness, bleeding, a bad taste, discharge or any feeling that it has moved.

Then the probe, and this matters. I take the **plastic or titanium probe**, not the steel one — a steel tip scratches the machined titanium surface, and a scratched surface is a plaque trap. Probing an implant is safe and necessary; the old advice never to probe is obsolete.

Whole mouth first. Plaque and bleeding scores for the whole dentition, because peri-implant health depends on her overall plaque control, and an implant is not assessed in isolation from the teeth beside it.

Six sites round the implant — mesiobuccal, buccal, distobuccal, and the three lingual equivalents — at a **light force of 0.2 to 0.25 newtons**, which I calibrate against my thumbnail until the nail bed just blanches. The seal round an implant is weaker than a periodontal attachment, so the same force penetrates further, and heavy probing produces a falsely deep reading and a bleeding point I have created myself.

As I go I record: the six probing depths, **bleeding on probing**, any **suppuration**, **mucosal recession**, and the **width of keratinised mucosa** on the buccal.

Mobility. Any mobility is significant. If there is doubt I would **remove the restoration and test the implant itself**, because a loose screw or a loose prosthesis is repairable and a **mobile implant is not** — that is loss of osseointegration and the implant comes out.

The prosthesis. Fit at the interface, contour and whether I can actually get a brush under it, screw integrity, wear, chipping, the occlusion, and I look and probe specifically for **residual cement**, which is a common and correctable cause of inflammation.

The radiograph. I compare a standardised periapical with the **baseline taken at fitting** — a single film today tells me nothing without that comparison. I would only take a new exposure if there is a clinical indication: increased probing depth, bleeding, suppuration or symptoms. The exposure has to be justified.

Diagnosis, using the 2017 criteria. Health — no bleeding, no suppuration, no bone loss beyond initial remodelling. **Peri-implant mucositis** — bleeding on gentle probing, no bone loss. **Peri-implantitis** — bleeding or suppuration, increased probing depth against baseline, and radiographic bone loss beyond initial remodelling.

Maintenance. Professional biofilm removal with **titanium or plastic instruments and glycine air-polishing**. Then the part that actually changes the outcome: **site-specific hygiene, individualised, and demonstrated**. I size an interdental brush to *this* embrasure — the right size is the one that meets slight resistance, not the one in the cupboard — and I put it in her hand and have her use it in front of the mirror. A single-tufted brush round the emergence, superfloss under any pontic, and a powered brush for the rest.

Records and review. I record the six depths, bleeding, suppuration, recession, keratinised width, the prosthesis findings, the radiographic comparison and the diagnosis; that I demonstrated the brushes and she performed them back to me; and I set a **risk-based recall — three to four months, because of the smoking and the periodontal history — otherwise three to six.**"

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 3 **selects the plastic or titanium probe** and says why · 4 six sites at a stated force of 0.2–0.25 N · 5 bleeding, suppuration, recession and keratinised width recorded

· 8 comparison with the baseline radiograph and justification of any new exposure · 9 the 2017 diagnostic criteria applied · 10 individualised brush sized and demonstrated, risk-based recall, recorded.

Accept these late or on prompting: 1 history and symptoms updated · 2 whole-mouth plaque and bleeding scores · 6 implant versus prosthesis mobility · 7 prosthesis fit, contour, cleansability and residual cement.

The commonest way five minutes is wasted here: lecturing on the difference between peri-implant and periodontal tissues instead of examining. The tray was laid out with a steel probe on it as a distractor, and the mark is for **reaching past it and saying why** in the first minute. The other loss is at the end: candidates who arrive at maintenance with 20 seconds left describe an interdental brush instead of sizing one and putting it in the patient's hand, and the demonstration is the mark, not the mention.

OSCE 21 — Atraumatic extraction and ridge preservation

Blueprint: minor augmentation; bone defects; graft and barrier materials (W + OS-P + SCE).

EQUIPMENT

model or pig jaw with a restorable-looking upper premolar; periotomes, luxators, forceps, curettes; xenograft, collagen plug, collagen membrane, sutures.

CANDIDATE INSTRUCTIONS

This upper first premolar is unrestorable. An implant is planned in 4 months. Extract the tooth and perform ridge preservation, describing each step.

DIRECTION OF THOUGHT

The whole point is preserving the buccal plate. Sequence: atraumatic extraction → debride → confirm walls → graft → seal → suture → instructions.

MODEL ANSWER

"I would confirm the tooth, the medical history and consent, including consent to the graft material and its bovine origin, then anaesthetise with buccal and palatal infiltration.

I would extract **atraumatically and flaplessly**: sever the periodontal ligament circumferentially with a periotome, then luxate with fine luxators applied to the ligament space rather than levering off the buccal plate, and deliver with forceps using rotational movements. This is a two-rooted premolar, so if it resists I would section it and remove the roots individually rather than force it.

I would then **debride the socket** thoroughly with curettes, remove all granulation tissue, irrigate with saline, and **probe the walls circumferentially to confirm the buccal plate is intact** — that determines whether this is an Elian Type 1, 2 or 3 socket and therefore what I do next.

Assuming an intact socket, I would place a **slowly resorbing deproteinised bovine bone mineral** into the socket, filling it to just below the crest without over-packing, because volume stability is what I need over four months.

I would seal it with either a **collagen plug or membrane tucked under the buccal and palatal soft tissue margins**, or a **free epithelialised punch graft** from the palate as a socket seal. I would not attempt full primary closure by advancing the buccal flap here, because that pulls keratinised tissue coronally and reduces vestibular depth.

Closure: a **figure-of-eight or crossed mattress suture** in 4/0 monofilament to retain the graft. If I had used a dense PTFE membrane I would deliberately leave it exposed and remove it at three to four weeks.

Post-operatively: written instructions, analgesia, chlorhexidine from day one, no brushing over the site, soft diet, no smoking, an interim bonded pontic rather than a denture if possible, review at one

week, and reassessment for implant placement at four months. I would record the graft material, its lot number and the socket classification."

EXAMINER CHECKLIST

1. Consent including graft origin
2. Periotome/luxator technique, no buccal levering
3. Sectioning if resistant
4. Debridement and irrigation
5. Probes and classifies the socket
6. Selects a slowly resorbing xenograft with justification
7. Barrier or socket seal
8. Appropriate suture technique
9. Interim prosthesis avoiding pressure
10. Instructions, review and records including lot number

Running this station to time

Time check. The model answer is 470 words, 3 minutes 21 seconds at 140 words per minute — but the instruction is *extract the tooth and perform ridge preservation, describing each step*, so the periotome, the luxators, the curettes, the graft and the suture all have to be handled while the commentary runs. About 120 words must go, and they go by **stating each step in a clause as the hand performs it**, with justification kept for three places only: why nothing levers off the buccal plate, why the socket walls are probed, and why the flap is not advanced.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Indicates the model, the periotomes, luxators, forceps, curettes, xenograft, collagen and sutures: "This upper first premolar is unrestorable. An implant is planned in four months. Extract the tooth and perform ridge preservation, describing each step."	Checklist 1, and it must be said **before an instrument is picked up**. Tooth and site confirmed, medical history checked, and **consent that specifically names the graft as deproteinised bovine bone mineral, of bovine origin**, with an alternative offered to a patient who declines on religious or personal grounds. The candidate should also state the organising principle in one line: *everything I do is to preserve the buccal plate*.
0:40–1:50	Silence. Watches the hands.	Checklists 2 and 3. Buccal and palatal infiltration; **flapless**; the **periotome passed circumferentially to sever the periodontal ligament**, then **fine luxators worked in the ligament space, never levering off the buccal plate**, then forceps with **rotational** rather than buccal movements. And the decision stated in advance: this is a **two-rooted premolar, so if it resists I section it and deliver the roots individually** rather than force it.
1:50–2:45	"Carry on."	Checklists 4 and 5. Socket **debrided of all granulation tissue with curettes and irrigated with saline**, then the walls **probed circumferentially to confirm the buccal plate is intact** — and the finding named as an **Elian Type 1 socket**, with the statement that a Type 2 or 3 (a missing or damaged buccal plate) changes the plan to a staged, flapped augmentation which is beyond a general dentist with an interest in implants and would be **referred**.

Time	The examiner does	What earns the mark
2:45–3:50	"What are you putting in it?" Then: "And how are you closing it?"	Checklists 6 and 7. A **slowly resorbing deproteinised bovine bone mineral**, chosen for **volume stability over four months** rather than rapid turnover, filled to just **below the crest and not over-packed**. Sealed with a **collagen plug or membrane tucked under the buccal and palatal margins**, or a **free epithelialised punch graft** from the palate. Explicitly **refuses to advance the buccal flap for primary closure**, because it drags keratinised tissue coronally and loses vestibular depth.
3:50–5:00	"Finish off."	Checklists 8, 9 and 10. A **figure-of-eight or crossed mattress suture in 4/0 monofilament** to retain the graft, not to close it; a **dense PTFE membrane left deliberately exposed and removed at three to four weeks** if that was the choice. Then **what the patient is told and given** — written instructions, analgesia, chlorhexidine from day one, no brushing over the site, soft diet, no smoking, and an **interim bonded pontic rather than a denture** so nothing presses on the graft. Then **review at one week** and reassessment at **four months**, and the **record: graft material, lot number and socket classification**.

The compressed answer — what a strong candidate says in five minutes

"Everything I do here is aimed at one thing — **preserving the buccal plate**.

Consent first. I confirm the tooth, check the medical history, and take consent that includes the graft: I would tell her plainly that the material is **deproteinised bovine bone mineral — of bovine origin** — and offer a synthetic or allogeneic alternative if she would rather not have it. That conversation happens before I anaesthetise, not after.

Anaesthesia. Buccal and palatal infiltration.

The extraction — atraumatic and flapless. No flap, because raising one strips the periosteum from a thin buccal plate and accelerates the resorption I am trying to prevent. I pass a **periotome circumferentially** to sever the periodontal ligament, working apically in small increments. Then **fine luxators in the ligament space**, rotated gently — **at no point do I lever off the buccal plate**, which is the single commonest way this operation is ruined in the first thirty seconds. Forceps last, with **rotational movements** rather than buccal pressure. This is a two-rooted premolar: if it resists, **I section it and deliver the two roots individually**, which is quicker and far safer than forcing it.

Then the socket. I **debride it completely** with curettes — all granulation tissue out, because residual soft tissue will not be replaced by bone — and irrigate with saline. Then I **probe every wall circumferentially** to confirm the buccal plate is intact. That is a decision point, not a formality: an intact plate with intact soft tissue is an **Elian Type 1** socket and I proceed. If the buccal plate is missing or the soft tissue is deficient — Type 2 or Type 3 — this becomes a staged augmentation with a flap and a fixed membrane, which is **beyond what I would do myself, and I would refer**.

The graft. Assuming Type 1, I place a **slowly resorbing deproteinised bovine bone mineral**, in increments, filled to **just below the crest and not over-packed**. I choose it deliberately for **volume stability** — I need this ridge to hold its dimension for four months, so I want a material that resorbs slowly, not one that turns over quickly.

The seal. A **collagen plug or a collagen membrane tucked under the buccal and palatal soft tissue margins**, or a **free epithelialised punch graft** taken from the palate as a socket seal. What I would **not** do is advance the buccal flap to get primary closure — that pulls keratinised tissue coronally, reduces vestibular depth, and leaves me a worse soft tissue bed for the implant than I started with.

Sutures. A **figure-of-eight or crossed mattress suture in 4/0 monofilament** across the socket, to retain the graft rather than to close the wound. If I had used a **dense PTFE membrane**, I would leave it deliberately exposed and remove it at **three to four weeks**.

Afterwards. Written instructions and analgesia; chlorhexidine from day one; no brushing over the site; soft diet; and **no smoking**, which matters more here than almost anywhere. For the gap, an **interim bonded pontic rather than a denture** — a denture presses on the graft, and pressure is exactly what it must not have.

Review and records. I see her at **one week**, and reassess for implant placement at **four months**. I record the socket classification, the technique, the graft material with its **lot number**, the barrier used, the suture, the instructions given and the consent to the bovine material."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 2 periotome and luxator technique with **no levering off the buccal plate** · 5 probes and classifies the socket, and names the limit at Type 2/3 · 6 slowly resorbing xenograft chosen for volume stability · 1 consent naming the bovine origin · 7 barrier or socket seal, without advancing the flap · 10 instructions, review at one week and four months, records with lot number.

Accept these late or on prompting: 3 sectioning the two-rooted premolar if it resists · 4 debridement and irrigation · 8 figure-of-eight or crossed mattress suture · 9 interim bonded pontic rather than a denture.

The commonest way five minutes is wasted here: a long preamble on socket healing — bundle bone, the Araújo and Lindhe model, horizontal and vertical resorption figures — before a single instrument is touched. It sounds authoritative and it costs 90 seconds, after which the graft goes in unnarrated and the consent, the socket classification and the records are never reached. The physiology is the reason for the operation, not the answer to the question.

OSCE 22 — Transcrestal sinus floor elevation

Blueprint: minor augmentation; limitations and timing (W + OS-P + SCE).

EQUIPMENT

maxillary model with 7 mm residual height at the 26; osteotomes, mallet, drills, graft material, CBCT printout.

CANDIDATE INSTRUCTIONS

This patient has 7 mm of residual bone height beneath a flat sinus floor at the UL6. Describe and, where possible, demonstrate a transcrestal sinus floor elevation with simultaneous implant placement, including how you would confirm the membrane is intact.

MODEL ANSWER

"First, indications. The ITI decision tree indicates a **transcrestal approach with simultaneous placement where residual height exceeds 6 mm and the sinus floor is horizontal**. Here I have 7 mm and a flat floor, so the criteria are met. Below 6 mm, or with an oblique floor or septa, a lateral window is indicated, and that is outside my scope. I would also confirm on the CBCT that the membrane is under 2 mm thick, there are no septa, no sinus pathology and the ostium is patent.

Technique: aseptic preparation, chlorhexidine rinse, buccal and palatal infiltration. **Mid-crestal incision** with a full-thickness flap for direct vision.

I prepare the osteotomy conventionally but stop **1 to 2 mm short of the sinus floor**, confirming depth with a depth gauge and, if needed, a periapical radiograph.

I then use **concave-tipped osteotomes of increasing diameter with gentle malleting** to green-stick fracture the floor. I place a small increment of graft material into the osteotomy and use the osteotome to push it upward — the graft, not the metal tip, elevates the membrane, which reduces the risk of tearing. I repeat this in small increments to gain the 3 to 4 mm I need.

I confirm the membrane is intact with the **Valsalva manoeuvre** — the patient pinches the nose and blows gently; I look for air bubbling in the osteotomy and for movement of the membrane. There should be neither. I also check by gentle probing with a blunt instrument.

I then place the implant, which needs to engage the residual 7 mm for primary stability. Because this is D3 or D4 bone, I would **under-prepare** the osteotomy and use a tapered implant. I would aim for at least 25 Ncm.

Closure with 4/0 monofilament interrupted sutures. **Prosthetic loading at approximately 3 months.**

Post-operatively: sinus precautions — no nose-blowing, sneeze with the mouth open, no straws, no smoking, no diving or flying if advised — plus antibiotics and consideration of a decongestant, analgesia and review at one week.

Complications I would warn about and be prepared for: **membrane perforation**, which is the commonest; **benign paroxysmal positional vertigo** from malleting; graft migration into the sinus; sinusitis; and, rarely, displacement of the implant into the sinus. If I perforated the membrane substantially, I would abandon the elevation, close, and re-plan at 3 to 6 months."

EXAMINER CHECKLIST

1. States the >6 mm / flat floor indication
2. Excludes sinus pathology and septa on CBCT
3. Osteotomy stopped 1–2 mm short
4. Osteotome technique with graft-mediated elevation
5. **Valsalva** to confirm integrity
6. Under-preparation for D3/D4
7. Primary stability target
8. Loading at ~3 months
9. Sinus precautions and medication
10. Names complications and the fallback plan

Running this station to time

Time check. The model answer is 533 words, 3 minutes 49 seconds at 140 words per minute — the longest of this group, and the instruction adds *demonstrate where possible*, so the osteotomes and mallet must be handled as well. Around 170 words have to go. They go by **compressing the complication list and the post-operative instructions to one clause each**, and by putting the indication and the competence limit in the first 45 seconds, where they cannot be lost.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Puts the maxillary model, the osteotomes, the mallet and the CBCT printout in front of the candidate: "This patient has 7 mm of residual bone height beneath a flat sinus floor at the UL6. Describe and, where possible, demonstrate a transcrestal sinus floor elevation with simultaneous implant placement, including how you would confirm the membrane is intact."	Checklists 1 and 2, and **the competence line, which must be drawn here and not left to the end**. The threshold stated as a rule — the ITI decision tree indicates a **transcrestal approach with simultaneous placement above 6 mm of residual height with a horizontal floor**; here it is 7 mm and flat, so the criteria are met — followed immediately by: **below 6 mm, or an oblique floor, or septa, means a lateral window, and a lateral window is outside my scope; I would refer.** Then the CBCT exclusions: **membrane under 2 mm, no septa, no sinus pathology, patent ostium**.

Time	The examiner does	What earns the mark
0:45–2:10	Silence. Watches the hands on the model.	Checklists 3 and 4. Aseptic preparation, chlorhexidine rinse, buccal and palatal infiltration, **mid-crestal incision with a full-thickness flap for direct vision**. Osteotomy prepared conventionally but **stopped 1 to 2 mm short of the floor**, confirmed with a depth gauge and, if needed, a periapical. Then **concave-tipped osteotomes of increasing diameter with gentle malleting** to green-stick fracture the floor, and the discriminating detail: **graft is placed into the osteotomy and pushed upward so that the graft, not the metal tip, elevates the membrane**, in small increments to gain the 3 to 4 mm required.
2:10–2:55	"How do you know the membrane is intact?"	Checklist 5, the question the station was built around. **Valsalva** described as an action, not a word: the **patient pinches the nose and blows gently** while the candidate watches the osteotomy for **bubbling and the membrane for movement — there should be neither**. Backed by **gentle probing with a blunt instrument**, and by direct vision down the osteotomy.
2:55–3:55	"Place the implant." Then: "When would you load it?"	Checklists 6, 7 and 8. The implant must **engage the residual 7 mm for primary stability**; this is **D3 or D4 bone**, so the candidate **under-prepares the osteotomy and selects a tapered implant**, and states a target of **at least 25 Ncm**. **Prosthetic loading at approximately 3 months.**
3:55–5:00	"What could go wrong, and what would you tell the patient?"	Checklists 10 then 9. **Membrane perforation, the commonest**; **benign paroxysmal positional vertigo from the malleting**; graft migration into the sinus; sinusitis; and rarely displacement of the implant into the sinus. Then the fallback, explicitly: **if I perforated the membrane substantially I would abandon the elevation, close, and re-plan at three to six months.** Then **what the patient is given** — sinus precautions in writing (**no nose-blowing, sneeze with the mouth open, no straws, no smoking, no diving and no flying if advised**), antibiotics, a decongestant considered, analgesia — the **records** (graft and lot number, elevation achieved, Valsalva result, insertion torque) and **review at one week**.

The compressed answer — what a strong candidate says in five minutes

"**The indication first, because it decides whether I do this at all.** The ITI decision tree indicates a **transcrestal elevation with simultaneous placement where the residual height is above 6 mm and the sinus floor is horizontal**. Here I have **7 mm and a flat floor**, so the criteria are met and this is within the scope of a general dentist with an interest in implants. **Below 6 mm, with an oblique floor, or with septa, the indication is a lateral window — and a lateral window is not within my competence. I would refer that patient to a specialist colleague.** On the CBCT I also confirm the **membrane is under 2 millimetres thick, there are no septa, no sinus pathology and the ostium is patent**; a blocked ostium or an active sinusitis is a contraindication until it is treated.

Preparation. Aseptic technique, chlorhexidine rinse, buccal and palatal infiltration, a **mid-crestal incision and a full-thickness flap** — I want direct vision of the crest for this, not a flapless approach.

The osteotomy. I prepare conventionally but **stop 1 to 2 millimetres short of the sinus floor**, checking with a depth gauge and, if I am not certain, a periapical radiograph.

The elevation. Concave-tipped osteotomes of increasing diameter, with gentle malleting, to green-stick fracture the floor. Then the important detail: I place a **small increment of graft into the osteotomy and push that upward with the osteotome**, so that **the graft elevates the membrane and not the metal tip**. That distributes the force hydraulically across the membrane and is what keeps it intact. I repeat in small increments to gain the **3 to 4 millimetres** I need.

Confirming the membrane. The **Valsalva manoeuvre**: I ask the patient to **pinch the nose and blow gently** while I watch the osteotomy. I am looking for **air bubbling up through the site, and for movement of the membrane** — there should be neither. I confirm with **gentle probing using a blunt instrument** and by looking directly down the osteotomy.

The implant. It has to gain its primary stability from the **residual 7 millimetres**, because the grafted portion contributes nothing at placement. This is **D3 or D4 bone**, so I **under-prepare the osteotomy**, use a **tapered implant**, and aim for **at least 25 newton centimetres**. If I cannot achieve that, I place it and allow submerged healing rather than pretend it is stable.

Closure and loading. 4/0 monofilament interrupted sutures. **Prosthetic loading at approximately three months.**

Complications I warn about and am prepared for. Membrane perforation — the commonest by a distance; **benign paroxysmal positional vertigo** from the malleting, which is why I support the head and mallet gently; graft migration into the sinus; sinusitis; and, rarely, displacement of the implant into the sinus. **If I perforated the membrane substantially, I would abandon the elevation, close the site, and re-plan at three to six months** — I would not graft over a large tear.

What the patient goes home with. Sinus precautions in writing: **no nose-blowing, sneeze with the mouth open, no straws, no smoking, no diving, and no flying if I have advised against it** — for two weeks. Antibiotics, a decongestant considered, and analgesia.

Records and review. I record the residual height, the elevation achieved, the graft material and its **lot number**, the **Valsalva result**, the insertion torque, and the instructions given. **Review at one week**, and a radiograph to confirm the graft position."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 the >6 mm and flat-floor indication, **with the lateral window named as outside scope** · 5 **Valsalva** described as an action and what is looked for · 3 osteotomy stopped 1–2 mm short · 4 graft-mediated elevation rather than the osteotome tip · 10 complications named with the abandon-and-re-plan fallback · 9 sinus precautions and medication.

Accept these late or on prompting: 2 CBCT exclusion of septa, pathology and a blocked ostium · 6 under-preparation for D3/D4 bone · 7 the 25 Ncm primary stability target · 8 loading at approximately three months.

The commonest way five minutes is wasted here: starting with anatomy — the Schneiderian membrane, the posterior superior alveolar artery, the ITI classification in full — and arriving at the osteotomes at three minutes, so the **Valsalva** and the complications are rushed and the fallback plan is never stated. The other, more serious loss is treating the 6 mm threshold as a piece of trivia rather than a boundary: candidates who describe transcrestal and lateral window techniques side by side, as though both were theirs to perform, forfeit the judgement mark that this station exists to test. Say where the line is, and say that you would refer across it.

OSCE 23 — Interpreting a cone beam CT for implant planning

Blueprint: interpret radiographic findings; assess need for investigations (W + SCE, practical interpretation).

EQUIPMENT

printed or on-screen CBCT cross-sections of a posterior mandible and an anterior maxilla, with a measuring tool.

CANDIDATE INSTRUCTIONS

Report these cone beam CT images and state how they affect your implant plan.

MODEL ANSWER

"I would report in a fixed order.

Preliminaries: type of image, date, patient identifiers, field of view, and whether the image quality is diagnostic. I note that the **whole volume must be examined and reported**, not just the region of interest, and that a written report goes in the notes.

Systematic review of the volume: dentition and restorations, periapical status, bone levels, the maxillary sinuses including membrane thickness, septa and ostia, the mandibular canals and mental foramina, the temporomandibular joints, the airway, and any incidental finding. If I found, for instance, an unexplained radiolucency or a calcified carotid atheroma, I would report it and act on it.

Region of interest, posterior mandible: I measure the **vertical height from the crest to the superior border of the canal**, the **buccolingual width at the crest and at 5 mm apically**, the **ridge inclination**, and I look specifically for a **lingual concavity** — the submandibular fossa — which is the classic cause of lingual plate perforation. I identify the mental foramen and any **anterior loop**. I plan an implant length that leaves **at least 2 mm** of clearance from the canal.

Region of interest, anterior maxilla: I measure ridge width, look for a **buccal concavity**, assess the position of the **incisive canal**, assess the nasal floor, and assess the buccal plate thickness and the bone level on the adjacent teeth.

Bone quality: I make an approximate assessment from the trabecular pattern and grey values, recognising that CBCT grey values are **not standardised Hounsfield units** and are approximations only.

Effect on the plan: the CBCT tells me implant length, diameter, angulation and position; whether augmentation is needed and of what kind; whether the case is Straightforward, Advanced or Complex; and therefore whether it is within my competence. If the measurements showed, for example, 5 mm above the canal with a marked lingual concavity, I would use a shorter implant, or reconsider the option entirely, rather than proceed."

EXAMINER CHECKLIST

1. Image identification and quality
2. States whole-volume reporting requirement
3. Systematic review of the volume
4. Measures vertical height to the canal
5. Measures ridge width at two levels
6. Identifies lingual/buccal concavity
7. Identifies mental foramen and anterior loop
8. States the 2 mm safety margin
9. Caveats on CBCT grey values
10. Translates findings into an implant plan and a complexity judgement

Running this station to time

Time check. The model answer is 449 words, 3 minutes 13 seconds at 140 words per minute — but there are two regions of interest to measure with the tool in hand, and every measurement has to be spoken as a number while it is taken. Roughly 120 words must go, and they go by giving the **systematic review of the whole volume one line per structure** so that the two minutes it would otherwise consume are spent on the measurements and the plan, which is where the marks specific to implant planning sit.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Displays the cross-sections of the posterior mandible and the anterior maxilla with the measuring tool: "Report these cone beam CT images and state how they affect your implant plan."	Checklists 1 and 2. **Image type, date, patient identifiers, field of view and whether the quality is diagnostic** — in three sentences — followed immediately by the statement that must not be left out: **the whole volume must be examined and reported, not only the region of interest, and a written report goes in the notes.** Candidates who go straight to the site lose both marks in the first ten seconds.

Time	The examiner does	What earns the mark
0:30–1:30	Silence. Lets the candidate run the survey.	Checklist 3, **one line per structure, delivered fast**. Dentition and restorations, periapical status, bone levels; the **maxillary sinuses including membrane thickness, septa and ostia**; the **mandibular canals and mental foramina**; the temporomandibular joints; the airway; and any incidental finding — with the explicit undertaking that an unexplained radiolucency or a **calcified carotid atheroma would be reported and acted on**, not merely noticed. Speed is the mark here, not detail.
1:30–3:00	"Now the posterior mandible. Show me your measurements."	Checklists 4, 5, 6, 7 and 8, with the tool actually on the image and each figure **spoken as a number**. **Vertical height from the crest to the superior border of the canal**; **buccolingual width at the crest and again 5 mm apically**; **ridge inclination**; and the specific search for a **lingual concavity — the submandibular fossa — the classic cause of lingual plate perforation**. The **mental foramen identified and the anterior loop measured**. Then the rule: an implant length that leaves **at least 2 mm of clearance from the canal**.
3:00–4:00	"And the anterior maxilla?" Then: "How reliable is the bone density on this?"	Checklists 5 and 6 again in the second region, then checklist 9. Ridge width, a **buccal concavity**, the position of the **incisive canal**, the nasal floor, buccal plate thickness and the bone level on the adjacent teeth. Then the caveat, stated plainly: bone quality is judged approximately from the **trabecular pattern and grey values**, and **CBCT grey values are not standardised Hounsfield units** — they vary with the machine, the field of view and the patient's position, and are approximations only.
4:00–5:00	"So what does all that mean for your plan?"	Checklist 10 and the close. The findings **translated into decisions** — implant length, diameter, angulation and position; whether augmentation is needed and of what kind; and the **complexity judgement: Straightforward, Advanced or Complex, and therefore whether it is within my competence or referred**. A worked instance — 5 mm above the canal with a marked lingual concavity means a shorter implant, or reconsidering the site, not proceeding. Then **records** (the written report dated and signed in the notes, with the justification for the scan), **what the patient is told** (the findings, any incidental finding and any medical referral), and **review** — the planning appointment deferred until the report and any referral are complete.

The compressed answer — what a strong candidate says in five minutes

"**I report in a fixed order.**

Preliminaries. This is a cone beam CT, taken on the date shown, of the named patient, with the field of view indicated, and I would state whether the image quality is **diagnostic** for the purpose for which it was justified. Two things follow from that: **the whole volume must be examined and reported, not just the site I am interested in**, and a **written report goes in the notes** over my signature.

Survey of the volume, one line each. Dentition and restorations, periapical status, alveolar bone levels. The **maxillary sinuses** — aeration, membrane thickness, any septum, the patency of the ostia. The

mandibular canals and mental foramina — course, symmetry, any bifid canal. The **temporomandibular joints**, the airway, and the cervical spine where it is in the volume. Any **incidental finding** — an unexplained radiolucency, or a **calcified carotid atheroma** — I would report, and **act on by written referral**, and record that I had done so.

Region of interest, posterior mandible. I measure the **vertical height from the crest to the superior border of the canal**, and I read that number off the scale rather than estimate it. I measure the **buccolingual width at the crest and again 5 millimetres apically**, because a ridge that is adequate at the crest can be knife-edged below it. I note the **ridge inclination**, which decides whether the implant can be restored where the tooth was. And I look specifically for a **lingual concavity — the submandibular fossa** — which is the classic cause of **lingual plate perforation and of the floor-of-mouth haematoma that follows it**; the cross-section is the only image that shows it. I identify the **mental foramen** and measure any **anterior loop**, which extends the danger zone mesially. Then I plan a length that leaves **at least 2 millimetres of clearance from the canal**.

Region of interest, anterior maxilla. Ridge width, and specifically a **buccal concavity** at the site, which will decide whether I need augmentation or an angulated approach. The position of the **incisive canal**, which can occupy the planned site at a central incisor. The nasal floor. And the **buccal plate thickness and the bone level on the adjacent teeth**, because the papillae depend on the neighbours' bone, not on the implant.

Bone quality. I make an approximate assessment from the **trabecular pattern and the grey values** — but I would qualify it: **CBCT grey values are not standardised Hounsfield units.** They vary with the machine, the field of view, the reconstruction algorithm and the patient's position, so they are an approximation and not a density measurement.

The effect on the plan. The scan gives me **implant length, diameter, angulation and three-dimensional position**; whether **augmentation** is needed and of what type; and the **complexity of the case — Straightforward, Advanced or Complex — and therefore whether it is within my competence or should be referred.** As an example: if this section showed **5 millimetres above the canal with a marked lingual concavity**, I would use a shorter implant, or reconsider the option entirely, rather than proceed on the height alone.

Records and follow-up. The full report **written, dated and signed in the notes**, with the justification for the exposure. I explain the findings to the patient, including any incidental finding and why I am writing to their doctor. And I defer the planning appointment until the report and any referral are complete."

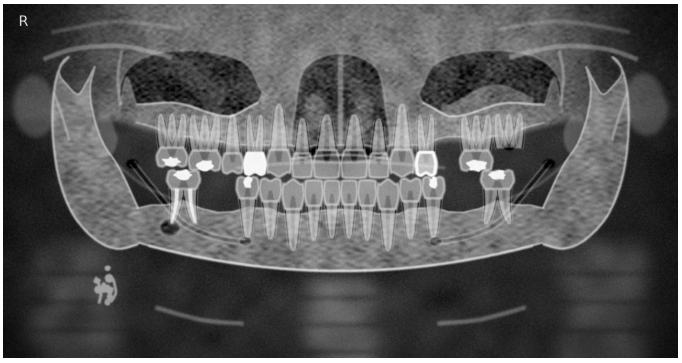
Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 2 whole-volume reporting stated explicitly · 4 vertical height measured to the canal · 6 lingual or buccal concavity identified · 8 the 2 mm safety margin · 9 grey values are not Hounsfield units · 10 findings translated into a plan and a complexity judgement.

Accept these late or on prompting: 1 image identification and quality · 3 the systematic review of the volume · 5 ridge width measured at two levels · 7 mental foramen and anterior loop.

The commonest way five minutes is wasted here: describing the images instead of measuring them. A candidate who narrates the trabecular pattern and the cortices for two minutes has produced a radiology essay and no numbers, and the four marks that belong to implant planning — the height, the two widths, the concavity and the 2 mm margin — are never spoken. The second loss is reporting only the site: the whole-volume statement takes four seconds and is worth a mark, and it is left out more often than any other item on the checklist.

OSCE 24 — Reporting a panoramic radiograph



A panoramic radiograph of the kind used at this station — simulated illustration, not a clinical image

Blueprint: interpret radiographic findings (W + SCE).

EQUIPMENT

a panoramic radiograph of a partially dentate patient with several incidental findings.

CANDIDATE INSTRUCTIONS

Provide a full radiographic report of this panoramic radiograph.

MODEL ANSWER

"Image type, date, and an assessment of quality — positioning, presence of the tongue against the palate, any ghost shadows or artefacts, and whether it is diagnostically acceptable.

Dentition: teeth present and absent, unerupted or impacted teeth, restorations, caries, periapical status, root morphology, retained roots.

Periodontal: generalised or localised bone loss, its pattern and severity, furcation involvement where visible, calculus.

Bone: trabecular pattern, any radiolucent or radiopaque lesion, cortical outlines.

Maxillary antra: floor outline, opacity, mucosal thickening, septa, any lesion.

Mandibular canals and mental foramina: course, symmetry, relationship to the crest, any bifid appearance.

Condyles and rami: outline, symmetry, flattening or erosion.

Other structures: hyoid, styloid processes, cervical spine, soft tissue calcifications including **carotid atheroma**, and the airway.

Then the region of interest for implant planning: apparent vertical bone height at the proposed sites, remembering the **magnification of approximately 20 to 30%, which is not uniform**, and that this image gives **no buccolingual information**.

Conclusion: a summary of the findings, the diagnoses they support, what I cannot determine from this image, and what further investigation I would justify — for example a small field-of-view CBCT if the canal position is critical, or periapicals for the periodontal or periapical assessment.

Finally, I would record the report in the notes with the date and my signature, and act on any incidental finding — for instance referring a patient with a calcified carotid atheroma to their general medical practitioner."

EXAMINER CHECKLIST

1. Image type, date, quality
2. Dentition
3. Periodontal
4. Bone and lesions

5. Antra
6. Canals and foramina
7. Condyles and other structures
8. Incidental findings identified and acted on
9. States magnification and lack of buccolingual data
10. Conclusion, further investigation, and written report in the notes

Running this station to time

Time check. The model answer in this part is only 241 words, 1 minute 43 seconds at 140 words per minute — but it is a skeleton of headings, not a spoken report, and a candidate who inflates each heading into a paragraph will spend all five minutes on the dentition and never reach the antra; the compression here is the reverse of the other stations — **one line per structure, delivered fast**, so that two full minutes remain for the incidental findings and the implant-planning conclusion.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Mounts the panoramic and says: "Give me a full radiographic report of this film."	Checklist 1. **Image type, date and quality** in three sentences — positioning, tongue against the palate, ghost shadows and artefacts, and an explicit statement of whether it is **diagnostically acceptable**. Never start with the teeth.
0:30–2:00	Silence. Lets the candidate run the sequence.	Checklists 2, 3 and 4, **one line each**. **Dentition** — teeth present and absent, unerupted or impacted, restorations, caries, periapical status, retained roots. **Periodontal** — pattern and severity of bone loss, furcations, calculus. **Bone** — trabecular pattern, any radiolucency or radiopacity, cortical outlines. Speed here is the mark, not detail.
2:00–3:00	Silence, or "keep going."	Checklists 5, 6 and 7, again one line each. **Antra** — floor outline, opacity, mucosal thickening, septa. **Mandibular canals and mental foramina** — course, symmetry, relation to the crest, any bifid appearance. **Condyles and rami** — outline, symmetry, flattening or erosion.
3:00–4:00	"Anything else on the film?" — the prompt that separates candidates.	Checklist 8. The structures outside the jaws: hyoid, styloid processes, cervical spine, the airway, and **soft tissue calcifications, above all a calcified carotid atheroma**, with the statement that it would be **acted on by referral to the general medical practitioner**. Checklist 9 follows — **magnification of roughly 20 to 30% and not uniform**, and **no buccolingual information**.
4:00–5:00	"So what does this mean for the implant plan?"	Checklist 10 — the conclusion: summary, diagnoses supported, **what cannot be determined from this image**, and the justified further investigation (small field-of-view CBCT if the canal position is critical; periapicals for the periodontal and periapical assessment). Then **records** (report written, dated and signed), **what the patient is told**, and **review**.

The compressed answer — what a strong candidate says in five minutes

"**This is a panoramic radiograph, taken on the date shown, of a partially dentate adult.** Before I report it I will comment on quality: the patient is positioned symmetrically with the occlusal plane correctly angled, there is no evidence of slumping or rotation, the tongue is against the palate so there is no air shadow over the apices, and I can identify the usual ghost shadows of the contralateral ramus and hard palate. I would state whether it is **diagnostically acceptable** for the purpose it was taken.

Dentition. I would state the teeth present and absent in FDI notation, any unerupted, impacted or supernumerary teeth, retained roots, the restorations present and their adequacy, any radiographic caries, and the periapical status of each standing tooth including root morphology.

Periodontal. The pattern and severity of alveolar bone loss — generalised or localised, horizontal or angular — expressed against root length, furcation involvement where it is visible, and any calculus deposits.

Bone. The trabecular pattern, cortical outlines, and any radiolucent or radiopaque lesion, which I would describe by site, size, shape, margin, internal structure and effect on adjacent structures.

Maxillary antra. The outline of each floor, symmetry of aeration, any generalised or localised opacity, mucosal thickening or retention pseudocyst, and any septum — which matters directly if sinus elevation is contemplated.

Mandibular canals and mental foramina. The course and symmetry of each canal, its relationship to the alveolar crest, the position of each mental foramen, and any bifid appearance or anterior loop.

Condyles and rami. Outline, symmetry, and any flattening, erosion or osteophyte formation.

Other structures — and this is where findings are missed. The hyoid, the styloid processes and any elongation, the cervical spine, the airway, and **soft tissue calcifications**. I would look specifically at the region of the carotid bifurcation, at the level of the third and fourth cervical vertebrae behind the mandibular angle, for a **calcified carotid atheroma** — a nodular radiopacity there is a marker of cerebrovascular risk, and I would **refer that patient to their general medical practitioner in writing** and record that I had done so. I would also look for sialoliths and tonsilloliths.

The region of interest for implant planning. I would assess the apparent vertical bone height between the crest and the antral floor or the mandibular canal at each proposed site — but I would qualify that in two ways. First, this image carries a **magnification of approximately 20 to 30%, and it is not uniform** across the film, so a measured height is not a true height without a calibration marker. Second, it gives **no buccolingual information at all** — no ridge width, no concavity, no inclination.

Conclusion. I would summarise the findings, state the diagnoses they support, state plainly **what I cannot determine from this image** — ridge width, the true buccolingual position of the canal, and the fine periapical and periodontal detail — and state the further investigation I would justify: **periapicals** for the periodontal and periapical assessment, and a **small field-of-view CBCT** limited to the planned site if the canal or antral position is critical to the plan, justified individually and with the dose accounted for.

Finally, records and follow-up. I would write the report in full in the notes, dated and signed, with the film's unique identifier; I would **explain the findings to the patient**, including any incidental finding and why I am writing to their doctor; and I would set the review — the further imaging arranged, the medical referral sent, and the planning appointment deferred until those results are back."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 image type, date and quality · 8 incidental findings identified **and acted on** · 9 magnification of 20–30% and the absence of buccolingual data · 10 conclusion, further investigation and a written report in the notes · 5 the antra · 6 the canals and mental foramina.

Accept these late or on prompting: 2 the dentition · 3 the periodontal assessment · 4 bone and lesions · 7 condyles and the remaining structures.

The commonest way five minutes is wasted here: tooth-by-tooth narration of the dentition. A candidate who names every restoration in FDI notation sounds thorough for 90 seconds and then never reaches the carotid atheroma, the magnification caveat or the conclusion — losing the four marks that are actually specific to implant planning. One line per structure, then spend the time where the marks are.

OSCE 25 — Obtaining consent for implant surgery

Blueprint: principles of valid consent; issues and complexity of informed consent; ensuring understanding (W + SCE, examined practically).

> **EQUIPMENT**
>
> a simulated patient (or examiner), a written information leaflet, a consent form.

CANDIDATE INSTRUCTIONS

This patient has agreed in principle to a single implant at the LR6. Obtain consent for the surgical phase.

MODEL ANSWER

As spoken

"Before we book the surgery I want to go through exactly what's involved, so you can decide with all the facts. Please stop me at any point.

What we're doing and why: you've lost the lower right first molar and we agreed an implant is the option that suits you because it doesn't involve preparing your other teeth. We also discussed a bridge, a denture, and leaving the gap — and you can still change your mind about any of those today.

The procedure: it's done under local anaesthetic in about an hour. I make a small incision in the gum, prepare a channel in the bone with cooled drills, place the implant, and close with stitches. There's a healing period of about three months before we make the crown, and you'll have stitches out in a week.

What to expect afterwards: swelling and discomfort for a few days, some bruising, and a soft diet.

The risks I need you to know about. Common and usually temporary: pain, swelling, bruising, bleeding, and infection which we treat with antibiotics. Less common: the implant not attaching to the bone — that happens in around two to five in a hundred, and if it does we remove it, let it heal and try again. Because this is the lower jaw, there is a nerve running below, and there's a small risk of bruising or damage to it, which can cause numbness or tingling in your lip and chin — usually temporary, occasionally permanent. I've planned the implant to leave a safety margin of at least two millimetres from that nerve, and I'll check the position with an X-ray during the procedure. Other risks are damage to the neighbouring tooth, and later problems — gum infection around the implant, which we call peri-implantitis, and mechanical problems like the screw loosening or the crown chipping.

What you have to commit to: cleaning around it every day with the brushes I'll show you, and coming for maintenance appointments every six months. Implants don't decay, but they can get gum disease, and without maintenance they can be lost.

Costs: here's the written estimate for the surgery and the crown, and here's what maintenance and any future component replacement is likely to cost.

Alternatives if it fails: we'd discuss a bridge or a denture, and I'd tell you honestly what the site can take.

Here's a written information sheet. I'd like you to take it home and come back — I don't want you to sign today. Before you go, can I ask you to tell me in your own words what we've agreed and what the main risks are, so I can be sure I've explained it clearly?

I'll record all of this in your notes, and we'll confirm your consent again on the day."

EXAMINER CHECKLIST

1. Restates diagnosis and why treatment is proposed
2. Restates alternatives including no treatment
3. Describes the procedure and timeline
4. General surgical risks
5. **Site-specific risk — nerve injury — with mitigation**
6. Failure rate quoted honestly
7. Late biological and mechanical risks, and the maintenance commitment
8. Written costs
9. Written information plus cooling-off period
10. **Checks understanding by asking the patient to summarise**, and states what will be recorded

Running this station to time

Time check. The model answer is 608 words, 4 minutes 20 seconds of uninterrupted speech at 140 words per minute — and a patient interrupts, asks about the nerve, and asks what it costs. Add the greeting, the leaflet, the estimate and the teach-back and the honest delivery is past seven minutes. Nothing can be

dropped, because each item is a limb of valid consent, so the compression is in phrasing: **one sentence per risk with the figure attached**, and the last minute reserved for the close.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–1:30	Sits as the patient who has agreed in principle to a single implant at the LR6: "So — are we going ahead with this, then?" May add: "How long does it take?"	Checklists 1, 2 and 3. **Why treatment is proposed** — the LR6 is missing, and an implant suits her because it does not involve preparing the adjacent teeth — and, in the same breath, the **alternatives including no treatment**: a bridge, a denture, and **leaving the gap**, with the explicit statement that she can still change her mind today. Both must be restated even though she has "already agreed"; consent is not inherited from a previous appointment. Then the procedure and the **timeline** in plain words and no jargon: local anaesthetic, about an hour, a small incision, a channel prepared in the bone with cooled drills, the implant placed, stitches; **stitches out at a week, about three months of healing before the crown**; then swelling, discomfort, bruising and soft diet in one sentence.
1:30–2:45	May ask: "Is it risky?"	Checklists 4, 5 and 6 — the highest-value stretch on the station. **Common and temporary**: pain, swelling, bruising, bleeding, infection treated with antibiotics. Then the **site-specific risk named as such**: this is the lower jaw, **a nerve runs below the site**, and injury can cause numbness or tingling of the lip and chin, **usually temporary, occasionally permanent** — immediately followed by the **mitigation**: the implant is planned to leave **at least 2 mm of clearance** and the position is **checked radiographically during the procedure**. Then the failure rate given honestly as a number: **around two to five in a hundred do not integrate**, and if that happens the implant is removed, the site heals, and it can usually be replaced.
2:45–3:40	May ask: "And once it's in, that's it?"	Checklist 7. The **late risks and the commitment together**: **peri-implantitis** explained as gum disease around the implant; mechanical problems — **screw loosening, crown chipping or wear**; then the obligation stated as a condition rather than a suggestion — **daily cleaning with the brushes she will be shown, and maintenance appointments every six months**. The line that earns the mark: *implants do not decay, but they can get gum disease, and without maintenance they can be lost.*
3:40–4:20	May ask: "What's it going to cost?"	Checklist 8. A **written estimate handed over** — the surgery and the crown itemised separately — **plus the ongoing cost**: maintenance appointments and the likely cost of replacing components in future. Verbal figures alone do not score; the estimate has to be in her hand and a copy in the notes.

Time	The examiner does	What earns the mark
4:20–5:00	"So shall I sign?" — the closing test.	Checklists 9 and 10. **Declines the signature today**: the **written information leaflet** is handed over and she is asked to **take it home and come back**, a **cooling-off period**. Then the **teach-back — "tell me in your own words what we have agreed and what the main risks are"** — which is the mark, not the offer to answer questions. Then **what is recorded**: the discussion, the alternatives offered, the risks named individually including the nerve, the estimate and leaflet given, that she was not asked to sign on the day, and that **consent will be confirmed again on the day of surgery**.

The compressed answer — what a strong candidate says in five minutes

"Before we book anything I want to go through exactly what is involved, so that you decide with all the facts. Please stop me at any point.

Why we are doing this. You have lost the lower right first molar. We agreed an implant because it replaces that tooth on its own and **does not involve cutting down the teeth either side**. We also talked about a bridge, a denture, and **simply leaving the gap** — a perfectly reasonable choice — and you can still change your mind today.

What the procedure involves. It is done under **local anaesthetic**, and takes about an hour. I make a small incision in the gum, prepare a channel in the bone with cooled drills, place the implant, and close it with stitches. **Stitches come out in a week**, and there is a healing period of **about three months** before we make the crown. Afterwards, expect swelling and discomfort for a few days, some bruising, and a soft diet.

The risks, and I want to be specific. Common and usually temporary: pain, swelling, bruising, bleeding, and infection, which we treat with antibiotics.

One risk is specific to this site. Because this is the lower jaw, **there is a nerve running below the implant**. There is a small risk of bruising it or damaging it, which would cause **numbness or tingling in your lip and chin — usually temporary, occasionally permanent**. What I do about it: I have planned the implant from the scan to leave **at least two millimetres of clearance from that nerve**, and I will **check the position with an X-ray during the procedure** before I finish.

Failure. In around **two to five in every hundred**, the implant does not attach to the bone. If that happens we remove it, let the site heal, and in most cases we can place another.

Then the longer-term risks. The gum around an implant can become infected — **peri-implantitis** — and it can lose bone. And mechanically, the **screw can loosen and the crown can chip or wear**, and components sometimes need replacing.

Which brings me to what you have to commit to. Cleaning around it **every day** with the brushes I will show you, and coming for **maintenance appointments every six months**. **Implants do not decay, but they can get gum disease** — and without that maintenance they can be lost. If that is not something you can commit to, a bridge or a denture may suit you better, and I would rather we said so now.

Costs. Here is the **written estimate** — the surgery and the crown are itemised separately. And here is what the **maintenance appointments and any future component replacement** are likely to cost — the part people are usually not told. **If it ever failed**, we would look again at a bridge or a denture.

Now, I am not going to ask you to sign today. Here is the **written information sheet** with everything we have covered. **Take it home, read it, and come back** — with any questions.

Before you go, could I ask you to tell me in your own words what we have agreed and what the main risks are? Not to test you — so that I can be sure I have explained it clearly.

And I will write all of this in your notes: what we discussed, the alternatives, the risks I have named including the nerve, that I gave you the estimate and the leaflet, and that you have taken them away to think about. **We will confirm your consent again on the day of the surgery.**"

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 5 **nerve injury named with the 2 mm mitigation and intra-operative radiograph** · 10 **teach-back asked for**, and what is recorded · 9 written information and a cooling-off period, no signature today · 6 failure rate given as a number · 7 late biological and mechanical risks with the maintenance commitment · 2 alternatives including no treatment.

Accept these late or on prompting: 1 the diagnosis and why treatment is proposed · 3 procedure and timeline · 4 the general surgical risks · 8 written costs including maintenance.

The commonest way five minutes is wasted here: describing the surgery in loving detail — the drill sequence, osseointegration, the healing abutment — and then delivering the risks as a rushed list with no numbers. Consent is tested on what the patient understands, not on what the candidate knows: the two marks that decide this station are the **nerve risk with its mitigation** and the **teach-back**, and the teach-back is the one most often lost, because candidates close with "any questions?" instead of asking the patient to say it back. Leave the last 40 seconds for it, always.

OSCE 26 — Consent for grafting materials

Blueprint: "importance of consent in the context of undertaking surgical procedures and the use of grafting materials and barriers" (W + SCE).

EQUIPMENT

samples of xenograft, allograft, alloplast, collagen membrane; a simulated patient.

CANDIDATE INSTRUCTIONS

Explain to this patient the graft materials you propose to use and obtain their consent.

MODEL ANSWER

"There isn't quite enough bone on the outer side of where the implant needs to go, so I plan to add some bone graft material and cover it with a membrane while it heals. I want to explain exactly what those materials are, because some people have strong views about them.

There are four kinds:

Your own bone, which I can collect from the site as I drill. It's the best biologically because it contains living cells, but there's only a small amount and it tends to shrink.

Human donor bone, from a regulated tissue bank, processed so there are no living cells. It works well, but some people are uncomfortable with human donor material.

Animal-derived bone, which is the one I'd normally recommend here. It's usually **from cattle, sometimes from pigs**. It's processed at high temperature so that only the mineral scaffold remains — no proteins and no cells. The reason I favour it is that it barely resorbs, so it holds the shape of the ridge for years. The theoretical risk of disease transmission is regarded as essentially eliminated by the processing, and there are no reported cases from these products, but I want you to know the origin.

Synthetic material, made in a laboratory. No animal or human origin at all, so it avoids those concerns; the trade-off is that it's less well documented in this particular application.

The **membrane** I'd use is also usually **collagen of porcine — that is, pig — origin**, though there are synthetic ones.

So: is there any religious, cultural, dietary or personal reason that would make animal-derived or human-derived material unacceptable to you? If so, we use your own bone with a synthetic material and a synthetic membrane, and I'd explain what that means for the predictability.

Whichever we choose, I'll record the material, the manufacturer and the batch number in your notes, and you'll get that on your implant card."

EXAMINER CHECKLIST

1. Explains why a graft is needed
2. Names all four categories
3. States bovine/porcine origin explicitly
4. States human donor origin explicitly
5. Explains processing and residual risk honestly
6. Explains the membrane and its origin
7. **Asks directly about religious, cultural, dietary or personal objections**
8. Offers an acceptable alternative pathway
9. Explains the consequence of choosing the alternative
10. States that material and lot number will be recorded and given to the patient

Running this station to time

Time check. The model answer is 460 words, 3 minutes 17 seconds at 140 words per minute — but four sample pots have to be picked up and shown, the patient interrupts, and checklist 7 is a **question**, which means the patient's answer and your response to it also have to fit inside the five minutes. Roughly 120 words must go, and they go by **naming each material in a single sentence with its origin attached** instead of explaining the biology of each one.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Sits as the patient, at a **planning appointment, not on the day of surgery**: "You mentioned something about needing extra bone?"	Checklist 1. **Why a graft is needed**, in plain words — there is not enough bone on the outer side of where the implant has to sit, so material is added and covered with a membrane while it heals. Then the framing sentence that buys the rest of the station: *some of these materials come from animals and some from human donors, people have strong views about that, and I would rather you knew now than on the day.* Raising origin **before the day of surgery** is itself the mark; a candidate who does this on the morning of the operation has given the patient no real choice.
0:40–2:10	Slides the four sample pots forward. Silence.	Checklists 2, 3 and 4 — the densest stretch. **All four categories named**, one sentence each, each with its origin said out loud in the patient's language: **your own bone**, collected from the site as I drill; **human donor bone** from a regulated tissue bank — the word **donor** must be used; **animal bone, from cattle, sometimes from pigs** — **cow** and **pig**, not "xenograft"; and **synthetic**, made in a laboratory with no animal or human origin at all. Naming the categories without naming those origins scores 2 and forfeits 3 and 4.
2:10–3:00	May ask: "Is it safe, though?"	Checklists 5 and 6. **Processing and residual risk, honestly**: the animal material is treated at high temperature so only the mineral scaffold remains — no cells, no proteins — the theoretical risk of disease transmission is regarded as essentially eliminated, there are no reported cases, **and I am telling you the origin anyway**. Then the **membrane**, which candidates forget: it too is usually **collagen of porcine — pig — origin**, though synthetic ones exist.

Time	The examiner does	What earns the mark
3:00–4:05	Waits. If the question is not asked, prompts: "Does it matter what it's made of?"	Checklists 7 and 8 — the two that carry the station. The question asked **directly and in full**: *is there any **religious, cultural, dietary or personal** reason that animal-derived or human-derived material would be unacceptable to you?* All four grounds should be offered, and **vegetarian or vegan diet** named among them, because a patient who would never raise religion will raise diet. Then the **alternative pathway offered without being asked twice**: your own bone with a synthetic substitute and a synthetic membrane.
4:05–5:00	"And if I did want the synthetic?"	Checklists 9 and 10. The **consequence of the alternative stated honestly** — synthetic material is less well documented in this application, autogenous bone is limited in volume and resorbs, so the result may be less predictable and I may need to stage it — said as information, not as pressure. Then **records**: the material, the manufacturer and the **batch number** in the notes, that the origin was discussed and which option was chosen, and the batch details given to the patient on an **implant card** they keep. Consent confirmed again on the day.

The compressed answer — what a strong candidate says in five minutes

"**Before we fix a date, there is something I want to go through.** There is not quite enough bone on the outer side of where your implant has to sit, so I plan to add some graft material and cover it with a thin membrane while it heals. I am raising it now rather than on the day, because **some of these materials come from animals or from human donors**, and people have strong views about that. You should decide with time, not in the chair.

There are four kinds.

Your own bone. I can collect a small amount from the site as I drill. Biologically it is the best, because it contains living cells — but there is only a little of it, and it tends to shrink.

Human donor bone. It comes from a regulated tissue bank and is processed so that no living cells remain. It works well. Some people are uncomfortable with the idea of donor material, and that is a perfectly reasonable thing to feel.

Animal-derived bone — and this is the one I would normally recommend here. It is **usually from cattle, sometimes from pigs**. It is processed at high temperature so that only the **mineral scaffold** is left: no cells, no proteins. The reason I favour it is that it barely resorbs, so it holds the shape of the ridge for years.

Synthetic material, made in a laboratory. **No animal or human origin at all**, so it avoids all of that. The trade-off is that it is less well documented in this particular situation.

And the membrane over the top — which people often do not think to ask about — **is also usually collagen of porcine, that is pig, origin**. There are synthetic membranes too.

On safety, honestly. The processing means the theoretical risk of disease transmission is regarded as essentially eliminated, and there are no reported cases from these products. I am not raising the origin because I think it is dangerous — I am raising it because it is **your** decision what goes into your body.

So let me ask you directly. Is there any religious, cultural or dietary reason — including if you are vegetarian or vegan — or any personal reason at all, that would make animal-derived or human-derived material unacceptable to you?

(*waits, and takes the answer at face value — no persuading*)

If the answer is yes, we have a route. We use **your own bone**, collected from the site, together with a **synthetic substitute and a synthetic membrane**. Nothing of animal or human origin at all.

And I owe you the consequence of that. The synthetic materials are less well documented for this job, and your own bone is limited in amount and shrinks a little as it heals, so the result may be **less predictable**

and I may need to build it in two stages rather than one. That is not a reason to change your mind — it is what you are choosing between.

Whichever we use, I will record the material, the manufacturer and the **batch number** in your notes, along with the fact that we discussed where it came from and what you chose. You will get those batch details on an **implant card to keep**, so any dentist anywhere knows exactly what is in there. And I will go over all of this with you again on the day, before we start."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 7 religious, cultural, dietary and personal objection asked about directly · 3 **bovine and porcine origin said in plain words** · 4 human donor origin stated · 8 acceptable alternative pathway offered · 2 all four categories named · 10 material, manufacturer and lot number recorded and given on an implant card.

Accept these late or on prompting: 1 why a graft is needed · 5 processing and residual risk explained honestly · 6 the membrane and its porcine origin · 9 the predictability consequence of the alternative.

The commonest way five minutes is wasted here: teaching the patient the difference between osteogenesis, osteoinduction and osteoconduction. The patient did not ask and the checklist does not reward it — the four categories score for their **origins**, not their mechanisms. The second loss is asking "are you happy with that?" instead of the actual question in checklist 7: a closed question the patient answers "yes" to is not an enquiry into objections, and the same silence swallows checklist 6, where candidates give a careful account of bovine bone mineral and then place a porcine collagen barrier over it without ever mentioning the pig.

OSCE 27 — Medical history interpretation and risk stratification

Blueprint: consider local and systemic risk factors; drug interactions (W + SCE).

EQUIPMENT

three medical history cards.

CANDIDATE INSTRUCTIONS

For each of these three patients, state the implant-relevant risks, what further information you need, and how you would modify your management.

Card 1: 64-year-old female; metastatic breast cancer; denosumab 120 mg monthly; tamoxifen; on chemotherapy. **Card 2:** 52-year-old male; type 2 diabetes, HbA1c 71 mmol/mol; smokes 15/day; treated periodontitis. **Card 3:** 40-year-old female; rheumatoid arthritis; methotrexate; prednisolone 5 mg; ibuprofen self-prescribed.

MODEL ANSWER

"**Card 1.** The critical fact is **denosumab 120 mg monthly** — that is the **oncology** dose, not the osteoporosis dose, in a patient with malignant bone disease. That places her in the **high MRONJ risk** group and **elective implant placement is contraindicated**. Chemotherapy adds immunosuppression. I would liaise in writing with her oncology team, avoid elective surgery, provide a prosthetic alternative, and manage any necessary extractions under the SDCEP high-risk pathway with specialist input.

Card 2. Three additive risk factors: **HbA1c of 71 mmol/mol is poorly controlled diabetes**; **smoking 15 a day** roughly doubles failure risk; and a **history of periodontitis** is the strongest risk factor for peri-implantitis. Together this is a high-risk profile. I would not place implants now. I would liaise with the diabetes team, deliver very brief smoking cessation intervention with referral, confirm periodontal stability against the EFP criteria, and re-assess. If the picture improves, I would reduce the scope, use screw-retained cleansable designs, consent explicitly to the elevated risk, and commit to three-monthly maintenance.

Card 3. Methotrexate means immunosuppression — meticulous asepsis, a lower threshold for antibiotic cover, and critically **NSAIDs must be avoided** because they increase methotrexate toxicity. She is **self-prescribing ibuprofen**, which is a genuine safety issue I would address and communicate to her rheumatologist. **Prednisolone 5 mg** long term impairs healing; steroid cover is generally unnecessary for minor oral surgery but I would confirm with her physician. Her **rheumatoid disease** may affect hand function and cervical spine positioning, so I would assess her dexterity directly and plan short, comfortable appointments. Analgesia would be paracetamol with codeine if needed."

EXAMINER CHECKLIST

1. Identifies oncology-dose denosumab as high risk
2. States elective implants contraindicated in Card 1
3. Identifies poor glycaemic control with a number
4. Quantifies smoking risk
5. Identifies periodontitis history as the peri-implantitis risk factor
6. States the plan to modify risk before treating
7. Identifies the methotrexate–NSAID interaction
8. Identifies the self-prescribing safety issue and acts on it
9. Addresses steroids and immunosuppression practically
10. Addresses dexterity and positioning

Running this station to time

Time check. The model answer is 431 words, 3 minutes 5 seconds at 140 words per minute — comfortable in the abstract, and almost always lost in practice, because the marks are not spread evenly across the three cards. **Card 1 carries two checklist points; cards 2 and 3 carry eight between them.** Nothing needs cutting so much as rationing: **a fixed 90 seconds per card**, timed, and no card opened until the one before it has been closed with a management decision.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:20	Lays out three history cards: "For each of these, the implant-relevant risks, what further information you need, and how you would modify your management."	Nothing scores yet, but the shape must be declared in one sentence — *for each card: the critical fact, the risk it creates, what I need to know, and what I do* — and the candidate should say they will take them in order and keep to time. A candidate who reads all three cards aloud before speaking has spent 40 seconds on transcription.
0:20–1:20	Card 1. 64-year-old, metastatic breast cancer, denosumab 120 mg monthly, tamoxifen, on chemotherapy.	Checklists 1 and 2. The critical fact is that **denosumab 120 mg monthly is the oncology dose, not the 60 mg six-monthly osteoporosis dose**, in a patient with malignant bone disease — say both figures, because the discrimination is the mark. That places her in the **high MRONJ risk group** and **elective implant placement is contraindicated** — stated as a decision, not as "I would be cautious". Chemotherapy adds immunosuppression. Then close it: written liaison with oncology, a **prosthetic alternative offered**, and any unavoidable extraction managed on the **SDCEP high-risk pathway with specialist input**. Ninety seconds, and move.

Time	The examiner does	What earns the mark
1:20–2:50	Card 2. 52-year-old, type 2 diabetes HbA1c 71 mmol/mol, 15 cigarettes a day, treated periodontitis.	Checklists 3, 4, 5 and 6 — the richest card on the station. **Three additive risks, each named with its number or its weight**: **HbA1c 71 mmol/mol is poorly controlled** — the target is under 58, and impaired healing and infection risk follow; **15 a day roughly doubles the failure risk** and worsens peri-implant outcomes; and a **history of periodontitis is the single strongest risk factor for peri-implantitis**. Then checklist 6, which is the one that scores: **I would not place implants now** — liaison with the diabetes team, **very brief smoking cessation advice with referral**, periodontal stability confirmed against the EFP criteria, and re-assessment. Only then: reduced scope, screw-retained cleansable design, explicit consent to elevated risk, three-monthly maintenance.
2:50–4:20	Card 3. 40-year-old, rheumatoid arthritis, methotrexate, prednisolone 5 mg, self-prescribed ibuprofen.	Checklists 7, 8, 9 and 10 — the card candidates never reach. **Methotrexate and NSAIDs must not be combined** — NSAIDs reduce methotrexate excretion and increase its toxicity — and she is **self-prescribing ibuprofen**, which is checklist 8 and requires an **action**: tell her to stop, explain why, and **write to her rheumatologist**. Then **prednisolone 5 mg long term impairs healing**; steroid cover is generally unnecessary for minor oral surgery but confirmed with her physician; **methotrexate means immunosuppression** — meticulous asepsis, a lower threshold for antibiotic cover. Then checklist 10: **hand function and cervical spine**, assessed directly, with short appointments and a design she can actually clean. Analgesia is **paracetamol, with codeine if needed**.
4:20–5:00	"How would you take this forward?"	The governance close. **What is written and to whom** — three letters: oncology, diabetes, rheumatology — **what each patient is told** (Card 1 that implants are not appropriate and why; Card 2 that treatment is deferred, not refused, with the targets stated; Card 3 to stop the ibuprofen today); and the **review interval** for each. Records: the risks identified, the advice given, the referrals made, and that consent will address the elevated risk explicitly.

The compressed answer — what a strong candidate says in five minutes

Q: Card 1. "The critical fact is **denosumab 120 milligrams monthly**. That is the **oncology dose** — the osteoporosis dose is 60 milligrams every six months — in a patient with **malignant bone disease**, and she is on chemotherapy as well. That puts her in the **high-risk group for medication-related osteonecrosis of the jaw**, and **elective implant placement is contraindicated**. I would not place implants and I would not defer the decision. I would liaise **in writing** with her oncology team, offer her a **prosthetic alternative**, and if an extraction becomes unavoidable manage it through the **SDCEP high-risk pathway** with specialist input. I would also want her dentition made as stable as possible and kept under close preventive review."

Q: Card 2. "Three risk factors, and they are **additive**. First, an **HbA1c of 71 mmol/mol** — the target is under 58, so this is **poorly controlled diabetes**, with impaired healing and increased infection risk. Second, **15 cigarettes a day**, which roughly **doubles** implant failure risk and worsens every peri-implant outcome. Third, and most important for the long term, a **history of periodontitis** — the **strongest known risk factor for peri-implantitis**. Together that is a high-risk profile, and my answer is that **I would not place implants now**."

Q: What would you need before you would? "Liaison with his diabetes team and a **meaningful improvement in glycaemic control**; **very brief smoking cessation intervention** with referral to a cessation service, and I would tell him plainly what smoking does to the odds; and confirmed **periodontal stability against the EFP criteria** with a period of successful maintenance behind it. If that picture improves, I would **reduce the scope** of what I plan, use **screw-retained, cleansable designs**, consent him **explicitly to the elevated failure and peri-implantitis risk**, and commit him to **three-monthly maintenance** as a condition of treatment, not a suggestion."

Q: Card 3. "The safety issue jumps out: she is taking **methotrexate** and **self-prescribing ibuprofen**. **NSAIDs reduce methotrexate excretion and increase its toxicity**, so ibuprofen must be avoided. I would tell her today to stop it, explain why in terms she will act on, arrange alternative analgesia, and **write to her rheumatologist** — she may have been doing this for months and nobody knows."

Q: And the rest of that history? "**Methotrexate is immunosuppressive**, so meticulous asepsis and a lower threshold for antibiotic cover. **Prednisolone 5 mg long term impairs healing**; steroid cover is generally unnecessary for minor oral surgery, but I would confirm that with her physician rather than assume. And her **rheumatoid disease itself matters practically** — hand and wrist function determine whether she can use an interdental brush at all, and cervical spine involvement determines how far back I can lay her. I would **test her dexterity in front of me**, not ask her about it, design something she can genuinely clean, and plan **short appointments with supported positioning**. Analgesia is **paracetamol, with codeine if needed**."

Q: How do you finish? "Three letters — oncology, diabetes, rheumatology — and a note of each in the records. Card 1 is told that implants are not appropriate for her and why, and offered the alternative. Card 2 is told treatment is **deferred, not refused**, with the targets he has to hit named. Card 3 is told to stop the ibuprofen **today**. And each is given a review date at which I reassess, with the risks, the advice and the referrals recorded."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **oncology-dose denosumab distinguished from the osteoporosis dose** · 2 elective implants contraindicated in Card 1 · 7 **methotrexate–NSAID interaction** · 8 the self-prescribing acted upon and communicated · 3 HbA1c quoted as a number against the target · 6 risk modified before treating, not alongside it.

Accept these late or on prompting: 4 smoking quantified as doubled risk · 5 periodontitis as the peri-implantitis risk factor · 9 steroids and immunosuppression handled practically · 10 dexterity and positioning assessed.

The commonest way five minutes is wasted here: delivering a lecture on the pathophysiology of MRONJ. Card 1 is worth **two marks out of ten** and is answered in four sentences; a candidate who spends two and a half minutes on bisphosphonate mechanisms and drug holidays reaches Card 3 with forty seconds left and loses the methotrexate interaction, the self-prescribing, the steroids and the dexterity — four easy marks. The second loss is hedging on Card 2: "I would consider whether implants were appropriate" does not score checklist 6, which is for **saying no, now, and naming what has to change first**.

OSCE 28 — A medical emergency during implant surgery

Blueprint: recognise clinical situations which are unsafe or could lead to an adverse event; patient safety (W + SCE).

EQUIPMENT

a manikin, emergency drugs box, oxygen, AED.

CANDIDATE INSTRUCTIONS

Ten minutes into implant surgery your patient becomes flushed, develops a widespread rash and wheeze, and their blood pressure falls. Manage this emergency.

MODEL ANSWER

"This is **anaphylaxis** until proved otherwise. I would work through the standard approach.

Stop treatment. Remove any potential trigger — in dentistry the commonest are **chlorhexidine, latex, antibiotics and, rarely, local anaesthetic**. Note what has been given and when.

Call for help immediately — dial 999, state anaphylaxis, and send a team member for the emergency drugs, oxygen and the AED.

Position: lay the patient flat with the legs raised, unless breathing is difficult, in which case sit them up; if pregnant, tilt to the left.

Adrenaline: 500 micrograms — 0.5 mL of 1:1000 — intramuscularly into the anterolateral thigh. Repeat after 5 minutes if there is no improvement.

Airway and breathing: high-flow oxygen through a non-rebreathe mask at 15 litres per minute.

Monitor pulse, blood pressure, oxygen saturation and level of consciousness continuously; be prepared for cardiac arrest and start CPR if there is no output.

Hand over to the paramedics with a clear account of what happened, what was given and at what time.

Afterwards: document the whole event contemporaneously with timings, ensure the patient is referred for allergy investigation, report the reaction through the Yellow Card scheme, inform the patient's general medical practitioner, replace the used emergency drugs, and hold a **significant event analysis** with the team.

I would also note that this is why the whole team must have annual medical emergency and immediate life support training, and why the emergency drugs and oxygen must be checked and in date."

EXAMINER CHECKLIST

1. Recognises anaphylaxis
2. Stops treatment and removes the trigger, naming likely dental triggers
3. Calls 999 and summons the team
4. Correct positioning
5. **Adrenaline 500 μg IM 1:1000, anterolateral thigh**
6. Repeat at 5 minutes
7. High-flow oxygen
8. Monitoring and readiness for CPR
9. Handover and documentation with timings
10. Yellow Card, allergy referral, drug replacement, significant event analysis

Running this station to time

Time check. The model answer is 372 words, 2 minutes 39 seconds at 140 words per minute — the shortest in this group, and the one candidates most often fail. The reason is that the clock that matters here is **not the OSCE clock but the patient's**: an examiner watching an anaphylaxis station is timing how many seconds elapse before **adrenaline is drawn and given**, and a beautifully ordered answer that reaches it at two minutes has let a simulated patient deteriorate. Speak it in **real time, in the order you would do it**, and hold the governance for the last ninety seconds.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:20 · *the patient's first twenty seconds*	Stands at the manikin: "Ten minutes into implant surgery your patient becomes flushed, develops a widespread rash and wheeze, and their blood pressure falls. Manage this."	Checklists 1, 2 and 3, all three inside twenty seconds and in this order. **"This is anaphylaxis until proved otherwise."** Named, out loud, first — differential diagnosis is not wanted and costs the patient time. **Stop treatment and remove the trigger**, naming the dental candidates: **chlorhexidine, latex, antibiotics and, rarely, local anaesthetic** — and note what was given and when. **Call 999, state the word anaphylaxis**, and send a named team member for the **emergency drugs, oxygen and the AED**. A candidate still taking a history at twenty seconds has already lost the station.

Time	The examiner does	What earns the mark
0:20–1:00 · *adrenaline in by one minute*	Silence. Hands over the drugs box.	Checklists 4 and 5 — the two that carry everything. **Position: flat with the legs raised**; if breathing is difficult, **sit them up**; if pregnant, **tilt to the left**. Then the single most important sentence in the station, said as a complete drug order: **"Adrenaline, 500 micrograms, that is 0.5 mL of 1 in 1000, intramuscularly into the anterolateral thigh."** Dose, concentration, route and site — all four, or the mark is partial. Ampoule checked and the **time of administration spoken aloud**, because that time is what the next decision hangs on.
1:00–2:15	May ask: "It's now been a few minutes and there's no improvement."	Checklists 6, 7 and 8. **Repeat adrenaline at 5 minutes** if there is no improvement — the station will end before the patient's five minutes are up, so this is stated as a timed plan with the clock running, not performed. **High-flow oxygen, 15 litres per minute through a non-rebreathe mask.** Then **continuous monitoring** — pulse, blood pressure, oxygen saturation, level of consciousness — and the readiness statement: **if there is no output, start CPR and use the AED**. Airway obstruction and cardiac arrest are the two ways this patient dies, and both must be named.
2:15–3:30	"The paramedics arrive."	Checklist 9. A **structured handover** — what happened, when it started, what has been given and **at what time**, the observations and their trend, and the suspected trigger. Then **contemporaneous documentation with timings**, which is the half of checklist 9 candidates drop: the exact times of onset, of the 999 call, of each adrenaline dose, of oxygen, and the observations at each point. The times are the record; a narrative without them is worth little.
3:30–5:00	"The patient is in hospital. What now?"	Checklist 10 — the aftermath, and the last row that everyone runs out of time for. **Referral for allergy investigation** so the trigger is identified, **Yellow Card report** of the reaction, the **general medical practitioner informed**, **used emergency drugs replaced and the box re-checked and dated**, and a **significant event analysis with the whole team**. Then what the patient is told and recorded, an **allergy alert on the records** and a warning card, and the standing requirement: **annual medical emergency and immediate life support training for the whole team**, with drugs and oxygen checked in date.

The compressed answer — what a strong candidate says in five minutes

"**This is anaphylaxis until proved otherwise.** Flushing, a widespread rash, wheeze and a falling blood pressure — I am not going to work through a differential; I am going to treat it.

Stop. Treatment stops, and I remove any potential trigger. In dentistry the likely ones are **chlorhexidine, latex, antibiotics and, rarely, local anaesthetic** — so the chlorhexidine goes, gloves and rubber items go, and I note **what has been given and at what time**.

Help, now. (*to the team*) 'Dial 999, say **anaphylaxis**. You — bring the emergency drugs, the oxygen and the AED.' I do not leave the patient.

Position. Flat, with the **legs raised**, to support the circulation. If breathing is the dominant problem I sit them up; if she were pregnant, I would **tilt her to the left**.

Adrenaline — and this is what saves her. Five hundred micrograms — 0.5 millilitres of 1 in 1000 — intramuscular, into the anterolateral thigh. I check the ampoule, give it through clothing if I must, and **say the time out loud so it is recorded**. If there is no improvement I **repeat it after five minutes**, and again at five-minute intervals.

Airway and breathing. High-flow oxygen, 15 litres a minute through a non-rebreathe mask. I listen for stridor as well as wheeze — upper airway swelling is what kills quickly.

Monitor continuously — pulse, blood pressure, oxygen saturation, level of consciousness — and I record them as I go. I am **prepared for cardiac arrest**: if there is no output, **CPR immediately and the AED on**.

Handover. When the paramedics arrive I give them a structured account: what happened and when it began, what has been given and **at what time**, the trend in her observations, and what I think the trigger was.

Afterwards — and this part is examined. I **document the whole event contemporaneously with timings**: onset, the 999 call, each dose of adrenaline, the oxygen, and every set of observations. I make sure she is **referred for allergy investigation**, because until the trigger is identified she is still at risk. I report the reaction through the **Yellow Card scheme**. I **inform her general medical practitioner**, put an **allergy alert on her records**, and make sure she is given a warning card.

Then the practice. The **used drugs are replaced immediately** and the box re-checked and dated, and I hold a **significant event analysis** with the whole team — what went well, what was slow, and what we change.

And the standing requirement behind all of it: the whole team, clinical and non-clinical, has **annual medical emergency and immediate life support training**, and the emergency drugs and oxygen are checked and in date. None of what I have just described works if the adrenaline is expired or nobody knows where it is."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 5 **adrenaline 500 μg, 1:1000, IM, anterolateral thigh** — all four elements · 1 anaphylaxis named immediately · 3 999 called, the word used, team summoned · 7 high-flow oxygen at 15 L/min · 4 flat with legs raised, with the two exceptions · 2 treatment stopped and the trigger removed, with the dental triggers named.

Accept these late or on prompting: 6 repeat at 5 minutes · 8 monitoring and readiness for CPR · 9 handover and timed documentation · 10 Yellow Card, allergy referral, drug replacement and significant event analysis.

The commonest way five minutes is wasted here: treating it as a viva. Candidates open by justifying the diagnosis — the differential of a vasovagal faint, the mechanism of type 1 hypersensitivity — and reach the adrenaline at two minutes, by which time the mark for **timely** administration is gone even though the drug is eventually named correctly. The other loss is a partial drug order: "I'd give adrenaline IM" scores less than the full **500 micrograms of 1 in 1000 into the anterolateral thigh**, and reaching for 1:10,000 or an intravenous route is a patient-safety failure, not a slip. Say the dose the way you would say it to a colleague about to draw it up.

OSCE 29 — Managing altered sensation after implant placement

Blueprint: strategies for managing complications; risk and adverse events (W + SCE).

EQUIPMENT

a diagram of the lower lip and chin, a probe, cotton wool, a periapical radiograph showing an implant close to the canal.

CANDIDATE INSTRUCTIONS

A patient telephones the day after placement of an LR5 implant reporting numbness of the lip. Describe your management.

MODEL ANSWER

"I would see the patient **today**, not at a routine review.

History: exact distribution; whether it is complete numbness (anaesthesia), reduced sensation (hypoaesthesia), tingling (paraesthesia) or unpleasant/painful sensation (dysaesthesia); the time of onset; and whether it has changed.

Examination and documentation: I would **map the affected area onto a diagram of the lip and chin with measurements**, photograph it, and perform objective tests — light touch with cotton wool, sharp/blunt discrimination, two-point discrimination and thermal testing — comparing with the contralateral side. I would record all of this and the date. This mapping repeated at each visit is how I will judge recovery.

Imaging: a periapical, and if inconclusive a small field-of-view CBCT, to determine whether the implant is **impinging on or within the canal**.

Action: - If the implant is impinging, **remove it or back it out immediately**. The evidence favours decompression within **24 to 36 hours**; delay reduces the chance of recovery. - If the implant is clear of the canal and the injury is likely from indirect trauma or haematoma, I would consider a short course of a **corticosteroid with an NSAID** — recognising the evidence is limited — and monitor.

Communication: explain honestly what has happened, apologise under **duty of candour**, describe what I am doing and the likely course, and give realistic information — many such injuries improve over weeks to months, but some are permanent.

Follow-up and referral: review at 1 week and 1 month with repeat mapping. If there is **no meaningful improvement**, refer to oral and maxillofacial surgery **early** — within weeks, not months — because the window for any surgical intervention is limited.

Governance: inform my indemnity provider, record everything contemporaneously, and undertake a significant event analysis reviewing my planning, my safety margin and whether CBCT was used."

EXAMINER CHECKLIST

1. Sees the patient the same day
2. Characterises the sensory disturbance correctly
3. **Maps and documents** with diagram and measurements
4. Objective sensory testing
5. Imaging to determine implant position
6. Removes/backs out an impinging implant within 24–36 h
7. Considers steroid/NSAID with appropriate caveat
8. Duty of candour and honest prognosis
9. Structured review and **early** referral if not improving
10. Indemnity notification, records, significant event analysis

Running this station to time

Time check. The model answer is 436 words, 3 minutes 7 seconds at 140 words per minute — but the diagram, the probe and the cotton wool are on the bench, and the examiner expects the mapping to be **shown**, not described, which costs a minute that the word count does not contain. Roughly 130 words must go, and they go from the sensory vocabulary: **name the four terms in one clause each** rather than defining them, and protect the two things this station is really about — the **24-to-36-hour window** and the **duty of candour**.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	"A patient telephones the day after placement of an LR5 implant, reporting numbness of the lip. Describe your management."	Checklists 1 and 2. The first sentence decides the station: **"I would see her today, not at her routine review."** That is checklist 1 and it must be unconditional — not "I would ask her to come in soon". Then the telephone history in one breath: **exact distribution**; whether it is **anaesthesia** — complete numbness — **hypoaesthesia** — reduced sensation — **paraesthesia** — tingling — or **dysaesthesia** — unpleasant or painful sensation; the **time of onset**, specifically whether it was present as the local anaesthetic wore off; and whether it has changed since.
0:30–1:50	Slides across the lip-and-chin diagram, the probe and the cotton wool. Says nothing.	Checklists 3 and 4 — the highest-value stretch, and it must involve the hands. **Map the affected area onto the diagram with measurements in millimetres**, photograph it, and **date and sign it** — this is the baseline against which every subsequent visit is judged, and the mark is for creating a record, not for observing a deficit. Then **objective testing, each named and each compared with the contralateral side**: **light touch with cotton wool, sharp–blunt discrimination, two-point discrimination, thermal testing** — said aloud as each is performed. "I would test the sensation" scores 4 at half value.
1:50–2:50	Hands over the periapical showing the implant close to the canal: "What does this change?"	Checklists 5 and 6 — the decision the station turns on. **A periapical, and if inconclusive a small field-of-view CBCT**, to establish whether the implant is **impinging on or within the canal**. Then the action, with the number attached: if it is impinging, **remove it or back it out immediately — the evidence favours decompression within 24 to 36 hours, and delay reduces the chance of recovery**. This is a same-day surgical decision and must be stated as one. If the implant is clear of the canal, the injury is more likely indirect — retraction, a drill overrun, an injection or a haematoma — and the management differs.
2:50–3:45	"And medically?"	Checklist 7, with its caveat, and the beginning of the honest conversation. A **short course of a corticosteroid with an NSAID** may be considered where the implant is clear of the canal — **and the candidate must say that the evidence for this is limited**, because the mark is for the caveat as much as the drug. Then the prognosis given honestly and without false comfort: **many such injuries improve over weeks to months, some do not, and some are permanent**.

Time	The examiner does	What earns the mark
3:45–5:00	"How do you follow her up, and what do you tell her?"	Checklists 8, 9 and 10 — follow-up and candour, the rows candidates run out of clock for. **Duty of candour discharged explicitly**: tell her honestly what has happened, **apologise**, say what is being done and what to expect — and record that the apology was given. Then the **structured review: one week and one month, with the mapping repeated onto a fresh diagram each time**, so recovery is measured rather than remembered. And the referral window: **if there is no meaningful improvement, refer to oral and maxillofacial surgery early — within weeks, not months**, because the window for surgical intervention closes. Finally: **indemnity provider informed, contemporaneous records, significant event analysis** on the planning, the safety margin and whether a CBCT should have been taken.

The compressed answer — what a strong candidate says in five minutes

"**I would see her today.** Not at her one-week review — today. Altered sensation the day after a lower implant is time-critical, and the single thing that most changes the outcome is how quickly it is assessed.

On the telephone I establish the essentials: exactly where she is numb; whether it is **complete numbness — anaesthesia**, **reduced sensation — hypoaesthesia**, **tingling — paraesthesia**, or **unpleasant or painful sensation — dysaesthesia**, which is the most troublesome; when it started, and in particular whether she was ever normal after the local anaesthetic wore off; and whether it has changed since.

When she arrives, I map it. I **draw the affected area onto a diagram of the lip and chin with measurements**, photograph it, and date and sign it. That map is the baseline — without it I will have no way of telling in six weeks whether she is better, and 'it seems a bit smaller' is not a record.

Then objective testing, each side compared: light touch with cotton wool, **sharp–blunt discrimination**, **two-point discrimination**, and **thermal testing**, always against the contralateral side. I record every result.

Imaging. A **periapical**, and if that is inconclusive a **small field-of-view CBCT**, for one purpose: to establish whether the implant is **impinging on, or within, the inferior alveolar canal**.

And then the decision, which is today's decision. If the implant **is** impinging, I **remove it or back it out immediately**. The evidence favours **decompression within 24 to 36 hours**, and every hour of delay reduces the chance of recovery. I would not wait to see whether it settles, and I would not wait for an opinion.

If the implant is clear of the canal, the injury is more likely from indirect trauma — retraction, a drill overrun, the injection itself, or a haematoma in the canal. There I would consider a **short course of a corticosteroid with an NSAID**, while being clear that **the evidence for this is limited**, and monitor closely.

Now what I tell her, and I would not delegate this. I explain honestly what has happened and why. I **apologise — this is the duty of candour**, and it is a professional obligation, not an admission that ends my career. I tell her what I am doing about it today. And I give her a realistic prognosis: **many of these injuries improve over weeks to months; some improve only partly; and some are permanent.** I would rather she heard that from me now than discovered it herself in three months.

Follow-up is structured, not 'come back if it's no better'. I review at **one week and at one month**, and at each visit I **repeat the mapping onto a fresh diagram** and repeat the objective tests, so recovery is measured. If there is **no meaningful improvement, I refer to oral and maxillofacial surgery early — within weeks, not months** — because the window in which surgical intervention can help is limited, and a late referral closes it.

And the governance. I **inform my indemnity provider** at the outset, and record everything **contemporaneously** — the maps, the test results, the imaging, the decision I made and why, the apology given and what she was told. Then a **significant event analysis**: was my planning adequate, did I leave a genuine 2 mm safety margin, and should I have had a CBCT before I drilled."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 6 **impinging implant removed or backed out within 24–36 hours** · 1 seen the same day · 3 **mapped and documented with a dated diagram and measurements** · 8 duty of candour, apology and honest prognosis · 9 review at one week and one month with **early** OMFS referral · 5 imaging to establish the implant's relation to the canal.

Accept these late or on prompting: 2 the four sensory terms used correctly · 4 the named objective tests against the contralateral side · 7 steroid and NSAID with the evidence caveat · 10 indemnity, contemporaneous records, significant event analysis.

The commonest way five minutes is wasted here: classifying the injury. Candidates deliver Seddon and Sunderland, neurapraxia through neurotmesis, and arrive at management with ninety seconds left — and the classification carries **no** checklist point here. The second and more damaging loss is softness about timing: "I would monitor and review in a week" reads as reasonable and forfeits checklist 6 outright, because the whole point of the station is that a numb lip the day after surgery is an **emergency with a 24-to-36-hour window**. The same softness costs checklist 8 — candidates describe explaining the complication and never actually apologise, and the examiner is listening for the words.

OSCE 30 — Harvesting particulate autogenous bone

Blueprint: use autogenous bone harvested from the oral cavity for minor augmentation (FGDP skill 7; W + OS-P + SCE).

EQUIPMENT

model of a posterior mandible; bone scraper, bone collector/trap, rongeur, chisel.

CANDIDATE INSTRUCTIONS

You require autogenous bone for a small buccal dehiscence. Demonstrate and describe how you would harvest it.

MODEL ANSWER

"I need only a small volume, so I would use **intraoral particulate sources** rather than a block — block harvesting from the ramus or symphysis is an advanced procedure with recognised morbidity and is outside my scope.

Sources, in order of preference here: 1. **Bone collected during the osteotomy** into a sterile bone trap in the suction line. This is free bone I am producing anyway. I would use a dedicated sterile trap, collect during the first drills before significant blood and saliva contamination, and use clinical judgement about its quality. 2. **A bone scraper** used on the **external oblique ridge** or the buccal cortex adjacent to the site. I extend the existing flap slightly, identify a flat cortical area away from the mental nerve and the canal, and draw the scraper in firm, controlled strokes **away from vital structures**, collecting the shavings in the scraper chamber. I take only cortical shavings, not a full-thickness block. 3. **A rongeur** on any sharp bony crest I am reducing anyway during alveoloplasty.

I would keep the harvested bone **moist in sterile saline** and use it promptly, placing it **directly against the implant surface** where its osteogenic cells are most valuable, then layering a deproteinised bovine bone mineral over it for volume stability.

Precautions: identify and avoid the **mental nerve** and the **inferior alveolar canal**; avoid perforating into the marrow space or the lingual plate; use irrigation with the scraper to avoid heat; and warn the patient about additional post-operative discomfort at the harvest site.

I would record the source, the approximate volume and the composite ratio used."

EXAMINER CHECKLIST

1. States that block harvesting is out of scope
2. Bone trap technique and its limitations
3. Bone scraper technique with direction of stroke
4. Correct donor site selection

5. Avoids mental nerve and canal
6. Keeps bone moist in saline
7. Places autogenous bone against the implant surface
8. Layers xenograft for volume stability
9. Warns patient about donor-site morbidity
10. Records source and materials

Running this station to time

Time check. The model answer is 396 words, 2 minutes 50 seconds at 140 words per minute — and the verb is *demonstrate and describe*, so three instruments have to be picked up, one has to be put back down, and the scraper has to be drawn across the model in the right direction while you talk. Budget **40 seconds for the scope statement and the instrument choice**, 90 seconds for the harvest itself, and keep the last minute for handling, layering and the record — the four marks that sit after the bone is out.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Presents the posterior mandible model and a bench carrying a **bone scraper, a bone collector, a rongeur and a chisel**: "You require autogenous bone for a small buccal dehiscence. Demonstrate and describe how you would harvest it."	Checklist 1, and it must come first, with the hands. **The chisel is put down and named as the block-harvesting instrument**: *I need a small volume, so I would use intraoral particulate sources — **block harvesting from the ramus or symphysis is an advanced procedure with recognised morbidity and is outside my scope**.* Naming the limit and declining the instrument is a mark, not a hedge, and it is lost by silence: a candidate who simply never touches the chisel has not scored checklist 1. Then the sources listed in order of preference, one clause each.
0:40–1:40	Watches the hands.	Checklist 2. **Bone collected during the osteotomy into a sterile trap in the suction line** — free bone that is being produced anyway. The mark includes the **limitations**, which candidates omit: a **dedicated sterile trap**, collection **during the early drills before significant blood and saliva contamination**, a small and unpredictable yield, and clinical judgement about whether what is in the trap is worth using.
1:40–2:50	May ask: "Show me on the model."	Checklists 3, 4 and 5 — the practical core. The **scraper on the external oblique ridge** or the buccal cortex adjacent to the site, reached by **extending the existing flap slightly** rather than opening a second site. A **flat cortical area is identified and named**, away from the mental foramen and the canal. Then the stroke itself, demonstrated: **firm, controlled strokes drawn away from vital structures**, **cortical shavings only, not a full-thickness block**, shavings collected in the scraper chamber, and **irrigation to avoid heat**. Checklist 5 must be said as the hand moves — **the mental nerve and the inferior alveolar canal identified and avoided**, and no perforation into the marrow space or through the lingual plate.

Time	The examiner does	What earns the mark
2:50–3:50	"You have your bone. What now?"	Checklists 6, 7 and 8, and these are worth as much as the harvest. The bone is kept **moist in sterile saline and used promptly** — never allowed to dry on the drape. It is placed **directly against the exposed implant surface**, because that is where its living osteogenic cells are worth having and there is not enough of it to waste anywhere else. Then **deproteinised bovine bone mineral layered outside it for volume stability**, because autogenous bone alone resorbs and will not hold the buccal contour. The composite is the answer; autogenous alone is not.
3:50–5:00	"Anything else before you close?"	Checklists 9 and 10 — what the patient is told and what is written. **Warn the patient in advance about donor-site morbidity**: additional soreness and swelling at the harvest site, which is a second surgical site even though it is through the same incision. Consent must have covered the **bovine origin of the layered xenograft** and the porcine membrane. Then **record the source of the autogenous bone, the approximate volume, the composite ratio, and the brand and lot numbers of the xenograft and membrane**, the lot numbers going onto the implant card. Review at one week for both the graft and the donor site.

The compressed answer — what a strong candidate says in five minutes

"**First, what I am not going to do.** (*putting the chisel back*) This is a **small buccal dehiscence** and I need only a small volume, so I would use **intraoral particulate sources**. **Block harvesting from the ramus or symphysis is an advanced procedure with recognised morbidity — donor-site pain, altered sensation, and in the symphysis pulpal damage — and it is outside my scope.** If I needed a block, I would refer.

My sources, in order of preference.

First, bone from the osteotomy itself. I collect into a **sterile bone trap in the suction line** — this is free bone I am producing anyway. I use a **dedicated sterile trap**, and I collect during the **first drills, before the field is heavily contaminated with blood and saliva**. The limitation is honest: the yield is small and unpredictable, and I use clinical judgement about whether what is in the trap is clean enough to place.

Second, and here for the volume I actually need, a bone scraper. I extend the existing flap slightly rather than opening a second site, and I work on the **external oblique ridge** or the buccal cortex just adjacent. I identify a **flat cortical area**, and — critically — I locate the **mental foramen** and know where the **inferior alveolar canal** runs before I start.

The stroke matters. (*demonstrating*) Firm, controlled strokes, **always drawn away from the vital structures**, never towards them, so that if the instrument runs on it runs into safe bone. I take **cortical shavings only — not a full-thickness block** — I do not perforate into the marrow space or through the lingual plate, and I irrigate as I go so the shavings are not heated. The shavings collect in the scraper chamber.

Third, a rongeur on any sharp bony crest I am reducing anyway during alveoloplasty — again, bone I would otherwise discard.

Handling. The harvested bone goes **straight into sterile saline and is kept moist**, and I use it promptly. Dried autogenous bone is dead autogenous bone, and its only advantage over a substitute is that it contains living cells.

Placement, and this is where the small volume earns its keep. The autogenous chips go **directly against the exposed implant surface** — that is where I want osteogenic and osteoinductive potential, and I have too little of it to spread thinly. **Over that I layer a deproteinised bovine bone mineral**, chosen for its very slow resorption, to hold the buccal volume — because autogenous bone alone would shrink and the contour would be lost. Then a collagen membrane, extended onto sound bone, and tension-free closure.

What the patient is told, beforehand. There is a **second surgical site**, even though it is through the same incision, and she should expect **additional soreness and swelling** there. And because I am layering a **bovine** substitute under a **porcine** collagen membrane, the origin of those materials must have been consented **before the day**, with a synthetic route offered if she objects.

What I record. The **source of the autogenous bone** — trap, scraper or rongeur — the **approximate volume**, the **ratio of autogenous to substitute**, and the **brand and lot numbers** of the xenograft and membrane, which go onto her implant card. And a review at one week to check both the graft site and the donor site."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **block harvesting named as out of scope, with referral** · 3 scraper technique with the **stroke drawn away from vital structures** · 5 mental nerve and canal identified and avoided · 7 **autogenous bone placed directly against the implant surface** · 8 xenograft layered outside it for volume stability · 6 kept moist in sterile saline and used promptly.

Accept these late or on prompting: 2 bone trap technique and its limitations · 4 donor site selection on the external oblique ridge · 9 donor-site morbidity warning · 10 source, volume, ratio and lot numbers recorded.

The commonest way five minutes is wasted here: a comparative lecture on autogenous bone versus substitutes — osteogenesis, osteoinduction, the gold standard, the resorption data — delivered before a single instrument has been touched. This is a *demonstrate* station; the marks are on the model, and the chisel is on the bench deliberately, so a candidate who neither uses it nor mentions it has quietly forfeited checklist 1. The second loss is stopping when the bone is out: four of the ten points — moist, against the implant, layered, recorded — all sit after the harvest, and a candidate who spends three minutes scraping never reaches them.

OSCE 31 — Membrane selection and placement

Blueprint: role of barriers in bone augmentation and their application during surgery, with limitations (W + SCE).

EQUIPMENT

a model with a grafted buccal dehiscence; collagen membrane, dPTFE membrane, titanium-reinforced membrane, tacks, scissors, tweezers.

CANDIDATE INSTRUCTIONS

Select and place an appropriate membrane over this graft, explaining your choice and the limitations of the material.

MODEL ANSWER

"This is a **contained buccal dehiscence** with the graft supported by the implant and the residual bone walls, so it is a space-maintaining situation. I would select a **native collagen membrane**.

Preparation: I trim the membrane so it extends **2 to 3 mm beyond the defect margins onto sound bone** in every direction. I would use dry tweezers and handle it minimally, because a collagen membrane becomes adherent and difficult to reposition once wet.

Placement: I tuck the palatal or lingual edge under the palatal flap and stabilise it, then drape it over the graft and tuck the buccal edge apically under the buccal flap. If the membrane were mobile I would fix it with **titanium tacks** or by trapping it under the cover screw.

Stability is essential — this is the second "S" of the PASS principles: primary closure, angiogenesis, space maintenance, stability. A mobile membrane allows soft tissue ingrowth and defeats the purpose.

Then closure: a **periosteal releasing incision** to advance the buccal flap for tension-free primary closure, with the suture line lying over sound bone rather than over the membrane edge, and horizontal mattress plus interrupted sutures in a monofilament or ePTFE.

Limitations I would state. A collagen membrane has **poor space-maintaining capacity** — it collapses into any non-supported defect, which is why it must be used with a supporting graft. Its

resorption is unpredictable, anywhere from about four weeks to six months. It is of **porcine or bovine origin**, so consent must cover that. A dense PTFE membrane maintains space better and tolerates exposure but must be removed; a titanium-reinforced membrane is best for large or vertical defects but that level of augmentation is **Complex and I would refer**.

Exposure, the commonest complication, follows tension at closure, a suture line over the membrane, denture pressure or smoking. If a collagen membrane exposes without infection, I manage with chlorhexidine, no brushing and close review; an exposed PTFE membrane I would remove early."

EXAMINER CHECKLIST

1. Correct membrane choice with justification
2. Trims to overlap 2–3 mm onto sound bone
3. Handles minimally and appropriately
4. Tucks and stabilises both edges
5. States fixation options
6. Cites PASS principles
7. Periosteal release and tension-free closure
8. Suture line off the membrane
9. States collagen limitations (space maintenance, resorption, origin/consent)
10. Exposure management differing by membrane type

Running this station to time

Time check. The model answer is 454 words, 3 minutes 15 seconds at 140 words per minute — but three membranes are laid out, one has to be chosen and trimmed with scissors, and it has to be tucked under two flaps on a model while you keep talking. Roughly 130 words must go, and they go by **stating each limitation as a clause with its consequence attached** rather than as a paragraph — "resorption is unpredictable, four weeks to six months, so I cannot rely on a fixed barrier period" is the whole of that mark.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Presents the model with the grafted buccal dehiscence and three membranes — **collagen, dPTFE and titanium-reinforced** — with tacks and scissors: "Select and place an appropriate membrane, explaining your choice and the limitations of the material."	Checklist 1, and it needs the defect read out loud before the hand moves. **This is a contained buccal dehiscence, supported by the implant and the residual bone walls — a space-maintaining situation** — therefore a **native collagen membrane**. The other two are picked up and put down with a reason each: **dPTFE** maintains space better and tolerates exposure but is non-resorbable and must be removed; **titanium-reinforced** is for large or vertical defects, and **that level of augmentation is Complex and I would refer**. Naming that limit is a mark, and it is only available here, at the moment the membrane is rejected.
0:40–1:40	Hands over the scissors and tweezers.	Checklists 2 and 3. The membrane **trimmed so that it extends 2 to 3 mm beyond the defect margins onto sound bone in every direction** — the figure said, and the overlap shown on the model. Then handling: **dry tweezers, handled minimally, positioned once**, because a native collagen membrane becomes adherent and effectively unrepositionable as soon as it is wet. Candidates who wet it "to make it drape" and then wrestle it into place have demonstrated the error the checklist is testing.

Time	The examiner does	What earns the mark
1:40–2:40	Silence. Watches the placement.	Checklists 4, 5 and 6. **Both edges tucked**: the palatal edge slid under the palatal flap first and stabilised, then the membrane draped over the graft and the **buccal edge tucked apically under the buccal flap**. Then **fixation options named** — **titanium tacks**, trapping it under the **cover screw**, or a periosteal suture — if the membrane is mobile. And the principle: **PASS — primary closure, angiogenesis, space maintenance, stability** — with its consequence, that **a mobile membrane permits soft tissue ingrowth and defeats the whole purpose**.
2:40–3:40	"Close it."	Checklists 7 and 8. A **periosteal releasing incision** to advance the buccal flap for **tension-free primary closure** — tension is the origin of the commonest complication here. Then the detail that separates candidates: the **suture line must lie over sound bone, not over the membrane edge**, because a suture line over the barrier is where dehiscence begins. **Horizontal mattress to evert and take the tension, plus interrupted sutures to appose**, in a **monofilament or ePTFE**.
3:40–5:00	"What are the limitations, and what if it exposes?"	Checklists 9 and 10, plus the record. **Collagen limitations, three, each with its consequence**: **poor space maintenance** — it collapses into any unsupported defect, hence the graft beneath it; **unpredictable resorption, roughly four weeks to six months**; **porcine or bovine origin, so consent must cover it**. Then **exposure managed by membrane type**: collagen exposed **without infection** — chlorhexidine, no brushing, no denture pressure, close review, accepting some graft loss; an **exposed PTFE membrane removed early**. Finally: **brand and lot numbers recorded** and given on the implant card, the patient told to **avoid pressure on the site, not to smoke, and to report anything showing through**, and **review at one week**, then before uncovering at four to six months.

The compressed answer — what a strong candidate says in five minutes

"**Reading the defect first.** This is a **contained buccal dehiscence**, and the graft is supported by the implant body and the residual bone walls — so this is already a **space-maintaining** situation and I do not need the membrane to hold space for me. That decides the choice.

I would use a native collagen membrane. (*putting the others down*) Not the **dense PTFE** — it maintains space better and tolerates exposure without infection, but it is non-resorbable and would require a second procedure to remove it, which this defect does not justify. And not the **titanium-reinforced** membrane: that is the right material for a **large or vertical defect**, and a defect that needs one is **Complex augmentation, which I would refer** rather than attempt.

Trimming. I trim it so that it **extends 2 to 3 millimetres beyond the defect margins onto sound bone in every direction**. Under-extension is the commonest error — the barrier has to sit on bone all the way round, or soft tissue simply enters at the edge.

Handling. Dry tweezers, minimal handling, and I place it once. Once a collagen membrane is wet it becomes adherent and cannot realistically be repositioned, so I decide where it is going before it goes there.

Placement. I tuck the **palatal edge under the palatal flap first** and stabilise it, then drape the membrane over the graft and **tuck the buccal edge apically under the buccal flap**. If it were mobile I would fix it — **titanium tacks**, or trapping the edge under the **cover screw**, or a periosteal suture.

Why stability matters — this is the second **S** of the **PASS principles: primary closure, angiogenesis, space maintenance and stability**. A **mobile membrane allows soft tissue ingrowth beneath it**, and then it is doing nothing at all.

Closure. A **periosteal releasing incision** at the base of the buccal flap so it advances without tension, and I test that the flap sits passively before I suture. The **suture line lies over sound bone, not over the membrane edge** — a suture line sitting on the barrier is precisely where a dehiscence starts. **Horizontal mattress sutures to take the tension and evert the edges, interrupted sutures to appose**, in **monofilament or ePTFE**, which harbour less plaque than braided silk.

The limitations I would state honestly. A collagen membrane has **poor space-maintaining capacity** — it collapses into any part of the defect that is not supported, which is exactly why it must be used with a graft under it. Its **resorption is unpredictable — anywhere from about four weeks to six months** — so I cannot count on a defined barrier period. And it is of **porcine or bovine origin**, which must be consented before the day of surgery, with a synthetic alternative offered if that is unacceptable.

Exposure is the commonest complication, and it follows **tension at closure, a suture line over the membrane, denture pressure or smoking**. Management depends on the material: a **collagen membrane exposed without infection** I manage conservatively — **chlorhexidine, no brushing over the site, no denture pressure, close review** — accepting some graft loss. An **exposed PTFE membrane I would remove early**, before the space beneath it is contaminated.

Records and review. Brand and lot number of graft and membrane in the notes and on her implant card. She is told to **avoid pressure on the site, not to wear a denture over it unless it has been relieved, not to smoke, and to telephone if she sees anything showing through the gum**. **Review at one week**, again at a month, and before uncovering at four to six months."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **collagen chosen because the defect is contained**, with the titanium-reinforced option referred · 2 trimmed to overlap **2–3 mm onto sound bone** · 4 both edges tucked and the membrane stabilised · 7 periosteal release and tension-free primary closure · 9 collagen's three limitations with consequences · 10 exposure managed differently for collagen and PTFE.

Accept these late or on prompting: 3 dry tweezers and minimal handling · 5 tacks, cover screw or periosteal suture as fixation · 6 the PASS principles cited by name · 8 suture line kept off the membrane.

The commonest way five minutes is wasted here: explaining guided tissue regeneration from first principles — cell exclusion, migration rates, the Nyman experiments — before the membrane has been picked up. The theory carries no checklist point here; **the trimming, the tucking and the closure do**. The second loss is finishing at the suture and never reaching exposure management, which is checklist 10 and the only place the difference between collagen and PTFE actually pays — and, alongside it, the titanium-reinforced membrane left untouched on the bench, since the mark for knowing that a vertical augmentation is Complex exists only if the candidate says so out loud.

OSCE 32 — Post-operative wound review and suture removal

Blueprint: postoperative care and advice; recognise unsafe situations (OS-P + SCE).

EQUIPMENT

a model or simulated patient with a healing implant/graft site; suture scissors, tweezers, mirror, probe.

CANDIDATE INSTRUCTIONS

This patient is attending one week after implant placement with simultaneous guided bone regeneration. Review the wound and remove the sutures.

MODEL ANSWER

"I would start with the patient rather than the wound: how have the last seven days been — pain, swelling, bleeding, any medication taken, any problems with the instructions, and are they smoking?

Examination: extra-orally for swelling, bruising, lymphadenopathy or any sign of spreading infection, and I would check **sensation of the lip and chin against the baseline** if this was a mandibular case. Intra-orally I look for: wound edge apposition, colour, any **dehiscence**, any **membrane exposure**, suppuration, granulation tissue, mobility of the healing abutment if one is present, and plaque accumulation. I would **not probe the implant** at this stage.

Suture removal: clean the area with chlorhexidine on a cotton pledget, lift each suture gently with tweezers, cut the strand **below the knot and close to the tissue** so that no external, contaminated portion is drawn through the wound, and withdraw it towards the wound rather than away from it. Count the sutures removed against the number recorded as placed.

Because this is a grafted site, I would consider leaving the sutures a further few days — to **10 to 14 days** — if healing is not yet secure.

Advice: may now brush gently around but not directly over the site with a soft brush; continue chlorhexidine for another week; continue soft diet; continue not smoking; and I would re-emphasise the warning signs.

Documentation: healing assessment, sutures removed and count, sensation check, advice given, and the next review — typically at 2 to 3 weeks then at 3 months for a grafted case.

If I found **membrane exposure**, I would manage it according to membrane type; if I found **suppuration or spreading infection**, I would drain, irrigate, consider antibiotics and review at 48 hours."

EXAMINER CHECKLIST

1. History of the week, including medication and smoking
2. Extra-oral examination
3. **Sensation check against baseline**
4. Systematic intra-oral wound assessment
5. Does not probe the implant at 1 week
6. Antiseptic before removal
7. Correct cutting technique below the knot
8. Counts sutures against the record
9. Considers delaying removal in a grafted site
10. Advice, documentation, next review, and a plan for exposure or infection

Running this station to time

Time check. The model answer is 417 words, 2 minutes 59 seconds at 140 words per minute — which is why this station looks easy and why full marks are expected on it. The time is not lost to length; it is lost to **silence**. Four of the ten points — not probing, the antiseptic wipe, where the cut is made, the count — are things the hands do in a second and the mouth must take five seconds to say. The rule here is **narrate every action as it happens, including the actions you decide not to take**.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Presents the healing site: "This patient is attending one week after implant placement with simultaneous guided bone regeneration. Review the wound and remove the sutures."	Checklist 1, and it comes **before the mirror goes in**. Start with the patient, not the wound: **pain and its trend, swelling, bleeding, what analgesia was actually taken and whether it worked, whether the antibiotics or chlorhexidine were completed, whether the post-operative instructions caused any difficulty — and whether they have smoked.** Smoking must be asked about by name; it is the single behaviour that most threatens a grafted site at one week, and "any problems?" does not elicit it.

Time	The examiner does	What earns the mark
0:45–1:45	Watches. Offers nothing.	Checklists 2 and 3. **Extra-oral first** — swelling, its character and whether it is still increasing, bruising, **lymphadenopathy**, trismus, and any sign of **spreading infection**. Then checklist 3, which candidates simply forget in the rush to the sutures: **sensation of the lip and chin tested and compared against the baseline recorded at placement**, because this was a mandibular case. Said aloud as it is done. A sensory deficit found at one week and not recorded against a baseline is a deficit nobody can date.
1:45–2:45	Silence.	Checklists 4 and 5. A **systematic intra-oral assessment, named item by item**: wound edge apposition, tissue colour, **dehiscence**, **membrane exposure**, suppuration, granulation tissue, mobility of the healing abutment if present, plaque accumulation, and any temporary prosthesis pressing on the site. Then the negative that scores: **"I would not probe the implant at this stage"** — said out loud, with the reason, that the seal at one week is immature and probing disrupts healing and tells me nothing useful. Leaving the probe on the tray silently does not score checklist 5.
2:45–3:50	Hands over tweezers and suture scissors.	Checklists 6, 7 and 8 — three marks in sixty seconds of hands, all of them narrated. **Clean the area with chlorhexidine on a cotton pledget first.** Then the technique, described as it is performed: **lift the suture gently with tweezers, cut the strand below the knot and close to the tissue**, so that **no external, contaminated portion of the suture is drawn through the wound**, and **withdraw it towards the wound rather than away from it** so the healing edges are not pulled apart. Then checklist 8: **count the sutures removed against the number recorded as placed**, and say the numbers. A suture left buried is a suture that will present as an abscess.
3:50–5:00	"Anything else?"	Checklists 9 and 10, including the closing decision. Because this is a **grafted site**, state the judgement explicitly: **if healing is not yet secure I would leave some or all of the sutures to 10 to 14 days** rather than remove them on schedule. Then **advice**: gentle brushing **around but not directly over** the site; **chlorhexidine for a further week**; soft diet continued; **not smoking**; **warning signs restated**. Then **documentation** — the healing assessment, the sensation check against baseline, **the number of sutures removed**, the advice given — and the **next review: 2 to 3 weeks, then 3 months for a grafted case**. Finally the two contingencies: **membrane exposure managed by membrane type**; **suppuration or spreading infection drained and irrigated, antibiotics considered, reviewed at 48 hours.**

The compressed answer — what a strong candidate says in five minutes

"**I start with her, not with the wound.** How have the last seven days been — how much pain, and is it settling or increasing? How much swelling, and has it peaked? Any bleeding? What painkillers did you

actually take, and did they work? Did you finish the chlorhexidine? Were the instructions manageable? **And have you smoked at all this week?** I ask that directly, because it is the thing most likely to lose this graft and nobody volunteers it.

Extra-orally, I look for swelling and whether it is still increasing, bruising, **lymphadenopathy**, trismus, and any sign of spreading infection. **And because this was a lower case, I test the sensation of the lip and chin and compare it with the baseline I recorded at placement** — light touch and sharp–blunt, both sides — and I record the result whether or not it has changed.

Intra-orally, systematically: wound edge apposition; the colour of the tissue; any **dehiscence**; any **membrane exposure**; suppuration; granulation tissue; mobility of the healing abutment if there is one; plaque accumulation; and whether any temporary denture is pressing on the site.

And one thing I deliberately do not do: I will not probe the implant at one week. The seal is immature, probing at this stage disturbs healing, and a reading now would tell me nothing I could use.

Suture removal. First I **clean the area with chlorhexidine on a cotton pledget** — the sutures have been sitting in plaque for seven days and I do not want to drag that into the wound.

Then, one at a time: I **lift the suture gently with tweezers**, and I **cut the strand below the knot, close to the tissue**, so that the portion that has been exposed to the mouth never passes through the wound. Then I **withdraw it towards the wound**, not away from it, so I am not pulling the healing edges apart.

And I count. Six were placed — I have removed six. If the count is short I look for a buried suture rather than assume, because a retained suture presents as an abscess in a fortnight.

Because this is a grafted site, I make a judgement rather than follow a rule. If the wound edges are not yet securely healed I would **leave some or all of the sutures a further few days — to 10 to 14 days** — and see her again, rather than remove them on the calendar.

Advice. She may now **brush gently around the site with a soft brush, but not directly over it**. **Continue chlorhexidine for another week.** Continue the soft diet. **Do not smoke.** And the warning signs again: increasing pain or swelling after this point, a bad taste, discharge, anything showing through the gum, or any change in the feeling of the lip — telephone, do not wait.

Documentation: the healing assessment item by item, the **sensation check against baseline**, **the number of sutures removed against the number placed**, the advice given, and the next review — **2 to 3 weeks, and then at 3 months because this site is grafted**.

And if I found something. Membrane exposure I manage by type — a collagen membrane exposed without infection conservatively, with chlorhexidine and no brushing; a PTFE membrane removed early. **Suppuration or spreading infection** I drain and irrigate, consider systemic antibiotics, and **review at 48 hours**, not at the next routine appointment."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 7 cut below the knot, withdrawn towards the wound, with the reason · 5 **states out loud that the implant will not be probed at one week** · 3 sensation checked against the placement baseline · 4 systematic wound assessment naming dehiscence and membrane exposure · 8 sutures counted against the record · 10 advice, documentation, review interval and the exposure/infection contingencies.

Accept these late or on prompting: 1 the week's history including analgesia and smoking · 2 extra-oral examination · 6 chlorhexidine before removal · 9 removal delayed to 10–14 days in a grafted site.

The commonest way five minutes is wasted here: working in silence. This is a full-marks station that candidates routinely score six on, because they clean the site, cut the sutures correctly and count them — and say none of it. The examiner marks what is **said**; a perfectly executed cut below the knot is invisible unless the candidate narrates where the blade is going and why. The second loss is the sensation check: fifteen seconds, checklist 3, and forgotten by almost everyone who heads straight for the scissors — as is checklist 9, where the mark is for **deciding** to leave a grafted site to 10 to 14 days rather than following the standard interval.

OSCE 33 — Second-stage (uncovering) surgery

Blueprint: one or two stage procedures; soft tissue management (W + OS-P + SCE).

EQUIPMENT

a model with two submerged implants — one with abundant keratinised tissue, one with 1 mm; blades, tissue punch, elevators, healing abutments.

CANDIDATE INSTRUCTIONS

Uncover these two implants. For each, select and justify your technique.

MODEL ANSWER

"Before I start: confirm the healing period is complete, check the site clinically, and take a periapical to confirm bone levels and to locate the cover screw.

Implant A, abundant keratinised tissue. I can afford to remove a small amount of tissue, so a **tissue punch** or a small crestal incision is appropriate. It is quick, needs no flap, and heals rapidly. I would locate the cover screw precisely — with a probe through the mucosa or from the radiograph — punch or incise, remove the tissue plug, expose and unscrew the cover screw, clean the internal connection, and place a healing abutment of appropriate height so it emerges through the mucosa without being over-contoured.

Implant B, 1 mm of keratinised tissue. A tissue punch is **contraindicated** here, because it excises the very tissue I need. I would use an **apically positioned flap**: a **crestal incision displaced palatally** so that the band of keratinised tissue is carried buccally, with two short vertical relieving incisions, a partial-thickness dissection buccally, and the flap sutured apically at the new position with periosteal sutures. That both uncovers the implant and increases the width of keratinised mucosa buccally.

If the tissue were also **thin**, I would consider a **connective tissue graft** at the same time — second stage is one of the two best moments to modify the phenotype, the other being placement.

In both cases: select a healing abutment of correct diameter and height to begin shaping the emergence, check it is fully seated radiographically if in doubt, hand-tighten, give hygiene instructions specific to a healing abutment, and record what was placed. I would allow **2 to 4 weeks of soft tissue maturation** before impressions."

EXAMINER CHECKLIST

1. Confirms healing period and takes a radiograph
2. Locates the cover screw before incising
3. Punch justified where KM is abundant
4. **Rejects punch where KM is scarce, with reason**
5. Palatally displaced crestal incision described
6. Apically positioned flap technique
7. Considers CTG for a thin phenotype and states the timing rationale
8. Cleans the connection and selects an appropriate healing abutment
9. Verifies seating
10. Hygiene advice, soft tissue maturation period, records

Running this station to time

Time check. The model answer is 420 words, 3 minutes exactly at 140 words per minute — but two implants have to be uncovered and the marks are not shared between them. **Implant A carries one checklist point; implant B carries four.** So the ration is deliberate: **50 seconds for the punch, a full 100 seconds for the apically positioned flap**, and the punch described in the past tense — chosen, justified, done — rather than performed slowly on a model where nothing is at stake.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:35	Presents the model with two submerged implants — **one with abundant keratinised tissue, one with 1 mm** — and blades, a tissue punch, elevators and healing abutments: "Uncover these two. For each, select and justify your technique."	Checklists 1 and 2, before any blade is picked up. **Confirm the healing period is complete** and the site is clinically quiet, and **take a periapical** — for bone levels and, specifically, **to locate the cover screw**. Then checklist 2 as an action: **locate the cover screw precisely before incising**, by probing through the mucosa and by reference to the radiograph. Cutting first and hunting afterwards is how keratinised tissue is wasted.
0:35–1:25	Indicates implant A.	Checklist 3, and briskly — this is one mark and it is not worth 90 seconds. **Keratinised tissue is abundant here, so I can afford to remove a small amount**: a **tissue punch**, or a small crestal incision, justified on exactly that ground — quick, no flap, rapid healing. Punch centred over the located cover screw, tissue plug removed, cover screw unscrewed, connection inspected. The justification is the mark; the technique is not.
1:25–3:10	Indicates implant B: "And this one?"	Checklists 4, 5, 6 and 7 — the stretch that decides the station. First the rejection, said explicitly: **"A tissue punch is contraindicated here, because with 1 mm of keratinised mucosa it excises the very tissue I need."** Then the alternative, described so the examiner can see it: an **apically positioned flap** — a **crestal incision displaced palatally**, so the existing band of keratinised tissue is carried **buccally** rather than split down the middle; **two short vertical relieving incisions**; a **partial-thickness dissection buccally**, leaving periosteum on the bone; the flap **sutured apically at its new position with periosteal sutures**. Then what that achieves: it uncovers the implant **and** widens the buccal band in one procedure. Then checklist 7: **if the tissue is also thin, a connective tissue graft at the same visit — because second stage is one of the two best moments to modify the phenotype, the other being placement.**
3:10–4:05	"Now finish both."	Checklists 8 and 9, which apply to both implants and are commonly dropped. **Clean the internal connection** of granulation tissue and debris before anything is seated. **Select a healing abutment of appropriate diameter and height** — tall enough to emerge through the mucosa, not so wide or so contoured that it blanches the tissue, the diameter chosen to begin shaping the emergence profile. Then checklist 9: **verify complete seating, radiographically if in doubt**, and **hand-tighten only** — a healing abutment is not torqued.

Time	The examiner does	What earns the mark
4:05–5:00	"What do you tell the patient, and when do you see them?"	Checklist 10. **Hygiene instruction specific to a healing abutment** — it is now exposed to the mouth, it collects plaque immediately, and it needs a soft brush and a single-tufted brush around the emergence from day one, demonstrated rather than described. Then the **soft tissue maturation period: 2 to 4 weeks before impressions**, longer where a flap has been repositioned or a graft placed. Then **records**: which technique was used at each implant and why, the **make, diameter and height of each healing abutment**, that seating was verified, the graft if one was placed, the advice given and the review date.

The compressed answer — what a strong candidate says in five minutes

"**Before I pick up a blade.** I confirm the healing period is complete and the sites are clinically quiet, and I **take a periapical** — to check the bone levels and, just as importantly, **to locate the cover screws**. Then I **find each cover screw through the mucosa with a probe** before I incise. I do not want to be searching for it after I have cut.

Implant A — abundant keratinised tissue. Here I can afford to lose a little tissue, so a **tissue punch** is appropriate, or a small crestal incision. It is quick, it needs no flap, and it heals in days. I centre the punch over the cover screw I have located, remove the tissue plug, unscrew the cover screw, and inspect the connection.

Implant B — one millimetre of keratinised tissue. Here a tissue punch is contraindicated, and this is the whole point of the second implant: a punch would **excise the only keratinised tissue this implant has**, and I would be creating a problem in order to save five minutes.

Instead, an apically positioned flap. I make a **crestal incision displaced palatally**, so that the existing band of keratinised tissue is carried **buccally on the flap** rather than divided. I add **two short vertical relieving incisions**, raise a **partial-thickness dissection buccally** — leaving periosteum on the bone for the flap to be sewn to — and **suture the flap apically in its new position with periosteal sutures**. That does two jobs at once: it **uncovers the implant** and it **increases the width of keratinised mucosa on the buccal**, which is where I want it for long-term maintenance.

And if the tissue were thin as well as narrow, I would place a **connective tissue graft at the same visit**. **Second stage is one of the two best moments to modify the soft tissue phenotype — the other is at placement** — and once the restoration is on, doing it becomes much harder and less predictable.

Then, at both implants. I **clean the internal connection** thoroughly — granulation tissue, debris, any bone that has grown over the screw — because anything left in there will compromise the fit of everything that follows.

I select a healing abutment for each of appropriate diameter and height: tall enough to **emerge through the mucosa** so the tissue heals around it rather than over it, and of a diameter that **begins to shape the emergence profile** without over-contouring and blanching the tissue. **I verify that it is fully seated — radiographically if there is any doubt** — and I **hand-tighten it**; a healing abutment is not torqued.

What she is told. There is now metal showing through the gum, and **it collects plaque from today**. A soft brush over it and a **single-tufted brush around the emergence**, demonstrated in her hand at the chair, not described. Chlorhexidine for the flap site for a week. And telephone if a healing abutment feels loose or comes out.

Timing. I allow **2 to 4 weeks of soft tissue maturation before impressions**, and longer at implant B, where the flap has been repositioned and must stabilise before I record anything.

And I record: the technique used at each implant and the reason, the **make, diameter and height of each healing abutment**, that seating was verified, the graft if placed, the hygiene instruction given, and the review."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 4 **punch rejected at implant B, with the reason stated** · 6 apically positioned flap described as a technique, not named · 5 **crestal incision displaced palatally** to

carry keratinised tissue buccally · 2 cover screw located before incising · 8 connection cleaned and a healing abutment of correct diameter and height chosen · 10 hygiene advice, 2–4 week maturation, records.

Accept these late or on prompting: 1 healing period confirmed and a radiograph taken · 3 punch justified where keratinised tissue is abundant · 7 connective tissue graft for a thin phenotype, with the timing rationale · 9 seating verified.

The commonest way five minutes is wasted here: performing implant A. It is the easy one, the punch is satisfying, and it is worth **one mark out of ten** — yet candidates spend ninety seconds on it and then describe implant B as "I'd raise a flap and reposition it apically", which is a summary, not a technique, and does not score checklists 5 and 6. The second loss is the final third: checklists 8, 9 and 10 all sit after the implants are uncovered, and a candidate who runs the surgery to 4 minutes 30 hands over a healing abutment with no diameter, no height, no seating check and no hygiene instruction. **Choosing and verifying what goes into the implant is half of this station.**

OSCE 34 — Selecting implant dimensions from the planning data

Blueprint: plan in detail and select the techniques for implant placement; bone quality and quantity (W + OS-P + SCE).

EQUIPMENT

CBCT cross-sections and a wax-up for three sites: an upper lateral incisor (5.8 mm space), a lower first molar (9 mm above the canal, 9 mm wide), and an upper second premolar (6 mm below the sinus, 7 mm wide).

CANDIDATE INSTRUCTIONS

Select an implant diameter and length for each site and justify your choices.

MODEL ANSWER

"**Upper lateral incisor, 5.8 mm mesiodistal space.** I need 1.5 mm to each adjacent root, so the maximum diameter is about 2.8 mm at the crest — which is tight. I would select a **3.3 mm narrow-diameter implant in a titanium-zirconium alloy** for strength, and accept slightly under 1.5 mm on one side only if the CBCT shows the roots diverge apically; otherwise I would discuss **orthodontic space adjustment** or an alternative prosthesis. Length around 10 to 12 mm depending on the nasal floor and the incisive canal. Position: shoulder 3 mm apical to the planned mid-facial margin, 1 to 2 mm palatal to the emergence of the adjacent teeth.

Lower first molar, 9 mm above the canal, 9 mm wide. I must leave **2 mm clearance from the canal**, so the maximum length is about 7 mm — I would choose a **short implant of 6 to 7 mm**, and compensate with a **wide diameter of 5.0 mm**, since diameter increases surface area and resistance to bending far more efficiently than length. Ridge width of 9 mm allows 5.0 mm with 2 mm of bone each side. Bone is likely D2, so I would drill conventionally and consider counter-sinking to avoid excessive torque.

Upper second premolar, 6 mm below the sinus, 7 mm wide. With only 6 mm and a flat floor, this sits exactly at the boundary of the ITI decision tree. My options are: a **short implant of 6 mm** with no sinus procedure; or a **transcrestal elevation** to gain 3 to 4 mm allowing an 8 to 10 mm implant — but the tree indicates a lateral window at 6 mm or less, so I would be cautious. Diameter 4.1 mm with 7 mm of width leaves about 1.5 mm each side. Bone will be D3 or D4, so I would **under-prepare**, use a tapered implant, and load conventionally at 3 to 6 months.

In every case, the position comes from the **wax-up**, not from where the bone happens to be."

EXAMINER CHECKLIST

1. Applies the 1.5 mm rule to the incisor space
2. Selects a narrow implant with a strength-appropriate material
3. Offers orthodontic/alternative option where space is marginal
4. Applies the 2 mm canal margin
5. Chooses a short, wide implant with a biomechanical justification
6. Checks ridge width against diameter plus 1.5–2 mm each side

7. Recognises the 6 mm sinus decision boundary
8. Modifies technique for D3/D4
9. Correct 3-D position for the anterior site
10. States that the wax-up drives position

Running this station to time

Time check. The model answer is 508 words, 3 minutes 38 seconds at 140 words per minute — the longest in this group, across three sites, and every mark on it is a **number**. That means the compression cannot come from the figures; it has to come from the reasoning around them. **Eighty seconds per site**, and at each one the shape is fixed: *the limiting anatomy, the subtraction, the implant I choose, the technique I modify* — four sentences, with the arithmetic said out loud rather than implied.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:25	Lays out the CBCT cross-sections and the wax-up for three sites: "Select an implant diameter and length for each, and justify your choices."	Checklist 10, said at the start and again at the end, because it frames everything: **the position comes from the wax-up, not from where the bone happens to be**. Then the method in one sentence — for each site, the limiting structure, the clearance, the arithmetic, the implant, the drilling modification.
0:25–1:45	Upper lateral incisor, **5.8 mm mesiodistal space**.	Checklists 1, 2, 3 and 9. **The subtraction spoken:** 1.5 mm to each root, so **5.8 minus 3.0 leaves about 2.8 mm at the crest** — tighter than any implant. Therefore a **3.3 mm narrow implant in titanium–zirconium alloy**, because **commercially pure titanium at that diameter is mechanically marginal in the anterior maxilla**; under 1.5 mm accepted **on one side only, and only if the roots diverge apically**. Otherwise checklist 3: **orthodontic space adjustment** or a different prosthesis. **Length 10 to 12 mm.** Then checklist 9: **shoulder 3 mm apical to the planned mid-facial margin, 1 to 2 mm palatal to the emergence of the adjacent teeth.**
1:45–3:00	Lower first molar, **9 mm above the canal, 9 mm wide**.	Checklists 4, 5 and 6 — the cleanest arithmetic on the station and the most often fudged. **2 mm clearance from the canal is mandatory**, so **9 minus 2 gives a maximum of 7 mm** — a **short implant of 6 to 7 mm**, stated as a choice and not an apology. Then the actual mark: **compensate with a wide diameter, 5.0 mm, because diameter increases surface area and resistance to bending far more efficiently than length does.** Then checklist 6 out loud: **9 mm of ridge minus 5.0 mm leaves 2 mm each side.** Bone likely **D2 — drill conventionally, consider counter-sinking to avoid excessive insertion torque**.
3:00–4:20	Upper second premolar, **6 mm below the sinus, 7 mm wide**.	Checklists 7, 8 and 6 again. **6 mm sits exactly on the ITI decision boundary** and must be named as such: at 6 mm or less with a flat floor, **the tree points towards a lateral window — Complex, and I would refer**. Within scope: a **short 6 mm implant with no sinus procedure**, or a **transcrestal elevation gaining 3 to 4 mm for an 8 to 10 mm implant**, with caution at exactly 6 mm. **Diameter 4.1 mm in a 7 mm ridge leaves about 1.5 mm each side.** Then checklist 8: **D3 or D4** bone — **under-prepare, tapered implant, omit the countersink, load conventionally at 3 to 6 months**.

Time	The examiner does	What earns the mark
4:20–5:00	"How does this get to the day of surgery?"	**Checklist 10 restated** — every position is dictated by the wax-up and the guide made from it. Then what is **recorded and ordered**: diameter and length for each site **with a back-up size either side on the day**, and a depth stop where the canal is close. Then **what each patient is consented to** — the orthodontic alternative at the incisor, nerve proximity and the 2 mm margin at the molar, sinus involvement and possible referral at the premolar — and the plan **re-confirmed against the guide and the radiograph on the day**.

The compressed answer — what a strong candidate says in five minutes

"**One principle before the numbers: the position comes from the wax-up, not from where the bone happens to be.** If the two conflict, I graft or I change the prosthesis.

Upper lateral incisor, 5.8 mm of space. I need **1.5 mm to each adjacent root**, so **5.8 minus 3 leaves 2.8 mm at the crest** — narrower than any implant I can place. So a **3.3 mm narrow-diameter implant in a titanium–zirconium alloy**, because 3.3 mm in commercially pure titanium is mechanically marginal here and the alloy gives the strength back. I would accept **slightly under 1.5 mm on one side only**, and only if the CBCT shows the roots **diverge apically** so the tight point is at the crest alone. If not, I would not force it — I would discuss **orthodontic space opening** or a **resin-bonded bridge**. **Length 10 to 12 mm**, depending on the nasal floor and incisive canal.

And the three-dimensional position, which matters more here than the dimensions: the **shoulder 3 mm apical to the planned mid-facial gingival margin**, and **1 to 2 mm palatal to the emergence of the adjacent teeth** — deep enough for an emergence profile, palatal enough to keep the buccal plate.

Lower first molar, 9 mm above the canal, 9 mm wide. The canal is the limit. **I must leave 2 mm of clearance, so 9 minus 2 gives me 7 mm — that is my maximum length**, and I would choose a **short implant of 6 to 7 mm**. I am not apologising for that: **I compensate with diameter, 5.0 mm**, because **diameter increases surface area and resistance to bending far more efficiently than length does**.

Checking the width: 9 mm of ridge minus a 5.0 mm implant leaves 2 mm each side. Bone here will be **D2**, so I drill **conventionally** and **consider counter-sinking** to avoid excessive insertion torque and crestal pressure necrosis.

Upper second premolar, 6 mm below the sinus, 7 mm wide. Six millimetres is exactly the ITI decision boundary, and I would say so: with a flat floor at 6 mm or less, **the tree points to a lateral window — and that is Complex; I would refer for it, not attempt it.**

Within my scope, two options. A **short 6 mm implant with no sinus procedure**; or a **transcrestal elevation to gain 3 to 4 mm**, allowing **8 to 10 mm** — but at exactly 6 mm with a flat floor I would be cautious, because that is where the tree stops recommending it. **Diameter 4.1 mm in a 7 mm ridge leaves about 1.5 mm each side.**

Bone here will be D3 or D4, so I change the technique: **under-prepare the osteotomy**, use a **tapered implant** for primary stability by compression, **omit the countersink**, and **load conventionally at 3 to 6 months**.

To finish. I record the diameter and length for each site, make sure a **back-up size either side is on the shelf on the day**, and use a **depth stop at the molar**. Each patient is consented to the issue at their own site — the space compromise and orthodontic alternative at the incisor, the nerve and the 2 mm margin at the molar, the sinus and possible referral at the premolar. And on the day I **re-confirm every one of these against the guide and the radiograph before I drill**."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 4 **2 mm canal clearance, 9 − 2 = 7 mm said as arithmetic** · 5 short 6–7 mm implant at 5.0 mm with the **bending-resistance reason** · 1 the 1.5 mm rule giving 2.8 mm · 7 **6 mm named as the sinus boundary**, lateral window referred · 2 3.3 mm titanium–zirconium with the strength reason · 10 the wax-up drives position.

Accept these late or on prompting: 3 orthodontic or alternative prosthesis · 6 ridge width against diameter plus 1.5–2 mm each side · 8 under-preparation and a tapered implant in D3/D4 · 9 shoulder 3 mm apical, 1–2 mm palatal.

The commonest way five minutes is wasted here: describing principles instead of doing the sums. "I would leave an adequate safety margin from the canal and select an implant of appropriate length" contains no mark at all; **"nine minus two is seven, so I am choosing a six"** contains two — this is a numbers station and the examiner is listening for 1.5, 2.8, 3.3, 7, 5.0, 6 and 4.1. The second loss is running the anterior site long: candidates spend two minutes on aesthetics and then give the premolar thirty seconds, forfeiting the 6 mm boundary and the D3/D4 modification for a discussion the checklist never asked for.

OSCE 35 — Preparing the surgical tray

Blueprint: infection control and aseptic technique; range of surgical components required (W + OS-P + SCE).

EQUIPMENT

a bench of instruments, some appropriate and some not; a surgical kit; implant packaging.

CANDIDATE INSTRUCTIONS

Lay out a surgical tray for a single implant placement with possible simultaneous guided bone regeneration. Name each item and its purpose. Identify anything missing.

MODEL ANSWER

"Working in the order of the operation.

Anaesthesia: syringe, needles, cartridges, topical. **Incision and reflection:** scalpel handle and blades — a 15 or 15c for the crestal incision — periosteal elevator, tissue forceps, retractors of appropriate width, and a Mitchell's trimmer or curette. **Osteotomy:** the implant system's surgical kit — round or precision drill, sequential twist drills with depth marks, direction indicators, depth gauge, countersink and tap if required, drill extensions; the **electric surgical motor with a sterile handpiece and sterile irrigation tubing**; and sterile saline. **Placement:** implant carrier/driver, **ratchet and torque wrench**, cover screws and healing abutments in a range of heights, and screwdrivers. **Augmentation:** bone scraper or trap, graft material, membrane, tacks and a tack driver, and rongeur. **Closure:** needle holder, suture scissors, tissue forceps, and sutures — I would have both a monofilament or ePTFE for the grafted site and a resorbable alternative. **General:** high-volume suction and a fine surgical sucker, gauze, a throat screen, saline-soaked gauze to keep the flap moist, and a sterile marker.

Missing or wrong items I would look for: an air-driven handpiece — remove it; steel curettes on the implant tray — remove them; an implant with a damaged pouch or past expiry — discard and record; non-sterile bur block; missing torque wrench; and no drill stop where the canal is close.

Before starting: confirm the correct patient, site and implant with a **surgical safety check** — patient identity, medical history, consent confirmed, site marked, implant system, diameter and length available with a back-up size, allergies, and radiographs displayed. I would also confirm the instrument and swab count with my assistant, and record the batch numbers."

EXAMINER CHECKLIST

1. Anaesthesia items
2. Incision and reflection instruments named with purpose
3. Correct drilling kit including direction indicators and depth gauge
4. **Electric motor with sterile tubing and saline**
5. Torque wrench and ratchet
6. Augmentation instruments and materials
7. Closure instruments and appropriate sutures
8. Identifies inappropriate items (air-turbine, steel instruments, expired implant)
9. **Surgical safety check** including back-up implant sizes

10. Counts and batch number recording

Running this station to time

Time check. The model answer is 414 words, 2 minutes 57 seconds at 140 words per minute — and this is a full-marks station that candidates routinely score seven on. The clock is not the enemy; **itemising is**. There are more than forty objects on that bench and only ten marks, so the discipline is **one clause of purpose per group of instruments, not per instrument** — "retractors of appropriate width, to protect the flap and give me vision" covers the group — and the last ninety seconds are reserved for what is **wrong**, what is **missing**, and the safety check.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–1:20	Indicates a bench of instruments, a surgical kit and implant packaging: "Lay out a tray for a single implant placement with possible simultaneous guided bone regeneration. Name each item and its purpose. Identify anything missing."	The method declared in one sentence while the hands start moving — **"I will work in the order of the operation, then what is wrong on this bench, what is missing, and the safety check"** — which saves thirty seconds later by stopping the examiner having to prompt three times. Then checklists 1 and 2, grouped and purposed. **Anaesthesia** — syringe, needles, cartridges, topical. **Incision and reflection** — scalpel handle with a **15 or 15c blade for the crestal incision**, **periosteal elevator** to raise a full-thickness flap cleanly off bone, **tissue forceps** to handle the flap by its edge rather than crush it, **retractors of appropriate width** for vision and flap protection, **Mitchell's trimmer** to clear the crest. The purpose is the mark; the name alone is half of it.
1:20–2:30	May ask: "And to prepare the site?"	Checklists 3, 4 and 5 — the densest stretch. **The implant system's own surgical kit**: round or precision drill, **sequential twist drills with depth marks**, **direction indicators** to check angulation against the adjacent teeth and the guide, **depth gauge**, countersink and tap if required, drill extensions. Then checklist 4 as a unit: the **electric surgical motor with a sterile handpiece and sterile irrigation tubing**, torque and speed set at the console, and **sterile saline** — never tap water. Then placement: **carrier and driver, ratchet, and torque wrench**, **cover screws and healing abutments in a range of heights**, screwdrivers.
2:30–3:20	Indicates the graft and sutures.	Checklists 6 and 7. **Augmentation** — bone scraper or trap, graft material, membrane, **tacks and a tack driver**, rongeur. **Closure** — needle holder, suture scissors, tissue forceps, and the suture **chosen** rather than merely present: **monofilament or ePTFE for the grafted site**, because braided material wicks plaque along a line that has to hold ten to fourteen days, with a resorbable alternative available. Then the general items in one clause: high-volume and fine surgical suction, gauze, throat screen, **saline-soaked gauze to keep the flap moist**, sterile marker.

Time	The examiner does	What earns the mark
3:20–4:15	"Is there anything here that should not be, or anything absent?" — and may not ask at all, in which case the candidate must raise it unprompted.	Checklist 8, and both halves. **Wrong:** the **air-driven handpiece** — removed, because it exhausts into the wound and risks surgical emphysema; **steel curettes on the implant tray** — removed, because steel scratches the titanium oxide layer; an implant in a **damaged or out-of-date pouch** — **discarded, and the discard recorded and reported**; a **non-sterile bur block**. **Missing:** the **torque wrench**, and **no drill stop where the canal is close**.
4:15–5:00	"Anything before you start?"	Checklists 9 and 10 — the closing marks, most often lost to the clock. A **surgical safety check** spoken as a checklist: **correct patient identity, medical history and allergies re-confirmed, consent confirmed, site marked and cross-checked against the radiographs on the screen, the implant system named, and the planned diameter and length available — with a back-up size either side on the shelf.** Then **an instrument and swab count agreed with the assistant before starting and repeated before closure**, and the **batch numbers of the implant, graft and membrane recorded** in the notes and onto the patient's implant card.

The compressed answer — what a strong candidate says in five minutes

"**I will work in the order of the operation.**

Anaesthesia: syringe, needles, cartridges, topical.

Incision and reflection: scalpel handle with a **15 or 15c blade** for the crestal incision; a **periosteal elevator** to raise the flap cleanly off bone in one plane; **tissue forceps**, so I handle the flap by its edge and do not crush it; **retractors of appropriate width** for vision and to protect the flap from the drill; a **Mitchell's trimmer** to clear the crest.

Osteotomy: the **implant system's own surgical kit** — round or precision drill to mark the point, **sequential twist drills with depth markings**, **direction indicators** so I can check angulation against the adjacent teeth and the guide before going deeper, a **depth gauge**, a countersink and tap if the bone is dense, drill extensions. The drive is an **electric surgical motor with a sterile handpiece and sterile irrigation tubing**, torque and speed set at the console — and **sterile saline**, never tap water.

Placement: the **implant carrier and driver**, the **ratchet**, and the **torque wrench**; **cover screws and healing abutments in a range of heights**, and the screwdrivers for them.

Augmentation, because this may be a simultaneous graft: a **bone scraper or sterile trap** for autogenous chips, the **graft material**, the **membrane**, **tacks and a tack driver**, a rongeur.

Closure: needle holder, suture scissors, tissue forceps, and — chosen deliberately — a **monofilament or ePTFE suture for the grafted site**, because braided material wicks plaque along a line that has to hold for ten to fourteen days. A resorbable alternative available.

General: high-volume and fine surgical suction, gauze, a **throat screen**, **saline-soaked gauze to keep the flap moist** while I drill, a sterile marker.

Now, what is wrong on this bench. The **air-driven handpiece comes off** — it exhausts into the surgical field and can drive air into the tissues, causing surgical emphysema; that is what the electric motor is for. The **steel curettes come off the implant tray** — steel scratches the titanium oxide layer; anything touching an implant is titanium, PEEK or carbon fibre. The **non-sterile bur block comes off**. And **this implant, with a damaged pouch and a passed expiry date, is discarded** — I open a replacement and **record the batch number and the reason for the discard, and report it as a device incident**.

And what is missing. There is **no torque wrench** — I cannot seat an abutment or verify insertion torque without one. And there is **no drill stop**, which I want here because the canal is close and a stop is the difference between a controlled depth and a nerve injury.

Before anything starts, the surgical safety check. Correct patient, confirmed by identity; medical history and **allergies** re-confirmed today; **consent confirmed and still valid**; **site marked and cross-checked against the radiographs on the screen**; implant system, **diameter and length available — and a back-up size either side on the shelf**, because the bone may not be what the scan promised.

And with my assistant: an instrument and swab count now, and the same count again before I close. At the end, the **batch numbers of the implant, the graft and the membrane go into the notes and onto her implant card**, so every device in her jaw is traceable."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 8 **air-turbine, steel instruments and the expired implant identified — and the torque wrench and drill stop named as missing** · 9 **surgical safety check including a back-up implant size** · 3 direction indicators and depth gauge named within the drilling kit · 4 electric motor with sterile tubing and sterile saline · 10 instrument and swab count, and batch numbers recorded · 5 ratchet and torque wrench.

Accept these late or on prompting: 1 anaesthesia items · 2 incision and reflection instruments with their purposes · 6 augmentation instruments and materials · 7 closure instruments with the suture chosen for a grafted site.

The commonest way five minutes is wasted here: naming forty instruments at one every four seconds and reaching four minutes with nothing said about what is wrong, what is missing, or the safety check — three of the ten marks, and the three that separate a competent tray from a competent surgeon. The second loss is treating "identify anything missing" as decoration: it is written into the candidate instruction, and the **absent torque wrench and the absent drill stop** are deliberate — as is the silence trap during the rejections, where lifting the air-turbine off the tray without saying **emphysema** scores nothing.

OSCE 36 — Torque, preload and the torque wrench

Blueprint: prosthetic components; fitting a prosthesis (OS-P).

EQUIPMENT

a torque wrench, abutment screws, a model with an implant analogue, an abutment.

CANDIDATE INSTRUCTIONS

Torque this abutment correctly, explaining the underlying mechanics and how you look after the instrument.

MODEL ANSWER

"First I confirm the **manufacturer's specified torque value** for this component — it is not a universal figure, and using another system's value is an error.

Mechanics. Tightening a screw generates **preload** — the tensile clamping force in the screw shank that holds the joint together and resists separation under load. Most of the torque I apply is consumed by **friction** at the threads and under the screw head; only a fraction becomes preload. That is why the manufacturer specifies a torque value calibrated for their component, screw material and surface coating — gold-coated screws, for instance, have lower friction and achieve higher preload for the same torque.

Settling, or embedment relaxation: the microscopic roughness of the mating surfaces flattens under the initial clamping load, so preload falls by roughly 2 to 10% within the first minutes. That is why I **re-torque after approximately ten minutes**.

Technique. Ensure the connection is clean and dry and the components fully seated — verified radiographically if deep. Engage the driver fully to avoid rounding the screw head. Apply the wrench **slowly and smoothly**, in a controlled arc, supporting the implant/abutment against rotational force, until the wrench signals the set value. Do not use a jerking motion and do not exceed the value 'for good measure'. Then re-torque at ten minutes.

Care of the instrument. Torque wrenches are precision devices: they must be **dismantled, cleaned and sterilised according to the manufacturer's instructions**, inspected for damage, wear and contamination, and **checked or calibrated at the recommended intervals** — many require annual calibration or replacement of the spring mechanism. A wrench dropped on the floor should be checked before reuse. I would keep a record of calibration as part of clinical governance.

Never use a torque wrench to force a non-seating component — that causes screw fracture, bone loss and pain."

EXAMINER CHECKLIST

1. Uses the manufacturer's value
2. Defines preload correctly
3. Explains friction consumption of torque
4. Explains settling and the 10-minute re-torque
5. Cleans and verifies seating first
6. Correct driver engagement and controlled application
7. Supports against rotational force
8. Does not exceed the specified value
9. Cleaning, sterilisation and **calibration record**
10. States never to force a non-seating component

Running this station to time

Time check. The model answer is 418 words, 2 minutes 59 seconds at 140 words per minute — but the wrench has to be picked up, engaged and turned while the mechanics are explained, and the mechanics are where candidates spend three minutes on physics that earns two marks. This is a **numbers station**: **quote the figures — the torque value, the fraction of it that becomes preload, the 2 to 10% settling loss, the ten minutes, the annual calibration** — and let the numbers carry the explanation instead of explaining around them.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Presents the model with an implant analogue, an abutment, screws and a torque wrench: "Torque this abutment correctly, explaining the underlying mechanics and how you look after the instrument."	Checklists 1 and 5, before the wrench is touched. **"First I confirm the manufacturer's specified torque value for this component."** Say a figure and its range — **typically 15 to 35 Ncm; 35 Ncm is common for a definitive abutment screw, 15 Ncm for some prosthetic screws, and a healing abutment is hand-tightened only** — then that **it is not a universal figure, and applying another system's value is an error**. Then checklist 5, with the hands: **connection cleaned and dry, components fully seated, seating verified radiographically if the margin is deep.**
0:40–1:50	"What are you actually doing to the screw?"	Checklists 2 and 3 — quoted, not derived. **Preload defined**: tightening generates **preload, the tensile clamping force in the screw shank** that holds the joint together and resists separation under function. Then the figure that makes the point: **most of the applied torque is consumed by friction — under the head and at the threads — and only around a tenth becomes preload.** That is why the value is calibrated for **their** component, screw alloy and coating, and why a **gold-coated screw, with lower friction, reaches a higher preload for the same torque** and carries its own number.

Time	The examiner does	What earns the mark
1:50–2:50	Hands over the wrench. Watches.	Checklists 6, 7 and 8, narrated as performed. **Engage the driver fully and axially** before any force is applied — a partially engaged driver rounds the head and turns a simple job into a retrieval. **Apply the wrench slowly and smoothly in a single controlled arc** until it signals the set value — **no jerking, no ratcheting up in stages**. **Support the implant and abutment against the rotational force**, so torque is not transmitted into the bone interface. And checklist 8, said explicitly: **stop at the value — no adding a little 'for good measure'**, because beyond it the screw is heading for its yield point.
2:50–3:40	"You have torqued it. Are you finished?"	Checklist 4, with both numbers. **Settling — embedment relaxation**: the microscopic roughness of the mating surfaces flattens under the initial clamping load, so **preload falls by roughly 2 to 10% within the first minutes**. Therefore **re-torque to the same value after approximately ten minutes**. This is the single most commonly omitted step in practice and it is a named checklist point; a candidate who torques once and moves to the occlusion has lost it.
3:40–5:00	"And the wrench itself?"	Checklists 9 and 10, plus the record. **The wrench is a precision measuring device**: **dismantled, cleaned and sterilised to the manufacturer's instructions**, **inspected for damage, wear and contamination**, and **checked or calibrated at the recommended interval, commonly annually**, with some designs requiring replacement of the spring. **A wrench dropped on the floor is checked before reuse.** The mark includes the **calibration record kept as part of clinical governance**. Then checklist 10 as an absolute: **never use a torque wrench to force a non-seating component** — forcing an interference causes screw fracture, crestal bone loss and pain. Finally: **record the component, the screw, the torque applied and the ten-minute re-torque**, tell the patient to report looseness or clicking immediately, and check the screw at review.

The compressed answer — what a strong candidate says in five minutes

"**First, the value.** I confirm the **manufacturer's specified torque for this exact component** — I look it up rather than remember it, because it is **not a universal figure**. In practice it is **somewhere between 15 and 35 newton centimetres**: **35 Ncm is common for a definitive abutment screw, 15 Ncm for some prosthetic and temporary screws**, and a **healing abutment is hand-tightened only**. Using another system's number is an error, not a shortcut.

Before I turn anything, I make sure the **connection is clean and dry** and the **components are fully seated** — and if the margin is deep I **verify seating radiographically**, because a component that is not down will not be made to go down by torque.

Now the mechanics. Tightening a screw generates **preload** — the **tensile clamping force in the screw shank** that holds the joint together and resists separation under function. Preload is what stops the screw loosening; loosening is a preload problem, not a screw problem.

But most of the torque I apply never becomes preload. The great majority — of the order of **ninety per cent** — is consumed by **friction**, under the head and along the threads, and **only around a tenth ends up as clamping force**. That is exactly why the value is calibrated for **their** component, screw alloy and

coating: a **gold-coated screw has lower friction and reaches a higher preload for the same torque**, so it carries a different number.

Technique. I **engage the driver fully and axially** — a partially engaged driver rounds the head and I end up retrieving a fragment. Then I apply the wrench **slowly and smoothly, in one controlled arc**, until it signals the set value. **No jerking, no ratcheting up in stages.** I **support the implant and abutment against the rotational force**, so I am not transmitting that torque into the bone interface. **And I stop at the value — nothing added for good measure**, because past that point I am heading for the yield of the screw.

Then I am not finished. The microscopic roughness of the mating surfaces **flattens under the clamping load — settling, or embedment relaxation** — and **preload falls by roughly 2 to 10% within the first few minutes**. So I **re-torque to the same value at about ten minutes**: the single most commonly skipped step, and the one that prevents a great deal of screw loosening.

Looking after the wrench. It is a **precision measuring instrument**, not a spanner. **Dismantled, cleaned and sterilised exactly as the manufacturer specifies, inspected for damage, wear and contamination** before each use, and **checked or calibrated at the recommended interval — commonly annually**, with some spring mechanisms replaced rather than recalibrated. **If it is dropped, it is checked before reuse.** And I keep a dated **calibration record** as part of clinical governance — an uncalibrated wrench delivers an unknown force.

One absolute. I never use a torque wrench to force a component that is not seating. If it will not go down, something is wrong — an interference, debris in the connection, a mismatched part, a non-passive framework — and forcing it fractures the screw, loads the crest and hurts the patient. I stop and find out why.

And I record the component, the screw, the **torque applied** and the **ten-minute re-torque**; I tell the patient to report looseness or clicking straight away; and I check the screw at review."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **manufacturer's value quoted as a figure, and stated not to be universal** · 4 **settling of 2–10% and re-torque at ten minutes** · 2 preload defined as tensile clamping force · 8 the value not exceeded · 9 sterilisation and a **dated calibration record** · 10 never force a non-seating component.

Accept these late or on prompting: 3 friction consuming most of the applied torque · 5 connection cleaned and seating verified first · 6 full driver engagement and a single smooth arc · 7 supporting against rotational force.

The commonest way five minutes is wasted here: treating it as a viva on screw mechanics — the helix angle, coefficients of friction, the settling literature — and arriving at the wrench with ninety seconds left, when checklists 5 through 10 are all **things the hands and the instrument cupboard do**, not physics. The second loss is stopping at the click: **checklist 4 lives ten minutes after the click**, and an answer that ends when the wrench signals has thrown away the one step that most reduces screw loosening in practice. Vagueness costs too — "I would use the recommended torque" is not checklist 1; **"typically 35 Ncm for this screw, but I would confirm it for this system"** is.

OSCE 37 — Retrieving a fractured abutment screw

Blueprint: managing complications relating to implant restorations (W + SCE).

EQUIPMENT

a model with an implant analogue containing a fractured screw fragment; ultrasonic scaler, explorer, screw removal kit, magnification.

CANDIDATE INSTRUCTIONS

This abutment screw has fractured within the implant. Describe and demonstrate your retrieval.

MODEL ANSWER

"First, **why did it fracture?** Almost always fatigue from repeated loosening, or overload from an eccentric occlusal contact, a cantilever or a non-passive fit. I must correct that, or I will simply repeat the problem.

Assessment: magnification and good illumination; assess how much of the fragment is exposed, whether it is loose or tight, and take a periapical to check the implant body for fracture and the bone levels.

Retrieval, escalating from least to most invasive: 1. **Ultrasonic scaler tip applied to the fragment in a counter-clockwise direction** at low power with light pressure. Vibration frequently walks a loose fragment out without any instrument entering the threads. 2. A **fine explorer or a sharp probe** engaged in the fragment's flute and rotated anticlockwise. 3. A **proprietary screw removal kit** — these have a guide that seats in the implant connection to keep the instrument centred, and a reverse-thread extractor or a small trephine that engages the fragment. Follow the kit instructions precisely. 4. If the internal threads are damaged, a **thread-repair tap** for that system may salvage the implant.

Throughout: protect the airway with a gauze screen and high-volume suction; irrigate to keep the site cool; and **never allow a rotary instrument to touch the internal threads**, because destroying them usually means losing the implant.

Afterwards: clean the connection, place a **new screw** — never reuse one — and torque to the manufacturer's value, re-torquing at ten minutes. Then **correct the cause**: adjust the occlusion, eliminate excursive contacts, reconsider a cantilever, verify passive fit, and provide a night guard if there is parafunction.

If retrieval fails, the options are to refer, or to accept that the implant is not restorable and plan removal — which I would explain honestly, along with the cost implications. I would record the whole episode and treat it as a significant event."

EXAMINER CHECKLIST

1. Asks why it fractured before retrieving
2. Magnification and radiographic assessment
3. **Ultrasonic counter-clockwise** as first line
4. Explorer technique
5. Proprietary kit with centring guide
6. Thread repair option
7. Airway protection and irrigation
8. Protects internal threads
9. New screw, correct torque, re-torque
10. Corrects the underlying cause; states the fallback and records it

Running this station to time

Time check. The model answer is 422 words, 3 minutes 1 second at 140 words per minute — but there are **four escalating retrieval methods** to describe and demonstrate, and candidates give each a paragraph. They do not need one. The shape here is a **ladder: one sentence per rung, in order, least invasive first**, with the reason for the order said once — and the ninety seconds saved goes on the two marks that bracket the retrieval, which are **why it broke** and **what stops it breaking again**.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:35	Presents the model with a fractured screw fragment in the analogue, an ultrasonic scaler, an explorer, a removal kit and magnification: "This abutment screw has fractured within the implant. Describe and demonstrate your retrieval."	Checklist 1, and it must come **before** any instrument is picked up. **"First — why did it fracture?"** Almost always **fatigue from repeated loosening**, or **overload from an eccentric occlusal contact, a cantilever, parafunction, or a non-passive fit**. Say the consequence out loud: *if I do not correct that, I will simply do this again in six months*. A candidate who starts with the ultrasonic tip has answered a different question and forfeited the first mark on the sheet.
0:35–1:15	Watches.	Checklist 2. **Magnification and good illumination** — loupes or a microscope, said as a requirement rather than a preference. Then the assessment itself: **how much of the fragment is exposed**, whether it is **loose or tight** (a fragment that unwound and is sitting free behaves completely differently from one still under tension), and a **periapical radiograph** to check the **implant body for fracture** and the crestal bone levels — because if the implant itself has fractured, none of what follows is worth doing.
1:15–2:45	"Show me."	Checklists 3, 4, 5 and 6 — the ladder, one rung per sentence, least to most invasive. **1. Ultrasonic scaler tip on the fragment in a counter-clockwise direction, low power, light pressure** — vibration frequently walks a loose fragment out with nothing entering the threads, and this is stated as **first line**. **2. A fine explorer or sharp probe engaged in the flute of the fragment and rotated anticlockwise. 3. A proprietary screw removal kit** — the detail that scores is **the guide that seats in the implant connection to keep the instrument centred**, with a reverse-thread extractor or small trephine, **instructions followed precisely**. **4. If the internal threads are damaged, a thread-repair tap for that system may salvage the implant.**
2:45–3:30	May ask: "What are you protecting?"	Checklists 7 and 8, said while the hands are working rather than as an afterthought. **Airway protection with a gauze screen and high-volume suction** — a 2 mm titanium fragment released under vibration is an inhalation risk and this is the patient-safety mark on the station. **Irrigation to keep the site cool.** And checklist 8 as an absolute: **never let a rotary instrument touch the internal threads** — destroying them usually means losing the implant, which converts a screw problem into an explantation.

Time	The examiner does	What earns the mark
3:30–5:00	"You have the fragment out. Now what?"	Checklists 9 and 10 — the part candidates run out of clock for. **A new screw — never the old one** — torqued to the **manufacturer's value** and **re-torqued at ten minutes**. Then **correct the cause**, item by item: **adjust the occlusion, eliminate excursive interferences, reconsider any cantilever, verify passive fit, provide a night guard if there is parafunction**. Then the fallback: **if retrieval fails, refer, or accept the implant is not restorable and plan removal** — explained honestly, **including the cost implications**. And the record: the episode documented, the cause identified and corrected, what the patient was told, treated as a **significant event**, and reviewed.

The compressed answer — what a strong candidate says in five minutes

"**Before I touch it — why did it break?** An abutment screw does not fracture spontaneously. It is almost always **fatigue from repeated loosening**, or **overload**: an eccentric occlusal contact, an excursive interference, a cantilever, parafunction, or a **framework that never fitted passively**. I have to find that and correct it, or I will be doing this again next year with a smaller fragment.

Assessment. Magnification and good illumination — I am not attempting this by naked eye. I look at **how much of the fragment is exposed**, and whether it is **loose or still tight**, because a fragment that has unwound comes out with vibration alone and a tight one will not. And a **periapical**, to check **the implant body itself is not fractured** and to see the bone levels — if the implant is fractured, I am planning removal, not retrieval.

Then I escalate from least invasive to most, and I do not skip rungs.

First, an ultrasonic scaler tip on the fragment, applied in a counter-clockwise direction, at low power and with light pressure. Vibration alone walks a great many fragments out, and nothing has entered the threads.

Second, a fine explorer or sharp probe, engaged in the flute of the fragment and **rotated anticlockwise**.

Third, a proprietary screw removal kit for this system. These have a **guide that seats into the implant connection and keeps the instrument centred** — that centring is the whole value of the kit, because it is what stops the extractor wandering into the threads. Then a reverse-thread extractor or a small trephine engages the fragment. I follow the kit instructions exactly rather than improvise.

Fourth, if the internal threads are damaged, a **thread-repair tap** for that system may still salvage the implant.

Throughout, two things I am protecting. The airway — a **gauze screen and high-volume suction**, because a small titanium fragment released under vibration is an inhalation or ingestion risk. And **the internal threads**: I **never let a rotary instrument touch them**. Destroying the threads turns a repairable screw fracture into a lost implant.

Once it is out. I clean the connection, and I place a **new screw** — never a reused one, because the old screw has already been through the loading that broke its partner. **Torqued to the manufacturer's value, and re-torqued at ten minutes**, because settling costs preload in the first few minutes and low preload is what caused this in the first place.

Then I correct the cause, which is the actual treatment. Adjust the occlusion — check the static contacts and eliminate excursive interferences on the implant. **Reconsider the cantilever** if there is one. **Verify the framework seats passively. Provide a night guard** if there is parafunction. If I do not do this, everything above is a repair, not a solution.

And if retrieval fails, I say so honestly. The options are to **refer to someone with the equipment and experience to try again**, or to accept the **implant is not restorable and plan its removal and the rebuild that follows** — and I would explain that plainly to the patient, **including what it costs**, rather than keep going until I have wrecked the threads.

I record the whole episode — the cause I identified, what I tried in what order, what I found, the new screw and its torque, the occlusal adjustment, what the patient was told — and I treat it as a **significant event**, and review the occlusion at the next appointment."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **asks why it fractured before retrieving** · 3 **ultrasonic, counter-clockwise, low power, as first line** · 8 internal threads protected from any rotary instrument · 10 **the underlying cause corrected**, with the fallback stated and recorded · 9 a **new** screw, torqued to the manufacturer's value and re-torqued at ten minutes · 7 airway protection and irrigation.

Accept these late or on prompting: 2 magnification and a periapical checking the implant body · 4 the explorer engaged in the flute · 5 proprietary kit with its centring guide · 6 thread-repair tap as a salvage.

The commonest way five minutes is wasted here: going straight for the kit. It is the impressive instrument on the bench, and a candidate who opens with it has skipped checklist 3, which is the first line, and checklist 1, which is the first mark. The second loss is finishing at the moment the fragment comes out — **checklists 9 and 10 sit after the retrieval**, and an answer that ends "and then I'd restore it" has thrown away the new screw, the re-torque, the occlusal correction and the record, along with the airway, which takes four words, is the only patient-safety mark on the sheet, and is routinely omitted because the model is not breathing.

OSCE 38 — Removing a cemented crown and residual cement

Blueprint: managing complications; peri-implant disease (W + SCE).

EQUIPMENT

a model with a cemented implant crown; crown removal instruments, ultrasonic, burs, titanium curettes, floss, air-polisher, periapical radiograph showing residual cement.

CANDIDATE INSTRUCTIONS

This radiograph shows residual cement around an implant with peri-implantitis. Explain and demonstrate how you would remove the crown and the cement.

MODEL ANSWER

"**Consent first**, and this is the crucial preliminary. I would tell the patient: 'To treat this properly I need to get underneath the crown, because I believe there is cement left behind driving the infection. The crown is cemented, so there is a real chance I will damage or destroy it and it will need remaking, with the cost that involves. If we don't do this, the infection continues and the implant may be lost.' I would record that discussion.

Crown removal, least destructive first: 1. **Ultrasonic vibration** applied to the crown margins, which frequently breaks a weak temporary cement seal. 2. A **crown removal instrument** — a sliding-hammer type or a Wamkey-style device — used cautiously, protecting the implant from excessive lateral force. 3. If those fail, **section the crown** with a fine diamond or carbide bur, cutting from occlusal to buccal, and expand it off the abutment. I would protect the implant shoulder and the soft tissue, use copious water and high-volume suction, and accept that the crown is being sacrificed.

Cement removal: - Irrigate and inspect under **magnification and good illumination**. - **Titanium or PEEK curettes and scalers** — never steel — worked circumferentially. - **Dental floss knotted and drawn around the abutment** in a see-saw motion is particularly effective at engaging cement in the sulcus. - **Glycine or erythritol air-polishing** delivered subgingivally to reach thread surfaces and residual particles. - If access is inadequate, a small flap may be needed for direct vision.

Verification: a **periapical radiograph** to confirm no radiopaque cement remains — while noting that a **non-radiopaque cement will not show**, which is why cement selection matters in the first place.

Then prevent recurrence: replace the restoration with a **screw-retained, cleansable design**; if cementation is unavoidable, keep the margin at or within 1 mm of the mucosal margin, use minimal

radiopaque cement and an extra-oral replica technique. Proceed with peri-implantitis therapy, and record everything."

EXAMINER CHECKLIST

1. **Consent to possible destruction of the crown, with cost**
2. Ultrasonic first
3. Crown removal instrument
4. Sectioning technique with implant and tissue protection
5. Magnification and illumination
6. Titanium/PEEK instruments only
7. Knotted floss technique
8. Glycine air-polishing
9. Radiographic verification with the non-radiopaque caveat
10. Converts to a screw-retained cleansable design and records

Running this station to time

Time check. The model answer is 444 words, 3 minutes 10 seconds at 140 words per minute — but the first mark on the sheet is a **conversation**, not a technique, and it cannot be rushed: the patient has to be told the crown may be destroyed and what that will cost, before a single instrument touches it. Budget **50 seconds for consent**, then run the two removals as ladders — **one sentence per rung** — and keep the last minute for verification, the non-radiopaque caveat and the redesign.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:50	Hands over the periapical showing residual cement and indicates the model: "This shows residual cement around an implant with peri-implantitis. Explain and demonstrate how you would remove the crown and the cement."	Checklist 1, and it is the crucial preliminary — **spoken to the patient, in patient language, before anything else**. Three elements, all required: **there is cement left behind and it is driving the infection, so I need to get underneath the crown**; **the crown is cemented, so there is a real chance I will damage or destroy it and it will need remaking — with the cost that involves**; and **if we do nothing the infection continues and the implant may be lost**. Then: **and I record that discussion.** A candidate who begins with the ultrasonic has forfeited the mark the station was built around.
0:50–2:00	"Show me."	Checklists 2, 3 and 4 — a ladder, least destructive first, one sentence per rung. **1. Ultrasonic vibration applied to the crown margins**, which frequently breaks a weak temporary cement seal without damaging anything. **2. A crown removal instrument** — a sliding-hammer type or a Wamkey-style device — used **cautiously, protecting the implant from excessive lateral force**, which is the qualifier that scores. **3. Section the crown** with a fine diamond or carbide bur, **cutting from occlusal to buccal and expanding it off the abutment**, with the **implant shoulder and the soft tissue protected**, **copious water and high-volume suction**, and the explicit acceptance that **the crown is being sacrificed** because the diagnosis matters more than the restoration.

Time	The examiner does	What earns the mark
2:00–3:15	"Now the cement."	Checklists 5, 6, 7 and 8 — four marks in seventy-five seconds, each a clause. **Irrigate and inspect under magnification and good illumination** — cement is submucosal and dark, and this is not a naked-eye task. **Titanium or PEEK curettes and scalers — never steel** — worked circumferentially round the abutment. **Dental floss, knotted, drawn around the abutment and worked in a see-saw motion**, which is particularly effective at engaging cement in the sulcus where an instrument cannot reach. **Glycine or erythritol air-polishing delivered subgingivally** to reach the thread surfaces and residual particles. And the escalation: **if access is inadequate, a small flap for direct vision**.
3:15–4:00	"How do you know you have it all?"	Checklist 9, and it is a two-part mark. **A periapical radiograph to confirm no radiopaque cement remains — and, in the same breath, that a non-radiopaque cement will not show at all**, so a clean film does not prove a clean sulcus. That caveat is why **cement selection matters in the first place**, and why a candidate who stops at "I'd take a radiograph to check" scores half.
4:00–5:00	"And to stop this happening again?"	Checklist 10, with records and review. **Convert to a screw-retained, cleansable design** — the definitive prevention. **If cementation is unavoidable**: **margin at or within 1 mm of the mucosal margin**, **minimal radiopaque cement**, and an **extra-oral replica technique** to extrude the excess outside the mouth. Then **proceed with peri-implantitis therapy** — non-surgical debridement and re-assessment, surgery only if that fails. Then **records**: the cement found and removed, the crown sectioned, what the patient was told about cost and prognosis, and the **review interval and new baseline radiograph** against which stability will be judged.

The compressed answer — what a strong candidate says in five minutes

"**Consent first, and this is the crucial preliminary.** I would say to her: *'To treat this properly I need to get underneath the crown, because I believe there is cement left behind that is driving the infection. The crown is cemented on, so there is a real chance I will damage or destroy it getting it off, and it would need remaking — with the cost that involves. If we don't do this, the infection continues and you may lose the implant.'* And I **record that conversation** before I start, because I am about to destroy something she paid for.

Crown removal, least destructive first.

One: ultrasonic vibration applied to the crown margins. A weak or temporary cement seal frequently breaks with vibration alone, and nothing is damaged.

Two: a crown removal instrument — a sliding-hammer or a Wamkey-style device — used **cautiously, protecting the implant from excessive lateral force**. I am mindful that I am applying force to an implant that already has bone loss around it.

Three: if those fail, I section it. A fine diamond or carbide bur, **cutting from occlusal to buccal**, then expanding the crown off the abutment. I **protect the implant shoulder and the soft tissue**, use **copious water and high-volume suction**, and I accept that the crown is being sacrificed. Getting to the cement matters more than saving the restoration.

Then the cement itself.

I **irrigate and inspect under magnification and good illumination**. Submucosal cement is dark and adherent and this is not a naked-eye job.

Titanium or PEEK curettes and scalers — never steel, because steel scratches the implant surface and creates the plaque trap I am trying to eliminate. Worked circumferentially, all the way round.

Knotted dental floss drawn around the abutment and worked in a see-saw motion. This is the technique people forget and it is remarkably effective — the knot engages cement in the sulcus where no instrument reaches.

Glycine or erythritol air-polishing, delivered subgingivally, to reach the thread surfaces and lift residual particles.

And if I cannot see or reach, I raise a **small flap for direct vision** rather than work blind.

Verification — with an important caveat. I take a **periapical radiograph to confirm no radiopaque cement remains**. **But a non-radiopaque cement will not show on any film**, so a clean radiograph does not prove a clean sulcus — which is exactly why the choice of cement mattered in the first place, and why I would only ever use a radiopaque one.

Then I prevent it recurring. I would **replace the restoration with a screw-retained, cleansable design**, which removes the problem entirely. **If cementation is unavoidable**, I keep the **margin at or within 1 mm of the mucosal margin** so excess is reachable, use the **minimum quantity of a radiopaque cement**, and seat the crown first on an **extra-oral replica of the abutment** so the excess is expressed outside the mouth.

And I carry on with the peri-implantitis itself — cause-related therapy, debridement, hygiene reinforcement and re-assessment at about a month, with surgery considered only if that fails.

I record what cement I found and how much, that the crown was sectioned, what she was told about the cost and about the prognosis, and I take a **new baseline radiograph** and set a **review** against which I will judge whether the disease has actually arrested."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **consent to possible destruction of the crown, with cost, before starting** · 9 radiographic verification **with the non-radiopaque caveat** · 6 titanium or PEEK instruments only · 10 converted to a screw-retained cleansable design, and recorded · 2 ultrasonic tried first · 8 glycine or erythritol air-polishing subgingivally.

Accept these late or on prompting: 3 crown removal instrument with lateral-force caution · 4 sectioning with implant and tissue protection · 5 magnification and illumination · 7 the knotted floss technique.

The commonest way five minutes is wasted here: starting with the bur. The candidate instruction says *explain and demonstrate*, and the explanation that carries the most marks is the one delivered **to the patient before anything is touched** — this is the only station in the group where destroying a restoration is part of the plan, and consent to that is checklist 1. The second loss is the radiograph: "I'd take a periapical to check the cement has gone" reads as complete and misses half the mark, because the clinical point is that **the cement you cannot see is the cement that keeps the disease going** — and the same premature ending costs checklist 10, which wants the **screw-retained redesign** and the records, not a tidy abutment.

OSCE 39 — Surgical management of peri-implantitis

Blueprint: appropriate management of common complications; problem solving of clinical outcomes (W + SCE).

EQUIPMENT

a model with a peri-implantitis defect around a posterior implant; flap instruments, titanium brush, curettes, saline, graft, membrane, burs.

CANDIDATE INSTRUCTIONS

Non-surgical therapy has failed at this posterior implant, which has a 5 mm circumferential intrabony defect. Describe the surgical management.

MODEL ANSWER

"First, confirm that the criteria for surgery are met — the ITI decision tree specifies **persistent suppuration and/or bleeding on probing with probing depths of 6 mm or more** after non-surgical therapy and re-assessment at about a month. I would also confirm that cause-related therapy has been done: risk factors modified, the prosthesis assessed and modified or replaced for cleansability, and any residual cement removed.

Consent must include the likely **soft tissue recession** and possible exposure of metal, the cost, the possibility that treatment does not arrest the disease, and that the implant may still need removal.

Surgery: 1. Local anaesthesia and antiseptic preparation. 2. **Full-thickness mucoperiosteal flap** with intrasulcular incisions and relieving incisions as needed for access, preserving as much keratinised tissue as possible. 3. **Complete removal of the inflamed granulation tissue** from the defect with curettes, so the defect morphology can be seen clearly. 4. **Decontamination of the exposed implant surface** — mechanically with a **titanium brush** or ultrasonic tips, chemically with saline, chlorhexidine or hydrogen peroxide, and/or with **Er:YAG laser irradiation**. No single method is clearly superior; the aim is thorough biofilm and calculus removal without damaging the surface excessively. 5. **Defect-specific decision.** This is a **circumferential intrabony defect — Type 1 CID** in the ITI classification, and the tree indicates that defects with an intrabony component are suitable for **reconstructive therapy**: a bone graft or substitute, with or without a membrane, and with or without a soft tissue graft. Had this been a supracrestal or combined semicircular defect in a non-aesthetic site, **resective therapy with bone recontouring, implantoplasty and an apically positioned flap** would be indicated instead. 6. **Adequate flap adaptation** and tension-free closure. 7. **Post-operative care:** antiseptics, analgesia, and peri-operative systemic antimicrobials where indicated; no brushing over the site; review at one week.

Then supportive therapy every 3 to 5 months, with professional biofilm removal and reinforcement of hygiene, and a new baseline radiograph against which to judge stability.

I would add that surgical management of peri-implantitis, particularly reconstructive therapy, is **Complex**, and unless I had specific training and experience I would refer for this phase having done the diagnosis and cause-related therapy myself."

EXAMINER CHECKLIST

1. Confirms surgical criteria from the decision tree
2. Confirms cause-related therapy completed first
3. **Consents for recession and possible failure**
4. Flap design preserving keratinised tissue
5. Complete granulation tissue removal
6. Names mechanical, chemical and laser decontamination
7. Correctly classifies the defect
8. Selects reconstructive therapy with justification, and states the resective alternative
9. Post-operative regimen
10. 3–5 monthly supportive therapy and a referral judgement

Running this station to time

Time check. The model answer is 496 words, 3 minutes 32 seconds at 140 words per minute — and the surgery itself, which is the part candidates enjoy, carries three of the ten marks. The other seven sit **before it** (criteria, cause-related therapy, consent) and **after it** (post-operative regimen, supportive therapy, the referral judgement). Roughly 150 words must go, and they go from the surgical narrative: **one clause per step of the operation**, so that the entry criteria at the start and the referral judgement at the end both survive.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:50	Presents the model with a peri-implantitis defect at a posterior implant: "Non-surgical therapy has failed at this implant, which has a 5 mm circumferential intrabony defect. Describe the surgical management."	Checklists 1 and 2, and they come before any incision is described. **The criteria, quoted from the ITI decision tree**: **persistent suppuration and/or bleeding on probing with probing depths of 6 mm or more**, after non-surgical therapy and **re-assessment at about one month**. Then checklist 2, which is the one candidates skip because the stem says non-surgical therapy has failed: **confirm cause-related therapy has genuinely been done — risk factors modified**, smoking addressed, glycaemic control addressed, the **prosthesis assessed and modified or replaced for cleansability**, and **any residual cement removed**. Operating on an implant under an uncleansable crown will fail whatever is done to the bone.
0:50–1:35	May ask, as the patient: "What will it look like afterwards?"	Checklist 3, which must be specific rather than general. Consent covers the **likely soft tissue recession and possible exposure of metal or threads**, the **cost**, the honest statement that **treatment may not arrest the disease**, and that **the implant may still ultimately need to be removed**. A patient consented only to "gum surgery" and presented afterwards with a visible thread has not been consented.
1:35–2:40	"Take me through the surgery."	Checklists 4, 5 and 6 — one clause each, not one paragraph each. **Local anaesthesia and antiseptic preparation. Full-thickness mucoperiosteal flap** with **intrasulcular incisions** and relieving incisions for access, **preserving as much keratinised tissue as possible** — the flap mark is the preservation, not the flap. **Complete removal of the inflamed granulation tissue** with curettes, **so the defect morphology can be seen**, which is what makes the next decision possible. Then **decontamination of the exposed implant surface, all three modalities named**: **mechanically with a titanium brush or ultrasonic tips, chemically with saline, chlorhexidine or hydrogen peroxide**, and/or **Er:YAG laser**, with the rider that **no single method is clearly superior**.
2:40–3:45	"What will you do with the defect?"	Checklists 7 and 8 — the decision the station is built on. **Classify it: a circumferential intrabony defect — Type 1 CID** in the ITI classification. Then the rule: **defects with an intrabony component are suitable for reconstructive therapy** — a **bone graft or substitute, with or without a membrane, with or without a soft tissue graft**. And the alternative must be stated, not merely known: a **supracrestal or combined semicircular defect in a non-aesthetic site** would instead indicate **resective therapy — bone recontouring, implantoplasty and an apically positioned flap**. Then **adequate flap adaptation and tension-free closure**.

Time	The examiner does	What earns the mark
3:45–5:00	"And afterwards?"	Checklists 9 and 10, and this is where the competence limit is named. **Post-operative regimen**: **antiseptics — chlorhexidine — analgesia, peri-operative systemic antimicrobials where indicated, no brushing over the site, review at one week.** Then **supportive therapy every 3 to 5 months** with professional biofilm removal and hygiene reinforcement, and a **new baseline radiograph** against which stability is judged thereafter. And the judgement that is a mark rather than a hedge: **surgical management of peri-implantitis, and reconstructive therapy in particular, is Complex — unless I had specific training and experience I would refer for this phase, having done the diagnosis, the risk factor control and the cause-related therapy myself.**

The compressed answer — what a strong candidate says in five minutes

"**First, do the criteria for surgery actually exist?** The ITI decision tree specifies **persistent suppuration and/or bleeding on probing with probing depths of 6 millimetres or more**, after non-surgical therapy and **re-assessment at about a month**. If that is the case here, surgery is indicated.

And second, has cause-related therapy really been done? I would check rather than assume: **risk factors modified** — smoking, glycaemic control, periodontal disease treated in the rest of the mouth; the **prosthesis assessed and modified or replaced so it can actually be cleaned**; and **any residual cement removed**. If the crown is over-contoured and I cannot get a brush under it, no amount of surgery on the bone will hold.

Consent, and it has to be specific. She must be told to expect **soft tissue recession, and that metal or threads may become visible afterwards**; what it **costs**; that **treatment may not arrest the disease**; and that **the implant may still have to be removed in the end**. That conversation is recorded.

The surgery. Local anaesthesia and antiseptic preparation. A **full-thickness mucoperiosteal flap** with **intrasulcular incisions**, relieving incisions where I need access, **preserving as much keratinised tissue as I possibly can** — I will want it at closure. Then **complete removal of the inflamed granulation tissue** with curettes, right down into the defect, because until it is out I cannot see the defect morphology and I cannot make the next decision.

Decontamination of the exposed implant surface. Mechanically with a **titanium brush** or ultrasonic tips; **chemically** with **saline, chlorhexidine or hydrogen peroxide**; and/or **Er:YAG laser irradiation**. I would say honestly that **no single method has been shown to be clearly superior** — the aim is thorough removal of biofilm and calculus without unnecessary damage to the surface.

Now the defect decides the treatment. This is a **circumferential intrabony defect — a Type 1 CID** in the ITI classification. The tree indicates that **defects with an intrabony component are suitable for reconstructive therapy**: a **bone graft or substitute, with or without a membrane, and with or without a soft tissue graft**. Had it instead been a **supracrestal or combined semicircular defect in a non-aesthetic site**, the indication would be **resective — bone recontouring, implantoplasty and an apically positioned flap** — and I would not attempt to reconstruct a defect that has no walls to hold the graft.

Then adequate flap adaptation and tension-free closure.

Post-operatively: chlorhexidine, analgesia, **peri-operative systemic antimicrobials where indicated**, **no brushing over the site**, and **review at one week**.

And then the part that determines whether any of this lasts: supportive therapy every 3 to 5 months, with professional biofilm removal and reinforcement of hygiene, and a **new baseline radiograph** taken at healing against which I judge stability from then on.

One thing I would say plainly. Surgical management of peri-implantitis — and reconstructive therapy especially — is Complex. Unless I had specific training and experience in it, **I would refer for this surgical phase**, having done the diagnosis, the risk factor control, the prosthesis assessment and the cause-related

therapy myself, and having consented the patient properly for what is coming. Referring the surgery is not the same as handing over the patient."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **criteria quoted — suppuration/BoP with PD $\geq$ 6 mm after re-assessment** · 2 cause-related therapy and prosthesis cleansability confirmed first · 7 the defect classified as **Type 1 CID, circumferential intrabony** · 8 reconstructive therapy chosen on that basis, **with the resective alternative named** · 3 consent covering recession, cost and possible failure · 10 **3–5 monthly supportive therapy, and the referral judgement stated**.

Accept these late or on prompting: 4 flap design preserving keratinised tissue · 5 complete granulation tissue removal · 6 mechanical, chemical and laser decontamination with the no-superior-method rider · 9 post-operative antiseptics, analgesia, antimicrobials and one-week review.

The commonest way five minutes is wasted here: starting at the incision. The stem says non-surgical therapy has failed, and candidates take that as permission to begin cutting — losing checklists 1 and 2, two of the easiest marks on the sheet, because they never state the threshold that justifies surgery or check that the crown can be cleaned. The second loss is decontamination: one mark, which attracts three minutes of comparative discussion about lasers, air abrasion and citric acid, starving the defect classification that follows. And **naming this as Complex and referring for the surgical phase is checklist 10 and a mark** — candidates who feel that referring looks weak talk their way through a reconstructive procedure they would not attempt in practice, and lose the point the examiner most wanted to hear.

OSCE 40 — Implantoplasty

Blueprint: management of compromised outcomes (W + SCE).

EQUIPMENT

a model with an implant with exposed threads; diamond and tungsten carbide burs, polishers, high-volume suction, rubber dam.

CANDIDATE INSTRUCTIONS

Describe and demonstrate implantoplasty on this exposed implant surface.

MODEL ANSWER

"**Indication:** exposed, roughened implant threads in a **non-aesthetic site** with a supracrestal or combined defect that cannot be reconstructed, where the aim is to render the exposed surface smooth and less plaque-retentive so that the patient can maintain it.

Contraindications: the aesthetic zone, where the metal will show; a narrow implant, where removing surface material risks weakening it to the point of fracture; and defects with an intrabony component that could be reconstructed instead.

Technique: 1. Flap access with granulation tissue removed and the defect visualised. 2. **Protect the surrounding tissues** — this is the critical step. I would use a rubber dam sheet or gauze packed around the site to catch titanium particles, and **high-volume suction held immediately at the bur**, with copious water coolant. 3. Reduce the exposed threads with a **fine-grit diamond or a tungsten carbide bur** under copious irrigation, working with light pressure, following the contour of the implant and avoiding notching. 4. **Progressively polish** with silicone polishers and polishing paste until the surface is smooth to a probe — the smoother the better, since surface roughness is exactly what I am trying to eliminate. 5. Irrigate copiously, remove the protective packing carefully with the particles it has caught, and inspect. 6. Apically position the flap so the smoothed surface is accessible for cleaning, and close.

Cautions to state: loss of implant diameter reduces fracture resistance, particularly in narrow implants; titanium particles disseminated into the tissues are a recognised concern; and heat generation must be controlled with irrigation. Implantoplasty does not regenerate bone — it is a **resective, maintenance-enabling** procedure.

Afterwards: hygiene instruction with an interdental brush that can reach the newly accessible surface, and **3- to 5-monthly supportive care**."

EXAMINER CHECKLIST

1. States indication (non-aesthetic, non-reconstructable)
2. States contraindications including narrow implants and aesthetic sites
3. Flap access and granulation removal
4. **Tissue protection against titanium particles**
5. High-volume suction at the bur
6. Appropriate burs with copious irrigation
7. Progressive polishing to a smooth finish
8. Irrigation and removal of debris
9. Apically positioned flap for access
10. States the limitations, and the maintenance regime

Running this station to time

Time check. The model answer is 401 words, 2 minutes 52 seconds at 140 words per minute — but the verb is *describe* ***and demonstrate***, and the demonstration alone (packing the site, holding the suction at the bur, reducing, then the whole polishing sequence) needs two of the five minutes. Roughly 120 words must go, and they go by **stating the indication and each contraindication in a single clause** so that the tissue-protection step gets the time the checklist gives it.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Presents the model with exposed threads and a tray carrying diamond and tungsten carbide burs, silicone polishers, a rubber dam sheet, gauze and high-volume suction: "Describe and demonstrate implantoplasty on this exposed implant surface."	Checklists 1 and 2, in clauses, not paragraphs. **Indication: exposed, roughened threads in a non-aesthetic site with a supracrestal or combined defect that cannot be reconstructed** — the aim is a smooth, maintainable surface, not new bone. **Contraindications: the aesthetic zone, a narrow implant, and any defect with an intrabony component that could be regenerated instead.** The competence clause belongs here and takes one sentence: **resective peri-implantitis surgery sits at the Advanced end of the SAC classification, and an aesthetic-zone case or one needing regeneration I would refer** — said once, then the candidate gets on with the procedure.
0:40–1:20	"Show me."	Checklist 3, with the hands and narrated. **Full-thickness flap raised for access**, the defect **visualised round its whole circumference**, and **all granulation tissue removed before a bur turns**. Implantoplasty performed on top of granulation tissue debrides nothing and simply sprays it with titanium; saying that earns the point, doing it silently may not.
1:20–2:30	Silence. Watches the hands and, in particular, where the suction tip is.	Checklists 4, 5 and 6 — **the heart of the station**. **Rubber dam sheet or gauze packed circumferentially** into the flap to catch titanium particles, placed *and* announced. **High-volume suction held at the bur tip**, not parked in the cheek. Then a **fine-grit diamond or tungsten carbide bur under copious water coolant**, light pressure, **following the contour of the implant, no notching**.

Time	The examiner does	What earns the mark
2:30–3:30	"How do you know when to stop?"	Checklists 7 and 8. **Progressive polishing** — coarse to fine silicone polishers, then polishing paste — until the surface is **smooth to a probe drawn across it**, because surface roughness is the whole target. Then **copious irrigation**, the **packing lifted out carefully with the particles it has trapped**, further irrigation, and the field inspected under suction.
3:30–5:00	"Close it, and tell me what happens next."	Checklists 9 and 10, plus records and review. Flap **apically positioned and sutured** so the smoothed surface is *accessible to a brush*, not buried. Then the limitations, plainly: **reduced diameter reduces fracture resistance, especially in a narrow implant**; **titanium particles in the tissues are a recognised concern**; **heat must be controlled by irrigation**; and **the procedure is resective — it does not regenerate bone**. Finally an **interdental brush sized to the newly accessible surface and demonstrated in the patient's hand**, **3- to 5-monthly supportive care**, a **new baseline radiograph**, and a record of the burs, the extent of reduction and the protection used.

The compressed answer — what a strong candidate says in five minutes

"**The indication first.** Implantoplasty is for **exposed, roughened threads in a non-aesthetic site**, where the defect is supracrestal or combined and **cannot be reconstructed**. I am not growing bone back — I am turning a plaque-retentive surface into one the patient can actually clean.

And where I would not do it. Not in the **aesthetic zone**, because I will expose grey metal under thin mucosa. Not on a **narrow implant**, because I am removing diameter from something already slender and I risk fracturing it. And not where there is a **contained intrabony component** that would respond to regeneration — I would reconstruct that rather than grind it away. I should be clear about the limit: **resective peri-implantitis surgery is Advanced on the SAC classification**. I would do this in a posterior site in a patient whose disease I had already controlled, and I would **refer an aesthetic-zone case, or a combined defect needing regeneration**, rather than start it.

Access. A full-thickness flap raised widely enough to see the whole circumference of the defect, and **every scrap of granulation tissue removed first**. Grinding on top of granulation tissue achieves nothing and drives particles into it.

Now the step that matters most — protection. Before any bur turns I **pack the site with a rubber dam sheet or gauze**, right round the implant and into the flap, so the titanium swarf is caught rather than scattered. I have **high-volume suction held at the bur tip**, not resting in the cheek, and **copious water coolant** running throughout.

Reduction. A **fine-grit diamond or a tungsten carbide bur**, light pressure, moving continuously, **following the contour of the implant** so I do not notch it and create a stress riser. Water the whole time: I am cutting titanium, and the heat goes straight into bone.

Then polishing, which is not the optional bit. Progressively finer **silicone polishers and polishing paste**, until the surface is **smooth to a probe drawn over it**. A half-polished surface is rougher than the machined implant I started with, so a rushed finish makes the problem worse rather than better.

Clearing up. Copious irrigation, then I **lift the packing out carefully with the particles it has trapped**, irrigate again, and inspect the field under suction before I close.

Closure. The flap is **apically positioned**, not replaced coronally — the entire point is that the smoothed surface is **reachable with an interdental brush**. Burying it defeats the procedure.

The honest limitations. I have **reduced the diameter, and therefore the fracture resistance, of the implant**. **Titanium particles disseminated into the tissues are a recognised concern** — which is exactly why I packed and suctioned. **Heat generation** must be controlled by irrigation. And **implantoplasty does not regenerate bone**: it is resective and maintenance-enabling, nothing more.

Afterwards. I **size an interdental brush to the newly accessible surface and have the patient use it in the mirror while I watch**, put her on **three- to five-monthly supportive care** with peri-implant probing at every visit, and take a **new baseline radiograph** to measure against in future. I record the indication, the burs used, the extent of reduction, the protection I used and what I told her — including that a smoothed implant still has to be cleaned every day, because the problem was never the metal, it was the biofilm."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 4 **tissue protection against titanium particles** · 5 **suction held at the bur** · 1 non-aesthetic, non-reconstructable indication · 2 contraindications, including narrow implants and the aesthetic zone · 7 progressive polishing to a probe-smooth finish · 10 limitations stated and 3- to 5-monthly maintenance.

Accept these late or on prompting: 3 flap access and granulation removal · 6 correct bur under copious irrigation · 8 irrigation and removal of the trapped debris · 9 apically positioned flap for access.

The commonest way five minutes is wasted here: delivering a viva on peri-implantitis surgery — access versus resective versus regenerative, the decontamination literature, the reconstructive evidence — and reaching the model with ninety seconds left, so the packing, the suction, the polishing and the apical flap, which are five of the ten marks, are never demonstrated at all. The second loss is picking up the bur without packing the site, because "it is only a model": checklist 4 is scored on the sheet being placed **and named**, and a candidate who protects nothing has thrown away the one step that distinguishes this procedure from vandalism.

OSCE 41 — Explantation

Blueprint: recognise compromised outcomes and suggest appropriate management (W + SCE).

EQUIPMENT

a model with an integrated implant; counter-torque device, piezo tips, trephines of various diameters.

CANDIDATE INSTRUCTIONS

This implant must be removed. Describe the techniques available and select one, explaining your reasoning.

MODEL ANSWER

"**First, confirm the decision.** Removal is indicated for mobility, implant fracture, progressive bone loss unresponsive to treatment, uncontrollable infection, bone loss beyond roughly half to two-thirds of the implant length, or a malposition that makes a cleansable, restorable prosthesis impossible. In the ITI framework, this is the point at which it becomes **'irrational to keep'** the implant.

Techniques, from least to most destructive:

1. **Counter-torque (reverse-torque) device.** A driver engages the internal connection and applies increasing anticlockwise torque — many devices generate 200 Ncm or more — until the implant unscrews. This is my **first choice** because it removes the implant with almost no bone loss, which is everything if the site is to be reconstructed. It works best where there is some bone loss already; a fully integrated implant in dense bone may resist it, and there is a risk of fracturing the implant or stripping the connection.
2. **Piezo-surgery** with fine tips used circumferentially in the interface, which is bone-sparing and avoids damage to adjacent roots and soft tissue, though it is slower.
3. **Trephine** of the smallest diameter that will pass around the implant, used with copious irrigation. This is the most destructive, since it sacrifices a cylinder of surrounding bone, and it also generates significant heat.
4. **Sectioning and elevation** where a fractured implant cannot be engaged.

Selection here: I would attempt **counter-torque first**, escalating to piezo and only then to a trephine.

Afterwards: thorough debridement of all granulation tissue, irrigation, and assessment of the residual defect. Then either allow **6 to 8 weeks of soft tissue healing** before staged grafting, or graft immediately if the defect is contained. Provide an interim prosthesis — ideally a **bonded pontic** rather than a denture, so nothing presses on the site.

Consent must cover: possible fracture of the implant during removal; the extent of bone loss and the likely need for grafting; the time to reconstruction; the possibility that a new implant is not possible; and the costs. In the **aesthetic zone**, explantation with ridge reconstruction is **Complex**, and I would refer."

EXAMINER CHECKLIST

1. Confirms the indication and uses the "irrational to keep" framing
2. Counter-torque as first choice with reasoning
3. States its limitations
4. Piezo-surgery as second line
5. Trephine described with its cost in bone and heat
6. Sectioning for fractured implants
7. Debridement and defect assessment
8. Healing/grafting timeline
9. Interim prosthesis avoiding pressure
10. Consent points and a referral judgement

Running this station to time

Time check. The model answer is 462 words, 3 minutes 18 seconds at 140 words per minute — so the words fit, and that is precisely the trap. The verb is *describe and select*, and the marks do not stop when the implant comes out: checklists 7 to 10 are debridement, timing, the interim prosthesis, consent and the referral judgement. Give each technique **one sentence and its cost in bone**, and spend the spare ninety seconds on what happens afterwards.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Presents a model with an integrated implant and a tray carrying a counter-torque device, piezo tips and trephines of several diameters: "This implant must be removed. Describe the techniques available and select one, explaining your reasoning."	Checklist 1, **before a hand goes near the tray**. The decision is confirmed first: **mobility, implant fracture, progressive bone loss unresponsive to treatment, uncontrollable infection, bone loss beyond roughly half to two-thirds of the implant length, or a malposition that makes a cleansable, restorable prosthesis impossible.** Then the phrase the examiner is listening for — in the ITI framework this is the point at which the implant becomes **"irrational to keep"**. A candidate who reaches for an instrument before saying why the implant is coming out has thrown away mark 1 in the first ten seconds.
0:40–1:50	"Which technique would you use, and why?"	Checklists 2 and 3, and the reasoning is the mark, not the choice. **Counter-torque — reverse-torque — first**: a driver engages the internal connection and applies increasing anticlockwise torque, **many devices generating 200 Ncm or more**, until the implant unscrews. The reason, said out loud: it removes the implant **with almost no loss of surrounding bone**, which is everything if the site is to be reconstructed. Then checklist 3 immediately, unprompted — **it works best where there is already some bone loss; a fully integrated implant in dense bone may resist it; and it risks fracturing the implant or stripping the connection.**

Time	The examiner does	What earns the mark
1:50–2:50	"And if it does not unscrew?"	Checklists 4, 5 and 6, stated as a **ladder from least to most destructive**, a clause each. **Piezo-surgery** with fine tips worked circumferentially in the interface — bone-sparing, and it spares adjacent roots and soft tissue, but it is slower. **Trephine** of the **smallest diameter that will pass around the implant**, with copious irrigation — the most destructive option, because it sacrifices a cylinder of surrounding bone and generates significant heat. **Sectioning and elevation** where a fractured implant cannot be engaged at all.
2:50–3:50	"The implant is out. What now?"	Checklists 7 and 8. **Debridement of all granulation tissue, irrigation, and assessment of the residual defect** — specifically whether it is contained or not, because that determines everything that follows. Then the timeline as a fork: **6 to 8 weeks of soft tissue healing before staged grafting, or immediate grafting if the defect is contained.**
3:50–5:00	"What do you tell the patient, and would you do this case yourself?"	Checklists 9 and 10. **Interim prosthesis: a bonded pontic, not a denture**, so that nothing presses on the healing site. Consent covering **fracture of the implant during removal, the extent of bone loss and the likely need for grafting, the time to reconstruction, the possibility that a replacement implant is not achievable, and the costs.** Then the judgement: **explantation with ridge reconstruction in the aesthetic zone is Complex, and I would refer.** Records: the reason for removal, the technique used, the defect found, the plan and the consent discussion.

The compressed answer — what a strong candidate says in five minutes

"**Before I remove anything, I confirm it should come out.** Removal is indicated for **mobility, implant fracture, progressive bone loss that has not responded to treatment, uncontrollable infection, bone loss beyond roughly half to two-thirds of the implant length**, or a **malposition that makes a cleansable, restorable prosthesis impossible**. In the ITI framework, this is the point at which the implant becomes **'irrational to keep'** — the treatment has stopped being worth its own morbidity.

Then I choose the least destructive technique that will work, because the bone I leave behind is the bone I have to rebuild in.

My first choice is a counter-torque device. A driver engages the internal connection and applies **increasing anticlockwise torque — many devices generate 200 Ncm or more** — until the implant unscrews. I choose it because it removes the implant **with almost no bone loss at all**, and if this site is going to be reconstructed, that is the whole game.

Its limitations, and I would state them before the examiner asks. It works best where there is **already some bone loss** to reduce the integrated surface area. A **fully integrated implant in dense bone may simply resist it**, and the forces involved risk **fracturing the implant or stripping the internal connection** — at which point I have lost my engagement and my options narrow.

If counter-torque fails, I escalate — piezo-surgery second. Fine tips used **circumferentially in the bone–implant interface**, which is bone-sparing and avoids damaging adjacent roots and soft tissue. It is slower, and I accept that.

A trephine is third and last, and I say why. The **smallest diameter that will pass around the implant**, under **copious irrigation** — because it **sacrifices a cylinder of surrounding bone** and **generates significant heat**, and heat necrosis compounds the defect I am about to graft. It is the technique I use when the others have failed, not the one I reach for first.

And where the implant is fractured and cannot be engaged at all, **sectioning and elevation**.

Once it is out, I am not finished. I **debride all granulation tissue thoroughly**, irrigate, and then **assess the residual defect** — its walls, its dimensions, and above all whether it is **contained**. That single judgement decides the timing: either **graft immediately if the defect is contained**, or allow **6 to 8 weeks of soft tissue healing and graft in a staged approach**.

In the meantime she needs a tooth, and it must not touch the site. I would provide a **resin-bonded pontic rather than a denture**, because a denture flange over a healing graft causes micromotion, dehiscence and membrane exposure — it is one of the commonest causes of early graft failure.

Consent has to be honest, because this is a failure being managed. I would cover: **the implant may fracture during removal**; **the extent of bone loss will not be fully known until it is out, and grafting is likely**; **the time to a definitive reconstruction, which is months, not weeks**; **the real possibility that a replacement implant is not achievable and we end up with a bridge, a denture or a gap**; and **the costs, including who bears them**.

And I would be clear about my own limit. Removing a posterior implant by counter-torque and grafting a contained defect is within my scope. **Explantation in the aesthetic zone with ridge reconstruction is Complex** — the site is unforgiving, the reconstruction is vertical and horizontal, and the aesthetic result of a second attempt is not something I would gamble. **I would refer that**, having made the diagnosis, taken the imaging and had the consent conversation myself.

I would record the reason for removal, the technique used, the defect found, the plan, the interim prosthesis and the consent discussion, and I would review the site before committing to a replacement."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 indications, with **"irrational to keep"** · 2 **counter-torque first, because it costs almost no bone** · 3 its limitations stated unprompted · 5 the trephine's cost in bone and heat · 7 debridement and whether the defect is contained · 10 consent points and the **aesthetic-zone referral**.

Accept these late or on prompting: 4 piezo as second line · 6 sectioning for a fractured implant · 8 the 6-to-8-week healing or immediate-graft fork · 9 a bonded pontic rather than a denture.

The commonest way five minutes is wasted here: treating all four techniques as equals and describing each in full — tip designs, trephine diameters, irrigation protocols — so that four minutes go on getting the implant out and forty seconds are left for the half of the checklist that follows it. The station says *select one, explaining your reasoning*: **the reasoning is bone preservation**, and once that sentence is said, the other three techniques are a clause each. The second loss is silence about scope. Naming the aesthetic zone as Complex and referring it is **checklist 10, a mark**; a candidate who describes a full ridge reconstruction they would never undertake has answered a different question.

OSCE 42 — Free gingival graft

Blueprint: soft tissue augmentation materials and techniques (W + SCE).

EQUIPMENT

a model with an implant site lacking keratinised mucosa; blades, template material, sutures, palatal stent.

CANDIDATE INSTRUCTIONS

This implant site has 1 mm of keratinised mucosa. Describe how you would perform a free gingival graft.

MODEL ANSWER

"**Indication:** to increase the **width of keratinised mucosa** to at least 2 mm, which is associated with less plaque, less inflammation, less recession and greater comfort during brushing. It is the most predictable procedure for that purpose — a connective tissue graft is for thickness and aesthetics, not primarily for width.

Recipient site preparation: 1. Anaesthetise. Make a **horizontal incision at the mucogingival junction**, then a **partial-thickness (split-thickness) dissection apically**, leaving the periosteum intact

on the bone — the periosteum is the vascular bed on which the graft will survive. 2. Remove or apically suture any mobile muscle fibres and loose tissue so the bed is immobile. 3. Make a **template** of the recipient bed in sterile foil or paper.

Donor site: 4. From the palate, between the distal of the canine and the mesial of the first molar, staying **at least 2 mm from the gingival margin** and superficial enough to avoid the **greater palatine bundle** — a shallow vault reduces the safe zone. 5. Outline with the template and harvest a graft of uniform thickness, typically **1 to 1.5 mm**, epithelialised. 6. Remove any fatty or glandular tissue from the deep surface.

Placement: 7. Transfer immediately, keep it moist in saline, and adapt it precisely to the bed. 8. Suture with fine (5/0 or 6/0) sutures at the coronal margin, plus **peripheral and cross-over (compressive) sutures** to hold the graft immobile against the periosteum. **Immobility is what determines survival**, since the graft initially survives by plasmatic circulation before revascularising at around 3 to 5 days.

Donor site care: 9. Haemostatic dressing and a **pre-made palatal stent or vacuum-formed plate**, since the donor site heals by secondary intention and is the most uncomfortable part of the procedure for most patients.

Post-operative: no brushing over the graft, chlorhexidine, soft diet, no denture over the site, analgesia, warning about donor-site bleeding, and review at one week.

Limitations to state: poor colour and texture match producing a visible 'patch', which is why it is generally avoided in the aesthetic zone; primary contraction of about 25 to 45%; and donor-site morbidity."

EXAMINER CHECKLIST

1. States the indication is KM **width**
2. Distinguishes from CTG
3. Split-thickness bed leaving periosteum
4. Immobilises the bed
5. Template used
6. Correct donor site and avoidance of the greater palatine bundle
7. Appropriate graft thickness and de-fatting
8. **Immobilisation sutures** and the plasmatic circulation rationale
9. Palatal stent and donor care
10. States limitations including colour mismatch and contraction

Running this station to time

Time check. The model answer is 470 words, 3 minutes 21 seconds at 140 words per minute — but it is built as nine numbered steps across three sites, and a candidate who narrates every step at equal weight will run to six minutes and finish somewhere in the palate. Around 120 words have to go, and they go by **giving each step a clause and reserving whole sentences for the three that carry the biology** — the intact periosteal bed, the immobilisation sutures, and the greater palatine bundle.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Presents a model with an implant site lacking keratinised mucosa, and a tray of blades, template material, sutures and a palatal stent: "This implant site has 1 mm of keratinised mucosa. Describe how you would perform a free gingival graft."	Checklists 1 and 2, and both must be said before any incision is described. **The indication is width of keratinised mucosa — to at least 2 mm** — associated with **less plaque, less inflammation, less recession and greater comfort when brushing.** Then the discrimination that earns mark 2: **the free gingival graft is the most predictable procedure for width; a connective tissue graft is for thickness and aesthetics.** Naming the right operation for the right deficiency is the first mark on the sheet.

Time	The examiner does	What earns the mark
0:40–1:50	"How do you prepare the recipient site?"	Checklists 3, 4 and 5. **Anaesthetise; horizontal incision at the mucogingival junction; partial-thickness — split-thickness — dissection apically, leaving the periosteum intact on the bone.** The reason is the mark: **the periosteum is the vascular bed on which the graft survives**, and a bed dissected down to bone loses the graft. Then **remove or apically suture mobile muscle fibres and loose tissue so the bed is immobile**, and **make a template of the bed in sterile foil or paper** — the template is a named checklist point and takes four words to earn.
1:50–2:50	"Now the donor site."	Checklists 6 and 7, with the anatomy quoted. **Palate, between the distal of the canine and the mesial of the first molar**, staying **at least 2 mm from the gingival margin** and **superficial enough to avoid the greater palatine bundle** — with the qualifier that discriminates: **a shallow vault reduces the safe zone**. Then **outline with the template**, harvest an **epithelialised graft of uniform thickness, typically 1 to 1.5 mm**, and **de-fat the deep surface** of fatty and glandular tissue.
2:50–3:50	"Place it."	Checklist 8 — the heaviest mark on the sheet. **Transfer immediately, keep moist in saline, adapt precisely to the bed.** Then the sutures, named: **fine 5/0 or 6/0 at the coronal margin, plus peripheral and cross-over compressive sutures** holding the graft immobile against the periosteum. And the rationale, out loud: **immobility determines survival, because the graft lives initially by plasmatic circulation and revascularises only at around 3 to 5 days** — a graft that moves in that window is a graft that sloughs.
3:50–5:00	"Anything else? And would you do this yourself?"	Checklists 9 and 10, plus the scope judgement. **Donor site: haemostatic dressing and a pre-made palatal stent or vacuum-formed plate**, because it heals by secondary intention and is the most uncomfortable part of the procedure. **Post-operative: no brushing over the graft, chlorhexidine, soft diet, no denture over the site, analgesia, a warning about donor-site bleeding, review at one week.** Then the limitations, which are a named mark: **poor colour and texture match producing a visible patch, hence generally avoided in the aesthetic zone; primary contraction of about 25 to 45%; donor-site morbidity.** And the line: a posterior site is within reach with training; **the aesthetic zone is not mine.**

The compressed answer — what a strong candidate says in five minutes

"**The indication first, because it decides which graft.** This site has 1 mm of keratinised mucosa and I want **at least 2 mm of width** — greater width is associated with **less plaque, less inflammation, less recession and a patient who can brush the site without discomfort.** The **free gingival graft is the most predictable procedure for width**. If the problem were **thickness, contour or colour stability, I would be describing a connective tissue graft instead** — they are not interchangeable.

Recipient site. Anaesthetise. A **horizontal incision at the mucogingival junction**, then a **split-thickness dissection apically that leaves the periosteum intact on the bone.** That intact periosteum is the whole

basis of the operation: it is the **vascular bed the graft survives on**. I then **remove or apically suture any mobile muscle fibres and loose tissue**, because a bed that moves is a bed that fails. Finally I make a **template of the prepared bed in sterile foil or paper**, so I harvest the right size once rather than trimming a graft that is drying out.

Donor site. The palate, **between the distal of the canine and the mesial of the first molar**, staying **at least 2 mm from the gingival margin** and **superficial enough to stay clear of the greater palatine neurovascular bundle** — and I would assess the vault first, because **a shallow palate brings the bundle closer and shrinks the safe zone.** I outline with the template and harvest an **epithelialised graft of uniform thickness, typically 1 to 1.5 mm**, then **remove fatty and glandular tissue from the deep surface**, which will not revascularise.

Placement, and this is where the operation is won or lost. I **transfer immediately, keep it moist in saline**, and adapt it precisely to the bed with no dead space beneath it. Then **fine 5/0 or 6/0 sutures at the coronal margin, plus peripheral and cross-over compressive sutures** that hold the graft down against the periosteum. **Immobility determines survival.** For the first days the graft lives by **plasmatic circulation** alone and only **revascularises at around 3 to 5 days** — any movement in that window shears the fragile vessels forming and the graft sloughs. A haematoma under the graft does the same thing.

The donor site needs as much care as the recipient. A **haemostatic dressing**, and a **pre-made palatal stent or vacuum-formed plate** — made before the day, not improvised afterwards — because the palate heals by **secondary intention** and is the part patients actually complain about.

Post-operatively: no brushing over the graft, **chlorhexidine** instead, a soft diet, **no denture over the site**, adequate analgesia, and a clear warning that the palate may bleed, with instructions on pressure and who to ring. **Review at one week.**

And I would state the limitations honestly, because the patient must consent to them. The graft brings palatal tissue to a buccal site, so it carries a **poor colour and texture match and can leave a visible pale patch** — which is why it is **generally avoided in the aesthetic zone**. It undergoes **primary contraction of roughly 25 to 45%**, so I harvest generously. And there is real **donor-site morbidity**.

On scope. A limited free gingival graft to gain keratinised width at a **posterior** implant site is achievable for a general dentist with appropriate training and mentoring. **Soft tissue augmentation in the aesthetic zone, or where the deficiency is combined with a hard tissue defect, is Advanced to Complex on the SAC classification, and I would refer it** — having made the diagnosis and identified the deficiency myself, which is the part that matters most.

I would record the indication, the width measured before and after, the donor site and graft dimensions, the sutures used, the stent issued and the instructions given."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **width of keratinised mucosa to at least 2 mm** · 2 **distinguished from a CTG, which is for thickness** · 3 split-thickness bed with **periosteum left intact** · 6 donor boundaries and **greater palatine bundle avoided** · 8 **compressive immobilisation sutures and the plasmatic-circulation rationale** · 10 limitations — colour mismatch, 25–45% contraction, donor morbidity.

Accept these late or on prompting: 4 mobile fibres removed so the bed is immobile · 5 a template taken of the bed · 7 graft 1–1.5 mm and de-fatted · 9 haemostatic dressing and a pre-made palatal stent.

The commonest way five minutes is wasted here: a long preamble on the biology of keratinised mucosa and the literature on whether 2 mm is necessary, so that the palate is reached at four minutes and checklists 8, 9 and 10 — the sutures, the stent and the limitations — are never said. The second loss is the opposite failure of nerve: this station is testing whether the candidate **recognises the deficiency, names the right operation and knows where their own line falls**, so a candidate who declines to describe the procedure at all because "I would refer" scores almost nothing. Describe it fully, then place the limit in a clause at the end.

OSCE 43 — Subepithelial connective tissue graft

Blueprint: soft tissue augmentation; gingival biotype influence (W + SCE).

EQUIPMENT

model, blades, sutures, palatal model.

CANDIDATE INSTRUCTIONS

Describe how you would harvest and place a subepithelial connective tissue graft to thicken the buccal tissue at an implant site.

MODEL ANSWER

"**Indication:** to increase soft tissue **thickness** and improve contour, colour stability and the position of the mucosal margin — phenotype modification. Best performed **at implant placement or at second stage**, rather than after the definitive crown is fitted.

Recipient site: a **split-thickness envelope or pouch** created from an intrasulcular or a small crestal incision, extended beyond the defect in all directions, with no vertical releasing incisions if possible so that blood supply is maximal. If coverage is also required, a **coronally advanced flap** with de-epithelialised papillae.

Donor site — two common techniques: - **Single-incision technique:** one horizontal incision about 2 mm apical to the gingival margin, parallel to the long axes of the teeth; a split-thickness dissection creates a pouch; the connective tissue is then dissected free from within and delivered through the single incision. Primary closure of the donor site is achieved, which gives less pain and better healing. - **Trapdoor technique:** parallel horizontal incisions with a flap raised, the graft harvested beneath, and the flap replaced.

Boundaries: between the distal of the canine and the mesial of the first molar; stay superficial to the **greater palatine bundle**; take a graft of 1 to 2 mm thickness; avoid the rugae anteriorly, which are thin and irregular.

Placement: transfer immediately, keep moist, position over the defect, and secure with **sling sutures** around the implant or the adjacent teeth, plus peripheral sutures, so it is immobile. Advance the overlying flap coronally without tension.

Prosthetic support: modify the provisional to a **concave subgingival contour** so it does not push the tissue apically during healing.

Donor care: haemostatic dressing, palatal stent, and warning about discomfort and bleeding.

Post-operative: no brushing, chlorhexidine, soft diet, no trauma to the site, and review at 1 and 2 weeks with sutures removed at 10 to 14 days.

Honest limitations: around **implants**, coverage procedures are considerably less predictable than around teeth; complete coverage of a buccally positioned implant is unlikely, though thickening the tissue and arresting further recession is achievable."

EXAMINER CHECKLIST

1. States the indication is thickness/phenotype
2. States optimal timing
3. Split-thickness envelope preferred, minimal releasing incisions
4. Describes single-incision or trapdoor harvest
5. Correct donor boundaries and bundle avoidance
6. Appropriate graft thickness
7. Immediate transfer, kept moist
8. **Sling and peripheral sutures for immobility**
9. Concave provisional contour
10. Donor care and honest statement of predictability around implants

Running this station to time

Time check. The model answer is 445 words, 3 minutes 11 seconds at 140 words per minute — but two harvest techniques are described in full, and candidates who narrate both blade by blade arrive at the sutures with a minute left. Around 100 words have to go: **describe the single-incision harvest properly and reduce the trapdoor to a single contrasting clause**, then spend what is saved on checklists 8, 9 and 10 — immobility, the provisional contour, and the honest statement about predictability round implants.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Presents a model, blades, sutures and a palatal model: "Describe how you would harvest and place a subepithelial connective tissue graft to thicken the buccal tissue at an implant site."	Checklists 1 and 2, both in the opening thirty seconds. **The indication is soft tissue thickness and contour — phenotype modification** — improving colour stability and the position and stability of the mucosal margin. Then the timing mark, which is easily lost by omission: **best performed at implant placement or at second-stage surgery, not after the definitive crown has been fitted**, when the tissue architecture is already established and the room to manipulate it has gone.
0:40–1:40	"How do you prepare the recipient site?"	Checklist 3. A **split-thickness envelope or pouch** raised from an **intrasulcular or small crestal incision**, **extended beyond the defect in every direction**, and — the clause that carries the mark — **with no vertical releasing incisions if it can be avoided, so that blood supply to the flap and graft is maximal.** If coverage as well as thickness is wanted, a **coronally advanced flap with de-epithelialised papillae.** Say "split-thickness" explicitly; "I raise a flap" is not mark 3.
1:40–2:50	"Harvest it."	Checklists 4, 5 and 6. The **single-incision technique** in detail: **one horizontal incision about 2 mm apical to the gingival margin, parallel to the long axes of the teeth**; a **split-thickness dissection creating a pouch**; the connective tissue **dissected free from within and delivered through that single incision** — with the reason, **primary closure of the donor site, so less pain and better healing.** The **trapdoor** in one clause: parallel horizontal incisions, flap raised, graft taken beneath, flap replaced. Then the boundaries: **distal of the canine to mesial of the first molar, superficial to the greater palatine bundle, graft 1 to 2 mm thick, avoiding the rugae anteriorly**, which are thin and irregular.
2:50–3:50	"Place it."	Checklists 7 and 8. **Transfer immediately and keep it moist**; position over the defect; secure with **sling sutures around the implant or the adjacent teeth plus peripheral sutures so that it cannot move**; and **advance the overlying flap coronally without tension.** Immobility and a tension-free cover are the two mechanical determinants of survival, and both must be said.

Time	The examiner does	What earns the mark
3:50–5:00	"Anything else, and how well does this work?"	Checklists 9 and 10. **Modify the provisional to a concave subgingival contour** so it does not push the newly thickened tissue apically during healing — the prosthetic step candidates forget entirely. **Donor care: haemostatic dressing, palatal stent, warning about discomfort and bleeding.** Post-operative: **no brushing, chlorhexidine, soft diet, review at 1 and 2 weeks, sutures out at 10 to 14 days.** Then the honesty mark: **coverage procedures around implants are considerably less predictable than around teeth; complete coverage of a buccally positioned implant is unlikely — thickening the tissue and arresting further recession is achievable.** And the scope clause: **in the aesthetic zone this is Advanced to Complex, and I would refer.**

The compressed answer — what a strong candidate says in five minutes

"**The indication is thickness, not width.** I am modifying the **phenotype** — increasing soft tissue thickness to improve **contour, colour stability and the position and long-term stability of the mucosal margin**. If I wanted **width of keratinised mucosa**, I would be describing a free gingival graft instead.

Timing matters as much as technique. This is **best done at implant placement or at second stage**, while the site is still open and the tissue can be positioned. **Doing it after the definitive crown is fitted is the least favourable moment** — the architecture is set, access is poor and the result is unpredictable.

Recipient site. I raise a **split-thickness envelope or pouch** from an **intrasulcular or small crestal incision**, **extended beyond the defect in all directions** so the graft is fully enclosed by vascular tissue. **I avoid vertical releasing incisions if I possibly can**, because they cut across the blood supply to the very flap that has to nourish the graft. If I also need coverage, then a **coronally advanced flap with de-epithelialised papillae**.

Harvest — I would use the single-incision technique. One horizontal incision about 2 mm apical to the gingival margin, parallel to the long axes of the teeth; a **split-thickness dissection creating a pouch**; then the connective tissue is **dissected free from within and delivered out through that single incision**. The advantage is that the donor site closes primarily, which means **less pain and better healing** — and palatal pain is what patients remember. The alternative is the **trapdoor technique** — parallel horizontal incisions, a flap raised, the graft harvested beneath and the flap replaced.

Boundaries, and I would state them before I cut. Between the distal of the canine and the mesial of the first molar. Superficial enough to stay clear of the greater palatine bundle — and a shallow vault brings it closer. A graft **1 to 2 mm thick**. And **anteriorly I avoid the rugae**, where the tissue is thin, irregular and of poor quality.

Placement. Transfer immediately, keep it moist, and position it over the defect so it is completely covered. Then **sling sutures around the implant or the adjacent teeth, plus peripheral sutures**, so the graft is **immobile** — it has no blood supply of its own for the first days and any movement destroys the vessels that are forming. And I **advance the overlying flap coronally without tension**; a flap under tension will dehisce and expose the graft.

Then the prosthetic step that is routinely forgotten. I modify the **provisional to a concave subgingival contour**. An over-contoured, convex provisional will press the tissue I have just thickened straight back apically during healing, and I will have done the operation for nothing.

Donor care: haemostatic dressing, a palatal stent, and an honest warning about discomfort and bleeding. **Post-operatively: no brushing over the site, chlorhexidine, soft diet, no trauma, review at one and two weeks, sutures removed at 10 to 14 days.**

And I would be honest about what this achieves, because the consent depends on it. Around **teeth**, root coverage with a connective tissue graft is highly predictable. Around **implants it is considerably less so** — the tissue attaches differently, there is no cementum or periodontal ligament to support coronal migration, and **complete coverage of a buccally positioned implant is unlikely.** What I can realistically offer is **thicker tissue, better contour and colour, and arrest of further recession.** If the implant is malpositioned,

no graft will fix that, and the honest options are to accept it, to modify the prosthesis, or to remove the implant.

On scope, thickening tissue at a posterior site is within reach with training. **In the aesthetic zone, soft tissue augmentation is Advanced to Complex, and I would refer** — having made the diagnosis, identified the phenotype and set the patient's expectations myself.

I would record the indication, the thickness before and after, the harvest technique and site, the graft dimensions, the sutures, the provisional contour and what the patient was told about predictability."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **thickness and phenotype modification, distinguished from width** · 2 **timing — at placement or second stage** · 3 split-thickness envelope with releasing incisions avoided · 5 donor boundaries and **greater palatine bundle** · 8 **sling and peripheral sutures for immobility, flap advanced without tension** · 10 **honest statement of predictability around implants**, with donor care.

Accept these late or on prompting: 4 single-incision versus trapdoor harvest · 6 graft 1–2 mm thick · 7 immediate transfer, kept moist · 9 concave provisional contour.

The commonest way five minutes is wasted here: describing both harvest techniques in equal detail, which eats ninety seconds and earns one mark, when checklist 4 is satisfied by naming the single-incision technique properly and the trapdoor in a clause. The second and more expensive loss is finishing at the sutures: **checklists 9 and 10 both live after the graft is placed** — the concave provisional and the honest statement about coverage round implants — and a candidate who has run out of clock will overclaim, promising the patient complete root coverage on an implant, which is both a lost mark and a consent failure.

OSCE 44 — Chairside pick-up of overdenture attachments

Blueprint: fitting a prosthesis; laboratory techniques; deliver a satisfactory restoration (OS-P).

EQUIPMENT

an edentulous mandibular model with two implant analogues and stud abutments; a denture, autopolymerising acrylic, block-out spacers, handpiece and burs, articulating paper.

CANDIDATE INSTRUCTIONS

Pick up these two stud attachment housings into this denture chairside. Describe each step.

MODEL ANSWER

"First I check the prerequisites: the implants are osseointegrated and the peri-implant tissues are healthy; the **abutments are the correct height** so the housing sits at the level of the mucosa without impinging; and the **denture is otherwise satisfactory** — extension, occlusion, vertical dimension and fit. If the denture is not satisfactory, I would remake it rather than commit attachments to a poor prosthesis.

Steps: 1. Torque the stud abutments to the manufacturer's value. 2. Place the **black processing inserts** (or equivalent) in the metal housings and seat the housings on the abutments. 3. Place **block-out spacers** — the white rubber washers or equivalent — around the base of each abutment, so acrylic cannot flow into the undercut beneath the housing. This is the step most often omitted, and omitting it locks the denture on. 4. **Relieve the fit surface of the denture** generously over each housing until it seats fully with clearance all round, verified with pressure-indicating paste. Also create a small vent hole through the lingual flange for each housing so excess acrylic and air can escape. 5. Check the denture seats fully and the **occlusion is correct** with the housings in place, before adding any acrylic. 6. Apply a little petroleum jelly to the mucosa and to the patient's lips. 7. Mix **autopolymerising acrylic** to a flowable consistency, place a small amount into each recess in the denture — not onto the housings — seat the denture, and have the patient **close into their normal occlusion** and hold, gently, without biting hard, until set. The occlusion, not finger

pressure, positions the denture. 8. Remove the denture with the housings captured, and inspect. 9. Remove the block-out spacers, trim and polish the excess acrylic, ensure no sharp edges and no acrylic inside the housing. 10. Replace the processing inserts with the **definitive retention inserts** of the appropriate strength — I would start with the lowest retention and increase only if the patient wants more. 11. Re-seat, re-check the occlusion and the extension, and confirm the patient can insert and remove the denture themselves — and have them demonstrate it. 12. **Instructions:** how to clean around the abutments with a single-tufted brush, how to clean the housings, that inserts wear and will need replacing, and that the denture will need relining as the ridge resorbs. Record the abutment heights, insert type and torque values."

EXAMINER CHECKLIST

1. Confirms prerequisites and denture quality
2. Torques abutments correctly
3. Processing inserts in housings
4. **Block-out spacers placed**
5. Denture relieved and vented, verified with pressure paste
6. Occlusion checked before acrylic
7. Patient closes into occlusion to seat
8. Trims, polishes, removes spacers
9. Definitive inserts selected, starting with lowest retention
10. Patient demonstrates insertion/removal; hygiene and maintenance instructions; records

Running this station to time

Time check. The model answer is 521 words, 3 minutes 43 seconds at 140 words per minute — and the verb is *pick up*, so twelve steps have to be performed while they are described, and hands are slower than speech. Roughly 170 words must go, and they go by **narrating each step in a clause as the hands do it**, keeping whole sentences for the two omissions that ruin the denture: the **block-out spacers** and the **generous relief**.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Presents an edentulous mandibular model with two analogues and stud abutments, a denture, autopolymerising acrylic, block-out spacers, a handpiece and articulating paper: "Pick up these two stud attachment housings into this denture chairside. Describe each step."	Checklists 1 and 2, before any component is picked up. **Prerequisites out loud: the implants are osseointegrated and the peri-implant tissues are healthy; the abutments are the correct height, so the housing sits at mucosal level without impinging; and the denture is otherwise satisfactory — extension, occlusion, vertical dimension and fit.** Then the judgement that earns the mark: **if the denture is not satisfactory I remake it rather than commit attachments to a poor prosthesis.** Then **torque the stud abutments to the manufacturer's value** — a figure, confirmed for the system, not "to the recommended torque".
0:45–1:45	Silence. Watches the hands.	Checklists 3 and 4 — the pass/fail pair. **Black processing inserts placed in the metal housings**, housings seated on the abutments. Then, said as it is done: **block-out spacers — the rubber washers — around the base of each abutment, so acrylic cannot flow into the undercut beneath the housing.** The candidate should name it as the step most often omitted and say the consequence: **acrylic in the undercut locks the denture onto the abutments, and it comes out with a bur or not at all.**

Time	The examiner does	What earns the mark
1:45–2:45	"Carry on."	Checklists 5 and 6. **Relieve the fit surface generously over each housing** with an acrylic bur **until the denture seats fully with clearance all round**, and **verify with pressure-indicating paste** — not by eye. **Cut a small vent hole through the lingual flange over each housing** so excess acrylic and air escape rather than lifting the denture. Then, before any acrylic is mixed: **seat the denture and check the occlusion with the housings in place.** Acrylic added to a denture that is not seating, or that is now propped open, sets the error permanently.
2:45–3:45	Hands over the acrylic.	Checklist 7, and the technique detail is the mark. **Petroleum jelly to the mucosa and the patient's lips. Autopolymerising acrylic mixed to a flowable consistency**, a **small amount placed into each recess in the denture — not onto the housings.** Seat the denture and have the patient **close into their normal occlusion and hold gently, without biting hard, until set.** Said explicitly: **the occlusion positions the denture, not finger pressure** — finger pressure seats it crooked and locks in a rocking base.
3:45–5:00	"It has set. Finish the job and tell the patient what they need to know."	Checklists 8, 9 and 10. **Remove the denture with the housings captured; inspect; remove the block-out spacers; trim and polish the excess; no sharp edges and no acrylic inside the housing.** Then **replace the processing inserts with the definitive retention inserts — starting with the lowest retention and increasing only if the patient wants more. Re-seat, re-check the occlusion and the extension, and have the patient insert and remove the denture themselves in front of you.** Then instructions and records: **single-tufted brush around each abutment, cleaning the housings, inserts wear and need replacing, the denture will need relining as the ridge resorbs**; record **abutment heights, insert type and torque values**, and set a review.

The compressed answer — what a strong candidate says in five minutes

"**Before I commit anything to this denture I check three things.** That the **implants are integrated and the peri-implant tissues are healthy**; that the **abutments are the correct height**, so each housing sits at the level of the mucosa without impinging on it; and that the **denture itself is satisfactory** — extension, fit, occlusion and vertical dimension. **If the denture is not good, I remake it.** There is no point building retention into a prosthesis that is already wrong; I will simply have made the patient's problem harder to take out.

First I torque the stud abutments to **the manufacturer's value for this system**, which I confirm rather than remember.

Then the housings. I place the **black processing inserts** in the metal housings and seat the housings on the abutments. The processing insert is a working part, not the final one.

Then the step this station is really about. I place **block-out spacers — the rubber washers — around the base of each abutment**, so that **acrylic cannot run into the undercut beneath the housing**. This is the step most commonly omitted, and omitting it **locks the denture onto the abutments**. If that happens I am cutting a patient's new overdenture off their implants at the fit appointment.

Now the denture. I **relieve the fit surface generously over each housing** with an acrylic bur — considerably more than looks necessary — **until the denture seats fully with clearance all round**, and I **verify**

that with pressure-indicating paste rather than trusting my eye. I also cut a **small vent hole through the lingual flange over each housing**, so excess acrylic and trapped air can escape instead of lifting the denture off the ridge.

Then I check the occlusion, with the housings in place, before any acrylic is mixed. If the denture is propped open by an unrelieved housing, adding acrylic makes that permanent.

Petroleum jelly to the mucosa and to the patient's lips, because autopolymerising acrylic is unpleasant on soft tissue and worse on a moustache.

Then the pick-up. I mix **autopolymerising acrylic to a flowable consistency** and place a **small amount into each recess in the denture — into the denture, not onto the housing.** I seat the denture and ask the patient to **close gently into their normal occlusion and hold, without biting hard, until it sets. The occlusion positions the denture, not my fingers.** If I hold it with finger pressure I will seat it off the ridge and lock in a rock.

When it has set I remove the denture with the housings captured and inspect it. I **remove the block-out spacers**, trim and polish the excess acrylic, and make sure there are **no sharp edges and no acrylic inside the housing** — acrylic in the housing means the insert will not seat and the retention will be wrong.

Then I exchange the processing inserts for the definitive retention inserts. I **start with the lowest retention available** and increase only if the patient wants more. A patient who cannot get their denture out will not persevere with it, and it is far easier to add retention later than to explain why they cannot manage.

Then I re-seat, re-check the occlusion and the extension, and I put the denture in the patient's own hands — they insert it and remove it in front of me until they can do it reliably.

Instructions. How to clean **around each abutment with a single-tufted brush**; how to clean the **fit surface and inside the housings**; that **the inserts wear and will need replacing** as routine maintenance, not as a failure; that the **denture will need relining as the ridge resorbs**; and to leave it out at night.

And I record the **abutment heights, the insert type and the torque values applied**, so that whoever sees this patient next can order the right parts, and I set a maintenance review."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 4 **block-out spacers placed, and why** · 5 **generous relief and venting, verified with pressure paste** · 7 **patient closes into occlusion to seat, not finger pressure** · 1 prerequisites, including remaking an unsatisfactory denture · 9 **definitive inserts, lowest retention first** · 10 patient demonstrates insertion and removal; hygiene, maintenance and records.

Accept these late or on prompting: 2 abutments torqued to the manufacturer's value · 3 processing inserts in the housings · 6 occlusion checked before acrylic · 8 spacers removed, trimmed and polished.

The commonest way five minutes is wasted here: discussing attachment systems — locators versus balls versus bars, retention values, resilience — when the instruction is *pick up these two housings*. That is a viva answer in a performance station, and none of it is on the checklist. The second loss is silent hands: candidates place the spacers and relieve the fit surface without a word, and an examiner cannot award checklists 4 and 5 for something that was merely done. **Say "block-out spacers, so acrylic cannot reach the undercut" as the fingers place them.** The third is finishing at the set: **checklists 8, 9 and 10 are all after the acrylic cures**, and the definitive inserts and the patient's own demonstration are worth more than any amount of preamble.

OSCE 45 — Adjusting and relining a denture over a surgical site

Blueprint: postoperative care; awareness of the difference in surgical management of the partially dentate and edentulous patient (OS-P + SCE).

EQUIPMENT

a partial denture, a model with a healing implant/graft site, pressure-indicating paste, acrylic burs, soft liner material.

CANDIDATE INSTRUCTIONS

This patient's partial denture overlies an implant site grafted 48 hours ago. Adjust and reline it appropriately.

MODEL ANSWER

"The principle is that **any pressure on a graft or a submerged implant causes micromotion, dehiscence and membrane exposure** — denture pressure is one of the commonest causes of early graft failure.

1. Confirm the patient has followed the instruction to leave the denture out for the first 24 to 48 hours.
2. Examine the site: wound apposition, any dehiscence or exposure, swelling.
3. **Identify the pressure areas** — seat the denture with **pressure-indicating paste** on the fit surface and ask the patient to close gently; the paste shows where it contacts the surgical site.
4. **Relieve generously** with an acrylic bur — considerably more than seems necessary, because swelling will change over the next few days. I would aim for clear space over the whole surgical field, and I would rather the denture be under-extended and slightly loose than press on the graft.
5. Re-check with paste until there is no contact over the site.
6. **Reline the remainder** of the fit surface with a **soft (tissue conditioner) liner** so the load is distributed onto the sound, non-surgical areas — never a hard reline over a healing site.
7. Check the **occlusion** and relieve any heavy contact that would transmit load through the denture onto the site.
8. Smooth and polish all adjusted surfaces so there are no sharp edges.
9. **Instructions:** wear as little as possible, particularly for the first two weeks; remove at night; clean the denture and the liner daily with a soft brush and cold water, not hot water, which distorts liners; and return immediately if it feels tight, rubs, or if anything appears through the gum.
10. Review at one week with the sutures, and plan to replace the soft liner every few weeks as healing progresses.

Better still: where possible I would have provided a **resin-bonded pontic** rather than a denture in the first place, since it removes the problem entirely — and in the aesthetic zone that is my preference."

EXAMINER CHECKLIST

1. States the biological rationale
2. Confirms 24–48 h denture-free period
3. Examines the wound
4. **Pressure-indicating paste** to locate contact
5. Generous relief anticipating swelling
6. Re-verifies with paste
7. Soft liner, not hard reline
8. Checks and adjusts the occlusion
9. Smooths and polishes
10. Instructions, review, liner replacement, and the bonded-pontic alternative

Running this station to time

Time check. The model answer is 454 words, 3 minutes 15 seconds at 140 words per minute — but the verb is *adjust and reline*, so paste has to go on, a bur has to run, paste has to go on again and a liner has to be mixed, all while it is explained. About 100 words must go, and they go by **stating the rationale once, in a sentence, and then narrating each adjustment in a clause as it is made** — the biology is one mark, and the hands earn six.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Presents a partial denture, a model with a healing implant and graft site 48 hours old, pressure-indicating paste, acrylic burs and soft liner material: "This patient's partial denture overlies an implant site grafted 48 hours ago. Adjust and reline it appropriately."	Checklists 1, 2 and 3, in that order and quickly. **The rationale first, in one sentence: any pressure on a graft or a submerged implant causes micromotion, dehiscence and membrane exposure, and denture pressure is one of the commonest causes of early graft failure.** Then **confirm she left the denture out for the first 24 to 48 hours as instructed** — a question, asked out loud. Then **examine the site before touching the denture**: wound apposition, any dehiscence or exposure of membrane or graft, swelling, and the sutures.
0:40–1:40	Silence. Watches the hands.	Checklist 4, and it must be paste, not eyesight. **Pressure-indicating paste applied to the fit surface**, the denture seated, the patient asked to **close gently**, and the denture removed and read: **the paste shows exactly where the denture contacts the surgical field.** Candidates who look at the fit surface and say "it is contacting here" have not earned mark 4.
1:40–2:40	"Carry on."	Checklists 5 and 6. **Relieve generously with an acrylic bur — considerably more than seems necessary**, with the reason said out loud: **swelling will change over the next few days**, so the relief has to anticipate it. The stated aim: **clear space over the whole surgical field**, and the explicit trade-off — **I would rather the denture were under-extended and slightly loose than pressing on the graft.** Then **re-check with paste, and again, until there is no contact at all over the site.** Mark 6 is a separate mark from mark 4: the verification has to happen twice.
2:40–3:40	"And the rest of the fit surface?"	Checklists 7 and 8, and the material choice is the mark. **Reline the remainder of the fit surface with a soft liner — a tissue conditioner — so that load is distributed onto the sound, non-surgical areas.** Said explicitly: **never a hard reline over a healing site.** Then the **occlusion**: check it and relieve any heavy contact, because load applied to the occlusal surface is transmitted through the denture base onto the site regardless of how well the fit surface has been relieved.
3:40–5:00	"Finish, and tell the patient what to do."	Checklists 9 and 10. **Smooth and polish every adjusted surface** so there are no sharp acrylic edges against a healing flap. Then instructions, specific: **wear it as little as possible, particularly for the first two weeks; out at night; clean the denture and the liner daily with a soft brush and cold water — not hot water, which distorts a liner; return immediately if it feels tight, rubs, or if anything appears through the gum.** Then **review at one week with the sutures, and replace the soft liner every few weeks as healing progresses.** Close with the judgement mark: **a resin-bonded pontic would have avoided the problem altogether, and in the aesthetic zone that is my preference.**

The compressed answer — what a strong candidate says in five minutes

"**The principle governs everything I am about to do. Any pressure on a graft or a submerged implant causes micromotion, dehiscence and membrane exposure** — and denture pressure is one of the commonest causes of early graft failure. So my aim is not to make this denture fit well. It is to make sure it **does not touch the surgical site at all**.

First I check she did as she was asked — that the denture was **left out for the first 24 to 48 hours.** Then I **examine the site before I examine the denture**: the wound apposition, any dehiscence, whether membrane or graft is exposed, the degree of swelling, and the sutures.

Then I find the pressure areas properly. I put **pressure-indicating paste on the fit surface**, seat the denture and ask her to **close gently** — gently, because a firm bite gives me a reading I have created. I take it out and read where the paste has been displaced: **that is where the denture is loading the surgical field**, and it is almost always more extensive than it looks.

Then I relieve, and I relieve generously. With an acrylic bur, **considerably more than seems necessary** — because the swelling will change over the next few days, and relief that is just adequate today will be pressure tomorrow. I aim for **clear space over the whole surgical field**. And I would say plainly: **I would far rather this denture were under-extended and a little loose for a fortnight than have it pressing on the graft.** A loose denture is an inconvenience; an exposed membrane is a failed graft and a lost site.

Then I check my work with the paste again, and I repeat it until there is **no contact whatsoever over the site**.

Then the reline — and the material choice matters. I reline the **remainder** of the fit surface with a **soft liner, a tissue conditioner**, so that the load is spread onto the **sound, non-surgical areas** and the denture is stable without recruiting the graft. **I would never place a hard reline over a healing site** — it sets rigid against tissue that is still changing shape, and it becomes a stamp on the graft.

Then the occlusion, which candidates forget. I check it and relieve any heavy or premature contact, because **load applied to the teeth is transmitted straight through the base onto the ridge**, however carefully I have relieved the fit surface.

Then I smooth and polish everything I have adjusted. Raw, sharp acrylic against a healing flap will ulcerate it.

Instructions, and they are as important as the adjustment. Wear it as little as you possibly can, especially for the first two weeks — for social situations and eating soft food, not all day. **Leave it out at night. Clean it and the liner daily with a soft brush and cold water — not hot water, which distorts the liner.** And **come back immediately if it feels tight, if it rubs, or if you can see anything coming through the gum.** I would give that in writing with a number to ring.

Review at one week, with the sutures, and I would plan to **replace the soft liner every few weeks** as healing and remodelling change the ridge — a liner left in place hardens, becomes colonised and stops being soft.

And I would say what I would have preferred. Where the span allows, **a resin-bonded pontic rather than a denture removes this problem entirely** — nothing rests on the site, nothing has to be relieved, and there is nothing to relieve wrongly. In the **aesthetic zone that is my routine choice**, and I would plan it before the surgery rather than adjusting a denture afterwards.

I would record the site findings, the areas relieved, the liner used, the occlusal adjustment, the instructions given and the review interval."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **pressure causes micromotion, dehiscence and membrane exposure** · 4 **pressure-indicating paste used to locate contact** · 5 **generous relief anticipating swelling**, loose accepted over loaded · 7 **soft liner, never a hard reline** · 8 occlusion checked and adjusted · 10 instructions, review, liner replacement and the **bonded-pontic alternative**.

Accept these late or on prompting: 2 the 24–48 hour denture-free period confirmed · 3 the wound examined first · 6 re-verified with paste · 9 smoothed and polished.

The commonest way five minutes is wasted here: a lecture on guided bone regeneration and the biology of membrane exposure, delivered while the denture sits untouched on the bench. The rationale is **one mark and one sentence**; the other nine are paste, bur, paste, liner, occlusion, polish and instructions. The second loss is timidity with the bur — a candidate who relieves conservatively and says "that should be enough" fails checklist 5 outright, because the mark is specifically for **over-relieving in anticipation of**

swelling. The third is stopping at the reline and never reaching the instructions, which is where the marks for hot water, night-time removal and the one-week review all sit.

OSCE 46 — Shade taking and communication

(*Blueprint: prescribe an appropriate type of restoration from the laboratory; evaluate laboratory techniques.*)

EQUIPMENT

shade guides (VITA classical and 3D-Master), a camera, a grey card, a model.

CANDIDATE INSTRUCTIONS

Take and communicate the shade for an upper central incisor implant crown.

MODEL ANSWER

"**Timing and conditions first**, because most shade errors are made here. - Take the shade **early in the appointment**, before the teeth dehydrate — dehydrated enamel appears lighter and higher in value, and can take 20 minutes or more to recover. - Use **natural daylight or a colour-corrected light source** of about 5500 K; avoid the operating light. - Remove bright lipstick, ask the patient to remove strongly coloured clothing or cover it with a grey bib, and work against a neutral background. - Have the patient at eye level and view for **5 seconds at a time**, resting the eye on a neutral grey between comparisons to avoid fatigue.

Method: 1. Determine **value (brightness) first** — squint or defocus to compare brightness; then **chroma (saturation)**; then **hue**. The 3D-Master guide is designed for this sequence. 2. Select the shade of the **cervical, middle and incisal thirds separately**, because a central incisor is rarely one shade. 3. Note **characterisation**: translucency of the incisal third, mamelons, white spots, craze lines, enamel cracks, surface texture, lustre, and any staining. 4. Record the **stump shade** — the colour of the underlying abutment — because it determines the opacity and thickness of ceramic required. This is essential with a titanium abutment.

Communication: 5. **Photographs** are the most valuable single tool: take images with the selected shade tab held **edge-to-edge in the same plane** as the tooth, with a grey card in frame, at least one polarised or cross-polarised image if available, and one with a black background for translucency. Include a full-face and a smile view. 6. **Written prescription** with shade, stump shade, characterisation, texture and lustre instructions, and the shade guide system used. 7. Consider a **custom shade tab** or, ideally, arrangement for the technician to see the patient — the most reliable method of all for a single anterior unit. 8. If in doubt, request a **bisque try-in** so adjustments can be made before glazing.

I would record the shade, the conditions and the guide used in the notes."

EXAMINER CHECKLIST

1. Takes shade early, before dehydration
2. Correct lighting and neutral surroundings
3. Patient preparation and viewing technique
4. **Value, then chroma, then hue**
5. Records shade by thirds
6. Notes characterisation and surface texture
7. Records **stump shade**
8. Photographs with tab in plane, grey card, black background
9. Written prescription and technician contact/custom tab
10. Bisque try-in option; records conditions

Running this station to time

Time check. The model answer is 454 words, 3 minutes 15 seconds at 140 words per minute — comfortable on paper, but this is a **precision station** and it is failed by vagueness, not by length. Nothing needs cutting;

what is needed is that every mark is carried by a **specific number or a named condition** — 5500 K, five seconds, value then chroma then hue, thirds, stump shade, grey card, black background — rather than by the phrase "I would take a careful shade".

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–1:00	Presents VITA classical and 3D-Master guides, a camera, a grey card and a model: "Take and communicate the shade for an upper central incisor implant crown."	Checklists 1, 2 and 3 — the conditions, and they come before the guide is picked up because **most shade errors are made here**. **Early in the appointment, before the teeth dehydrate** — dehydrated enamel appears lighter and higher in value and takes **20 minutes or more** to recover, so a shade taken after rubber dam or a long preparation is wrong. **Natural daylight or a colour-corrected source of about 5500 K; the operating light off. Remove bright lipstick, cover strongly coloured clothing with a grey bib, neutral background, patient at eye level.** And the viewing technique: **look for 5 seconds at a time, resting the eye on neutral grey between comparisons to avoid retinal fatigue.**
1:00–2:00	"Take it."	Checklists 4 and 5 — the sequence, in the right order and named. **Value — brightness — first**, comparing by **squinting or defocusing**, because the eye discriminates brightness better than hue at low resolution. **Then chroma, the saturation. Then hue.** State that **the 3D-Master guide is arranged for exactly that sequence.** Then mark 5: **select separately for the cervical, middle and incisal thirds**, because **a central incisor is never one shade**. A single tab quoted for the whole tooth loses two marks at once.
2:00–2:50	"Anything else you would record?"	Checklists 6 and 7. **Characterisation: incisal translucency, mamelons, white spots, craze lines, enamel cracks, surface texture, lustre and staining** — recorded on a diagram, not in prose. Then the mark candidates most often miss entirely: **the stump shade — the colour of the underlying abutment — because it determines the opacity and thickness of ceramic the technician must build. This is essential with a titanium abutment**, which will shine grey through thin ceramic and thin tissue.
2:50–4:00	"How do you communicate it?"	Checklists 8 and 9. **Photographs are the single most valuable tool**, and the technique is the mark: **the selected tab held edge-to-edge, in the same plane as the tooth**, a **grey card in frame**, at least one **polarised or cross-polarised image** if available, and one against a **black background for translucency**, plus a **full-face and a smile view**. Then the **written prescription: shade by thirds, stump shade, characterisation, texture and lustre, and the shade guide system used** — because a number means nothing without the guide it came from. And **a custom shade tab, or ideally the technician seeing the patient**, which is the most reliable method for a single anterior unit.

Time	The examiner does	What earns the mark
4:00–5:00	"And if you are not confident?"	Checklist 10 and the records. **Request a bisque try-in**, so shade, characterisation and texture can be adjusted **before glazing** — after glazing the options are staining or a remake. Then **record the shade, the stump shade, the guide used, the lighting conditions and the time of day** in the notes, so that if the crown returns wrong the variables can be traced, and any second attempt is made under the same conditions. Tell the patient the shade may be reviewed at try-in.

The compressed answer — what a strong candidate says in five minutes

"**The conditions first, because that is where shades are lost.**

Timing. I take the shade **at the start of the appointment, before the teeth dehydrate.** Dehydrated enamel goes **lighter and higher in value**, and it takes **20 minutes or more to recover** — so a shade taken after isolation is a shade taken from a tooth that no longer exists.

Light. Natural daylight or a colour-corrected source at about 5500 K. The operating light is off — it is too bright and the wrong colour temperature.

Surroundings. Bright lipstick removed, strongly coloured clothing covered with a **grey bib**, and a **neutral background** behind the patient. The patient sits at **my eye level**.

And the eye itself. I look for **five seconds at a time** and then rest my eye on **neutral grey** between comparisons — hold a comparison longer than that and retinal fatigue shifts what I see.

Now the method, in a fixed order. Value first — brightness. I **squint or defocus** to compare, because at low resolution the eye reads brightness far better than it reads hue, and value is the dimension that is unforgivable to get wrong. **Then chroma — the saturation. Then hue.** The **3D-Master guide is built for that sequence**, which is why I prefer it here to the classical guide.

Then I stop treating the tooth as one colour. I select separately for the **cervical, middle and incisal thirds** — a central incisor is a gradient, and one tab across the whole crown is how a technician is given an opaque, flat, monolithic tooth.

Then characterisation, which is what makes the crown belong in the mouth: **translucency of the incisal third, mamelons, white spots, craze lines, enamel cracks, surface texture, lustre and any staining.** I map these on a diagram.

Then the stump shade — and this is the one that separates a tooth crown from an implant crown. I record **the colour of the underlying abutment**, because it dictates the **opacity and thickness of ceramic** the technician needs. **With a titanium abutment this is essential**: titanium under thin ceramic and a thin phenotype shows as grey at the margin, and no amount of shade matching in the body of the crown rescues it. If the stump shade is a problem, the answer is a **ceramic abutment or a titanium base with a ceramic superstructure**, and that is a conversation to have now, not at fit.

Communication. Photographs are the most valuable single tool. I take images with the **selected tab held edge-to-edge, in the same plane as the tooth** — a tab angled towards the light photographs as a different colour. A **grey card in frame** for white balance. At least one **polarised or cross-polarised image** to remove surface reflection and show the underlying dentine, and one against a **black background** to show incisal translucency. Plus a **full-face and a smile view**, so the technician sees the tooth in the face it has to sit in.

A written prescription with all of it: shade **by thirds**, the **stump shade**, the characterisation, texture and lustre instructions, and — crucially — **which shade guide system I used**, because 'A2' and '2M2' are not interchangeable and a number without its guide is useless.

Best of all, for a single upper central, I would arrange either a **custom shade tab** or, ideally, **for the technician to see the patient directly.** That is the most reliable method there is, and for a single anterior unit next to a natural central it is worth the appointment.

And if there is any doubt, I request a bisque try-in, so shade, characterisation and surface texture can be adjusted **before glazing** — once it is glazed my options are surface stain or a remake.

I record the shade, the stump shade, the guide used, the lighting and the time of day, so that the whole thing can be reproduced or investigated if the crown comes back wrong. And I tell the patient the shade will be confirmed at try-in."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **shade taken early, before dehydration** · 2 **5500 K colour-corrected light, neutral surroundings** · 4 **value, then chroma, then hue** · 7 **stump shade recorded — essential with titanium** · 8 photographs with **tab edge-to-edge in plane, grey card, black background** · 9 written prescription **naming the guide system**.

Accept these late or on prompting: 3 patient preparation and the 5-second viewing rule · 5 shade selected by thirds · 6 characterisation and surface texture mapped · 10 bisque try-in offered; conditions recorded.

The commonest way five minutes is wasted here: an excursion into colour science — the Munsell dimensions, metamerism, the physiology of cone response — which earns nothing, because every one of the ten marks is a **practical condition or a piece of paperwork**. The second loss is quoting a single tab. "A2" for an upper central is a wrong answer as well as an incomplete one: the marks are for **thirds, characterisation and stump shade**, and a candidate who names one tab and moves on has forfeited three checklist points in five seconds. The third is treating photography as an optional extra — for an implant crown next to a natural central, checklist 8 is the mark the whole aesthetic result actually depends on.

OSCE 47 — Facebow and jaw registration for an implant case

Blueprint: role of jaw registration when restoring implants (W + SCE, examined practically).

EQUIPMENT

a facebow, a semi-adjustable articulator, record bases, wax rims, bite registration material.

CANDIDATE INSTRUCTIONS

Record a facebow and jaw registration for a patient having a full-arch implant-supported prosthesis planned.

MODEL ANSWER

"**Why:** without a facebow, casts are mounted arbitrarily and the arc of closure in the articulator does not match the patient's, so occlusal contacts developed in the laboratory will be wrong when the prosthesis is fitted — particularly when the vertical dimension is being altered.

Facebow: 1. Explain to the patient and seat them upright with the head supported. 2. Locate and mark the **reference points** — usually the two posterior points at the external auditory meati or the condylar hinge axis, and an anterior reference point such as the orbitale or a point 43 mm from the incisal edge, depending on the system. 3. Attach the **bite fork** to the maxillary arch with a thin, rigid registration material recording only cusp tips — an over-thick fork record introduces error. 4. Seat the earpieces or condylar rods, position the nasion relator or anterior reference pin, tighten the clamps **in the correct sequence** without displacing the assembly, and check the bow is level and symmetrical. 5. Remove as a unit and transfer to the articulator to mount the maxillary cast.

Jaw registration: 6. Use **rigid, accurately fitting record bases** — for a full-arch implant case, ideally **screwed to the implants or to multi-unit abutments**, which eliminates the commonest source of error, the displacement of a resilient mucosa or a flexible base. 7. Adjust the wax rim for **lip support, incisal level at rest and on smiling, the occlusal plane parallel to the interpupillary line anteriorly and the ala-tragus line posteriorly, and the midline**, marking each on the rim. 8. Establish the **vertical dimension**: measure freeway space by the marks-and-gauge method with phonetic confirmation, and adjust the rim so freeway space is about 2 to 4 mm. 9. Record the **horizontal relationship** in centric relation, guiding the mandible gently, with the rims not in contact — use a low-viscosity registration material or notches with a small amount of material, so the bases are not displaced. 10. Verify by re-seating the record and checking it fits without rocking; take a second record and confirm the two agree. 11. Record the **shade, mould and any patient preferences**, photograph the rim in situ, and send with a full written prescription."

EXAMINER CHECKLIST

1. Explains the purpose of the facebow
2. Correct reference points located and marked
3. Thin, rigid bite fork record
4. Correct assembly, levelling and tightening sequence
5. **Rigid record bases, screwed to implants where possible**
6. Rim adjusted for lip support, incisal level, occlusal plane, midline
7. Vertical dimension determined with freeway space measurement
8. Centric relation record with bases not displaced
9. **Verified with a duplicate record**
10. Shade, photographs and written prescription

Running this station to time

Time check. The model answer is 497 words, 3 minutes 33 seconds at 140 words per minute — but a facebow has to be assembled, seated, levelled and tightened while it is explained, and then a whole jaw registration follows it. Roughly 140 words must go, and they go by **giving the purpose one sentence and then naming each reference point, material and check in a clause as the hands reach for it.** This is a precision station: the marks are in the named points and the named materials, not in the theory.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Presents a facebow, a semi-adjustable articulator, record bases, wax rims and bite registration material: "Record a facebow and jaw registration for a patient having a full-arch implant-supported prosthesis planned."	Checklist 1, in one sentence and then move. **Without a facebow the maxillary cast is mounted arbitrarily, so the arc of closure in the articulator does not match the patient's; occlusal contacts developed in the laboratory will be wrong at fit — and the error is magnified whenever the vertical dimension is being altered**, which in a full-arch case it always is. Then explain to the patient and **seat them upright with the head supported.**
0:40–1:50	Silence. Watches the hands assemble the bow.	Checklists 2, 3 and 4, narrated as performed. **Locate and mark the reference points**, named: **the two posterior points — the external auditory meati, or the condylar hinge axis** — and **an anterior reference point, the orbitale or a point 43 mm from the incisal edge, depending on the system.** Then the material detail that is the mark: **the bite fork is attached to the maxillary arch with a thin, rigid registration material recording cusp tips only** — an **over-thick fork record introduces error into every subsequent step.** Then **seat the earpieces or condylar rods, position the nasion relator or anterior reference pin, tighten the clamps in the correct sequence without displacing the assembly, and check the bow is level and symmetrical** before removing it as a unit to mount the cast.

Time	The examiner does	What earns the mark
1:50–2:50	"Now the jaw registration."	Checklist 5, the heaviest mark in the second half, and it is specific to implants. **Rigid, accurately fitting record bases — and for a full-arch implant case, ideally screwed to the implants or to multi-unit abutments.** The reason must be said: **it eliminates the commonest source of error in registration, displacement of a resilient mucosa or flexure of a poorly fitting base.** A candidate who describes shellac bases on an edentulous ridge has missed the point of the station. Then checklist 6: the rim adjusted for **lip support, incisal level at rest and on smiling, occlusal plane parallel to the interpupillary line anteriorly and the ala-tragus line posteriorly, and the midline** — each **marked on the rim** rather than remembered.
2:50–4:00	"How do you determine the vertical and horizontal relationships?"	Checklists 7 and 8. **Vertical: measure freeway space by the marks-and-gauge method with phonetic confirmation, and adjust the rim so freeway space is about 2 to 4 mm. Horizontal: record in centric relation, guiding the mandible gently, with the rims not in contact** — using a **low-viscosity registration material, or notches with a small amount of material**, so that closing onto the record does not displace the bases. The phrase "rims not in contact" is where the mark sits.
4:00–5:00	"Are you confident in that record?"	Checklists 9 and 10. **Verify: re-seat the record and check it fits without rocking, then take a second, independent record and confirm the two agree.** A registration taken once is unverified. Then the transfer: **record the shade, mould and the patient's preferences, photograph the rim in situ**, and **send everything with a full written prescription** — the articulator type and settings, the reference points used, and what is being asked for. Copy retained, and the patient told what happens next.

The compressed answer — what a strong candidate says in five minutes

"**Why a facebow at all.** Without one, the maxillary cast is mounted **arbitrarily** in the articulator, so the **arc of closure on the instrument does not match the patient's.** Any occlusion developed in the laboratory will then be wrong in the mouth — and the discrepancy grows with every millimetre I change the **vertical dimension**, which in a full-arch implant case I always do. So this is not a formality; it is the reason the prosthesis will need one occlusal adjustment rather than five.

Facebow. I explain it to the patient — it looks alarming — and seat them **upright with the head supported.**

Reference points, located and marked. Posteriorly, the **external auditory meati** with an earbow, or the **condylar hinge axis** if the system uses one. Anteriorly, an **orbitale pointer, or a point 43 mm from the incisal edge**, depending on the system I am using.

The bite fork. I attach it to the maxillary arch with a **thin, rigid registration material that records cusp tips only.** An over-thick or soft fork record is the classic error — it flexes, the whole transfer rotates, and the mounting is wrong from the start.

Then I assemble carefully. Seat the **earpieces or condylar rods**, position the **nasion relator or anterior reference pin**, and **tighten the clamps in the correct sequence — without displacing anything I have already set.** I check the bow is **level and symmetrical**, then remove it **as a single unit** and transfer it straight to the articulator to mount the maxillary cast.

Jaw registration — and here the implant case differs from a denture case. I use **rigid, accurately fitting record bases**, and for a full-arch implant case I want them **screwed to the implants or to multi-unit**

abutments. That **eliminates the single commonest source of error**: a base rocking on resilient mucosa or flexing under closing pressure. Everything I record afterwards is only as good as the base it was recorded on.

Then I adjust the rim to the face, and mark each thing on the wax as I set it: **lip support**; **incisal level at rest and on smiling** — I want to see how much tooth and how much gingiva shows; the **occlusal plane parallel to the interpupillary line anteriorly and the ala-tragus line posteriorly**; and the **midline**, taken from the face rather than from what is left of the ridge.

Vertical dimension. I measure **freeway space** using the **marks-and-gauge method**, confirm it **phonetically**, and adjust the rim so freeway space is about **2 to 4 mm**. In an implant case I also need to know that the resulting restorative space is enough for the components and the material I am prescribing — if it is not, I need to know now, not at framework try-in.

Horizontal relationship. I record in **centric relation**, guiding the mandible gently and unforced, **with the rims not in contact** — I use a **low-viscosity registration material, or notches with a small amount of material** — so that the act of closing does not displace the bases and build a shift into the record.

Then I verify it, because a single record is an assumption. I **re-seat the record and check it fits without rocking**, then **take a second, independent record and confirm the two agree.** If they do not, I take both again; I do not average them.

Finally, the information the technician needs. I record the **shade, the mould and the patient's own preferences**, I **photograph the rim in situ** — full face and smile — and I send everything with a **full written prescription** stating the articulator and its settings, the reference points used, the vertical dimension established and exactly what I am asking for at the next stage.

I keep a copy in the notes, and I tell the patient what the next appointment involves."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 the purpose — arc of closure, magnified by a change in vertical dimension · 2 **posterior and anterior reference points named and marked** · 3 **thin, rigid bite fork record of cusp tips only** · 5 **rigid bases screwed to the implants or multi-unit abutments** · 8 **centric relation recorded with the rims not in contact** · 9 **verified with a duplicate record**.

Accept these late or on prompting: 4 correct assembly, levelling and tightening sequence · 6 rim adjusted for lip support, incisal level, occlusal plane and midline · 7 freeway space of 2–4 mm, measured and phonetically confirmed · 10 shade, photographs and the written prescription.

The commonest way five minutes is wasted here: debating whether a facebow is necessary at all, or comparing arbitrary with kinematic hinge axis location — an interesting argument that earns one mark and costs ninety seconds, when checklist 1 is satisfied by a single sentence about the arc of closure. The second loss is answering as though this were a complete denture: the mark that is unique to this station is **checklist 5, record bases screwed to the implants**, and a candidate who never mentions screwing the base down has answered a prosthetics question rather than an implant one. The third is stopping at the centric record — **checklist 9 is the duplicate**, and it takes eight words.

OSCE 48 — Oral hygiene instruction for an implant patient

Blueprint: identify appropriate maintenance care regimes; patient-centred maintenance programmes (W + SCE).

EQUIPMENT

a model with an implant crown, an implant bridge and an overdenture; a range of interdental brushes, single-tufted brush, superfloss, powered brush, mirror.

CANDIDATE INSTRUCTIONS

Give this patient individualised oral hygiene instruction for their implant restorations.

MODEL ANSWER

"I would make this **specific, demonstrated and checked**, not generic.

First, assess: disclose plaque so the patient can see where it actually is; record a plaque score; assess dexterity and whether they can reach the site; and ask what they currently use and when.

Then explain, briefly, why: 'Implants don't decay, but the gum and bone around them can get infected in the same way as around teeth, and that's the main reason implants are lost. The good news is it's almost entirely preventable with daily cleaning.'

Then instruct, site by site:

Single crown: a soft or powered toothbrush twice daily; and the important one — an **interdental brush sized to the space**. I would try several sizes on the model and in the mouth, select the largest that passes without forcing, and **have the patient insert it themselves in the mirror** while I watch. A **single-tufted brush** angled into the sulcus around the emergence.

Implant bridge or splinted units: the same, plus **superfloss or a floss threader** passed under the pontic and drawn buccolingually in a shoe-shine motion; interdental brushes through the embrasures where they fit.

Overdenture: remove the denture to clean; clean **around each abutment with a single-tufted brush**; clean the **fit surface of the denture, especially inside the housings**, with a denture brush and mild soap or denture cleaner; leave the denture out at night in water.

Adjuncts: a powered toothbrush is helpful, particularly with reduced dexterity; a water flosser can help but does not replace mechanical interdental cleaning; **chlorhexidine is for short-term use only**, not routine, because of staining and taste disturbance.

Then check: have the patient demonstrate everything back to me, correct their technique, disclose again to show improvement, and give **written instructions with the specific brush sizes noted**.

Then set the recall: risk-based — 3 to 4 months for a patient with a history of periodontitis, smoking or previous peri-implant disease; otherwise 6 months. I would record the sizes prescribed, the advice given and the plaque score, so I can measure change at the next visit."

EXAMINER CHECKLIST

1. Discloses and records a plaque score first
2. Assesses dexterity and current routine
3. Explains the rationale in plain language
4. **Sizes an interdental brush to the site**
5. Single-tufted brush technique
6. Superfloss under pontics
7. Overdenture-specific advice including housings
8. Correct advice on adjuncts and chlorhexidine
9. **Patient demonstrates back**; technique corrected
10. Written instructions with sizes, risk-based recall, and recorded baseline

Running this station to time

Time check. The model answer is 470 words, 3 minutes 21 seconds at 140 words per minute — but there are **three different prostheses on the model**, brushes have to be tried and sized in real time, and the patient has to demonstrate back. About 120 words must go, and the shape that makes it fit is a **fixed allocation: forty seconds per prosthesis**, with the assessment, the check-back and the recall taking the minutes at either end. This is a taught skill, not a talk.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:50	Presents a model carrying an implant crown, an implant bridge and an overdenture, with interdental brushes of several sizes, a single-tufted brush, superfloss, a powered brush and a mirror: "Give this patient individualised oral hygiene instruction for their implant restorations."	Checklists 1, 2 and 3 — and 1 comes before any advice. **Disclose the plaque so the patient can see where it actually is, and record a plaque score as a baseline.** Then **assess dexterity and reach**, and **ask what they currently use and when** — advice given before that question is generic advice, which is what this station exists to catch. Then the rationale in **plain language, in the patient's words**: *"Implants don't decay, but the gum and bone around them can become infected in the same way as around teeth, and that is the main reason implants are lost — and it is almost entirely preventable with daily cleaning."* No "peri-implant mucositis" to the patient.
0:50–1:50	Watches. Says nothing.	Checklists 4 and 5 at the **single crown**, and mark 4 is the heaviest on the sheet. A **soft or powered brush twice daily**, then the **interdental brush sized to this space**: **try several sizes, on the model and in the mouth, and select the largest that passes without forcing** — then **put it in the patient's hand and have them insert it themselves in the mirror** while the candidate watches. Sizing it is the mark; recommending one is not. Then the **single-tufted brush angled into the sulcus around the emergence profile**, demonstrated at the same time.
1:50–2:35	"And the bridge?"	Checklist 6. The same regime, **plus superfloss or a floss threader passed under the pontic and drawn buccolingually in a shoe-shine motion** — described as a movement, not as a product. **Interdental brushes through the embrasures where they fit**, with the honest addition that if a brush will not fit, the prosthesis is not cleansable and that is a design problem to be addressed, not a hygiene failure to be blamed on the patient.
2:35–3:20	"And the overdenture?"	Checklist 7, which is the one candidates omit altogether. **Take the denture out to clean. Clean around each abutment with a single-tufted brush. Clean the fit surface of the denture, and especially inside the housings**, with a denture brush and mild soap or a denture cleaner. **Leave it out at night, in water.** The housings are the specific mark: plaque and debris inside them are invisible to the patient and never mentioned by candidates.

Time	The examiner does	What earns the mark
3:20–5:00	"Anything else, and when will you see them?"	Checklists 8, 9 and 10. **Adjuncts:** a **powered brush**, particularly with reduced dexterity; a **water flosser can help but does not replace mechanical interdental cleaning; chlorhexidine is short-term only**, not routine, because of staining and taste disturbance. Then the check that defines the station: **the patient demonstrates everything back, technique is corrected, and the plaque is disclosed again to show the improvement.** Then **written instructions with the specific brush sizes written on them**, a **risk-based recall — 3 to 4 months with a history of periodontitis, smoking or previous peri-implant disease, otherwise 6 months** — and the **sizes, advice and plaque score recorded**, so change can be measured next time.

The compressed answer — what a strong candidate says in five minutes

"**I would not give any advice until I know what is actually happening**, so I start by **disclosing the plaque** — she sees it, I see it, and we are both looking at the same thing rather than at my opinion. I **record a plaque score** as a baseline. Then I **assess her dexterity and whether she can physically reach these sites**, and I ask **what she uses now, how often, and in what order** — because I am modifying a routine that already exists, not issuing a new one.

Then, briefly, why. *'Implants don't decay, but the gum and bone around them can get infected in the same way as around natural teeth, and that is the main reason implants are lost. The good news is it's almost entirely preventable with daily cleaning.'* That is the whole rationale, in the patient's language.

Then I teach it, site by site, with the aids in her hand.

The single crown. A **soft or powered toothbrush twice daily** — and then the important one, an **interdental brush sized to this particular space.** I try several sizes: **the right one is the largest that passes without forcing**, and I find that by trying, not by guessing. Then I **put it in her hand and have her insert it herself, in the mirror, while I watch** and correct the angle. I add a **single-tufted brush angled into the sulcus around the emergence profile**, which is where the biofilm actually sits.

The bridge. The same, plus **superfloss or a floss threader passed under the pontic and drawn back and forth buccolingually — a shoe-shine motion**, not a saw. **Interdental brushes through the embrasures wherever they fit.** And if a brush will not fit anywhere, I say so: that is a **prosthesis design problem**, and the answer is to modify or remake the restoration, not to ask her to try harder.

The overdenture. It comes out to be cleaned. Around **each abutment with a single-tufted brush**, following the contour. Then the **denture itself — the fit surface, and especially inside the housings**, with a denture brush and **mild soap or a denture cleaner**, because debris inside a housing is invisible to her and is exactly where the mucosa becomes inflamed. **Out at night, stored in water.**

Adjuncts, with honest limits. A **powered toothbrush** helps, particularly if dexterity is reduced. A **water flosser can be a useful addition but does not replace mechanical interdental cleaning** — it is not a substitute for the brush. And **chlorhexidine is for short-term use only** — a fortnight after surgery, or during an acute episode — **not routinely**, because of staining and taste disturbance.

Then the part that turns advice into instruction. I have her **demonstrate all of it back to me** — the brush, the tuft, the floss, the denture — and I correct the technique in her hand rather than describing it again. Then I **disclose a second time**, so she can see the difference she has just made. If she cannot do it, the answer is a different aid, not a louder explanation.

Then I write it down. Written instructions with the actual brush sizes on them, so she buys the right ones and so the next clinician knows what was prescribed.

And I set a risk-based recall — 3 to 4 months if there is a history of periodontitis, smoking or previous peri-implant disease, **otherwise 6 months.** I **record the sizes prescribed, the advice given, that she demonstrated it back, and the plaque score**, so at the next visit I can measure whether any of this worked rather than repeat it."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **disclosed and a plaque score recorded first** · 4 **interdental brush sized by trying, largest that passes without forcing** · 9 **patient demonstrates back and technique is corrected** · 7 overdenture — abutments, **inside the housings**, out at night · 10 written instructions **with the sizes**, risk-based recall, baseline recorded · 3 the rationale given in plain language.

Accept these late or on prompting: 2 dexterity and current routine assessed · 5 single-tufted brush at the emergence · 6 superfloss under the pontic in a shoe-shine motion · 8 adjuncts, and **chlorhexidine short-term only**.

The commonest way five minutes is wasted here: dismissing it as the easy station and delivering a list — "toothbrush, interdental brushes, superfloss, powered brush, six-monthly recall" — in forty seconds, then standing back. That list is a **description of aids and scores almost nothing**, because seven of the ten marks are for **assessing, sizing, demonstrating, checking back, writing down and recording.** The station is testing whether the candidate can *teach*. The second loss is talking to the examiner instead of to the patient: peri-implant mucositis and biofilm maturation are the wrong register here, and a candidate who never puts a brush into the patient's hand cannot score checklists 4 or 9 however fluent the explanation. The third is giving one regime for three prostheses — **the overdenture housings are a separate mark**, and the word "individualised" in the instruction is the examiner telling you so.

OSCE 49 — Taking a baseline periapical radiograph

Blueprint: interpret radiographic findings; assess treatment outcomes and monitoring parameters (OS-P + SCE).

EQUIPMENT

a phantom head, film/sensor holders (paralleling), an X-ray set (may be non-functional), lead apron.

CANDIDATE INSTRUCTIONS

Take a baseline periapical radiograph of a newly restored implant at the LR6, describing your technique and justification.

MODEL ANSWER

"**Justification first.** A baseline radiograph at the fitting of the prosthesis is the reference against which every future bone level measurement will be judged, and without it the 2017 diagnostic criteria for peri-implantitis cannot be applied properly. I would record that justification in the notes, and confirm that I am acting as referrer, IR(ME)R practitioner and operator, and that I am trained to do so.

Technique — the paralleling technique, because reproducibility is the whole point: 1. Confirm the patient's identity, the site, and check for pregnancy where relevant. 2. Explain the procedure and gain consent. 3. Select the correct **holder with a beam-aiming device** — a posterior mandibular holder here — and assemble it with the sensor or film. 4. Position the sensor **parallel to the long axis of the implant**, well away from the crown so the whole implant and the crestal bone on both sides are included, using cotton wool rolls for comfort if needed. 5. Align the **beam-aiming ring** so the central ray is perpendicular to both the sensor and the implant long axis. The key indicator of correct angulation with an implant is that the **implant threads appear crisp and symmetrical on both sides** — if the threads are blurred, the beam is not perpendicular and bone levels cannot be measured reliably. 6. Use **rectangular collimation**, which reduces dose substantially compared with a round collimator. 7. Select the correct exposure factors for the region and the patient. 8. Ensure everyone else is outside the controlled area; expose. 9. Assess the image quality against the practice's quality assurance criteria. If it is non-diagnostic because of positioning, that is a justified repeat — but repeated errors trigger a quality assurance review. 10. **Report the whole image** in writing, measure the bone level mesially and distally relative to the implant shoulder or a known reference such as the thread pitch, and record those measurements as the baseline. 11. Store the image in the patient record, and note the settings and holder used so the view can be **reproduced** at future reviews.

Future radiographs would be at **one year, then only when clinically indicated** — increased probing depth, bleeding, suppuration or symptoms."

EXAMINER CHECKLIST

1. Justification stated and recorded, with IR(ME)R roles
2. Identity, site, pregnancy check, consent
3. Correct paralleling holder and beam-aiming device
4. Sensor parallel to the implant long axis, whole implant and crest included
5. Beam perpendicular; **crisp symmetrical threads as the quality criterion**
6. Rectangular collimation
7. Correct exposure factors and radiation safety
8. Image quality assessment
9. **Written report and recorded baseline measurements**
10. Records settings for reproducibility and states the follow-up regime

Running this station to time

Time check. The model answer is 500 words, 3 minutes 34 seconds at 140 words per minute — but a holder has to be assembled and positioned on a phantom head while it is described, and the marks at the end are written ones. Roughly 140 words must go, and they go by **compressing the regulatory preamble into two sentences and giving each positioning step a clause**, so that the last ninety seconds are free for the report, the baseline measurements and the follow-up regime, which are checklists 9 and 10 and are the ones candidates never reach.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:50	Presents a phantom head, paralleling holders, an X-ray set which may be non-functional, and a lead apron: "Take a baseline periapical radiograph of a newly restored implant at the LR6, describing your technique and justification."	Checklists 1 and 2, and the justification comes before the equipment is touched. **This film is the reference against which every future bone level measurement is judged; without it the 2017 diagnostic criteria for peri-implantitis cannot properly be applied.** Then the governance clause: **the justification is recorded in the notes, and I confirm that I am acting as referrer, IR(ME)R practitioner and operator, and that I am trained to do so.** Then mark 2 in one breath: **identity confirmed, site confirmed, pregnancy check where relevant, procedure explained and consent obtained.**
0:50–2:00	Silence. Watches the hands assemble the holder.	Checklists 3 and 4, narrated as performed. **The paralleling technique, and the reason must be said: reproducibility is the entire point of a baseline.** A **posterior mandibular holder with a beam-aiming device**, assembled with the sensor or film. Then the positioning: **the sensor placed parallel to the long axis of the implant, well away from the crown, so that the whole implant and the crestal bone on both the mesial and the distal are on the film** — with a cotton wool roll for comfort if needed. A film that clips the apex or misses the crest on one side is a film that cannot be a baseline.

Time	The examiner does	What earns the mark
2:00–3:00	"Take it."	Checklists 5, 6 and 7 — and mark 5 is the one that is unique to implants. **Align the beam-aiming ring so the central ray is perpendicular to both the sensor and the implant long axis**, then the quality criterion, said out loud: **the implant threads must appear crisp and symmetrical on both sides — if the threads are blurred, the beam is not perpendicular and bone levels cannot be measured reliably.** Then **rectangular collimation, which substantially reduces dose compared with a round collimator**; **exposure factors selected for the region and the patient**; and **everyone else outside the controlled area before exposure.**
3:00–3:50	"You have the image."	Checklist 8. **Assess the image quality against the practice's quality assurance criteria** — and the discriminating clause: **a non-diagnostic film caused by positioning is a justified repeat, but repeated errors trigger a quality assurance review**, not a habit of retaking. A candidate who says "I would retake it if it was not good" without the QA framing has half the mark.
3:50–5:00	"What do you do with it?"	Checklists 9 and 10, where the station is won. **Report the whole image in writing** — not just the implant. Then the measurement that makes it a baseline: **the bone level mesially and distally relative to the implant shoulder, or against a known reference such as the thread pitch, recorded as numbers.** Then **store the image in the record and note the settings, holder and positioning used so the view can be reproduced at review.** Finally the regime: **one year, then only when clinically indicated — increased probing depth, bleeding, suppuration or symptoms.** Tell the patient why the film was taken and when the next one is due.

The compressed answer — what a strong candidate says in five minutes

"**Justification first, because no exposure happens without it.** This film is taken at the fitting of the prosthesis, and it is the **reference against which every future bone level measurement will be judged.** Without it I cannot properly apply the **2017 diagnostic criteria for peri-implantitis**, because those criteria depend on **bone loss beyond initial remodelling measured against a baseline** — and a single film years later, with nothing to compare it to, tells me very little. I record that justification in the notes, and I am clear that for this exposure I am acting as **referrer, IR(ME)R practitioner and operator**, and that I am trained and entitled to do so.

Then the routine checks: confirm the patient's identity and the site, check for pregnancy where relevant, explain the procedure and obtain consent.

Technique: the paralleling technique, and I choose it precisely because reproducibility is the whole point. A bisecting-angle film distorts length and cannot be repeated in the same geometry, so it is useless as a baseline.

I select a **posterior mandibular holder with a beam-aiming device** and assemble it with the sensor. I position the **sensor parallel to the long axis of the implant**, sitting **well away from the crown**, so that **the whole implant and the crestal bone on both the mesial and the distal are included** — cotton wool rolls if the patient needs the comfort. A film that shows the implant but clips the crest on one side is not a baseline; it is a wasted exposure.

Then the beam. I align the **beam-aiming ring so the central ray is perpendicular to both the sensor and the implant long axis.** And here is the quality criterion I actually use, which is specific to implants: **the**

threads should appear crisp and symmetrical on both sides of the implant. If the threads are **blurred or the pitch looks different mesially and distally**, the beam is not perpendicular, and I cannot measure bone levels from that film reliably — so I would reposition rather than accept it.

Dose. Rectangular collimation, which reduces the dose substantially compared with a round collimator. **Exposure factors appropriate to the region and the patient.** Everyone else **outside the controlled area** before I expose.

Then I assess the image against the practice's quality assurance criteria. If it is **non-diagnostic because of my positioning, that is a justified repeat** — but I would record it, because **repeated positioning errors trigger a quality assurance review**, and a pattern of retakes is a training issue, not a run of bad luck.

Then the part that makes this a baseline rather than just a radiograph. I **report the whole image in writing** — the implant, the adjacent teeth, the bone, the sinus and inferior dental canal as visible, and any incidental finding — not only the region of interest. Then I **measure the bone level mesially and distally relative to the implant shoulder**, or against a known reference such as the **thread pitch**, and **record those measurements as numbers in the notes.** A baseline that is not measured is not a baseline.

And I make it reproducible. I **store the image in the patient record and note the exposure settings, the holder and the positioning used**, so that at review the same view can be taken again and the two films can genuinely be compared.

Follow-up regime: a further radiograph at one year, and after that only when clinically indicated — increased probing depth against baseline, bleeding on probing, suppuration or symptoms. I do not take annual films as routine, because every exposure must be justified on its own.

I tell the patient why this film was taken, what it will be used for, and when the next one is due."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **justification recorded, with the IR(ME)R roles named** · 3 **paralleling holder with a beam-aiming device, chosen for reproducibility** · 5 **crisp, symmetrical threads as the angulation criterion** · 6 rectangular collimation · 9 **written report of the whole image and recorded baseline measurements** · 10 settings recorded for reproducibility, and the one-year-then-indicated regime.

Accept these late or on prompting: 2 identity, site, pregnancy check and consent · 4 sensor parallel to the long axis with the whole implant and crest included · 7 exposure factors and controlled area · 8 image quality assessed against QA criteria.

The commonest way five minutes is wasted here: a full recitation of IR(ME)R — duty holders, employer's procedures, dose constraints, referral criteria — which is worth exactly one mark and can be said in two sentences, after which the candidate has ninety seconds left for a positioning task and no time at all for the report. The second loss is treating the exposure as the end of the station: **checklists 9 and 10 both happen after the image appears**, and the **measured baseline** is the reason the film was taken at all. The third is generic quality criteria — "no cone cutting, no elongation" is true of any periapical; the mark here is the implant-specific one, **symmetrical threads**, and it takes one clause.

OSCE 50 — Writing a referral letter

Blueprint: assess the need for referral; recognise the complexity of the case; communicate the planned treatment to the wider team (W + SCE).

EQUIPMENT

a case summary with clinical photographs, radiographs and a CBCT report; blank letterhead.

CANDIDATE INSTRUCTIONS

This patient requires bilateral lateral window sinus elevation, which is beyond your scope. Write the referral letter, reading it aloud as you write.

MODEL ANSWER

"I would structure it under headings so the receiving clinician can find what they need quickly.

Header: date; my name, practice address, GDC number and contact details; the patient's full name, date of birth, address and contact details; and the addressee.

Reason for referral, in the first line: 'I would be grateful for your opinion and treatment regarding bilateral sinus floor elevation and implant placement in the posterior maxilla.' The receiving clinician should not have to read three paragraphs to find out what I want.

Presenting complaint and the patient's expectations: what she has asked for, in her words, and what I have already told her is and is not achievable.

Relevant medical history: conditions, medications with doses, allergies, smoking status in pack-years, and the results of any liaison I have already undertaken.

Dental and social history: attendance, periodontal history and current status, previous implant experience.

Clinical findings: extra-oral and intra-oral findings, periodontal status with plaque and bleeding scores, occlusal findings, the edentulous span, ridge dimensions, keratinised tissue, and the opposing dentition.

Investigations: 'I enclose a panoramic radiograph dated…, periapicals of…, clinical photographs, a small field-of-view CBCT dated… with my report, and mounted study casts with a diagnostic wax-up.' **Sending the images matters** — it prevents repeat exposure and is an IR(ME)R consideration.

My assessment: 'Residual bone height is 4 mm at the 16 and 3 mm at the 17 with a flat sinus floor and no septa. On the ITI decision tree this indicates a lateral window approach, which falls outside my scope of practice.'

What I have already discussed and consented: the options presented — including the shortened dental arch and no treatment — the risks I have covered, and that she understands the referral is for an opinion.

What I am requesting: opinion only, or treatment; and **who will do what afterwards** — 'I am happy to undertake the restorative phase and to provide ongoing maintenance, or to refer the whole case, whichever you prefer.'

Urgency and any access needs: routine; interpreter not required; wheelchair access needed.

Closing: my direct contact details, and an offer to discuss.

I would keep a copy in the notes, tell the patient what I have written and what to expect, and set a reminder to follow up if I have not heard within a reasonable period — because the referral is my responsibility until it is picked up."

EXAMINER CHECKLIST

1. Full identifiers for patient and referrer
2. **Reason for referral in the first line**
3. Patient expectations in their own words
4. Medical history with medications and smoking
5. Dental, periodontal and social history
6. Structured clinical findings
7. **Encloses existing images to avoid repeat exposure**
8. States the assessment and why it exceeds scope, with a framework reference
9. States what has been consented and precisely what is requested, including shared-care arrangements
10. Copy retained, patient informed, follow-up arranged

Running this station to time

Time check. The model answer is 554 words, 3 minutes 57 seconds at 140 words per minute — and this is the one station where the words cannot be spoken at speaking speed, because the instruction is *reading it aloud as you write*, and a hand writes at perhaps 25 words a minute. At least 200 words must go. The shape that works: **say each heading aloud and give it one line of content, dictating verbatim only the two sentences that carry marks — the opening line of the letter and the assessment.** Write in headed notes, not prose.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:40	Hands over a case summary with photographs, radiographs and a CBCT report, and blank letterhead: "This patient requires bilateral lateral window sinus elevation, which is beyond your scope. Write the referral letter, reading it aloud as you write."	Checklists 1 and 2, and mark 2 must be the **first line of the letter**, not the third paragraph. Header: **date; my name, practice address, GDC number and contact details; the patient's full name, date of birth, address and contact details; the addressee.** Then, dictated verbatim: **"I would be grateful for your opinion and treatment regarding bilateral sinus floor elevation and implant placement in the posterior maxilla."** Said out loud as the reason it comes first: **the receiving clinician should not have to read three paragraphs to find out what is being asked.**
0:40–1:40	Watches the pen.	Checklists 3, 4 and 5, one line each under spoken headings. **Presenting complaint and expectations — in her own words** — and what has already been said to her about what is and is not achievable. **Relevant medical history: conditions, medications with doses, allergies, smoking status in pack-years, and the outcome of any liaison already undertaken. Dental and social history: attendance pattern, periodontal history and current status, previous implant experience.** Three headings, three lines, sixty seconds.
1:40–2:40	Silence.	Checklists 6 and 7. **Clinical findings, structured:** extra-oral and intra-oral, **periodontal status with plaque and bleeding scores**, occlusal findings, the edentulous span, ridge dimensions, keratinised tissue and the opposing dentition. Then the enclosures, and the reason is the mark: **"I enclose a panoramic dated..., periapicals of..., clinical photographs, a small field-of-view CBCT dated... with my report, and mounted study casts with a diagnostic wax-up." Sending the images prevents a repeat exposure, and that is an IR(ME)R consideration**, not a courtesy.
2:40–3:50	"Why are you referring?"	Checklist 8 — the heaviest mark, and it must be a **quantified statement with a framework attached**, dictated verbatim: **"Residual bone height is 4 mm at the 16 and 3 mm at the 17, with a flat sinus floor and no septa. On the ITI decision tree this indicates a lateral window approach, which falls outside my scope of practice."** A referral that says only "this is beyond me" earns a fraction of the mark; the numbers and the named framework earn all of it.

Time	The examiner does	What earns the mark
3:50–5:00	"Anything else before you sign it?"	Checklists 9 and 10. **What has been discussed and consented** — the options presented **including the shortened dental arch and no treatment**, the risks covered, and that she understands this is a referral for an opinion. Then **precisely what is being requested — opinion only, or treatment — and who does what afterwards: "I am happy to undertake the restorative phase and provide ongoing maintenance, or to refer the whole case, whichever you prefer."** Then **urgency and access needs**, contact details and an offer to discuss. Finally the governance: **keep a copy in the notes, tell the patient what has been written and what to expect, and set a reminder to follow up if nothing is heard — the referral remains my responsibility until it is picked up.**

The compressed answer — what a strong candidate says in five minutes

"**I would set it out under headings**, so the receiving clinician can find what they need without reading the whole thing.

Header: today's date; **my name, practice address, GDC number and contact details**; the patient's **full name, date of birth, address and contact details**; and the addressee by name.

Then the first line of the letter, because that is where the reason goes: *'I would be grateful for your opinion and treatment regarding bilateral sinus floor elevation and implant placement in the posterior maxilla.'* Nobody should have to read three paragraphs to work out what I want.

Presenting complaint and expectations, in **her own words** — what she has asked for, and what I have already told her is and is not achievable, so that the receiving clinician is not blind-sided by an expectation I created.

Relevant medical history: conditions, **medications with doses**, allergies, **smoking in pack-years**, and the result of any liaison I have already had with her general medical practitioner or specialist.

Dental and social history: attendance pattern, **periodontal history and current periodontal status**, previous implant experience and how it went.

Clinical findings, structured: extra-oral and intra-oral findings; **periodontal status with plaque and bleeding scores**; occlusal findings; the edentulous span; **ridge dimensions**; keratinised tissue; and the opposing dentition.

Investigations, enclosed rather than described: *'I enclose a panoramic radiograph dated…, periapicals of the…, clinical photographs, a small field-of-view CBCT dated… with my written report, and mounted study casts with a diagnostic wax-up.'* **Sending the images matters** — it spares the patient a **repeat exposure**, which is an **IR(ME)R consideration**, and it stops the receiving clinician starting the assessment from nothing.

My assessment, with the numbers: *'Residual bone height is 4 mm at the 16 and 3 mm at the 17, with a flat sinus floor and no septa. On the ITI decision tree this indicates a lateral window approach, which falls outside my scope of practice.'* That is the sentence the whole letter exists for. **I am not referring because I am uncertain; I am referring because I have measured the site, applied a recognised framework, and identified where my competence ends.** A lateral window is **Complex** — a general dentist with an interest in implants should recognise it, plan up to it, and hand it over.

What I have already discussed and consented: the options I presented — **including the shortened dental arch and no treatment at all** — the risks I have covered, the costs discussed, and that **she understands this is a referral for an opinion**, not a booking for surgery.

What I am requesting, precisely, and who does what afterwards: *'I am happy to undertake the restorative phase and to provide ongoing maintenance, or to refer the whole case, whichever you prefer.'* Ambiguity about shared care is how patients end up with nobody responsible for the maintenance.

Urgency and access needs: routine; no interpreter required; wheelchair access needed.

Closing: my **direct contact details** and an offer to discuss the case by telephone.

Then the governance, which is part of the referral and not an afterthought. I **keep a copy in the notes**; I **tell the patient what I have written, what to expect and roughly how long it will take**; and I **set a reminder to follow up if I have not heard within a reasonable period** — because **the patient remains my responsibility until the referral is picked up**, and a letter that vanished into an unopened inbox is a failure of care, not of the post."

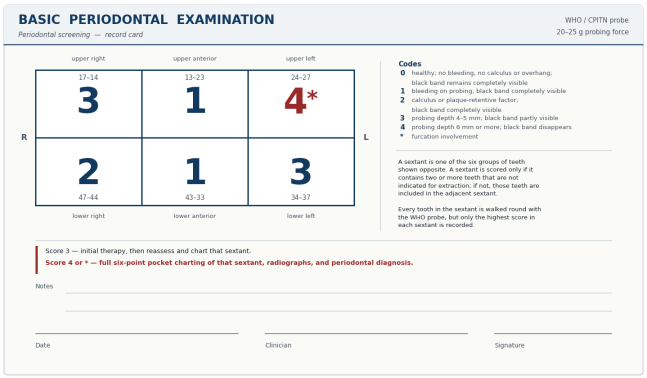
Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 2 **the reason for referral in the first line** · 8 **quantified assessment with the ITI framework and the scope statement** · 7 **existing images enclosed, to avoid repeat exposure** · 9 what has been consented and **precisely what is requested, including shared care** · 10 **copy retained, patient informed, follow-up arranged** · 1 full identifiers for patient and referrer, including the GDC number.

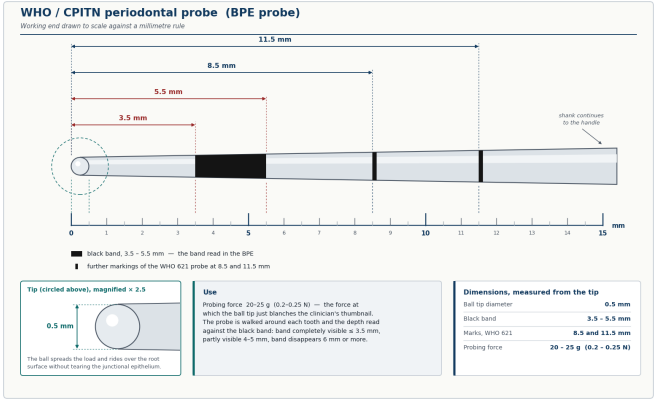
Accept these late or on prompting: 3 expectations in the patient's own words · 4 medical history with medications and pack-years · 5 dental, periodontal and social history · 6 structured clinical findings.

The commonest way five minutes is wasted here: writing a beautifully composed narrative letter in flowing prose, so that at four minutes the pen is still in the medical history and the assessment, the request and the governance — checklists 8, 9 and 10, the three heaviest — are never written. The instruction says *reading it aloud as you write*, which is an invitation to **speak the headings and write in notes**. The second loss is the vague reason: "I would be grateful for your opinion regarding this patient" scores nothing, because it names neither the procedure nor the site. The third is stopping at the signature. **A referral is not complete when it is sent** — the copy, the conversation with the patient and the diarised follow-up are a full checklist mark, and they are the governance dimension the examiner is looking for.

OSCE 51 — Interpreting a BPE chart, and the CPITN/WHO probe



A completed BPE chart of the kind you are handed at this station — simulated illustration, not a clinical image



The WHO/CPITN (BPE) probe, drawn to scale against a millimetre rule — simulated illustration, not a clinical image

Blueprint: patient examination; overall dental health in relation to the provision of dental implants; record keeping.

EQUIPMENT

a completed BPE chart on card or screen, a **WHO/CPITN (BPE) probe**, a UNC-15 or other graduated probe, a plastic or titanium peri-implant probe, a blank six-point charting form.

CANDIDATE INSTRUCTIONS

You are shown a completed Basic Periodontal Examination for this patient. The examiner will ask you about the findings, about the instrument used to obtain them, and about what follows. Answer verbally; you are not asked to examine a patient.

DIRECTION OF THOUGHT

This is a discussion station, not a performance station. You will not be asked to walk a probe round a manikin and call out numbers — you will be *given* the numbers and asked what they mean, what you would do next, and to describe the instrument and the technique **in words**. That is a harder test of precision, not an easier one: everything you would normally do with your hands has to be said accurately out loud, including the millimetre markings on the probe and the force in grams or newtons. Fix the order: **read the chart → highest code and what it obliges → define every code → the probe → the force → the technique → the limitations → what it means for the implant.**

The chart you are shown.

Sextant	UR	UA	UL
Upper	3	1	4*
Lower	2	1	3

MODEL ANSWER

As a discussion

Q Tell me what this chart shows.

"The highest score is a **4 with an asterisk in the upper left sextant**, so that is what drives management for the whole mouth. There is a code 3 upper right and lower left, a code 2 lower right, and code 1 in both anterior sextants. In summary: generalised gingival inflammation, a plaque-retentive factor or calculus in one sextant, moderate pocketing in two, and **deep pocketing with either furcation involvement or 7 mm or more of attachment loss in the upper left**."

Q Define each of the codes for me.

"**0** — probing depth under 3.5 mm, the black band fully visible, no calculus or defective margin, and no bleeding after probing. **1** — the same, but with **bleeding after probing**. **2** — the same depth, but **supra- or subgingival calculus or a defective restoration margin** is present. **3** — probing depth **3.5 to 5.5 mm**, so the black band is **partially visible**. **4** — probing depth **6 mm or more**, so the black band **disappears completely**."

Q And the asterisk?

"The asterisk is added to **any** code where there is **furcation involvement, or recession plus pocket depth totalling 7 mm or more of attachment loss** from the cemento-enamel junction. It is not a score in itself — it is a modifier, and it can sit on a code 1 just as easily as on a code 4."

Q What does each code oblige you to do?

"**0** — no treatment beyond continued prevention. **1** — oral hygiene instruction. **2** — oral hygiene instruction plus removal of calculus and correction of any plaque-retentive factor. **3** — as for 2, then **six-point charting of that sextant before and after initial therapy**, with review. **4, or any code with an asterisk — six-point charting of the entire dentition**, radiographs as clinically indicated, a full diagnosis with stage and grade, and consideration of referral. So this patient needs full-mouth charting and radiographs, not simply a scale and polish."

Q Which probe was used, and describe it to me.

"A **WHO probe**, also called the **CPITN probe** or BPE probe. It has a **0.5 mm diameter ball at the tip**, and a **black band running from 3.5 mm to 5.5 mm** from the tip. The extended version — sometimes called the WHO 621 — carries **two further markings at 8.5 mm and 11.5 mm**, so the second band runs 8.5 to 11.5 mm. The shank is fine and the working end is lightweight, and it is colour-coded so the band can be read at a glance."

Q Why does it have a ball end?

"Three reasons. It **stops the tip penetrating the junctional epithelium** and over-reading the pocket. It lets me **feel subgingival calculus and defective margins** as the ball passes over them, which is how code 2 is detected. And it is more comfortable and less likely to traumatise the sulcus base than a sharp tip."

Q What force do you use, and how do you know you are using it?

"A **light force of 20 to 25 grams**, which is **about 0.2 to 0.25 newtons**. I calibrate it by pressing the probe against my **thumbnail until the tip just blanches the nail bed** — that is roughly the right force. Too much force pushes the probe through the inflamed junctional epithelium and over-reads the depth; too little under-reads it and misses bleeding."

Q Take me through how you would actually use it.

"I divide the mouth into **six sextants** — 17–14, 13–23, 24–27, and the three matching lower sextants; third molars are excluded unless they are functioning in place of a second molar. A sextant is only scored if it contains **two or more teeth**; if it has one tooth or none, that tooth is scored within the adjacent sextant and the sextant is recorded as **X**. Within each sextant I insert the probe **parallel to the long axis of the tooth**, gently to the base of the sulcus, then **walk it** in small vertical steps around the whole circumference, covering **at least six sites per tooth** — mesiobuccal, mid-buccal, distobuccal and the three matching lingual sites. I keep the ball in contact with the root surface throughout. I then record **only the highest score in that sextant**, and once a sextant scores 4 I may move on, since nothing higher can be found — although I would still look for the asterisk."

Q What are the limitations of the BPE?

"It is a **screening tool, not a diagnosis**. It records only the worst site in a sextant, so it hides the distribution and the number of affected sites; it cannot be used to monitor response to treatment; it gives no information on recession or attachment loss unless the asterisk is triggered; and it does not by itself let you stage and grade under the 2017 classification. It also **cannot be used around implants**."

Q Why not around implants?

"Because the peri-implant tissues are different — no periodontal ligament, no perpendicular collagen insertion into the surface, and a weaker connective tissue seal — so the probe penetrates further for the same force, and 'normal' probing depths differ from tooth to tooth depending on how deeply the implant was placed. Peri-implant probing is a separate examination: **six sites per implant, a plastic or titanium probe, a gentle force of about 0.2 to 0.25 newtons**, recording depth, bleeding, suppuration, recession and the width of keratinised mucosa, and **compared against the baseline recorded at prosthesis fitting** rather than against an absolute number."

Q What would the six-point chart record?

"For each tooth, **probing depth at six sites** — mesiobuccal, mid-buccal, distobuccal, mesiolingual, mid-lingual, distolingual — using a graduated probe such as a **UNC-15**, rounded to the nearest whole millimetre; **recession**, so that clinical attachment loss can be calculated; **bleeding on probing** and **suppuration**; **mobility** by the Miller grades; **furcation** with a Nabers probe and the Hamp grades; and a **plaque score**."

Q What does this chart mean for a proposed implant?

"That implants are not on the table yet. A history of periodontitis is the **strongest patient-level risk factor for peri-implantitis**, so this patient must be diagnosed, staged and graded under the 2017 classification, treated through the EFP stepwise approach, and shown to be **stable** first. Stability means **probing depths of 4 mm or less, no bleeding at those 4 mm sites, and full-mouth bleeding under 10%**. If the upper left sextant does not

respond, or if that asterisk turns out to be a Grade C picture, the whole plan may change — including whether the tooth I am replacing should be replaced at all."

Q What would you record?

"The six sextant scores with the asterisk, the date, the probe used, and the fact that the findings were explained to the patient; then the full chart, the diagnosis with stage and grade, the treatment plan, and the review interval."

EXAMINER CHECKLIST

1. Reads the chart accurately and identifies the highest score as the driver
2. **Codes 0–4 defined correctly, including the band position for each**
3. **The asterisk defined correctly as a modifier**
4. States the action obliged by each code, including full-mouth charting for 4/*
5. **Names the WHO/CPITN probe and states the 0.5 mm ball and the 3.5–5.5 mm band**, ideally the 8.5/11.5 mm markings
6. Explains why the tip is ball-ended
7. **States the force in grams or newtons and how it is calibrated**
8. Describes sextants, the two-tooth rule, walking the probe and recording the highest score
9. States the limitations of the BPE and that it is not used around implants, with the peri-implant alternative
10. Relates the findings to implant risk and stability criteria, and states what is recorded

See also **OSCE 20** (peri-implant assessment and maintenance) and **OSCE 48** (oral hygiene instruction).

Running this station to time

Time check. The model answer in this part runs to 1,186 words, which at 140 words per minute is 8 minutes 28 seconds of unbroken speech before the examiner has asked a single question — so roughly 550 words, nearly half of it, has to go, and it goes by compressing the definitions into single clauses rather than by dropping any of the ten checklist headings.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Puts the completed BPE chart in front of the candidate and asks: "Tell me what this chart shows."	Checklist 1. The candidate names the **4* in the upper left as the highest score and says it drives management for the whole mouth**, then reads the other five sextants without laboured description. Anything longer than 30 seconds here is time stolen from the probe.
0:30–2:00	"Define each of the codes for me." Then: "And the asterisk?" Then: "What does each code oblige you to do?"	Checklists 2, 3 and 4. Codes 0 to 4 defined with the **band position** for each (fully visible, partially visible, disappears), the asterisk defined as a **modifier and not a score**, and the obliged action for each code, finishing with **full-mouth six-point charting plus radiographs for a 4 or any asterisk**.
2:00–3:30	Hands the candidate the WHO probe. "Which probe is this, describe it, why the ball end, what force, and how would you use it?"	Checklists 5, 6, 7 and 8. **0.5 mm ball and 3.5–5.5 mm black band** (8.5/11.5 mm on the WHO 621); three reasons for the ball end; **20–25 g or 0.2–0.25 N with the thumbnail calibration**; sextants, the two-tooth rule and X, walking the probe, six sites, highest score only.
3:30–4:30	"What are the limitations of the BPE, and can you use it round an implant?" Then: "What does this chart mean for the implant she wants?"	Checklists 9 and 10. Screening tool not a diagnosis, worst site only, cannot monitor treatment, no staging and grading; **not used around implants**, with the peri-implant alternative; history of periodontitis as the **strongest patient-level risk factor for peri-implantitis** and the numerical **stability criteria**.

Time	The examiner does	What earns the mark
4:30–5:00	"What would you record, what have you told the patient, and when will you see her?"	Checklist 10, second half — **records** (six scores with asterisk, date, probe used, six-point chart, diagnosis with stage and grade), **what the patient is told** (that implants wait until the periodontium is stable), and the **review interval**. These three are lost more often than any other mark on this station.

The compressed answer — what a strong candidate says in five minutes

Q: Tell me what this chart shows. "The highest score is a **4 with an asterisk in the upper left**, and that drives management for the whole mouth. With a 3 upper right and lower left, a 2 lower right and 1 anteriorly, this is generalised gingival inflammation, calculus in one sextant, moderate pocketing in two, and deep pocketing with furcation or 7 mm of attachment loss upper left."

Q: Define the codes. "**0** — under 3.5 mm, black band fully visible, no calculus, no bleeding. **1** — the same, but bleeding after probing. **2** — the same depth with supra- or subgingival calculus or a defective margin. **3** — 3.5 to 5.5 mm, band partially visible. **4** — 6 mm or more, band gone completely. The **asterisk is a modifier, not a score**: furcation involvement, or recession plus pocket totalling 7 mm or more of attachment loss. It can sit on a 1 as easily as on a 4."

Q: What does each code oblige you to do? "0, prevention only. 1, oral hygiene instruction. 2, plus calculus removal and correction of the plaque-retentive factor. 3, as for 2, then **six-point charting of that sextant before and after initial therapy**. 4, or any asterisk — **six-point charting of the whole dentition**, radiographs, diagnosis with stage and grade, and consideration of referral. So she needs full-mouth charting, not a scale and polish."

Q: Describe this probe. "A **WHO probe**, also called the CPITN or BPE probe: a **0.5 mm ball at the tip** and a **black band from 3.5 to 5.5 mm**; the WHO 621 carries further markings at 8.5 and 11.5 mm. The ball end stops the tip penetrating the junctional epithelium, lets me feel subgingival calculus and defective margins — which is how code 2 is detected — and is atraumatic."

Q: What force, and how do you know? "**20 to 25 grams, about 0.2 to 0.25 newtons.** I calibrate against my thumbnail until the nail bed just blanches. Too much over-reads through inflamed tissue; too little misses bleeding."

Q: How would you use it? "**Six sextants** — 17 to 14, 13 to 23, 24 to 27 and the three matching lower ones; third molars excluded unless functioning as second molars. A sextant is scored only if it has **two or more teeth**; otherwise that tooth is scored in the adjacent sextant and the sextant recorded as **X**. Probe parallel to the long axis, gently to the base of the sulcus, **walked** in small steps around the whole circumference, six sites per tooth, recording **only the highest score** per sextant."

Q: Limitations? "It is a **screening tool, not a diagnosis** — worst site only, so it hides distribution; it cannot monitor treatment; it gives no attachment loss unless the asterisk triggers; and it does not stage and grade. It is **not used around implants**: no periodontal ligament and a weaker connective tissue seal, so the probe penetrates further for the same force. Peri-implant probing is separate — six sites, a plastic or titanium probe, 0.2 to 0.25 newtons, recording depth, bleeding, suppuration, recession and keratinised width **against the baseline taken at fit**."

Q: What does this mean for her implant? "Implants are not on the table yet. Previous periodontitis is the **strongest patient-level risk factor for peri-implantitis**, so she needs diagnosis with stage and grade, stepwise therapy, and demonstrated **stability — pockets 4 mm or less, no bleeding at those sites, full-mouth bleeding under 10%** — first."

Q: Records and review? "The six scores with the asterisk, the date, the probe used, and that I explained the findings and told her the implant waits until she is stable; then the six-point chart with recession, bleeding, mobility, furcation and plaque score, the diagnosis with stage and grade, the plan, and review after initial therapy."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 2 codes 0–4 with band position · 3 the asterisk as a modifier · 4 action obliged by each code, including full-mouth charting for 4/* · 5 WHO probe named, 0.5 mm ball and 3.5–5.5 mm band · 7 force in grams or newtons and how it is calibrated · 10 implant risk, stability criteria and what is recorded.

Accept these late or on prompting: 1 reading the chart and naming the highest score as the driver · 6 why the tip is ball-ended · 8 sextants, two-tooth rule, walking the probe, highest score only · 9 limitations of the BPE and why it is not used around implants.

The commonest way five minutes is wasted here: narrating the chart sextant by sextant at the start, as though the examiner cannot read it, and then re-deriving each code from first principles instead of stating it in one clause — the candidate reaches the probe at 3 minutes and loses the force, the technique and the whole of the implant and records section.

OSCE 52 — Buccally tilted 24 implant with a high smile line: discussion with the technician

Blueprint: select the appropriate abutment; difference between screw- and cement-retained restorations and their application; **communicate the planned treatment to the wider team**; prescribe an appropriate type of restoration from the laboratory. Marked W + OS-P + SCE.

EQUIPMENT

a working cast with an implant analogue at the **24** angled markedly buccally; a screw-retained trial crown whose access hole emerges through the buccal surface; a selection of abutments including an **angulated screw channel (ASC) abutment**, a straight titanium abutment, an angled abutment and a titanium base; photographs of the patient smiling showing a **high smile line**; a laboratory prescription form; a telephone or a role-player acting as the technician.

CANDIDATE INSTRUCTIONS

An implant was placed at the UL4. The implant is tilted buccally. The patient has a high smile line. The laboratory has returned a screw-retained crown whose access channel emerges through the buccal surface. Discuss the problem and the possible solutions with the technician, and record your decision.

DIRECTION OF THOUGHT

The problem is a **surgical error being presented as a prosthetic problem**. Do not start with "I'll cement it." Start by naming the problem, then run through the options in order of preference, and only then arrive at cementation — with the caveats. This station rewards a structured conversation with a colleague: greet, describe the problem, state what you have observed, ask for the technician's view, agree a plan, confirm it in writing.

MODEL ANSWER

As a conversation with the technician

"'Hello — I've got the UL4 crown you made. Can I talk you through a problem? The implant is tilted buccally, and the screw access is coming out through the buccal surface of the crown, right in the middle of the visible face. The patient has a high smile line, so I can see the whole crown and about 2 mm of gingiva when she smiles. A composite-filled access hole on the labial surface is going to be visible and will discolour over time, so I don't think we can deliver it as it is.'

Then I would set out the options with him.

Option 1 — an angulated screw channel abutment. My **first choice**. These allow the screw channel to be redirected, typically by up to about 25 to 30 degrees depending on the system, so the access can be moved palatally while the restoration stays **screw-retained**. That keeps retrievability and eliminates cement entirely. I would ask him to check whether our system offers an ASC solution for this platform and whether the angulation is within range.

Option 2 — a 'screwmentable' or hybrid restoration. The crown is cemented **extra-orally in the laboratory** onto a titanium base, and the whole assembly is then delivered as a **screw-retained** unit. All the excess cement is removed on the bench where it can be seen, so there is no submucosal cement, and the restoration remains retrievable. This is my preferred fallback if the ASC angulation is insufficient.

Option 3 — a cement-retained crown on a custom angled abutment. This solves the aesthetic problem and is the classic answer, but it introduces the risk that concerns me most: **residual submucosal cement is a well-documented cause of peri-implantitis**. If we go this way, I would ask

him for: - a **custom CAD-CAM abutment**, not a stock one, so I control the emergence profile and the margin; - the **margin placed at or no more than 1 mm below the mucosal margin**, following the gingival scallop, and ideally **supramucosal on the palatal aspect** where it will not show; - an **adequate abutment height and taper** for retention — around 5 mm — because a short abutment plus a weak cement means the crown will come off; - a **Ti-base with a ceramic superstructure** rather than a plain titanium abutment, because with a high smile line and possibly thin tissue, a titanium abutment can produce grey show-through.

Option 4 — accept the buccal access hole and mask it. I would reject this with a high smile line.

Then: 'Which of those can you do with this system? If the ASC works, that's what I'd like. If not, let's do the extra-oral cementation on a Ti-base so it comes back as a screw-retained unit.'

Then I would confirm it in writing on a new laboratory prescription, stating the implant system, platform and connection; the abutment type and material; the margin position; the retention type; the shade and stump shade; the occlusal scheme; and the date required. I would keep a copy in the notes.

If I do end up cementing, I would use a **radiopaque cement**, apply it only to the coronal half of the intaglio, use an **extra-oral replica abutment** to extrude the excess before seating, remove excess meticulously with floss and scalers, and **take a periapical radiograph to verify no residual cement**.

And I would record the underlying lesson. The real problem here is implant angulation, which is a planning and surgical issue. I would record it as a **significant event**, review whether a surgical guide derived from a diagnostic wax-up was used, and audit my own use of restoratively driven planning."

EXAMINER CHECKLIST

1. Names the problem clearly and states why the crown cannot be delivered as made
2. Structured, courteous professional exchange with the technician, inviting their view
3. **Angulated screw channel abutment offered first, with the angulation limit**
4. Extra-oral cementation on a Ti-base ("screwmentable") described
5. States the cement risk explicitly
6. If cementing: **custom abutment, margin ≤1 mm subgingival, adequate height**
7. Ti-base/ceramic to avoid grey show-through with a high smile line
8. Rejects a labial access hole and says why
9. **Written laboratory prescription with all required detail, copy retained**
10. Identifies the underlying planning/surgical cause and the governance response

Extended by practice stations **OSCE 12** (abutment selection), **OSCE 13** (screw versus cement) and **OSCE 18** (laboratory prescription).

Running this station to time

Time check. The model answer is 985 words, 7 minutes 2 seconds at 140 words per minute — two minutes over the station before the technician has said a word. Around 280 words must go, and the format does most of the cutting for you: this is a **conversation**, so **each option gets one sentence and its single decisive reason, and the technician's replies carry the alternatives.** Only the candidate's own words spend the clock, so ask, then listen.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:50	Sets out a cast with the 24 analogue tilted markedly buccally, a screw-retained trial crown with the access hole emerging through the buccal surface, an abutment selection, smile photographs showing a high smile line, and a prescription form. A role-player or a telephone: "Discuss the problem and the possible solutions with the technician, and record your decision."	Checklists 1, 2 and 8, all in the opening exchange. **Greet the technician by name, say what you are ringing about, and describe the problem before asking for anything**: the implant is tilted buccally, the **screw access emerges through the middle of the visible buccal surface**, and the patient has a **high smile line — the whole crown and about 2 mm of gingiva show.** Then mark 8, said now rather than saved for last: **a composite-filled access hole on the labial surface will be visible and will discolour, so we cannot deliver it as it is.** Then the courtesy that is a mark: **"Can I talk you through it and hear what you think?"**
0:50–1:50	As the technician: "So what would you like to do?"	Checklist 3 — the first choice, with its number. **An angulated screw channel abutment. The channel can be redirected, typically by up to about 25 to 30 degrees depending on the system**, moving the access **palatally while the restoration stays screw-retained** — which keeps **retrievability and eliminates cement entirely.** Then the two-way part: **ask him to check whether the system offers an ASC solution for this platform, and whether the angulation needed is within range.** A candidate who announces the answer without asking the technician anything loses mark 2.
1:50–2:45	As the technician: "We don't have an ASC option on that platform, and the angle looks like more than thirty degrees."	Checklists 4 and 5. The fallback, named: **a 'screwmentable' or hybrid restoration — the crown cemented extra-orally in the laboratory onto a titanium base, and the whole assembly delivered as a screw-retained unit.** The reason is the mark: **all excess cement is removed on the bench, where it can be seen, so there is no submucosal cement, and the restoration stays retrievable.** Then mark 5 stated explicitly: **residual submucosal cement is a well-documented cause of peri-implantitis** — which is why cementation in the mouth is third, not first.
2:45–3:50	As the technician: "And if we do end up cementing it in the mouth?"	Checklists 6 and 7, and these must be said in **laboratory-prescription language**, because they are instructions, not opinions. **A custom CAD-CAM abutment, not a stock one**, so the emergence profile and margin are controlled. **The margin at or no more than 1 mm below the mucosal margin, following the gingival scallop, and supramucosal on the palatal aspect. Adequate abutment height and taper for retention — around 5 mm.** And mark 7, tied to this patient: **a titanium base with a ceramic superstructure rather than a plain titanium abutment, because with a high smile line and possibly thin tissue a titanium abutment can produce grey show-through.**

Time	The examiner does	What earns the mark
3:50–5:00	As the technician: "Fine — put it in writing." Then, as examiner: "And how did this happen?"	Checklists 9 and 10. **Agree the plan out loud — "ASC if the system allows it, otherwise extra-oral cementation on a Ti-base so it comes back screw-retained"** — then **a new written laboratory prescription stating the implant system, platform and connection; the abutment type and material; the margin position; the retention type; the shade and stump shade; the occlusal scheme; and the date required. Copy retained in the notes.** Then the governance mark: **the real problem is implant angulation, which is a planning and surgical failure presenting as a prosthetic one** — record it as a **significant event**, review whether a surgical guide derived from a diagnostic wax-up was used, and **audit my own restoratively driven planning.**

The compressed answer — what a strong candidate says in five minutes

You: "Hello — it's about the UL4 crown you sent back. Can I talk you through a problem and get your view? **The implant is tilted buccally**, and the **screw access is coming out through the middle of the buccal surface of the crown.** The patient has a **high smile line** — she shows the whole crown and about 2 mm of gum when she smiles. **A composite-filled access hole on the labial face is going to be visible and it will discolour**, so I don't think we can deliver it as it stands."

Technician: "Agreed. What do you want to do?"

You: "**My first choice is an angulated screw channel abutment.** Those let us **redirect the channel — typically up to about 25 to 30 degrees depending on the system** — so the access comes out **palatally** and the crown stays **screw-retained.** That keeps it **retrievable and takes cement out of the equation altogether.** Can you check whether we have an **ASC solution for this platform**, and whether the angle we need is within range?"

Technician: "There's no ASC for that platform, and I'd say we're past thirty degrees anyway."

You: "Then my fallback is a **'screwmentable' — a hybrid.** You **cement the crown extra-orally, on the bench, onto a titanium base**, and send it back as a **single screw-retained assembly.** The advantage is that **all the excess cement is cleaned off where you can see it**, so **nothing is left submucosally** — and **residual submucosal cement is a well-documented cause of peri-implantitis.** It also stays retrievable, which matters on a tilted implant I may need to get back into."

Technician: "That we can do. What if you'd rather cement it in the mouth on an angled abutment?"

You: "It solves the aesthetics, and it's the classic answer, but it brings back the cement risk — so if we go that way I'd want it prescribed tightly. **A custom CAD-CAM abutment, not a stock one**, so I control the emergence profile and the margin. **The margin at, or no more than 1 mm below, the mucosal margin, following the gingival scallop — and supramucosal on the palatal aspect** where it won't show, so I can actually see and remove the excess. **Adequate height and taper for retention — around 5 mm** — because a short abutment plus a weak cement means it comes off in six months. And **a Ti-base with a ceramic superstructure rather than plain titanium**, because **with a high smile line and possibly thin tissue, titanium shows through as grey at the margin.**"

Technician: "And you don't want to just mask the access hole?"

You: "No — **not with this smile line.** That's the one option I'd reject outright. It would be reasonable on a lower molar; on an upper premolar she shows in full, it's a visible patch of composite that stains."

Technician: "Right. Send it to me in writing."

You: "I will. **A new laboratory prescription**, stating the **implant system, platform and connection**; the **abutment type and material**; the **margin position**; the **retention type — screw-retained via the Ti-base**; the **shade and the stump shade**; the **occlusal scheme**; and the **date required.** I'll **keep a copy in the notes**, and I'll ring you if anything changes at try-in."

And afterwards, to myself and in the record: "This is a surgical and planning problem being presented as a prosthetic one. The angulation is the actual fault. I'd log it as a **significant event**, check whether a

surgical guide derived from a diagnostic wax-up was used and, if not, why not, and **audit my own use of restoratively driven planning** — because the prosthetic workaround has costs, and the way to avoid needing one is to place the implant where the crown needs it to be."

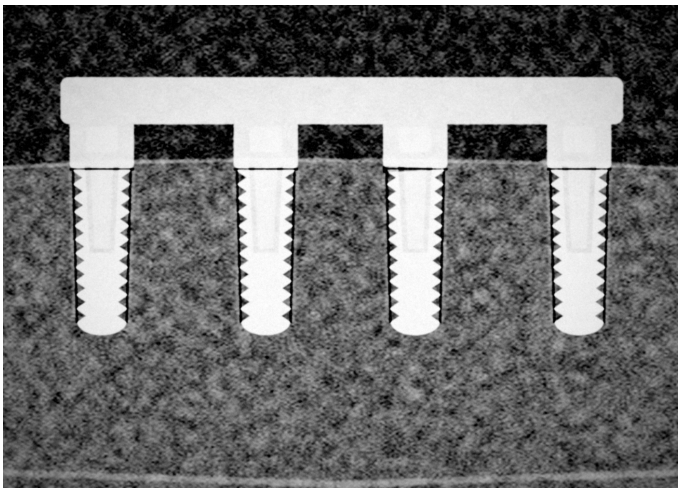
Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 **names the problem and why the crown cannot be delivered** · 3 **ASC offered first, with the 25–30 degree limit** · 4 **extra-oral cementation on a Ti-base, delivered screw-retained** · 5 **submucosal cement stated as a cause of peri-implantitis** · 9 **written prescription with the full component detail, copy retained** · 10 the underlying **planning and surgical cause**, logged as a significant event.

Accept these late or on prompting: 2 courteous structured exchange that invites the technician's view · 6 custom abutment, margin ≤1 mm subgingival, ~5 mm height · 7 Ti-base and ceramic to avoid grey show-through · 8 the labial access hole rejected, with the reason.

The commonest way five minutes is wasted here: opening with "I'd just cement it." That is the fourth-best answer given first, and it forfeits marks 3, 4 and 5 in one sentence. The second loss is monologue: this is scored as a **professional exchange**, so a candidate who lectures the technician for four minutes without once asking **"can you do that on this platform?"** or **"what would you suggest?"** loses checklist 2 outright, and never gets the reply that would have moved the conversation on. The third is talking about the prescription instead of dictating it — the mark is for the **actual items**: system, platform, connection, abutment type and material, margin position, retention, shade and stump shade, occlusal scheme, date. And the fourth is ending politely at the agreed plan: **checklist 10 lives after the phone call**, and a candidate who never names the implant angulation as the real fault has treated the symptom and missed the governance mark entirely.

OSCE 53 — Framework try-in with a transfer jig: what if it does not fit



Periapical view at the framework try-in — note the interface at one abutment — simulated illustration, not a clinical image

Blueprint: describe how to manage a mis-fitting prosthesis or framework; evaluate the laboratory techniques involved in implant reconstruction.

EQUIPMENT

a model with four implant analogues; a cast metal framework; a **verification (transfer) jig**; a set of crowns; screws and a screwdriver; a periapical radiograph showing a wedge-shaped gap at one abutment; probes.

CANDIDATE INSTRUCTIONS

You are trying in a metal framework and crowns for an implant-supported prosthesis. A transfer jig has been provided. Demonstrate and describe the try-in, and describe what you would do if the framework does not fit.

DIRECTION OF THOUGHT

Two halves: **how to verify fit**, and **what to do when it fails**. Use the jig first — that is why it is on the table and it is a named checklist item. Be explicit that you never force a framework with a torque wrench.

MODEL ANSWER

"**The purpose of a verification or transfer jig** is to check that the master cast reproduces the implant positions in the mouth accurately, *before* an expensive framework is made. It is usually made of pattern resin or a rigid material joining copings, **sectioned and re-joined** to compensate for polymerisation shrinkage.

Step 1 — verify the cast with the jig. Remove the healing abutments or provisional. Seat the jig in the mouth. It should seat **passively and completely** on all implants with no rocking, and each screw should engage without resistance. I check visually and with a fine probe at each interface. If the jig does not seat, the **master cast is wrong** and there is no point trying the framework — I would take a new splinted open-tray impression.

Step 2 — try in the framework. Seat it by hand, hand-tighten one screw only, and then verify passivity: - **Visual and tactile:** inspect each interface and run a fine probe around each margin. - **The one-screw, or Sheffield, test:** tighten **only the most distal screw** and examine the fit at the opposite end. A visible gap or rocking means the framework is not passive. **I repeat the test from the other end**, because a framework can appear to fit from one direction and not the other. - **Radiographically:** a paralleling periapical at each abutment. The classic sign of misfit is a **wedge-shaped radiolucent gap** between framework and implant. - **Patient symptoms:** pressure, tightness or pain as the screws are tightened is a warning sign and I would stop.

Step 3 — check the rest. Contour and cleansability, interocclusal clearance for the ceramic, the occlusal scheme, the shade, and the position of the screw access channels.

If it does not fit. First, **I do not force it.** Torquing a non-passive framework transmits static stress to the implants and components, and causes screw loosening or fracture, marginal bone loss and pain.

Then I identify the cause: - **Interference** — soft tissue or bone over a platform, or an incompletely seated component. Remove it and re-try. - **A component error** — the wrong analogue, or an analogue rotated in the cast. - **Impression distortion** — unsplinted copings, a coping that moved, a flexible tray, or a coping not fully seated at the time of impression. - **Cast inaccuracy** — expansion or damage. - **Laboratory distortion** — during casting, soldering or ceramic firing. A framework can be passive before firing and not afterwards.

The remedies, in escalating order: 1. Remove any interference and re-try. 2. **Section the framework at the point of misfit, lute the sections together intra-orally with pattern resin, and return it for soldering or laser welding**, then re-verify. This is the standard remedy for a framework that is accurate in its parts but distorted overall. 3. If the error is in the impression or the cast, **make a new verification jig and take a new splinted open-tray impression**, and have the framework remade.

Then the professional part. I would explain the delay to the patient honestly and apologise for it, provide or adjust a provisional so they are not inconvenienced, **discuss the case with the technician to establish where the error arose** so it does not recur, and record everything. I would not criticise the laboratory in front of the patient."

EXAMINER CHECKLIST

1. Explains the purpose of the verification/transfer jig
2. **Uses the jig first and states what a non-seating jig means**
3. Sections-and-rejoins principle understood
4. **One-screw (Sheffield) test performed and repeated from both ends**
5. Radiographic verification and the wedge-shaped gap
6. Checks contour, cleansability, clearance, occlusion, shade, access position
7. **States explicitly that the framework must not be forced, with the consequences**
8. Lists causes systematically
9. Section-and-solder remedy, and the new-impression remedy
10. Patient communication, provisional, technician discussion, records

Extended by practice stations **OSCE 14** (implant impression), **OSCE 16** (fitting the prosthesis) and **OSCE 17** (the mis-fitting framework).

Running this station to time

Time check. The model answer in this part is 556 words, 3 minutes 58 seconds at 140 words per minute — which leaves barely a minute for the examiner and none for actually handling the jig, the framework and the screwdriver on the bench; about 100 words of the causes list has to be compressed into a single sweep so that the demonstration is not sacrificed to the commentary.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–1:15	Points to the model with four analogues, the framework, the jig and the periapical: "Demonstrate and describe the try-in, and tell me what you would do if it does not fit." May add: "How is that jig made?"	Checklists 1, 2 and 3. The candidate **picks up the jig first, not the framework**, states its purpose — verifying that the master cast reproduces the implant positions before an expensive framework is made — describes it in one sentence as rigid pattern resin joining copings, **sectioned and re-joined** to compensate for polymerisation shrinkage, and says that if the **jig does not seat, the cast is wrong and the framework is not tried at all**.
1:15–2:45	Hands over the framework and the screwdriver. Silence, then the periapical.	Checklists 4, 5 and 6. Seats by hand; **one-screw (Sheffield) test on the most distal screw, then repeated from the other end**; visual and fine-probe check at every interface; paralleling periapical with the **wedge-shaped radiolucent gap**; patient symptoms as a warning; then contour, cleansability, interocclusal clearance, occlusal scheme, shade and screw access position.
2:45–3:45	"It doesn't fit. What now?"	Checklists 7 and 8. **"I do not force it"** said explicitly, with the consequences — static stress, screw loosening and fracture, marginal bone loss, pain — then the causes run through **systematically**: interference, component error, impression distortion, cast inaccuracy, laboratory distortion during casting, soldering or firing.
3:45–4:30	"So what do you actually do about it?"	Checklist 9. The escalating remedies: remove interference and re-try; **section at the point of misfit, lute intra-orally with pattern resin, return for soldering or laser welding, and re-verify**; and if the error is in the impression or cast, a new verification jig, a new splinted open-tray impression, and a remade framework.
4:30–5:00	"Anything else?"	Checklist 10 — **records** of the try-in, the tests done and the decision; **what the patient is told** (an honest explanation of the delay and an apology, with the provisional adjusted so they are not inconvenienced); and **review** with the technician to establish where the error arose, without criticising the laboratory in front of the patient.

The compressed answer — what a strong candidate says in five minutes

"I will start with the jig, not the framework.

The purpose of a verification or transfer jig is to confirm that the master cast reproduces the implant positions in the mouth accurately, *before* an expensive framework is made. It is rigid pattern resin joining copings, **sectioned and re-joined** to compensate for polymerisation shrinkage.

Step one — verify the cast. I remove the healing abutments, seat the jig, and check it sits **passively and completely** on all four implants with no rocking, each screw engaging without resistance. I check visually and with a fine probe at every interface. If the jig does not seat, **the master cast is wrong**, and there is no point trying the framework — I would take a new splinted open-tray impression there and then.

Step two — try in the framework. I seat it by hand and hand-tighten one screw only. Then I verify passivity four ways. **Visually and tactilely**, inspecting each interface and running a fine probe round each margin. With the **one-screw or Sheffield test** — tightening only the most distal screw and examining the fit at the opposite end; a visible gap or rocking means it is not passive, and I **repeat the test from the other end**, because a framework can appear to fit from one direction and not the other. **Radiographically**, with a paralleling periapical at each abutment: the classic sign of misfit is a **wedge-shaped radiolucent gap** between framework and implant. And by the **patient's symptoms** — pressure, tightness or pain as I tighten is a warning, and I stop.

Step three. Contour and cleansability, interocclusal clearance for the ceramic, the occlusal scheme, the shade, and where the screw access channels emerge.

If it does not fit, the first thing is that I do not force it. Torquing a non-passive framework transmits static stress to the implants and components, and causes screw loosening or fracture, marginal bone loss and pain.

Then I find the cause, systematically. Interference — soft tissue or bone over a platform, or a component not fully seated; remove it and re-try. **A component error** — the wrong analogue, or an analogue rotated in the cast. **Impression distortion** — unsplinted copings, a coping that moved or was not fully seated, or a flexible tray. **Cast inaccuracy** — expansion or damage. **Laboratory distortion** during casting, soldering or ceramic firing; a framework can be passive before firing and not afterwards.

The remedies, in escalating order. First, remove any interference and re-try. Second — and this is the standard remedy for a framework that is accurate in its parts but distorted overall — **section it at the point of misfit, lute the sections together intra-orally with pattern resin, and return it for soldering or laser welding**, then re-verify with the same tests. Third, if the error lies in the impression or the cast, make a **new verification jig, take a new splinted open-tray impression, and have the framework remade**. I would not adjust my way out of a misfit by grinding the intaglio.

Then the professional part. I explain the delay to the patient honestly and apologise for it, and I adjust or replace the provisional so they leave with something comfortable and are not inconvenienced. I **discuss the case with the technician** to establish where the error arose so it does not recur — but I would not criticise the laboratory in front of the patient. And I **record** it all: the jig result, which tests I performed and from which end, the radiographic findings, the cause I identified, the decision taken, what I told the patient, and the review appointment for the re-try."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 2 uses the jig first and states what a non-seating jig means · 4 one-screw (Sheffield) test performed and repeated from both ends · 7 states explicitly that the framework must not be forced, with the consequences · 5 radiographic verification and the wedge-shaped gap · 9 section-and-solder remedy and the new-impression remedy · 10 patient communication, provisional, technician discussion and records.

Accept these late or on prompting: 1 the purpose of the verification jig · 3 the sections-and-rejoins principle · 6 contour, cleansability, clearance, occlusion, shade and access position · 8 the systematic list of causes.

The commonest way five minutes is wasted here: reciting the full causes list as a lecture before the examiner has asked what to do about it, and — worse — reaching for the framework and the torque wrench before the jig, which forfeits the two marks the examiner put the jig on the table to test.

OSCE 54 — Chairside provisional implant crown



The instruments and materials laid out for this station — simulated illustration, not a clinical image

Blueprint: demonstrate the fitting of a prosthesis; plan the type of prosthesis and translate this into the definitive restoration; different techniques of restoring implants.

EQUIPMENT

Equipment — laid out on the table in front of you: a **study model with a diagnostic wax-up** of the 11; **silicone putty**; a **temporary implant abutment** (titanium or PEEK) with its screw and screwdriver; **Protemp or another bis-acryl provisional material** with mixing tips; **flowable composite**; **PTFE tape**; **articulating paper**; a **polishing bur kit** — acrylic burs, discs, silicone wheels, polishing paste; **temporary luting cement (temp bond)**; **dental floss**; and a model carrying an implant analogue at the **11** with matured soft tissue.

CANDIDATE INSTRUCTIONS

An implant has been placed at the 11 and is ready to be provisionalised. Using the instruments and materials on the table, ***explain to the examiner, step by step, how you would make the provisional crown.*** *You are not asked to make it — describe the procedure verbally, say which of these items you would use and which you would not, and explain why.*

DIRECTION OF THOUGHT

This is a discussion station: everything you would do with your hands must be said out loud, in order, naming the material each time. Three things carry the marks. First, the sequence — index, abutment, block out, form, remove before set, contour, polish, seat, check, seal, review. Second, the reason the provisional exists at all: it is the **instrument that shapes the peri-implant soft tissue**, and its subgingival contour is a deliberate design decision, not an accident of the material. Third — and this is where the tray is testing you — **not every item on the table should be used**. Say which you reject and why. Work through the list before you start: *"I would use the wax-up and putty to make an index, the temporary abutment, PTFE, the bis-acryl, flowable, the polishing kit, articulating paper and floss. I would not use the temporary cement, and I will explain why."*

MODEL ANSWER

"The purposes of an implant provisional are to restore appearance and function temporarily, to protect the site, and above all to **develop and support the peri-implant soft tissue and the emergence profile**, which I will then copy into the definitive restoration.

Steps: 0. **Make the index first, before anything goes near the mouth.** I take the **study model with the diagnostic wax-up** and load **silicone putty** over the wax-up and two teeth either side, to make a **rigid palatal or labio-palatal index**. I trim it so it seats on the adjacent teeth only, and cut a small vent over the 11 so that excess material and air can escape. The wax-up is the shape I have already agreed with the patient, so the index transfers an approved contour rather than an invented one. An alternative index is a **vacuum-formed matrix** made on the same wax-up. 1. **Verify the implant** is integrated and the soft tissue healthy. Remove the healing abutment and clean the internal connection. 2. **Select a temporary abutment** — titanium or PEEK — of the correct platform, and check it seats fully. 3. **Block out the screw channel** with **PTFE tape** so that no resin enters the screw head, and so I can retrieve it. 4. **Protect the soft tissue and the interface**: I place a small amount of rubber dam sheet or PTFE around the abutment collar, or apply a separating medium,

so that resin does not lock into the undercut around the abutment or contact the mucosa. 5. **Form the crown.** Either load a **vacuum-formed matrix taken from the diagnostic wax-up** with bis-acryl and seat it over the abutment; or, in an immediate case, relieve the **patient's own natural crown**, hollow it, and lute it to the abutment; or free-hand with flowable composite and a matrix strip. 6. **Remove before full set** — unscrew and remove the assembly while the resin is still rubbery, then allow it to bench-cure fully outside the mouth. Removing it prevents locking into undercuts and prevents the exotherm from damaging the tissue. 7. **Add resin to the subgingival portion and shape the emergence profile.** This is the critical step. Below the mucosal margin I create a **flat or slightly concave (under-contoured) profile**, which supports the tissue without pushing it apically; a convex, over-contoured subgingival form causes recession and inflammation. Above the margin I develop the correct emergence to support the papillae. 8. **Finish and polish to a very high lustre** — any roughness subgingivally becomes a plaque trap and a source of inflammation. I polish with discs, silicone wheels and a polishing paste. 9. **Seat and check.** Hand-tighten only, or torque to the manufacturer's provisional value. Check the proximal contacts with floss, check the emergence supports the papillae, and check the **occlusion — for an immediate provisional, it must be completely free of contact in intercuspation and in all excursions**; for a conventionally loaded provisional, light contact is acceptable. 10. **Seal the access** with PTFE and composite so it can be removed and modified. 11. **Instructions and review.** Soft diet, no incising with the tooth, hygiene around the provisional, and review in 2 to 3 weeks so I can **add or remove resin incrementally** to refine the emergence over the following 8 to 12 weeks. 12. When the profile is mature, I take a **customised impression coping** copying the provisional's subgingival contour, so the definitive restoration reproduces it exactly.

What I would use from this tray, and how. The **wax-up and putty** for the index. The **temporary abutment** and its screw. **PTFE** for the block-out and, at the end, for the access seal. **Bis-acryl** (**Protemp**) through the index to form the crown, and **flowable composite** to add and refine the subgingival contour incrementally. The **polishing kit** — acrylic bur, disc, silicone wheel, then paste — because a rough subgingival surface is a plaque trap and the single commonest cause of an inflamed provisional site. **Articulating paper** to confirm the occlusion: for an immediately placed and immediately restored anterior implant the crown must be **completely free of contact in intercuspation and in every excursive movement**, and I check protrusion and both lateral excursions, not just tapping. **Floss** to check the proximal contacts and, just as importantly, to check that I can clean under and around the emergence — if floss shreds, the surface is not polished, and if it will not pass, the contact is too tight.

What I would not use, and why. The **temporary cement**. This is a **screw-retained** provisional, and that is a deliberate choice in an anterior implant: it is retrievable without stressing the interface, it can be removed repeatedly to add resin as the tissue matures, and — decisively — **it eliminates residual subgingival cement**, which is a well-documented cause of peri-implant inflammation and bone loss and is very difficult to detect radiographically once it is deep. Cement excess is far harder to remove around an implant than around a tooth because there is no periodontal ligament and the margin is often deeply subgingival. If the emergence angle made a screw channel impossible — a labially inclined implant with the channel emerging through the incisal edge — I would either use an angulated screw-channel abutment or accept a cemented provisional on a customised abutment with a **supragingival or barely subgingival margin**, and I would still extra-orally cement it onto the abutment and clean it outside the mouth wherever possible.

I would record the abutment used, the material, the occlusal arrangement and the review plan."

EXAMINER CHECKLIST

1. States the purpose, including **soft tissue conditioning**, and verifies integration and tissue health first
2. **Putty or vacuum-formed index taken from the diagnostic wax-up**, and says why the wax-up is the source of the contour
3. Correct **temporary abutment** selected, seated fully, connection cleaned
4. **PTFE block-out of the screw channel** and protection of the tissue and undercuts from resin
5. Forms the crown in **bis-acryl through the index** and **removes it before full set**, with the reason (undercuts, exotherm)
6. **Concave or flat subgingival emergence profile, with the reason**, refined incrementally in flowable
7. **High polish**, naming the sequence of burs, wheels and paste
8. **Occlusion checked with articulating paper** in intercuspation *and* all excursions, and kept out of contact where required; **floss** used for contacts and cleansability
9. **Screw-retained, and explicitly rejects the temporary cement**, giving residual subgingival cement as the reason, with the angulated-screw-channel alternative

10. Access sealed with PTFE and composite; instructions, incremental refinement plan and **customised impression coping** to transfer the profile

Extended by practice stations **OSCE 16** (fitting the prosthesis) and **OSCE 14** (implant impression).

Running this station to time

Time check. The model answer in this part runs to about 1,050 words — roughly 7½ minutes at 140 words per minute before the examiner has asked anything — so about a third has to go. It goes by **naming each step in one clause instead of justifying it**, and by keeping the full justification for only three places: the source of the contour, the subgingival profile, and the refusal of the temporary cement. This is a **discussion station**: the candidate is not making the crown, they are talking the examiner through it while the materials sit on the table between them.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Indicates the tray and says: "An implant is in at the 11 and is ready to be provisionalised. Talk me through how you would make the provisional."	Checklist 1. The candidate states the **purpose** first — appearance and function, protection, and above all **conditioning the peri-implant soft tissue and the emergence profile that the definitive crown will copy** — and confirms **integration and tissue health** before touching anything.
0:30–1:30	Silence, then: "Where does the shape come from?"	Checklists 2 and 3. **Putty index taken from the diagnostic wax-up** on the study model, trimmed to seat on the adjacent teeth, vented — with the reason: *the wax-up is a contour already agreed with the patient*. Then the **temporary abutment** selected, seated fully, connection cleaned.
1:30–3:00	"Carry on — how do you actually form it?"	Checklists 4 and 5. **PTFE into the screw channel** and around the abutment collar; **bis-acryl loaded into the index** and seated; **removed before full set** with both reasons — locking into undercuts, and exotherm on the mucosa; bench-cured outside the mouth.
3:00–4:00	"Now the part below the gum." Then: "How would you finish it?"	Checklists 6 and 7. **Flat or slightly concave subgingival profile**, added incrementally in flowable, with the reason that an over-contoured convex profile pushes the tissue apically and causes recession; then the **polishing sequence named** — acrylic bur, disc, silicone wheel, paste — because a rough subgingival surface is a plaque trap.
4:00–4:30	"Which of these would you not use?" — ask it if the candidate has not already volunteered it.	Checklist 9, the discriminating mark. **Screw-retained, so the temporary cement is not used**, because of **residual subgingival cement** and how hard it is to detect and remove around an implant; with the angulated-screw-channel or customised-abutment alternative if the channel emerges through the incisal edge.
4:30–5:00	"Finish off."	Checklists 8 and 10. **Articulating paper** — out of contact in intercuspation *and* all excursions for an immediately restored implant; **floss** for the contacts and for cleansability; **access sealed with PTFE and composite**; then **what is recorded**, **what the patient is told** (soft diet, do not incise with it), and **review at two to three weeks** for incremental refinement, ending with the **customised impression coping** that transfers the matured profile.

The compressed answer — what a strong candidate says in five minutes

"**The provisional is not a stop-gap; it is the instrument that shapes the peri-implant soft tissue**, and the definitive crown will copy whatever shape it creates. So before anything else I would confirm the implant is integrated, the mucosa is healthy, and that I have an agreed target shape.

That shape comes from the diagnostic wax-up. I would load **silicone putty** over the wax-up on the study model and two teeth either side, trim the index to seat on the adjacent teeth only, and cut a small vent over the 11 so excess and air escape. A vacuum-formed matrix on the same wax-up does the same job. The point is that the contour is one the patient has already seen and approved, not one I invent at the chair.

Then the abutment. I remove the healing abutment, clean the internal connection, and seat the correct-platform **temporary titanium or PEEK abutment**, checking it is fully home. I block the screw channel with **PTFE tape** so no resin reaches the screw head, and place PTFE or a separating medium around the collar so resin does not lock into the undercut or contact the mucosa.

Forming it. I load the index with **bis-acryl — Protemp — and seat it**, then **remove the whole assembly before the material has fully set**, while it is still rubbery. Two reasons: resin locks into undercuts if it sets in the mouth, and the exotherm damages the mucosa. It bench-cures outside the mouth.

The subgingival contour is the critical step. Below the mucosal margin I add **flowable composite** to build a **flat or slightly concave profile** that supports the tissue without pushing it apically. A convex, over-contoured subgingival form is what causes recession and inflammation. Above the margin I develop the emergence that supports the papillae.

Then I polish it to a high lustre — acrylic bur, disc, silicone wheel, then paste. Any roughness subgingivally is a plaque trap, and an inflamed provisional site is usually a badly polished one.

Seating and checking. Hand-tighten or torque to the manufacturer's provisional value. **Articulating paper**: if this implant was placed and restored immediately, the crown must be **completely free of contact in intercuspation and in every excursion** — I check protrusion and both lateral movements, not just tapping. **Floss** for the proximal contacts, and to confirm it will pass and not shred, which tells me the contact and the polish are both right. I seal the access with **PTFE and composite** so I can retrieve it.

One thing on this tray I would not use: the temporary cement. This is a **screw-retained** provisional deliberately — retrievable without stressing the interface, removable repeatedly so I can add resin as the tissue matures, and, decisively, **no residual subgingival cement**, which is a well-recognised cause of peri-implant inflammation and bone loss and is very difficult to detect or remove once it is deep, because there is no periodontal ligament and the margin is often well below the tissue. If the screw channel emerged through the incisal edge I would use an **angulated screw-channel abutment**, or accept a cemented provisional on a customised abutment with a supragingival margin, cemented extra-orally where possible.

Finally, I would record the abutment, the material, the occlusal arrangement and the plan; tell the patient soft diet, not to bite into anything with it, and how to clean around it; and review at **two to three weeks** to refine the emergence incrementally over the next two to three months. When the profile is mature I take a **customised impression coping** that copies it, so the definitive crown reproduces the shape exactly."

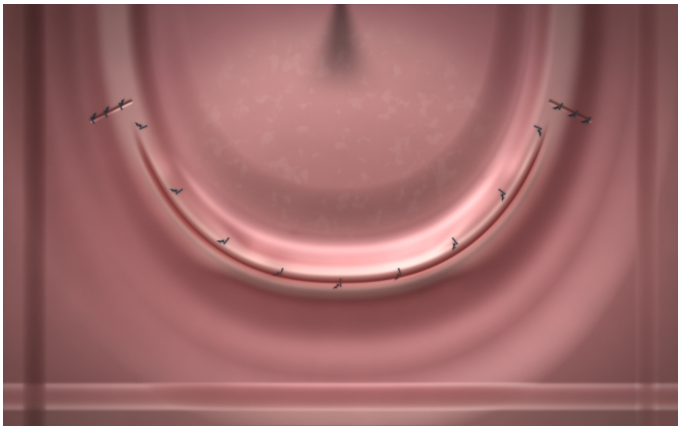
Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 1 purpose, including soft-tissue conditioning · 2 index taken from the diagnostic wax-up, with the reason · 5 forms in the index and **removes before full set**, with the reason · 6 **flat or concave subgingival profile**, with the reason · 9 **screw-retained; rejects the temporary cement** because of residual cement · 10 records, patient instructions, review and the customised impression coping.

Accept these late or on prompting: 3 abutment selection and seating · 4 PTFE block-out and tissue protection · 7 the polishing sequence by name · 8 articulating paper in excursions and floss for contacts and cleansability.

The commonest way five minutes is wasted here: describing the material rather than the procedure — a candidate who spends 90 seconds on the merits of bis-acryl against PMMA and then rushes the subgingival contour has spent the time on the least examinable part of the station. The other failure is never mentioning the temporary cement at all: the tray was laid out with a distractor on it, and saying nothing about it is treated as not having noticed.

OSCE 55 — Post-operative instructions after multiple implant placement



The surgical site at the end of the appointment — simulated illustration, not a clinical image

Blueprint: suggest appropriate and adequate postoperative care and advice including the use of post-operative painkillers and antibiotics. Marked OS-P + SCE.

EQUIPMENT

a simulated patient or the examiner acting as the patient; a blank post-operative instruction sheet; a prescription pad.

CANDIDATE INSTRUCTIONS

You have just placed four implants in the mandible in a single visit. Give the patient their post-operative instructions.

DIRECTION OF THOUGHT

This is a **spoken** station: plain language, no jargon, in a fixed order, ending with warning signs and a contact number. **Multiple implants changes the emphasis** compared with a single implant: more swelling and bruising, more analgesia, a longer flap and more sutures, a greater risk of bleeding, more attention to the denture, and — because the mandible is involved — an explicit airway warning.

MODEL ANSWER

As spoken to the patient

"That all went to plan — four implants are in. This was a bigger procedure than a single implant, so I want to go through what to expect, and I'll give you all of this in writing too.

Bleeding. Some oozing today is normal and can look worse than it is because it mixes with saliva. If it bleeds, bite firmly on the gauze I've given you for **20 minutes without checking it**. Don't rinse at all today.

Swelling and bruising. Because we've worked on both sides, expect **more swelling than for a single implant**, and expect bruising that may track down towards your jawline or neck over a few days — that's normal and it settles. Swelling is usually worst at **two to three days** and then improves. Use a cold pack on the outside, **20 minutes on and 20 off**, for the first day, and sleep with an extra pillow.

Pain. Take **ibuprofen 400 mg with food, up to four times a day**, and **paracetamol 1 gram up to four times a day** — you can take them together and that combination works better than either alone. **Take the first dose now, before the numbness wears off.** Avoid aspirin.

Medication I've prescribed. [If given] Finish the whole antibiotic course. Start the **chlorhexidine mouthwash tomorrow, twice a day for two weeks** — hold it in your mouth gently, don't swill hard.

Cleaning. Brush all your other teeth normally, but **do not brush over the stitches** until I say so. Keeping the rest of your mouth clean matters more than usual.

Eating and drinking. Soft food for the first week — you've got sutures across a wide area, so nothing that needs biting or chewing hard. Nothing hot today. Keep well hydrated, and don't use a straw.

Your denture. Leave it out completely for the next **48 hours**. Then bring it in and I'll adjust and reline it so it doesn't press on the implants. **Do not wear it until I've done that** — pressure on a healing implant is one of the main causes of failure. And leave it out at night from now on.

Activity. No smoking at all — smoking is the single biggest thing that will make these fail. No alcohol, and no strenuous exercise or heavy lifting for at least three days, because that can restart the bleeding.

Numbness. Your lip and chin will be numb from the anaesthetic for a few hours. **If any numbness is still there tomorrow, ring me** — that's important and I need to know straight away.

Contact me if: the bleeding doesn't stop with 20 minutes of firm pressure; the pain gets *worse* after day three rather than better; you get a temperature or the swelling spreads towards your eye or down your neck; a stitch comes away and something feels rough; or you feel unwell.

And this one is an emergency — call 999: if you develop swelling under your tongue, difficulty swallowing, or any difficulty breathing.

Here is the written sheet, my number, and the out-of-hours number. I'll see you in a week to take the stitches out, and I'll ring you this evening to check how you are."

EXAMINER CHECKLIST

1. Bleeding advice with a specific duration of pressure
2. Swelling and bruising, **anticipating a greater response after multiple implants**, with cold pack and head elevation
3. Specific analgesia with drug, dose, frequency and **pre-emptive timing**
4. Antibiotic and chlorhexidine instructions
5. Hygiene — no brushing over the surgical site
6. Diet, extended to reflect a larger surgical field; no straws
7. **Denture out for 48 hours, then relieved and soft-relined; not worn until adjusted**
8. Smoking, alcohol, exercise
9. **Numbness instruction and the airway emergency warning**
10. Written instructions, contact numbers, review appointment and a same-day follow-up call

See also **OSCE 10** (post-operative care after a single implant with a graft) and **OSCE 45** (denture relief and relining).

Running this station to time

Time check. The model answer in this part is 544 words, 3 minutes 53 seconds of pure speech at 140 words per minute — but a patient interrupts, and a candidate who greets, checks understanding, hands over the sheet and answers one question will overrun five minutes comfortably; nothing here can be cut for content, so the compression is in phrasing, and the three items that must survive intact are the **airway warning**, the **analgesia doses** and the **contact arrangements**.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–1:00	Sits as the patient, four implants just placed in the mandible: "So what happens now?"	Checklists 1 and 2. **Bleeding** — bite firmly on the gauze for **20 minutes without checking**, no rinsing today. **Swelling and bruising** — explicitly **more than after a single implant**, bruising tracking to the jawline or neck, worst at two to three days, cold pack **20 minutes on and 20 off**, extra pillow.
1:00–2:00	Listens. May ask: "What can I take for the pain?"	Checklist 3, the highest-value minute on the station. **Ibuprofen 400 mg with food up to four times a day and paracetamol 1 g up to four times a day, taken together**, with the **first dose now, before the numbness wears off**, and avoid aspirin. Checklist 4 follows — finish the antibiotic course, chlorhexidine from tomorrow, twice daily for two weeks.

Time	The examiner does	What earns the mark
2:00–3:00	May ask: "Can I eat? Can I wear my denture?"	Checklists 5, 6 and 7. No brushing over the stitches, brush everything else normally; soft food for a week, nothing hot today, no straws; and the mark examiners most often find missing — **denture left out completely for 48 hours, then relieved and soft-relined, and not worn at all until adjusted**, because pressure on a healing implant is a leading cause of failure.
3:00–4:00	May ask: "When can I have a cigarette?"	Checklists 8 and 9. **No smoking at all**, named as the single biggest cause of failure; no alcohol; no strenuous exercise or heavy lifting for three days. Then **numbness** — normal for a few hours, but **ring me if any numbness is still there tomorrow** — followed immediately by the **airway warning: swelling under the tongue, difficulty swallowing or breathing means call 999**.
4:00–5:00	"Is that everything?"	Checklist 10 and the close. The **warning signs** that prompt a call, then **records** (that instructions were given and understood, and the prescription written), **what the patient is told in writing** (the sheet, the practice number and the out-of-hours number), and **review** — sutures out in a week, plus a **phone call this evening**.

The compressed answer — what a strong candidate says in five minutes

"That all went to plan — all four implants are in. This was a bigger procedure than a single implant, so let me talk you through what to expect, and you will have all of it in writing as well.

Bleeding. A bit of oozing today is normal, and it looks worse than it is because it mixes with saliva. If it bleeds, bite firmly on the gauze I have given you for **twenty minutes without checking it**. Do not rinse at all today.

Swelling and bruising. Because we have worked on both sides, expect **more swelling than you would get with one implant**, and expect bruising that may spread down towards your jawline and neck over a few days. That is normal and it settles. Swelling is usually worst at **two to three days** and then improves. Hold a cold pack against the outside of your face, **twenty minutes on and twenty minutes off**, for the first day, and sleep with an extra pillow tonight.

Pain. Take **ibuprofen four hundred milligrams with food, up to four times a day**, and **paracetamol one gram, up to four times a day**. You can take them together, and together they work better than either on its own. **Take your first dose now, before the numbness wears off** — do not wait for it to start hurting. Please avoid aspirin.

Medicines I have given you. Finish the whole course of antibiotics. Start the mouthwash **tomorrow, twice a day for two weeks** — hold it in your mouth gently, do not swill it hard.

Cleaning. Brush all your other teeth normally, but **do not brush over the stitches** until I tell you it is safe. Keeping the rest of your mouth clean matters more than usual just now.

Eating. Soft food for the first week — you have stitches right across the front, so nothing you have to bite or chew hard. Nothing hot today. Drink plenty, but **no straws**.

Your denture. Leave it out completely for the **next forty-eight hours**. Then bring it in to me and I will ease it and put a soft lining in so it does not press on the implants. **Please do not wear it until I have done that** — pressure on a healing implant is one of the main reasons they fail. And from now on, leave it out at night.

Smoking, alcohol and exercise. No smoking at all — smoking is the single biggest thing that would make these fail. No alcohol, and no hard exercise or heavy lifting for at least three days, because that can start the bleeding again.

Numbness. Your lip and chin will feel numb from the injection for a few hours. **If any numbness is still there tomorrow morning, ring me** — that is important and I need to know straight away.

Ring me if the bleeding does not stop after twenty minutes of firm pressure; if the pain gets **worse** after day three instead of better; if you get a temperature, or the swelling spreads towards your eye or down your neck; if a stitch comes away and something feels rough; or if you simply feel unwell.

And this one is an emergency — call 999 if you get swelling under your tongue, difficulty swallowing, or any difficulty breathing.

Here is the written sheet with all of that on it, my number, and the out-of-hours number. I will **ring you this evening** to see how you are, and I will see you **in a week to take the stitches out**. Is there anything there you would like me to go over again?"

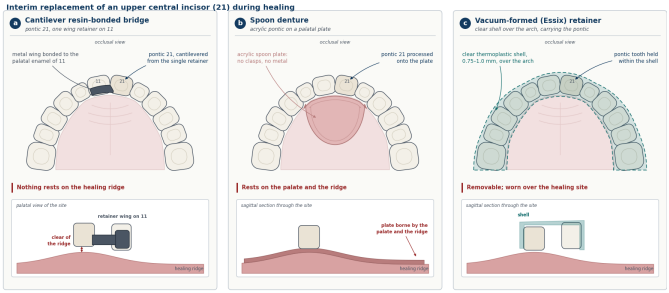
Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 3 analgesia by drug, dose, frequency and pre-emptive timing · 9 numbness instruction and the airway emergency warning · 7 denture out for 48 hours, then relieved and soft-relined, not worn until adjusted · 10 written instructions, contact numbers, review and the same-day call · 8 smoking, alcohol and exercise · 1 bleeding advice with a specific duration of pressure.

Accept these late or on prompting: 2 swelling and bruising with the greater response after multiple implants, cold pack and elevation · 4 antibiotic and chlorhexidine instructions · 5 no brushing over the surgical site · 6 diet for a larger surgical field and no straws.

The commonest way five minutes is wasted here: drifting into jargon and explanation — osseointegration, primary stability, why the flap was raised — instead of instructions, so the airway warning and the review arrangements arrive after the bell. The patient does not need the physiology; they need what to do tonight, what to take, and who to ring.

OSCE 56 — Three interim prostheses: resin-bonded bridge, spoon denture and Essix retainer



The three interim prostheses — simulated illustration, not a clinical image

Blueprint: differentiate between the treatment options for tooth replacement; decipher the differences between removable and fixed prostheses; plan the type of prosthesis during the initial planning.

EQUIPMENT

examples of a **resin-bonded bridge**, a **spoon denture**, and an **Essix (vacuum-formed) retainer with a pontic tooth**; a model of an anterior edentulous space; photographs.

CANDIDATE INSTRUCTIONS

Describe these three interim prostheses. Give the advantages and disadvantages of each, and state which you would use for a patient awaiting an implant at the UR1 following extraction and ridge preservation.

DIRECTION OF THOUGHT

Describe → compare → **choose, and justify the choice against the clinical situation**. The decisive clinical criterion in an implant case is **whether the prosthesis loads the healing site**. Say that early and use it as your organising principle.

MODEL ANSWER

"**1. Resin-bonded bridge (adhesive bridge / bonded pontic).** A pontic supported by one or more thin retainer wings bonded to the enamel of an adjacent tooth — for a single anterior unit, the evidence favours a **cantilever design with a single retainer**, because fixed-fixed designs debond at one retainer and allow undetected caries.

Advantages: **fixed**, so the patient is never without a tooth; excellent aesthetics; **applies no pressure to the healing ridge**, which is its decisive advantage over the other two; comfortable; no palatal coverage; good speech; minimal or no tooth preparation, confined to enamel; well documented, with good survival for single anterior units. *Disadvantages:* needs a suitable abutment tooth with sufficient sound enamel and adequate occlusal clearance; may debond, particularly in deep overbite or parafunction; metal retainer may cause slight greying of the abutment tooth; higher cost and requires laboratory work, so not immediate; unsuitable where there is heavy occlusal loading on the pontic; and it may need modification when the implant is restored.

2. Spoon denture. A simple acrylic removable prosthesis with the pontic supported by a 'spoon'-shaped palatal acrylic plate that does not engage the teeth with clasps.

Advantages: cheap and quick; easily made and easily added to or modified; no preparation of any tooth; useful when several teeth may be lost sequentially; can be relieved over a graft. *Disadvantages:* **poor retention and stability** — it relies on the palatal vault and muscular control, so it can dislodge in function and speech; may **traumatise the healing site**, so it must be generously relieved; palatal coverage can cause gagging and affects taste and speech; unsuitable in the mandible; poor patient acceptance; and, if under-relieved, it presses directly on the ridge, which is exactly what must be avoided over a graft.

3. Essix retainer (vacuum-formed retainer with a pontic). A clear thermoplastic vacuum-formed appliance covering the arch, with a denture tooth or composite added at the missing site.

Advantages: very quick and cheap, often same-day; **excellent immediate aesthetics**; no palatal coverage; retentive because it grips the whole arch; easily relieved over the surgical site; can double as a retainer; useful as a short-term aesthetic solution after extraction. *Disadvantages:* **full occlusal coverage**, so there is no tooth contact when it is worn and it cannot be worn while eating; wears, cracks and discolours; not durable beyond a few months; hygiene is poor if worn continuously; can cause occlusal changes and intrusion if worn full-time for a long period; and, as with the spoon denture, it can press on the ridge if not relieved.

My choice for this case. For a UR1 site after extraction and ridge preservation, my first choice is the **resin-bonded bridge or bonded pontic**, because it is the only one of the three that **applies no load to the healing ridge**, it is fixed so the patient does not have to remove anything, and the aesthetics are the best of the three — which matters in an upper central incisor.

If a bonded bridge is not possible — an unsuitable abutment tooth, a deep overbite, cost, or the need for something on the day of surgery — I would provide an **Essix retainer with a pontic** immediately, generously relieved over the site, and warn the patient not to wear it while eating; and I would consider converting to a bonded pontic once the site has settled.

I would reserve the **spoon denture** for situations where several teeth are being lost sequentially and I need something I can add to repeatedly, and I would relieve it heavily and warn the patient about pressure.

Whatever I use, I would **review it at one week**, check for pressure on the site with pressure-indicating paste, and adjust it as swelling and healing change the contour."

EXAMINER CHECKLIST

1. Describes the RBB correctly, including cantilever design
2. RBB advantages and disadvantages
3. Describes the spoon denture correctly
4. Spoon denture advantages and disadvantages
5. Describes the Essix retainer correctly
6. Essix advantages and disadvantages, **including that it cannot be worn while eating**
7. **Identifies loading of the healing site as the decisive criterion**
8. Makes a clear choice with justification
9. States a fallback and the circumstances for it

10. Review, pressure check and adjustment

See also **OSCE 45** (denture relief and relining) and SOE cases 1 and 2.

Running this station to time

Time check. The model answer in this part is 636 words, 4 minutes 33 seconds at 140 words per minute — which sounds survivable until you add the examiner's instruction, the handling of three appliances and the inevitable "why?", at which point the honest delivery time is well past 7 minutes; the fix is a **fixed 60-second frame per prosthesis** — what it is, its one decisive advantage, its one decisive disadvantage — then 60 seconds for the choice, then the close.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:30	Places the three appliances and the anterior model in front of the candidate: "Describe these three, give advantages and disadvantages, and tell me which you would use for a UR1 awaiting an implant after ridge preservation."	Nothing is marked yet, but the candidate must **state the organising principle in one sentence** — "the decisive criterion in an implant case is whether the prosthesis loads the healing site" — which pre-earns checklist 7 and buys permission to be brief about everything else.
0:30–1:30	Holds up the resin-bonded bridge. Says nothing further.	Checklists 1 and 2. **What it is** (pontic on a thin wing bonded to enamel, **cantilever with a single retainer** for a single anterior unit) · **decisive advantage** (fixed, and **no load on the healing ridge**) · **decisive disadvantage** (needs a sound enamel abutment with clearance; debonds in deep overbite or parafunction; laboratory time so not same-day).
1:30–3:30	Slides the spoon denture forward, then the Essix retainer — 60 seconds each, on the same frame.	Checklists 3, 4, 5 and 6. **Spoon:** acrylic pontic on a clasp-free palatal plate · cheap, quick, added to repeatedly · **poor retention, and it presses on the healing ridge unless heavily relieved**; palatal coverage, gagging, unsuitable in the mandible. **Essix:** clear vacuum-formed appliance with a pontic at the site · same-day, cheap, excellent immediate aesthetics, easily relieved · **full occlusal coverage, so it cannot be worn while eating**; wears, cracks, discolours, not durable beyond months.
3:30–4:30	"So which would you use for this patient, and why?"	Checklists 7, 8 and 9. Loading named again as the **decisive criterion**, a **clear single choice** (resin-bonded bridge or bonded pontic) with justification tied to the UR1 and the ridge preservation, and a **named fallback with the circumstances that trigger it** (Essix on the day of surgery if the abutment is unsuitable, the overbite is deep, or cost forbids).
4:30–5:00	"How will you follow this up?"	Checklist 10 — **records** (which appliance, the relief provided, the instructions given), **what the patient is told** (do not eat in the Essix; report any pressure or ulceration), and **review at one week with pressure-indicating paste and adjustment** as swelling settles.

The compressed answer — what a strong candidate says in five minutes

"Before I describe them, the criterion that decides this case: in an implant patient the question is **whether the prosthesis loads the healing site**. I will use that to organise all three.

First, the resin-bonded bridge. A pontic carried on one or more thin metal wings bonded to the enamel of an adjacent tooth; for a single anterior unit the evidence favours a **cantilever with a single retainer**,

because fixed-fixed designs debond at one end and allow caries to go undetected. Its decisive advantage is that it is **fixed and applies no pressure at all to the healing ridge** — the patient is never without a tooth, the aesthetics are the best of the three, there is no palatal coverage, and preparation is minimal and confined to enamel. Its decisive disadvantage is that it **needs a suitable abutment tooth** with sound enamel and occlusal clearance; it may debond, particularly with a deep overbite or parafunction; the metal wing can grey the abutment slightly; and it needs laboratory time, so it is not available on the day of surgery.

Second, the spoon denture. A simple acrylic removable prosthesis with the pontic carried on a spoon-shaped palatal plate that engages no clasps. Its decisive advantage is that it is **cheap, quick and easily added to** — useful where teeth are being lost sequentially — with no preparation of any tooth. Its decisive disadvantage is **poor retention and stability**: it relies on the palatal vault and muscle control, so it moves in function and speech, and unless generously relieved it **presses directly on the healing ridge**, which is exactly what must be avoided over a graft. The palatal coverage also causes gagging and affects taste and speech, and it is unsuitable in the mandible.

Third, the Essix retainer. A clear thermoplastic vacuum-formed appliance covering the whole arch, with a denture tooth or composite at the missing site. Its decisive advantage is that it is **same-day, cheap and aesthetically excellent immediately**, retentive because it grips the whole arch, and easily relieved over the surgical site. Its decisive disadvantage is that it gives **full occlusal coverage, so there is no tooth contact when it is in and it cannot be worn while eating**; it wears, cracks and discolours, it is not durable beyond a few months, hygiene is poor if worn continuously, and it too can press on the ridge if not relieved.

My choice. For a UR1 after extraction and ridge preservation, my first choice is the **resin-bonded bridge or bonded pontic**, because it is the only one of the three that **applies no load to the healing ridge**, it is fixed so nothing is taken in and out over the graft, and the aesthetics are the best of the three — which matters most at an upper central incisor.

My fallback. If a bonded bridge is not possible — an unsuitable abutment tooth, a deep overbite, the cost, or the need for something on the day of surgery — I would fit an **Essix retainer with a pontic** immediately, generously relieved over the site, and tell the patient not to wear it while eating; I would convert to a bonded pontic once the site has settled. I would reserve the **spoon denture** for a patient losing several teeth sequentially where I need something I can keep adding to, and I would relieve it heavily.

Whatever I fit, I record which appliance was provided, the relief given and the instructions; I tell the patient to report any pressure, ulceration or looseness, and that this is temporary until the implant is restored; and I **review at one week, check the fit surface against the site with pressure-indicating paste, and adjust it** as the swelling and the ridge contour change."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 7 loading of the healing site identified as the decisive criterion · 8 a clear choice with justification · 1 resin-bonded bridge described correctly including the cantilever design · 6 Essix disadvantages, including that it cannot be worn while eating · 9 fallback named with the circumstances · 10 review, pressure check and adjustment.

Accept these late or on prompting: 2 the rest of the RBB advantages and disadvantages · 3 the spoon denture described correctly · 4 spoon denture advantages and disadvantages · 5 the Essix described correctly.

The commonest way five minutes is wasted here: exhausting every listed advantage and disadvantage of the resin-bonded bridge in the first 2 minutes, so the Essix gets 20 seconds and the choice, the fallback and the review — four of the ten marks, and the four that actually distinguish a clinician from a textbook — are never reached. One decisive advantage and one decisive disadvantage each, and move.

OSCE 57 — Severe periodontal bone loss and furcation involvement at 16 and 17

Blueprint: interpret radiographic findings; assess the need for referral on the basis of clinical findings; arrive at a prognosis of the remaining teeth and the impact on the treatment options for tooth replacement.

> **EQUIPMENT**
>
> a periapical radiograph of the **16 and 17** showing **severe horizontal and angular bone loss**; **axial CBCT slices** showing **furcation involvement** of both teeth; a Nabers probe; a periodontal chart.

CANDIDATE INSTRUCTIONS

Report these images. Discuss the management of these teeth and the effect of your findings on the replacement of the teeth.

DIRECTION OF THOUGHT

Three parts, and the third carries most of the marks: **report** → **manage the teeth** → **what this means for replacement**. The furcation finding is the hinge: grade II–III furcation involvement in maxillary molars carries a poor prognosis, and the interradicular bone that is being destroyed is exactly the bone an implant would need.

MODEL ANSWER

"**Report.** A periapical of the upper right molar region. I would state the image type and its diagnostic quality, then report the whole image before the region of interest.

At the **16 and 17** there is **severe bone loss** — I would quantify it as a proportion of root length, and describe whether it is horizontal or angular and whether it is generalised or localised. I would look at the furcation regions, the periodontal ligament space, the periapical status of both teeth, any existing restorations and caries, the **relationship to the maxillary sinus floor** and whether the sinus is pneumatised, and the bone level on the adjacent 15.

The **axial CBCT slices** confirm **furcation involvement**. In the maxilla each molar has three roots and therefore three furcation entrances — **buccal, mesiopalatal and distopalatal** — and cross-sectional imaging lets me see which are involved and how far through. I would state that CBCT is more sensitive than a periapical for furcation involvement, but that it must have been **individually justified and the whole volume reported**.

Clinically I would need: full six-point charting with recession so I can calculate attachment loss; **furcation grading with a Nabers probe** using the Hamp classification — grade I under 3 mm horizontally, grade II more than 3 mm but not through-and-through, grade III through-and-through; mobility; bleeding and suppuration; sensibility testing of both teeth; assessment for cracks; plaque scores; and a full-mouth BPE to establish whether this is localised or generalised. I would then assign a **stage and grade** under the 2017 classification.

Management of the teeth. The options are: - **Non-surgical periodontal therapy** and reassessment — always the first step, whatever the eventual plan. - **Retain with periodontal surgery** — furcation surgery, tunnel preparation or root resection. Grade III furcation with mobility carries a poor long-term prognosis; root resection in maxillary molars is a Complex procedure with modest survival, and I would not offer it as a general dentist with an interest in implants. - **Extraction** of one or both teeth. - **Referral** to a periodontist for an opinion before extracting, which I would consider if there is any prospect of retention.

My likely conclusion, with severe bone loss and confirmed furcation involvement in both molars, is that **both teeth have a poor to hopeless prognosis** — but I would not extract before completing cause-related therapy, reassessing, and having the conversation with the patient about the alternatives.

The effect on replacement — and this is the important part.

1. **The bone that is being lost is the bone an implant would need.** Furcation involvement means the interradicular bone is destroyed, so after extraction I should expect a **deficient site, not a simple socket**.
2. **Immediate implant placement is contraindicated** here — there is active infection, and the interradicular septum that would provide primary stability in a molar socket is exactly what the disease has destroyed.
3. **The sinus.** The posterior maxilla pneumatises after tooth loss, and periodontal bone loss reduces the residual height before I even start. I should measure the residual height beneath the sinus on the CBCT. If it is **more than 6 mm with a flat floor**, a transcrestal elevation with simultaneous placement is possible; **6 mm or less, or an oblique floor**, indicates a **lateral window**, which is outside my scope and would mean referral.
4. **Ridge preservation at the time of extraction** is therefore particularly valuable here, and I would section the molars and remove the roots individually to preserve the buccal and palatal plates.

5. **Timing.** These are large sockets in infected sites; I would allow **at least four to six months** before reassessing for implant placement.
6. **The periodontal condition governs everything.** A history of periodontitis is the **strongest patient-level risk factor for peri-implantitis**, so implants should not be planned until the periodontal condition is **stable** by the EFP criteria and the patient has demonstrated compliance with maintenance.
7. **And replacement may not be necessary.** If the 15 and the opposing teeth are sound, a **shortened dental arch to the second premolar** is a legitimate and well-evidenced outcome that avoids sinus surgery altogether. I would present that seriously, alongside a partial denture and implants.

On the SAC classification, implant replacement here would be at least **Advanced**, and Complex if a lateral window or vertical augmentation were required — in which case I would refer, having completed the periodontal treatment and the planning myself."

EXAMINER CHECKLIST

1. Structured report of the whole image, then the region of interest
2. Quantifies bone loss and describes its pattern
3. States the three maxillary furcation entrances and grades with a Nabers probe/Hamp
4. States the CBCT justification and whole-volume reporting requirement
5. Clinical assessment listed, with staging and grading
6. Management options for the teeth, with a reasoned prognosis and appropriate referral
7. **Links furcation destruction to the deficient future implant site**
8. **Rules out immediate placement and gives the reason**
9. Sinus assessment against the 6 mm / flat floor threshold, with the referral implication
10. Ridge preservation, timing, the periodontal stability precondition, and **offers the shortened dental arch**

Extended by SOE case 22 (the furcation-involved molar) and practice stations **OSCE 23** and **OSCE 24** (CBCT and panoramic reporting).

Running this station to time

Time check. The model answer is 1,025 words, 7 minutes 19 seconds at 140 words per minute — nearly two and a half minutes over. Around 330 words must go, and the only reliable way is a **fixed allocation across the three parts: 1 minute 40 for the report, 1 minute for the teeth, 2 minutes for the replacement.** Nothing is done with the hands here, so every second is speech: **findings as a list of clauses, and full sentences reserved for the three inferences — furcation to deficient site, no immediate placement, and the sinus threshold.**

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–1:00	Hands over a periapical of the 16 and 17 showing severe horizontal and angular bone loss, with axial CBCT slices, a Nabers probe and a periodontal chart: "Report these images. Discuss the management of these teeth and the effect of your findings on the replacement of the teeth."	Checklists 1 and 2. **Name the image type and its diagnostic quality**, then **report the whole image before the region of interest** — the structure is itself a mark. Then **quantify: severe bone loss at the 16 and 17, as a proportion of root length, horizontal or angular, localised or generalised.** Covered in the same sweep: **furcation regions, periodontal ligament space, periapical status, restorations and caries, the relationship to the maxillary sinus floor and whether it is pneumatised, and the bone level on the adjacent 15.**

Time	The examiner does	What earns the mark
1:00–1:40	Points at the CBCT slices. "And these?"	Checklists 3 and 4, and both are specific facts. **The axial slices confirm furcation involvement**, and the anatomy must be named: **each maxillary molar has three roots and therefore three furcation entrances — buccal, mesiopalatal and distopalatal** — and cross-sectional imaging shows which are involved and how far through. Then the governance clause candidates skip: **CBCT is more sensitive than a periapical for furcation involvement, but it must have been individually justified and the whole volume reported.**
1:40–2:40	"What else would you need, and what would you do with these teeth?"	Checklists 5 and 6. **Clinically: six-point charting with recession, so attachment loss can be calculated; furcation grading with a Nabers probe by Hamp — grade I under 3 mm horizontally, grade II more than 3 mm but not through-and-through, grade III through-and-through; mobility, bleeding, suppuration, sensibility testing, cracks, plaque scores and a full-mouth BPE; then a stage and grade under the 2017 classification.** Then the options as a ladder: **non-surgical therapy and reassessment first, whatever the eventual plan; retention with furcation surgery, tunnelling or root resection — Complex in maxillary molars, with modest survival, and not something I would offer; extraction; or referral to a periodontist before extracting.** Conclusion, reasoned: **poor to hopeless prognosis for both — but no extraction before cause-related therapy, reassessment and a conversation with the patient.**
2:40–4:20	"And what does all this mean for replacing them?"	The four marks the station is actually built on. **Checklist 7: the bone being lost is the bone an implant would need — furcation involvement means the interradicular bone is destroyed, so after extraction expect a deficient site, not a simple socket. Checklist 8: immediate placement is contraindicated — there is active infection, and the interradicular septum that would give primary stability in a molar socket is exactly what the disease has destroyed. Checklist 9, with the numbers: measure residual height beneath the sinus on the CBCT — more than 6 mm with a flat floor allows transcrestal elevation with simultaneous placement; 6 mm or less, or an oblique floor, indicates a lateral window, which is outside my scope and means referral.**

Time	The examiner does	What earns the mark
4:20–5:00	"So what would you offer her?"	Checklist 10, all four parts, quickly. **Ridge preservation at extraction, sectioning the molars and removing the roots individually to preserve the plates. At least four to six months before reassessing for placement**, because these are large sockets in infected sites. **The periodontal precondition: a history of periodontitis is the strongest patient-level risk factor for peri-implantitis, so nothing is planned until the condition is stable by the EFP criteria and compliance is demonstrated.** And the option that must be offered seriously: **a shortened dental arch to the second premolar**, alongside a partial denture and implants. Close on **SAC: at least Advanced, Complex if a lateral window or vertical augmentation is needed — in which case I refer, having done the periodontal treatment and the planning myself.**

The compressed answer — what a strong candidate says in five minutes

Q: Report the periapical. "A periapical of the upper right molar region — I would state its diagnostic quality first, and **report the whole image before the region of interest**. At the **16 and 17 there is severe bone loss**, which I would **quantify as a proportion of root length** and describe as **horizontal or angular**, and as **localised or generalised** once I have a full-mouth chart. I would also look at the **furcation regions, the periodontal ligament space, the periapical status of both teeth, existing restorations and caries, the relationship to the maxillary sinus floor and whether it is pneumatised, and the bone level on the adjacent 15.**"

Q: And the CBCT slices? "They **confirm furcation involvement.** Each maxillary molar has three roots and therefore **three furcation entrances — buccal, mesiopalatal and distopalatal** — and cross-sectional imaging tells me which are involved and how far through. **CBCT is more sensitive than a periapical for furcation involvement**, but it must have been **individually justified**, and I am obliged to **report the whole volume**, not only the teeth I am interested in."

Q What would you need clinically, and how would you manage the teeth? "Six-point charting with recession, so I can calculate attachment loss rather than pocket depth alone. Furcation grading with a Nabers probe, Hamp classification — grade I under 3 mm horizontally, grade II more than 3 mm but not through-and-through, grade III through-and-through. Mobility, bleeding, suppuration, sensibility testing, cracks, plaque scores and a full-mouth BPE, then a **stage and grade under the 2017 classification.**

Management begins with non-surgical periodontal therapy and reassessment — that comes first whatever the eventual plan, and I would not extract anything before it. Then the options are **retention with furcation surgery, tunnel preparation or root resection**; **extraction**; or **referral to a periodontist for an opinion before extracting**, which I would take if there were any real prospect of retention. **Root resection in maxillary molars is Complex, with modest survival, and I would not offer it as a general dentist with an interest in implants.** My likely conclusion, with **severe bone loss and confirmed furcation involvement in both molars, is a poor to hopeless prognosis** — but reached after cause-related therapy and a conversation, not from a radiograph."

Q: And the effect on replacement? "**This is the part that matters, and the furcation finding is the hinge. The bone that is being destroyed is exactly the bone an implant would need.** Furcation involvement means the **interradicular bone has gone**, so when these teeth come out I should expect a **deficient site, not a simple socket.**"

Q: Would you place immediately? "**No — immediate placement is contraindicated here.** There is **active infection**, and the **interradicular septum that would provide primary stability in a molar socket is precisely what the disease has destroyed.** Placing into that is how an implant ends up mobile at second stage."

Q: What about the sinus? "The **posterior maxilla pneumatises after tooth loss**, and periodontal bone loss has already reduced the residual height before I start. I would **measure the residual height beneath the sinus on the CBCT**. **More than 6 mm with a flat floor** — a **transcrestal elevation with simultaneous**

placement is feasible. **6 mm or less, or an oblique floor** — that indicates a **lateral window**, which is **outside my scope, and I would refer.**"

Q: So what is your plan? "**Ridge preservation at the time of extraction**, which is particularly valuable here — **sectioning the molars and removing the roots individually** to preserve the buccal and palatal plates. Then **at least four to six months** before I reassess for placement, because these are large sockets in infected sites. And **the periodontal condition governs everything**: a **history of periodontitis is the strongest patient-level risk factor for peri-implantitis**, so no implant is planned until the condition is **stable by the EFP criteria** and she has demonstrated compliance with maintenance.

And replacement may not be necessary at all. If the 15 and the opposing teeth are sound, a **shortened dental arch to the second premolar is a legitimate, well-evidenced outcome** that avoids sinus surgery altogether — and I would put that to her seriously, alongside a partial denture and implants, not as a consolation.

On the SAC classification, implant replacement here is at least **Advanced**, and **Complex** if a lateral window or vertical augmentation is needed — in which case I refer, having completed the periodontal treatment and the planning myself."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 7 **furcation destruction means a deficient future implant site** · 8 **immediate placement ruled out, with the septum and the infection as the reasons** · 9 **the 6 mm / flat floor threshold and the referral implication** · 3 **three maxillary furcation entrances, graded with a Nabers probe by Hamp** · 10 ridge preservation, timing, **periodontal stability precondition and the shortened dental arch** · 2 bone loss quantified as a proportion of root length.

Accept these late or on prompting: 1 whole image reported before the region of interest · 4 CBCT justification and whole-volume reporting · 5 the clinical assessment list, with staging and grading · 6 management options with a reasoned prognosis and referral.

The commonest way five minutes is wasted here: an exhaustive radiographic report. Four of the ten marks are in the first two parts and six are in the third, but the film is in the candidate's hand and the temptation is to describe every restoration and every millimetre for three minutes, arriving at "and I'd probably take them out" as the clock stops. **The instruction contains three verbs — report, discuss the management, and discuss the effect on replacement — and the third is the one the station was written for.** The second loss is treating the furcation as a periodontal finding only: the mark is for saying out loud that **the interradicular bone is the implant site**, which converts a periodontal diagnosis into a reconstructive problem. The third is never offering the **shortened dental arch** — candidates assume the examiner wants implants, when the mark is for putting a no-surgery option on the table and meaning it.

OSCE 58 — Buccal abscess on a lower implant-supported bridge, with no bone loss

Blueprint: suggest appropriate management of complications relating to the provision of implant restorations; recognise clinical situations which are unsafe or could lead to an adverse event; discuss appropriate management of common complications.

EQUIPMENT

a model or photograph of a **mandibular implant-supported bridge** with a **buccal swelling and a sinus/fistula**; a medical history card stating **no relevant medical history**; a periapical radiograph. **Additional written information given during the station:** ***there is no bone loss.***

CANDIDATE INSTRUCTIONS

This patient has a lower implant-supported bridge and presents with a buccal abscess. The medical history is clear. Discuss your management.

DIRECTION OF THOUGHT

Build a differential *before* the extra information arrives, then **say out loud how the "no bone loss" statement changes it**. That single sentence is the point of the station: it removes advanced peri-implantitis from the top

of your list and pushes you towards causes that do not destroy bone — a foreign body, residual cement, an adjacent tooth, a leaking screw joint, or a non-cleansable prosthesis.

MODEL ANSWER

"**First, the patient.** Is she systemically unwell — fever, malaise, lymphadenopathy, trismus, difficulty swallowing or breathing? A spreading infection in the floor of the mouth is an airway emergency and takes priority over everything else. Assuming she is systemically well, I proceed to assessment.

History. When did it start, is it getting worse, is there pain, discharge or a bad taste, has it happened before, and does it come and go? When was the bridge fitted, by whom, is it screw- or cement-retained, and have there been previous problems such as a loose screw? What maintenance has she had? Any food packing or difficulty cleaning? Medical history and smoking — the card says the history is clear.

Examination. Extra-oral swelling, lymph nodes, temperature. Intra-orally: the site, size and character of the swelling; whether it is fluctuant; whether there is a **sinus tract**, which I would **trace with a gutta-percha point and radiograph**; **peri-implant probing at six sites with a plastic probe at gentle force**, recording depth, bleeding and suppuration; **mobility of the prosthesis and, separately, of each implant** — I would remove the bridge if it is retrievable, which is the single most useful step in this station; the **cleansability of the bridge**, the emergence profile and the embrasures; any residual cement; and **sensibility testing and periodontal probing of the adjacent natural teeth**.

Radiographs. A standardised paralleling periapical of the implants **and the adjacent teeth**, compared with any baseline. A small field-of-view CBCT if the periapical is inconclusive.

Differential diagnosis before the extra information. Peri-implantitis with abscess formation; **residual submucosal cement**; a **foreign body** or chronic food impaction under a pontic; an **endodontic or periodontal lesion of an adjacent natural tooth** draining buccally; **retrograde peri-implantitis** from residual pathology at the site; a **retained root fragment** or residual cyst; a **fractured implant or abutment**, or a **loose or fractured screw** allowing bacterial leakage through the microgap; and a soft tissue lesion unrelated to the implants.

Now, given that there is no bone loss — that changes my differential substantially. Peri-implantitis by definition requires **bone loss beyond initial crestal remodelling**, so if the bone levels are stable against baseline, this is **not established peri-implantitis**. What remains, in order of likelihood:

1. **Residual submucosal cement or another foreign body**, producing a localised soft tissue abscess without yet having caused bone loss. Note that **a non-radiopaque cement will not appear on the radiograph**, so absence on the film does not exclude it.
2. **Chronic food impaction beneath a pontic or in an inaccessible embrasure**, particularly with an over-contoured or ridge-lap prosthesis.
3. **An odontogenic infection from an adjacent natural tooth** — a periapical abscess or a periodontal abscess — draining buccally and appearing to arise from the implant. This is why I test the adjacent teeth.
4. **A loose or fractured abutment screw**, allowing bacterial colonisation of the internal connection and leakage through the microgap, which can produce recurrent soft tissue infection with little or no bone change.
5. **Retrograde peri-implantitis** or residual pathology, though I would expect an apical radiolucency.
6. A **soft tissue lesion** — a mucocele, a fistula from another source, or rarely a neoplasm; anything that does not resolve requires biopsy or referral.

Immediate management. Establish **drainage** — through the sulcus with a probe, or by incision of a fluctuant swelling under local anaesthetic — and irrigate. **Remove the prosthesis** if it is screw-retained; this gives access, allows me to inspect for cement and to check the screws, and often resolves the problem in itself. **Antibiotics are not the primary treatment** and I would prescribe them only if there were systemic signs, spreading infection or immunocompromise; drainage and removal of the cause is the treatment. Analgesia. Review at 48 hours.

Definitive management, according to cause. - Cement or foreign body: meticulous removal under magnification with titanium or PEEK instruments, knotted floss drawn circumferentially, and glycine air-polishing; then **replace or modify the prosthesis to a cleansable, screw-retained design**

so it cannot recur. - **Food impaction / poor contour:** recontour or remake with open, cleansable embrasures and a non-ridge-lap pontic form; re-instruct with correctly sized interdental brushes and superfloss. - **Adjacent tooth:** treat the tooth — endodontic or periodontal — as indicated. - **Loose or fractured screw:** remove the prosthesis, replace the screw with a new one, inspect for component or implant fracture, **identify and correct the cause** (occlusion, cantilever, non-passive fit), torque to the manufacturer's value and re-torque at ten minutes. - **Persistent or unexplained:** consider microbiological sampling, further imaging, and referral; **any lesion that does not resolve requires biopsy or specialist opinion.**

Then prevention and governance. Establish a **new baseline radiograph** after treatment; place her on **3- to 5-monthly supportive therapy** with peri-implant probing at every visit; record the diagnosis, the cause, what was done and what was said; and, if this arose from the original prosthetic design, treat it as a significant event and review my own use of screw retention and cleansable contours."

EXAMINER CHECKLIST

1. Excludes systemic illness and airway compromise first
2. Structured history including retention type and previous problems
3. Examination including peri-implant probing with a plastic probe, sinus tracing, and **separate assessment of implant versus prosthesis mobility**
4. **Tests the adjacent natural teeth**
5. Radiographs of implants and adjacent teeth, compared with baseline
6. Comprehensive differential offered
7. **Explicitly revises the differential when told there is no bone loss, and states why peri-implantitis is excluded**
8. **Removes the prosthesis** and establishes drainage; antibiotics reserved for systemic/spreading infection
9. Cause-specific definitive management, including redesign for cleansability
10. New baseline radiograph, supportive therapy interval, records and governance

Extended by practice stations **OSCE 38** (removing a cemented crown and residual cement), **OSCE 37** (fractured screw retrieval) and SOE cases 4 and 40.

Running this station to time

Time check. The model answer is 1,136 words, 8 minutes 7 seconds at 140 words per minute — the longest overrun in the section, with more than three minutes to lose. Nothing is done with the hands, so all of it is speech, and the cut is made by **naming each differential in a single clause rather than a sentence each**, and by holding whole sentences for one thing only: **what changes when the examiner says there is no bone loss.** That revision is the station.

The five-minute run sheet

Time	The examiner does	What earns the mark
0:00–0:45	Presents a photograph or model of a mandibular implant-supported bridge with a buccal swelling and a sinus, a medical history card stating no relevant history, and a periapical: "This patient has a lower implant-supported bridge and presents with a buccal abscess. The medical history is clear. Discuss your management."	Checklist 1, and it must come first even though the card says the history is clear. **Is she systemically unwell — fever, malaise, lymphadenopathy, trismus, difficulty swallowing or breathing?** With the reason: **a spreading infection in the floor of the mouth is an airway emergency and takes priority over everything else.** Then move on: **assuming she is systemically well, I proceed to assessment.** A candidate who opens with the differential has lost mark 1 in the first sentence.

Time	The examiner does	What earns the mark
0:45–2:00	Silence.	Checklists 2, 3 and 4, as lists of clauses. **History: onset, progression, discharge, bad taste, whether it has recurred; when the bridge was fitted and by whom; screw- or cement-retained; previous loose screws; maintenance; food packing; smoking. Examination: swelling, nodes, temperature; whether it is fluctuant; a sinus tract traced with a gutta-percha point and radiographed; peri-implant probing at six sites with a plastic probe at gentle force; the cleansability, emergence profile and embrasures of the bridge; any residual cement.** And the two discriminating marks: **mobility of the prosthesis and, separately, of each implant — removing the bridge if it is retrievable, which is the single most useful step here** — and **sensibility testing and periodontal probing of the adjacent natural teeth.**
2:00–2:50	"What are you thinking?"	Checklists 5 and 6. **A standardised paralleling periapical of the implants and the adjacent teeth, compared with any baseline; a small field-of-view CBCT if it is inconclusive.** Then the differential, **six named causes in one breath**: **peri-implantitis with abscess formation; residual submucosal cement; a foreign body or chronic food impaction under a pontic; an endodontic or periodontal lesion of an adjacent tooth draining buccally; retrograde peri-implantitis or a retained root or cyst; a loose or fractured screw or component allowing leakage through the microgap; and a soft tissue lesion unrelated to the implants.**
2:50–3:50	Hands over a written card: **"There is no bone loss."**	Checklist 7 — the heaviest single mark in the station, scored for **saying out loud what the new information changes. Peri-implantitis by definition requires bone loss beyond initial crestal remodelling; if the bone levels are stable against baseline, this is not established peri-implantitis** — and it comes off the top of the list. Then the revised order: **residual cement or another foreign body first, with the trap named — a non-radiopaque cement will not appear on the film, so absence on the radiograph does not exclude it**; then **food impaction under an over-contoured or ridge-lap pontic**; then **an adjacent tooth draining buccally**; then **a loose or fractured screw, which can cause recurrent soft tissue infection with little or no bone change**; then retrograde pathology; then a soft tissue lesion.

Time	The examiner does	What earns the mark
3:50–5:00	"So what do you do?"	Checklists 8, 9 and 10. **Establish drainage — through the sulcus with a probe, or by incision of a fluctuant swelling under local anaesthetic — and irrigate. Remove the prosthesis if it is screw-retained**, which gives access, allows inspection for cement, lets the screws be checked, and often resolves it. **Antibiotics are not the primary treatment — only for systemic signs, spreading infection or immunocompromise.** Analgesia, review at 48 hours. Then **cause-specific definitive care, including redesign to a cleansable, screw-retained prosthesis.** Close with **a new baseline radiograph, 3- to 5-monthly supportive therapy with peri-implant probing at every visit, the diagnosis and cause recorded, and a significant event review if the prosthetic design caused it.**

The compressed answer — what a strong candidate says in five minutes

Q: How do you begin? "**With the patient, not the implant.** Is she **systemically unwell** — fever, malaise, lymphadenopathy, **trismus, difficulty swallowing or breathing**? A **spreading infection in the floor of the mouth is an airway emergency** and outranks everything else in this station. The card says her medical history is clear; assuming she is also systemically well, I proceed."

Q: What do you ask, and what do you examine? "**When it started, whether it is worsening, discharge, bad taste, and whether it has happened before.** Then the prosthesis: **when the bridge was fitted and by whom, whether it is screw- or cement-retained, and whether there have been previous problems such as a loose screw.** Then **maintenance, food packing, whether she can actually clean it**, and smoking.

Extra-orally, **swelling, nodes and temperature.** Intra-orally: the **character of the swelling and whether it is fluctuant**, and **if there is a sinus tract I would trace it with a gutta-percha point and radiograph it** — that alone often identifies the source. **Peri-implant probing at six sites with a plastic probe at gentle force**, recording depth, bleeding and suppuration. Then two things separately: **the mobility of the prosthesis, and the mobility of each implant** — and **if the bridge is retrievable I would remove it**, which is the single most useful step available to me. I would assess the **cleansability, emergence profile and embrasures** and look specifically for **residual cement**. And I would **sensibility test and periodontally probe the adjacent natural teeth** — because the source may not be an implant at all."

Q: Radiographs? "A **standardised paralleling periapical of the implants and the adjacent teeth, compared with any baseline**, and a **small field-of-view CBCT** if that is inconclusive."

Q: Your differential? "**Peri-implantitis with abscess formation; residual submucosal cement; a foreign body or chronic food impaction under a pontic; an endodontic or periodontal lesion of an adjacent natural tooth draining buccally; retrograde peri-implantitis, a retained root fragment or a residual cyst; a loose or fractured screw or component allowing bacterial leakage through the microgap; and a soft tissue lesion unrelated to the implants.**"

Q: [Handed the card] There is no bone loss. "**That changes my differential substantially, and I would say so. Peri-implantitis by definition requires bone loss beyond initial crestal remodelling** — so if the bone levels are stable against baseline, **this is not established peri-implantitis**, and it drops off the top of my list. What remains, in order:

First, residual submucosal cement or another foreign body, producing a localised soft tissue abscess before it has caused any bone change. And the trap: **a non-radiopaque cement will not show on the film, so its absence on the radiograph does not exclude it.**

Second, chronic food impaction beneath a pontic or in an inaccessible embrasure — particularly with an **over-contoured or ridge-lap** prosthesis.

Third, an odontogenic infection from an adjacent natural tooth — periapical or periodontal — **draining buccally and appearing to arise from the implant.** That is exactly why I tested those teeth.

Fourth, a loose or fractured abutment screw, allowing colonisation of the internal connection and **leakage through the microgap**, which can produce recurrent soft tissue infection **with little or no bone change at all**.

Fifth, retrograde peri-implantitis or residual pathology, though I would expect an apical radiolucency. **Sixth**, a soft tissue lesion — a mucocele, a fistula from elsewhere, rarely a neoplasm — and **anything that does not resolve requires biopsy or referral.**"

Q: Immediate management? "**Drainage** — through the sulcus with a probe, or **incision of a fluctuant swelling under local anaesthetic** — and irrigation. **Remove the prosthesis if it is screw-retained**: it gives me access, lets me inspect for cement and check the screws, and frequently resolves the problem in itself. **Antibiotics are not the primary treatment** — I would prescribe only for systemic signs, spreading infection or immunocompromise. **Drainage and removal of the cause is the treatment.** Analgesia, and **review at 48 hours.**"

Q And definitively? "By cause. Cement or foreign body: meticulous removal under magnification with titanium or PEEK instruments, knotted floss drawn circumferentially, glycine air-polishing — then modify or remake the prosthesis to a cleansable, screw-retained design so it cannot recur. Food impaction or poor contour: remake with open embrasures and a non-ridge-lap pontic, and re-instruct with sized interdental brushes. Adjacent tooth: treat the tooth. Loose or fractured screw: replace it, inspect for component fracture, correct the cause — occlusion, cantilever or non-passive fit — and re-torque at ten minutes. Persistent or unexplained: microbiology, further imaging, **biopsy or referral.**

Then a **new baseline radiograph after treatment**, **3- to 5-monthly supportive therapy with peri-implant probing at every visit**, the diagnosis, cause and advice recorded — and **if this arose from the original prosthetic design, a significant event review** of my own use of screw retention and cleansable contours."

Marking priority when the clock beats the candidate

Ask for these first (the six that carry the station): 7 **the differential explicitly revised on "no bone loss", with peri-implantitis excluded by definition** · 8 **prosthesis removed and drainage established; antibiotics reserved** · 4 **the adjacent natural teeth tested** · 3 plastic-probe probing, sinus traced, **implant mobility assessed separately from prosthesis mobility** · 6 a comprehensive differential offered before the extra information · 9 cause-specific management, **including redesign for cleansability**.

Accept these late or on prompting: 1 systemic illness and airway excluded first · 2 structured history including retention type and previous problems · 5 radiographs of implants and adjacent teeth against baseline · 10 new baseline, supportive therapy interval, records and governance.

The commonest way five minutes is wasted here: answering the word "abscess" with a peri-implantitis protocol — surgical debridement, decontamination, implantoplasty, regeneration — before the extra information arrives, and then failing to unwind it when it does. **Checklist 7 is not scored for silently changing course; it is scored for saying "that changes my differential, and here is why."** The second loss is forgetting that the source may be a tooth: **checklist 4 is a discrete mark**, and the adjacent teeth are the commonest thing a candidate never mentions. The third is reaching for antibiotics — amoxicillin as the opening move is a marked-down answer, because **drainage and removal of the cause is the treatment**. The fourth is stopping at drainage: **checklists 9 and 10 are the redesign, the new baseline, the recall interval and the significant event**, and a bridge that cannot be cleaned will simply present again.

PART 7

Structured oral: 51 clinical scenarios

Three set-pieces to prepare first, then 51 cases worked across the four marked domains with a full examiner question flow.

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Where the marks are in the SOE 421

How the fifteen minutes is spent, and what the examiner is holding 421